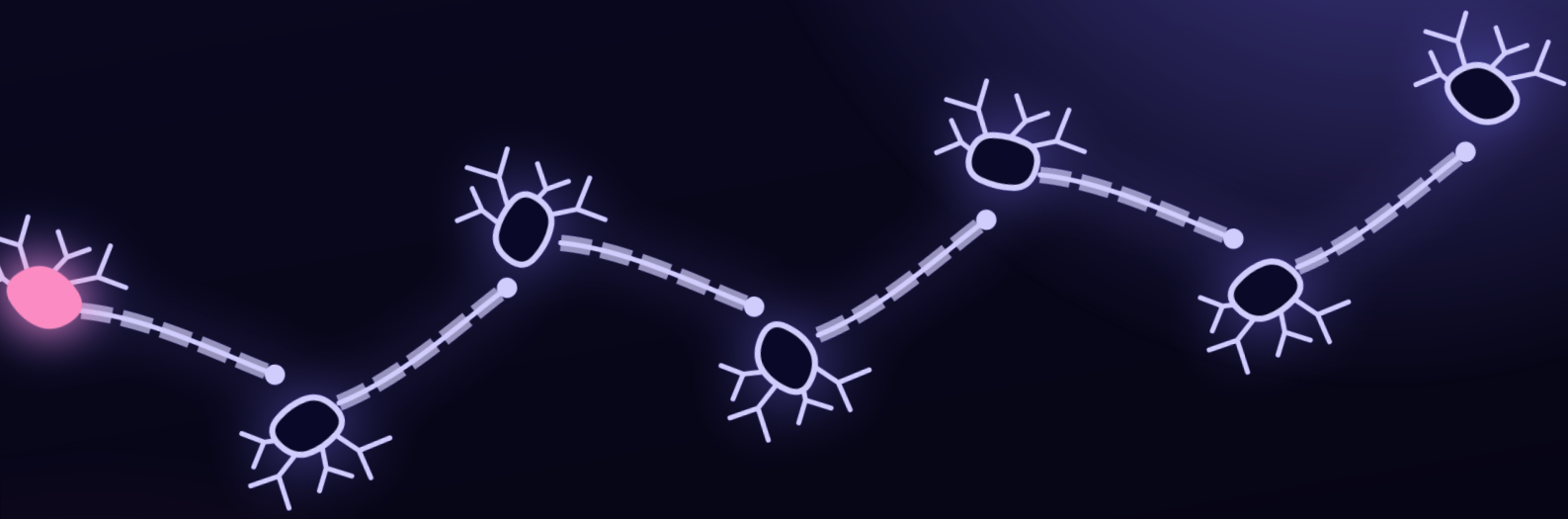
Clinical *Conditions*

REVIEW ARTICLES
LAST UPDATED 13 NOVEMBER 2026

INSIDE

Physiology
Strokes and lesions
Epilepsy
Demyelinating
Eye
Ear
Dementia
Movement disorders
Special topics
Pharmacology



FROM STROKE TO MOVEMENT DISORDERS

Neurology

Trace a chain of neurons through stroke, seizures, multiple sclerosis, the eye and ear, dementia, and movement disorders.

For educational use only. This is a study aid, not a clinical reference.

Contents

PHYSIOLOGY

1 Neurons and action potentials: structure, channels, and conduction

2 Synapses and neurotransmitters: release, receptors, and clearance

3 Ascending tracts: dorsal columns, spinothalamic tracts, and spinocerebellar tracts

4 Descending tracts: corticospinal, corticobulbar, and brainstem motor tracts

5 Basal ganglia pathways: direct, indirect, and the role of dopamine

6 Pain pathways and descending modulation: from nociceptor to cortex and back

7 Olfactory, optic, and vestibulocochlear nerves: the special senses

8 Oculomotor, trochlear, and abducens nerves: moving the eyes

9 Trigeminal and facial nerves: face sensation, mastication, and expression

10 Glossopharyngeal, vagus, accessory, and hypoglossal nerves: the lower cranial nerves

11 The respiratory center: rhythm, chemical control, and failures of drive

12 The cardiovascular center: the baroreflex and its failures

STROKES AND LESIONS

13 Stroke syndromes: localizing the lesion from the cortex to the spinal cord

14 Acute ischemic stroke: embolic, thrombotic, and low-flow injury and the race to reperfuse

15 Intracerebral hemorrhage: causes by location, and limiting hematoma growth

16 Subarachnoid hemorrhage: aneurysmal rupture and the threats that follow

EPILEPSY

17 Seizures: lowered thresholds, the patterns they produce, and stepwise control

DEMYELINATING

18 Multiple sclerosis: immune demyelination and lesions scattered in space and time

19 Neuromyelitis optica spectrum disorder: an antibody-mediated astrocytopathy of the optic nerves, spinal cord, and area postrema

20 MOG antibody-associated disease: an antibody-mediated demyelinating disease distinct from MS and NMOSD

EYE

21 Glaucoma: aqueous outflow, intraocular pressure, and optic nerve damage

22 Cataract: opacification of the crystalline lens

23 Vision loss and red eye: telling eye conditions apart by presentation

EAR

24 Hearing loss: conductive, sensorineural, and mixed

25 Vertigo: timing, nystagmus, and hearing as guides to the cause

DEMENTIA

26 Alzheimer disease: amyloid plaques, tau tangles, and progressive neuronal loss

27 Frontotemporal dementia: frontal and temporal degeneration of behavior and language

28 Dementia with Lewy bodies: α-synuclein, fluctuating cognition, and antipsychotic sensitivity

MOVEMENT DISORDERS

29 Motor signs and syndromes: localizing weakness, abnormal movements, and nerve lesions

30 Parkinson disease: α-synuclein, dopamine neuron loss, and dopaminergic treatment

31 Atypical parkinsonism: multiple system atrophy, progressive supranuclear palsy, and corticobasal degeneration

32 Huntington disease: CAG repeat expansion, striatal degeneration, and chorea

SPECIAL TOPICS

33 Localizing a neurologic lesion: from signs to level and side

34 Ruling out mimics in stroke, seizure, multiple sclerosis, Alzheimer disease, and Parkinson disease

35 Choosing imaging and EEG in neurology, and the signs to look for

PHARMACOLOGY

36 Antithrombotic drugs and their reversal in cerebrovascular disease

37 Lowering blood pressure and intracranial pressure after brain hemorrhage

38 Antiseizure drugs: mechanisms, choice by seizure type, and hazards

39 Immunotherapy for multiple sclerosis, NMOSD, and MOGAD

40 Drugs for spasticity, bladder dysfunction, orthostatic hypotension, and sleep behavior

41 Drugs for dementia, myasthenia, and mood in neurological disease

42 Drugs for Parkinson disease, chorea, and tremor

43 Analgesics and anesthetics acting on the pain pathway

44 Drugs for glaucoma, macular degeneration, and conjunctivitis, and those that cause angle closure

45 Drugs for vertigo and Ménière disease

46 Drugs as causes of neurological disease

Neurons and action potentials: structure, channels, and conduction

Neurons receive input on dendrites, integrate it at the axon hillock, and conduct all-or-none action potentials to their terminals. The resting potential of about −70 mV reflects potassium leak and the Na/K-ATPase, and voltage-gated sodium and potassium channels drive depolarization and repolarization. Refractory periods limit firing rate and keep conduction one-way, and myelin allows fast saltatory conduction. Demyelination, sodium channel mutations, and sodium channel toxins disrupt this process, and sodium channel blockers exploit it. Here, we review neuron structure, membrane channels, the action potential, and their failures.

Key points

- The resting potential is about −70 mV, close to the potassium equilibrium potential, because potassium leak channels dominate at rest.
- At threshold, about −55 mV, voltage-gated sodium channels open, producing an all-or-none spike.
- Sodium channel inactivation and potassium efflux repolarize the membrane, and inactivation causes the absolute refractory period.
- Myelin from oligodendrocytes and Schwann cells allows saltatory conduction between nodes of Ranvier.
- Lidocaine and phenytoin block voltage-gated sodium channels.

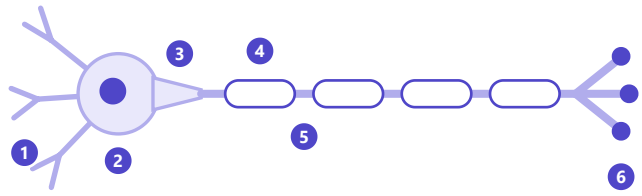


Fig. 1 | A neuron: dendrites, 1, receive input; the soma, 2, holds the nucleus; the axon hillock, 3, starts the action potential; myelin, 4, insulates the axon; nodes of Ranvier, 5, regenerate the signal; and axon terminals, 6, release neurotransmitter. 1, Dendrites; 2, Soma; 3, Axon hillock; 4, Myelin; 5, Nodes of Ranvier; 6, Axon terminals.

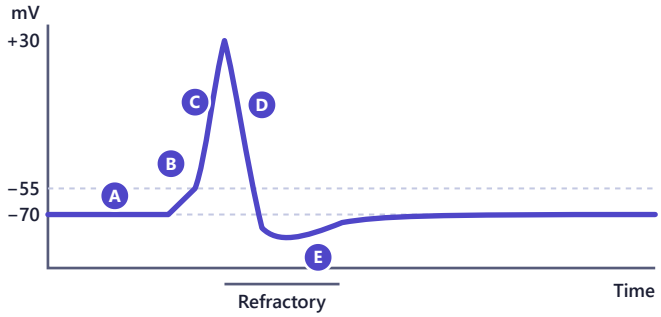


Fig. 2 | A neuronal action potential plotted as voltage over time: A, resting at minus 70 millivolts; B, a stimulus reaches threshold at minus 55; C, depolarization as sodium enters; D, repolarization as potassium leaves; E, hyperpolarization before the return to rest. The refractory period spans the spike and hyperpolarization. A, Resting potential; B, Threshold; C, Depolarization; D, Repolarization; E, Hyperpolarization.

Structure

Dendrites receive afferent signals, the soma holds the nucleus, and the axon hillock initiates the action potential, which the axon carries to terminals that release neurotransmitter. Myelin, from oligodendrocytes in the CNS and Schwann cells in the periphery, insulates the axon, leaving nodes of Ranvier packed with voltage-gated sodium channels. Most neurons, including motor neurons, are multipolar; some are bipolar; and sensory neurons of the dorsal root ganglia are pseudounipolar.

Glial cells support neurons. Astrocytes buffer potassium, take up neurotransmitter, and with their foot processes help form the BBB; microglia, of monocyte origin, are the immune cells of the CNS; oligodendrocytes myelinate many CNS axons each, whereas each Schwann cell myelinates one PNS axon; and ependymal cells line the ventricles, with choroid epithelium regulating CSF. The BBB consists of capillary endothelial cells joined by tight junctions, supported by pericytes, astrocyte foot processes, and the basal lamina.

Channels and the action potential

Potassium leak channels make the resting membrane most permeable to potassium, so the resting potential of about −70 mV lies near the potassium equilibrium potential, and the Na/K-ATPase pumps 3 sodium out and 2 potassium in to maintain the gradients. When summed input reaches threshold, about −55 mV, voltage-gated sodium channels open and sodium rushes in, an all-or-none depolarization. Sodium channels then inactivate and voltage-gated potassium channels open, repolarizing the membrane, and slow potassium channel closure causes hyperpolarization. At the terminal, voltage-gated calcium channels trigger transmitter release, and ligand-gated channels generate postsynaptic potentials.

During the absolute refractory period, inactivated sodium channels prevent another spike; during the relative refractory period, a stronger stimulus is needed. These periods set the maximum firing rate and keep conduction one-way. In unmyelinated axons, depolarization spreads to adjacent membrane slowly, whereas in myelinated axons current jumps from node to node, saltatory conduction, more than 10 times faster.

When it fails

Multiple sclerosis destroys oligodendrocytes, MOGAD targets myelin, and NMOSD destroys astrocytes through antibodies against AQP4, with secondary demyelination, slowing or blocking conduction. Tetrodotoxin blocks sodium channels, causing numbness and weakness, and sodium channel mutations such as SCN1A raise excitability and cause epilepsy. After an axon is cut, Wallerian degeneration breaks down the axon and myelin distal to the cut, starting within hours, and regrowth, at about 1 mm/day, begins only after it is complete; within hours, chromatolysis swells the cell body and nucleus and breaks down Nissl bodies.

Pharmacology

Local anesthetics such as lidocaine block voltage-gated sodium channels in nerves, so pain signals fail to conduct. Antiseizure drugs such as phenytoin, carbamazepine, and lamotrigine reduce sodium influx in neurons, lowering excitability.

Relationships to other topics

Synapses and neurotransmitters. The action potential ends at the terminal, where calcium entry converts the electrical signal into chemical transmission.

Synapses and neurotransmitters: release, receptors, and clearance

At chemical synapses, an action potential opens voltage-gated calcium channels, and calcium-activated SNARE proteins fuse vesicles to release neurotransmitter. Transmitters act on fast ionotropic or slower metabotropic receptors, producing excitatory or inhibitory postsynaptic potentials that sum at the axon hillock, and are cleared by reuptake or enzymes. Glutamate is the main excitatory and GABA the main inhibitory transmitter, with acetylcholine, dopamine, norepinephrine, and serotonin modulating specific systems. Here, we review synaptic transmission, the major transmitters, their failures, and drugs that target them.

Key points

- Calcium entry activates SNARE proteins, which fuse vesicles and release transmitter.
- Ionotropic receptors are fast ion channels; metabotropic receptors act through G proteins.
- Glutamate is the main excitatory transmitter, GABA the main inhibitory transmitter in the brain, and glycine inhibitory in the cord.
- Acetylcholine is cleared by AChE; small-molecule transmitters are cleared by reuptake.
- Benzodiazepines increase GABA-A channel opening, and cholinesterase inhibitors raise acetylcholine.

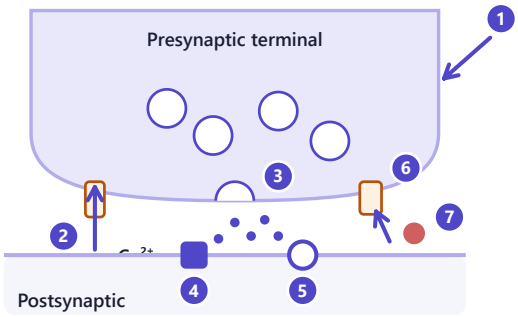


Fig. 1 | A chemical synapse: 1, an action potential reaches the terminal; 2, voltage-gated calcium channels open; 3, vesicles fuse and release neurotransmitter; 4, it binds ionotropic receptors and, 5, metabotropic receptors; it is cleared by, 6, reuptake or, 7, enzymes. 1, Action potential; 2, Calcium channels; 3, Vesicle fusion; 4, Ionotropic receptors; 5, Metabotropic receptors; 6, Reuptake; 7, Enzymes.

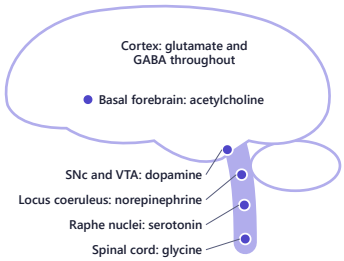


Fig. 2 | Where the main transmitters come from, on a side view of the brain: glutamate and GABA throughout the cortex; acetylcholine from the basal forebrain; dopamine from the SNc and ventral tegmental area; norepinephrine from the locus coeruleus; serotonin from the raphe nuclei; and glycine in the spinal cord.

Synaptic transmission

An action potential depolarizes the terminal, opening voltage-gated calcium channels, and calcium activates SNARE proteins that fuse vesicles with the membrane. Transmitter binds ionotropic receptors, ion channels that act quickly, or metabotropic receptors, G protein-coupled receptors with slower, longer effects. Excitatory postsynaptic potentials depolarize the next neuron and inhibitory potentials hyperpolarize it, as chloride enters through GABA-A receptors, and inputs sum at the axon hillock. Transmitter is cleared by reuptake into the terminal or glia, for small molecules only, or by enzymes such as AChE. Electrical synapses pass current directly through gap junctions with almost no delay.

Neurotransmitters

Glutamate, acting on AMPA and NMDA receptors, is the main excitatory transmitter and underlies plasticity and memory. GABA, acting on GABA-A chloride channels and GABA-B G protein-coupled receptors, is the main inhibitory transmitter in the brain, and glycine is inhibitory mainly in the cord. Acetylcholine acts on

nicotinic ion channels and muscarinic M1–M5 receptors at muscle, autonomic synapses, and the basal forebrain. Dopamine from the SNc and ventral tegmental area acts on D1 and D2 families, norepinephrine from the locus coeruleus drives arousal, and serotonin from the raphe nuclei modulates mood, sleep, and gut motility.

When it fails

Loss of cholinergic neurons contributes to memory decline in Alzheimer disease, and death of SNc dopamine neurons leaves too little dopamine in Parkinson disease. Low serotonin, with low plasma levels of its precursor tryptophan, is linked to depression. Glutamate excitotoxicity contributes to both, seizures arise when excitation outweighs GABA inhibition, and in hepatic encephalopathy, ammonia enhances GABA-A chloride flow. Increased presynaptic striatal dopamine function is the final common pathway to psychosis in schizophrenia.

At the neuromuscular junction, myasthenia gravis targets postsynaptic nicotinic AChR or MuSK, causing fatigable weakness that worsens through the day, with ptosis and diplopia in 85% and thymoma in 10%; pyridostigmine, a cholinesterase inhibitor, is followed by immunosuppression. Lambert-Eaton syndrome targets presynaptic voltage-gated calcium channels, reducing ACh release, and causes proximal weakness, autonomic dysfunction, and reduced reflexes that improve with repeated contraction, often with small cell lung cancer. In Huntington disease, striatal GABA neurons die.

Pharmacology

Benzodiazepines increase how often GABA-A chloride channels open, causing sedation and seizure control. Cholinesterase inhibitors such as donepezil block acetylcholine breakdown, botulinum toxin blocks acetylcholine release at the neuromuscular junction, and SSRIs block serotonin reuptake.

Relationships to other topics

Neurons and action potentials. The action potential arriving at the terminal opens the calcium channels that start release.

Basal ganglia pathways. Dopamine acts on D1 and D2 striatal neurons to balance the direct and indirect pathways.

Ascending tracts: dorsal columns, spinothalamic tracts, and spinocerebellar tracts

Ascending tracts carry sensation from the body to the thalamus and cortex or to the cerebellum. The dorsal column-medial lemniscus carries fine touch, vibration, and conscious proprioception, ascends on the same side, and crosses in the medulla. The spinothalamic tract carries pain, temperature, and crude touch and crosses in the cord within 1–2 segments of entry. The spinocerebellar tracts carry unconscious proprioception to the cerebellum on the same side. Where each crosses determines the side of deficits after a lesion. Here, we review their routes and failures.

Key points

- The dorsal columns ascend on the same side and cross in the medulla as the medial lemniscus.
- Spinothalamic fibers cross in the anterior white commissure 1–2 segments above entry.
- A cord hemisection causes same-side dorsal column loss and opposite-side pain and temperature loss.
- Spinocerebellar tracts end in the cerebellum on the same side, so their lesions cause same-side ataxia.

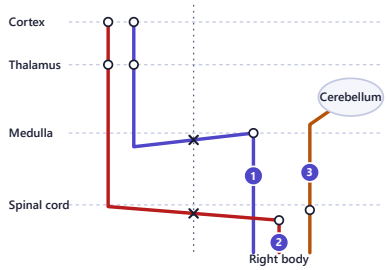


Fig. 1 | Ascending tracts from the right side of the body. 1, the dorsal column-medial lemniscus ascends on the same side and crosses in the medulla. 2, the spinothalamic tract crosses in the cord, one to two segments above entry. Both reach the left thalamus and cortex. 3, the dorsal spinocerebellar tract stays on the same side and enters the cerebellum. 1, Dorsal column-medial lemniscus; 2, Spinothalamic tract; 3, Spinocerebellar tracts.

Routes

In the dorsal column-medial lemniscus, large DRG fibers ascend on the same side, gracilis from below T6 and cuneatus from T6 up, to the nucleus gracilis and cuneatus in the medulla. Second-order internal arcuate fibers cross and ascend as the medial lemniscus to the VPL thalamus, which projects to S1. The pathway carries fine touch, vibration, conscious proprioception, and two-point discrimination.

In the spinothalamic tract, thin Aδ and C fibers synapse in the dorsal horn, and second-order fibers cross in the anterior white commissure 1–2 segments above entry. The lateral tract carries sharp pain and temperature, with sacral fibers most lateral, and the anterior tract carries crude touch; both reach the VPL thalamus and S1, insula, and cingulate cortex.

The dorsal spinocerebellar tract arises in the Clarke nucleus (C8–L3) and ascends uncrossed through the inferior cerebellar peduncle, carrying leg and trunk information. The ventral tract arises from spinal border cells, crosses twice, and enters through the superior cerebellar peduncle, so both end on the same side, carrying unconscious proprioception from muscle spindles and tendon organs.

Dermatome landmarks localize sensory levels: C6 supplies the thumb, C7 the middle finger, T4 the nipple, T10 the umbilicus, L4 the anterior knee and medial malleolus, and S1 the lateral malleolus.

When it fails

A cord hemisection, Brown-Séquard syndrome, causes same-side dorsal column loss and weakness with opposite-side pain and temperature loss. Posterior cord lesions and dorsal column loss from B12 deficiency or tabes dorsalis cause vibration and position loss with a positive Romberg test, and syringomyelia interrupts crossing spinothalamic fibers, causing cape-like pain and temperature loss. Medial medullary syndrome damages the medial lemniscus, causing opposite-side vibration and position loss, and spinocerebellar loss, as in Friedreich ataxia, causes same-side ataxia.

Relationships to other topics

Descending tracts. Lesions often hit both: in a hemisected cord, the corticospinal tract adds same-side weakness to the sensory pattern.

Descending tracts: corticospinal, corticobulbar, and brainstem motor tracts

Descending tracts carry motor commands to cranial nerve nuclei and the spinal cord. The lateral corticospinal tract crosses at the pyramids and drives skilled distal movement, and the corticobulbar tract reaches cranial nerve nuclei mostly bilaterally. Brainstem tracts set posture and tone: the rubrospinal tract favors upper limb flexion, and the vestibulospinal and pontine reticulospinal tracts favor extension, whereas the medullary reticulospinal tract inhibits extensors. Lesions above or below the pyramidal decussation and the red nucleus produce predictable weakness and posturing. Here, we review these tracts and their failures.

Key points

- About 90% of corticospinal fibers cross at the pyramidal decussation, so lesions above it cause opposite-side weakness.
- Corticobulbar input is mostly bilateral, and the upper face receives input from both hemispheres.
- The rubrospinal tract crosses in the midbrain and favors upper limb flexion.
- Lesions above the red nucleus cause decorticate posturing; lesions below it cause decerebrate posturing.

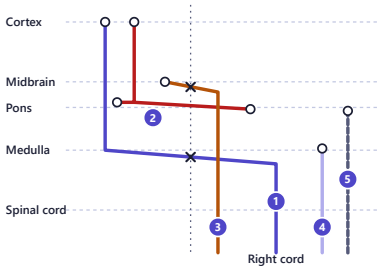


Fig. 1 | Descending tracts from the left cortex and brainstem. 1, the lateral corticospinal tract crosses at the pyramids to the right cord. 2, the corticobulbar tract reaches cranial nerve nuclei on both sides. 3, the rubrospinal tract crosses in the midbrain. 4, the vestibulospinal and, 5, reticulospinal tracts descend on their own side. 1, Lateral corticospinal; 2, Corticobulbar; 3, Rubrospinal; 4, Lateral vestibulospinal; 5, Reticulospinal.

Corticospinal and corticobulbar tracts

The lateral corticospinal tract arises from M1, the premotor cortex and SMA, and S1, and descends through the posterior limb of the internal capsule, cerebral peduncle, basis pontis, and pyramid; 90% of fibers cross at the pyramidal decussation to reach ventral horn neurons for distal limbs, driving skilled movement. The anterior corticospinal tract, about 10%, stays uncrossed until its segment and serves axial and proximal muscles on both sides. The corticobulbar tract arises from the face area of the motor cortex, passes through the genu of the internal capsule, and reaches the motor nuclei of CN V, VII, and IX–XII, mostly bilaterally; because the upper face receives input from both hemispheres, a cortical lesion spares the forehead.

Brainstem tracts

The rubrospinal tract arises from the red nucleus, which receives cortical and cerebellar input, crosses in the midbrain, and favors upper limb flexion. The lateral vestibulospinal tract, from the lateral vestibular nucleus, and the pontine reticulospinal tract descend uncrossed and excite antigravity extensors, whereas the medullary reticulospinal tract, mainly uncrossed, inhibits extensors, facilitates flexors, and contributes to autonomic and breathing control. A lesion above the red nucleus causes decorticate posturing, with flexed arms and extended legs, and a lesion below it causes decerebrate posturing, with extension of all limbs.

When it fails

Capsular or brainstem lesions above the decussation cause opposite-side spastic hemiparesis, and cord lesions below it cause same-side spasticity, hyperreflexia, and a Babinski sign. Corticobulbar lesions cause supranuclear facial palsy, sparing the forehead, and bilateral lesions cause pseudobulbar palsy, with dysarthria, dysphagia, and a brisk jaw jerk.

Relationships to other topics

Ascending tracts. Combined lesions localize precisely: a cord hemisection adds same-side corticospinal weakness to the sensory pattern.

Basal ganglia pathways. The basal ganglia shape movement by acting on the thalamus and cortex, which then act through the corticospinal tract.

Basal ganglia pathways: direct, indirect, and the role of dopamine

The basal ganglia select and scale movement through two opposing striatal output pathways. In the direct pathway, D1 medium spiny neurons inhibit the GPi and SNr, releasing the thalamus to drive movement; in the indirect pathway, D2 neurons inhibit the GPe, freeing the STN to excite the GPi and restrain movement. Dopamine from the SNc excites D1 and inhibits D2 neurons, so both actions favor movement. Dopamine loss causes parkinsonism, whereas indirect-pathway loss and STN lesions cause chorea and ballismus. Here, we review the circuit, its failures, and drugs that act on it.

Key points

- The GPi and SNr tonically inhibit the VA and VL thalamus.
- The direct pathway (D1) reduces that inhibition; the indirect pathway (D2, GPe, STN) increases it.
- Dopamine excites D1 and inhibits D2 neurons, favoring movement through both pathways.
- Dopamine loss in Parkinson disease causes bradykinesia; indirect-pathway loss in Huntington disease and STN lesions cause hyperkinesia.

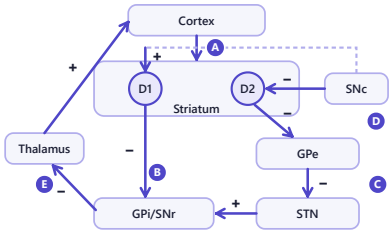


Fig. 1 | The basal ganglia loop. A, the cortex excites the striatum. B, in the direct pathway, D1 neurons inhibit the GPi and SNr, releasing the thalamus. C, in the indirect pathway, D2 neurons inhibit the GPe, freeing the STN to excite the GPi, which inhibits the thalamus. D, dopamine from the SNc excites D1 and inhibits D2 neurons, so both changes favor movement. E, the thalamus excites the cortex. A, Cortex; B, Direct pathway; C, Indirect pathway; D, Dopamine; E, Thalamus.

Circuit

The striatum, made of the caudate and putamen, contains GABA medium spiny neurons that receive cortical and dopamine input. The GPi and SNr form the output and tonically inhibit the VA and VL thalamus, which relays to the motor cortex. In the direct pathway, D1 neurons inhibit the GPi and SNr, releasing the thalamus and allowing movement. In the indirect pathway, D2 neurons inhibit the GPe, which then inhibits the STN less, and the STN, the loop's only glutamate nucleus, excites the GPi, restraining movement. Dopamine from the SNc excites D1 and inhibits D2 neurons, so both actions favor movement.

When it fails

In Parkinson disease, loss of SNc dopamine weakens the direct pathway and strengthens the indirect pathway, increasing GPi inhibition and causing bradykinesia, and dopamine-blocking drugs cause parkinsonism by blocking D2 receptors. In Huntington disease, indirect-pathway neurons die first, reducing GPi inhibition and causing chorea, and an STN lesion removes the GPi's excitation, releasing the thalamus and causing hemiballismus.

Pharmacology

Levodopa crosses the BBB and becomes dopamine, with carbidopa blocking its peripheral conversion, and dopamine agonists such as pramipexole stimulate receptors directly. MAO-B and COMT inhibitors slow dopamine and levodopa breakdown. VMAT2 inhibitors such as tetrabenazine block vesicular loading of dopamine, reducing chorea, and D2 antagonists such as haloperidol reduce hyperkinesia but can cause parkinsonism.

Relationships to other topics

Synapses and neurotransmitters. Dopamine acts through D1 and D2 receptors, members of the metabotropic receptor family.

Descending tracts. The basal ganglia act through the thalamus and cortex, which drive movement through the corticospinal tract.

Pain pathways and descending modulation: from nociceptor to cortex and back

Pain begins when noxious stimuli open TRP channels on the free endings of Aδ and C fibers, which synapse in laminae I and II of the dorsal horn. The lateral spinothalamic tract crosses in the cord and relays through the VPL to S1, conveying where and how intense a stimulus is, while medial pathways reach the reticular formation, PAG, intralaminar thalamus, cingulate, and insula to carry its emotional and visceral weight. The dorsal horn is a gate: touch input and descending signals from the PAG, RVM, and locus coeruleus inhibit transmission through enkephalin interneurons and presynaptic opioid receptors. Here, we review nociceptors, the ascending and descending pathways, their failures, and the analgesics that act on them.

Key points

- Aδ fibers carry fast, well-localized first pain; C fibers carry slow, poorly localized pain.
- The lateral spinothalamic tract crosses in the anterior white commissure and relays in the VPL to S1.
- The PAG drives serotonergic raphe neurons in the RVM and noradrenergic locus coeruleus neurons, which inhibit the dorsal horn.
- Opioid receptors are Gi-coupled: they open potassium channels and close N-type calcium channels, cutting substance P release.
- Opioids, paracetamol, and SNRIs strengthen descending inhibition; NSAIDs, gabapentinoids, and ketamine act at the periphery and dorsal horn.

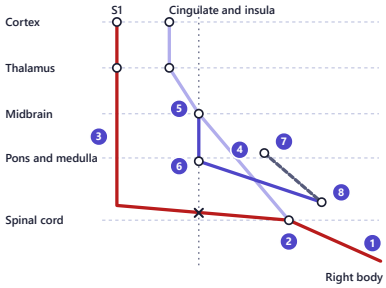


Fig. 1 | Pain pathways from the right side of the body. 1, a nociceptor enters 2, the dorsal horn. 3, the lateral spinothalamic tract crosses in the cord and ascends to the thalamus and primary somatosensory cortex. 4, medial pathways ascend to the periaqueductal gray, medial thalamus, cingulate, and insula. Descending modulation runs from 5, the periaqueductal gray, through 6, the rostral ventromedial medulla, and from 7, the locus coeruleus, to 8, inhibitory interneurons in the dorsal horn. 1, Nociceptor; 2, Dorsal horn; 3, Lateral spinothalamic tract; 4, Medial pathways; 5, PAG; 6, RVM; 7, Locus coeruleus; 8, Inhibitory interneurons.

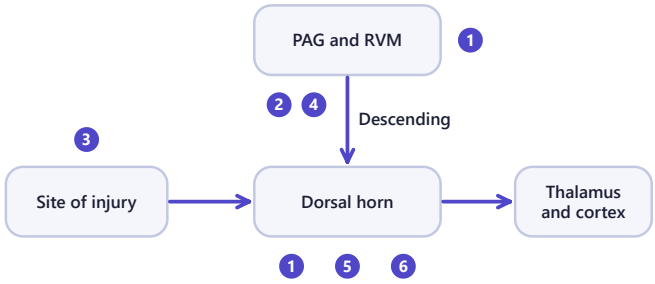


Fig. 2 | Six numbered drug sites. 3, NSAIDs act at the site of injury. 1, opioids, 5, gabapentinoids, and 6, ketamine act in the dorsal horn, and opioids also act in the periaqueductal gray and rostral ventromedial medulla. 2, paracetamol, and 4, SNRIs and TCAs, strengthen descending inhibition. 1, Opioids; 2, Paracetamol; 3, NSAIDs; 4, SNRIs and TCAs; 5, Gabapentinoids; 6, Ketamine.

Nociceptors

Nociceptors are free nerve endings whose cell bodies sit in the DRG, or the trigeminal ganglion for the face. Lightly myelinated Aδ fibers, with small receptive fields, carry the fast, sharp, well-localized first pain and release glutamate, while unmyelinated C fibers, with large receptive fields, carry slow, poorly localized pain that conveys intensity and release neuropeptides such as substance P. They are activated by histamine, NGF, substance P, CGRP, potassium, serotonin, acetylcholine, ATP, lactic acid, low pH, and arachidonic acid, and by temperature extremes, high pressure, and tissue damage. Silent nociceptors respond only after inflammation sensitizes them and cause rest pain in arthritis. The brain has no nociceptors, so awake craniotomy is painless, and because nociception is detection and pain is experience, either can occur without the other.

Somatic pain from skin is sharp or burning and well localized, while pain from joints, tendons, and bone is aching and less localized. Visceral pain is carried by C fibers and poorly localized. Referred pain arises when visceral and somatic afferents converge on the same spinal segments, as when MI pain spreads to the neck, shoulder, and back. Neuropathic pain follows nerve damage, with spontaneous firing or altered conduction; allodynia is pain from a normally harmless stimulus, and hyperalgesia is an exaggerated response to a painful one.

Ascending pathways

Aδ fibers end in lamina I and C fibers in lamina II, the substantia gelatinosa. The lateral, neospinothalamic tract crosses in the anterior white commissure and relays in the VPL to S1, conveying location and intensity. The medial paleospinothalamic and archispinothalamic pathways project bilaterally to the reticular formation, PAG, and intralaminar thalamus, then to the cingulate, insula, and hypothalamus, carrying emotional and visceral responses. From the face, the trigeminal ganglion projects to the spinal trigeminal nucleus in the medulla, which relays through the VPM to S1.

Descending modulation

In the gate control theory of Melzack and Wall (1965), the substantia gelatinosa acts as a gate. Aβ and Aδ input excites inhibitory interneurons that damp dorsal horn projection neurons, which is why rubbing a wound dulls sharp pain, and cortical centers also control the gate, which depression holds open more often.

Ascending signals and opioids activate the PAG, which drives raphe neurons of the RVM with glutamate. These serotonergic neurons, the main endogenous modulator and the main target of supraspinal opioids, activate enkephalin interneurons in the dorsal horn, whose opioid receptors on nociceptive terminals open potassium channels and block calcium entry, reducing substance P release and suppressing C fiber input. A parallel route from the PAG to the locus coeruleus releases norepinephrine onto presynaptic α2 receptors. Opioids in the PAG inhibit local GABA interneurons, releasing the descending pathway, and the PAG can also facilitate pain, contributing to chronic pain.

μ receptors bind β-endorphin and endomorphins and mediate analgesia, euphoria, respiratory depression, constipation, and dependence; κ receptors bind dynorphins and mediate spinal analgesia, diuresis, and dysphoria; and δ receptors bind enkephalins and mediate analgesia and decreased gastric motility. All three are Gi-coupled, opening potassium channels and closing N-type calcium channels.

When it fails

Lateral medullary syndrome damages the spinal trigeminal nucleus and spinothalamic tract, causing crossed pain and temperature loss on the same side of the face and the opposite side of the body. Because spinothalamic fibers have already crossed, cord hemisection causes pain and temperature loss on the opposite side below the lesion, and spinothalamic damage in multiple sclerosis also causes this loss.

Chronic pain, including neuropathic pain and painful attacks, affects 80–100% of patients with NMOSD. CRPS follows minor injury or surgery with severe limb pain, allodynia, and changes in sweating and blood flow, as inflammation sensitizes nociceptive neurons and sympathetic drive sustains the pain. In central sensitization, peripheral and dorsal horn changes lower the threshold and amplify pain, so chronic pain becomes a disease in itself.

Pharmacology

Opioids such as morphine and fentanyl act on μ receptors in the PAG, RVM, dorsal horn, and peripheral afferents, opening potassium channels and closing N-type calcium channels, and in the PAG they inhibit GABA interneurons to release descending inhibition. Paracetamol inhibits central COX and strengthens descending serotonergic inhibition, giving analgesia and antipyresis with little anti-inflammatory effect. SNRIs and TCAs, such as duloxetine and amitriptyline, block serotonin and norepinephrine reuptake in the spinal cord, strengthening descending inhibition and relieving neuropathic pain, and duloxetine also treats fibromyalgia.

NSAIDs such as ibuprofen, aspirin, and celecoxib decrease prostaglandin synthesis by inhibiting COX-1 and COX-2, with aspirin binding irreversibly and celecoxib selecting COX-2. Gabapentin and pregabalin bind the α2δ-1 subunit of voltage-gated calcium channels, reducing excitatory transmitter release and relieving neuropathic pain, and ketamine blocks NMDA receptors noncompetitively, giving analgesia and dissociative anesthesia.

Relationships to other topics

Ascending tracts. The lateral spinothalamic tract is the pain and temperature route among the ascending tracts, crossing in the cord rather than the medulla.

Synapses and neurotransmitters. Descending modulation works through serotonin, norepinephrine, glutamate, and enkephalin, and opioids act presynaptically to cut release.

Olfactory, optic, and vestibulocochlear nerves: the special senses

Cranial nerves I, II, and VIII carry smell, sight, hearing, and balance. Olfactory receptor neurons pass through the cribriform plate to the olfactory bulb and reach cortex without a thalamic relay. The optic pathway crosses only its nasal fibers at the chiasm, so lesions produce characteristic field defects, and CN II forms the afferent limb of the pupillary light reflex. CN VIII carries hearing, which becomes bilateral above the cochlear nuclei, and vestibular input to the eye, spinal, and cerebellar pathways. Here, we review these nerves and their failures.

Key points

- Smell reaches cortex without a thalamic relay, and olfactory receptor neurons regenerate.
- Only nasal fibers cross at the chiasm, so chiasmal lesions cause bitemporal hemianopia and tract lesions homonymous hemianopia.
- CN II is the afferent and CN III the efferent limb of the pupillary light reflex.
- Hearing becomes bilateral above the cochlear nuclei, so one-sided deafness localizes to the ear, nerve, or nucleus.

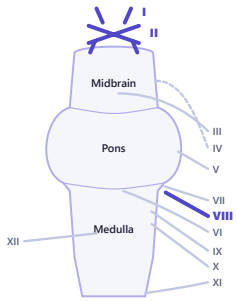


Fig. 1 | The brainstem from the front with the olfactory, optic, and vestibulocochlear nerves highlighted. Brainstem from the front; IV leaves from the back

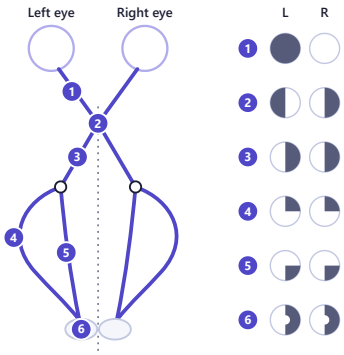


Fig. 2 | The visual pathway from above, with lesions on the left side and the field each one leaves, left and right eye: 1, the left optic nerve blinds the left eye; 2, the chiasm causes bitemporal hemianopia; 3, the left optic tract causes right homonymous hemianopia; 4, the left Meyer loop causes right upper quadrantanopia; 5, the left parietal radiation causes right lower quadrantanopia; 6, the left occipital cortex causes right homonymous hemianopia, with the macula spared. 1, Optic nerve; 2, Optic chiasm; 3, Optic tract; 4, Meyer loop (temporal); 5, Parietal radiation; 6, V1.

Olfactory and optic nerves

Olfactory receptor neurons, which regenerate for life, pass through the cribriform plate to synapse on mitral cells in the olfactory bulb, which project to the piriform cortex, amygdala, and entorhinal cortex without a thalamic relay. Ammonia tests the trigeminal nerve rather than smell. Retinal ganglion cells form the optic nerve, a CNS tract myelinated by oligodendrocytes, which passes through the optic canal to the chiasm, where nasal fibers cross. Each optic tract carries the opposite visual field to the LGN, and radiations reach V1 in the calcarine cortex, the lower field through Meyer loop in the temporal lobe. Optic nerve lesions blind one eye, chiasmal lesions cause bitemporal hemianopia, tract and V1 lesions cause homonymous hemianopia, and Meyer loop lesions cause an upper quadrantanopia. Pupil fibers leave before the LGN for the pretectum, and CN III constricts both pupils, so a damaged optic nerve produces an RAPD.

Argyll Robertson pupils constrict to a near target but not to light and mark neurosyphilis.

Vestibulocochlear nerve

Hair cells of the cochlea and vestibular labyrinth synapse on bipolar neurons of the spiral and Scarpa ganglia, whose axons pass through the internal acoustic meatus with CN VII to the cochlear and vestibular nuclei. Hearing becomes bilateral at the trapezoid body and ascends through the lateral lemniscus, inferior colliculus, and MGN to the Heschl gyrus, so one-sided deafness localizes to the ear, nerve, or cochlear nucleus. Vestibular nuclei project through the MLF for the VOR, to the vestibulospinal tracts, and to the cerebellum.

When it fails

Smell is lost early in Parkinson disease and after head injury. Optic neuritis demyelinates CN II, causing painful vision loss with an RAPD, and strokes behind the chiasm cause homonymous hemianopia. Vestibular schwannoma causes one-sided hearing loss and tinnitus, and vestibular neuritis causes vertigo with an abnormal head impulse.

Relationships to other topics

Oculomotor, trochlear, and abducens nerves. CN III forms the efferent limb of the pupillary light reflex, and the vestibular nuclei drive eye movements through the MLF.

Oculomotor, trochlear, and abducens nerves: moving the eyes

Cranial nerves III, IV, and VI move the eyes. CN III supplies four extraocular muscles and the levator, and its parasympathetic fibers constrict the pupil; CN IV supplies the superior oblique and is the only nerve to leave the back of the brainstem; CN VI supplies the lateral rectus. Their palsies produce characteristic deviations, and the pattern of pupil involvement in CN III palsy separates compression from ischemia. The MLF links the abducens and oculomotor nuclei for conjugate gaze. Here, we review these nerves and their failures.

Key points

- CN III palsy turns the eye down and out with ptosis and, if compressive, a dilated pupil.
- A PComm aneurysm dilates the pupil early, whereas a diabetic infarct spares it.
- CN IV palsy causes hypertropia and extorsion, with a head tilt away from the weak side.
- CN VI palsy causes esotropia and is a false localizing sign of raised ICP.
- An MLF lesion causes INO, as in MS.

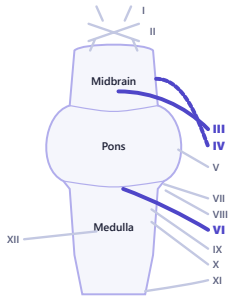


Fig. 1 | The brainstem from the front with the oculomotor, trochlear, and abducens nerves highlighted; the trochlear nerve leaves from the back. Brainstem from the front; IV leaves from the back

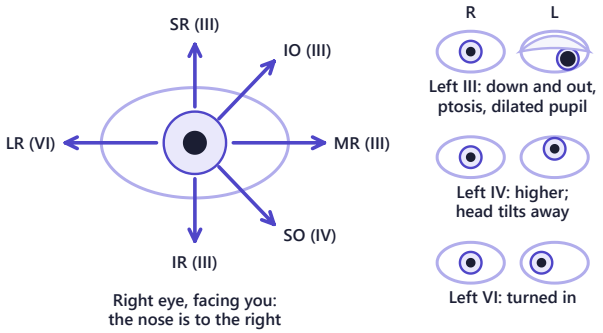


Fig. 2 | Actions of the extraocular muscles on a right eye facing the viewer: the lateral rectus (VI) abducts, the medial rectus (III) adducts, the superior rectus (III) elevates, the inferior rectus (III) depresses, the superior oblique (IV) depresses the adducted eye, and the inferior oblique (III) elevates the adducted eye. Beside it, left nerve palsies: III leaves the eye down and out with ptosis and a dilated pupil, IV leaves it higher with the head tilted away, and VI leaves it turned in.

Eye muscles

The lateral rectus (VI) abducts, the medial rectus (III) adducts, the superior rectus (III) elevates, and the inferior rectus (III) depresses. The superior oblique (IV) depresses and intorts, especially when adducted, the inferior oblique (III) elevates and extorts when adducted, and the levator (III) lifts the lid.

Sympathetic fibers reach the eye through 3 neurons: from the hypothalamus to the ciliospinal center at C8–T2, out at T1 to the superior cervical ganglion, and along the internal carotid artery to the iris dilator and Müller muscle. A lesion anywhere along the path, as in lateral medullary syndrome, Pancoast tumors, or carotid dissection, causes Horner syndrome: ptosis, miosis, and anhidrosis.

The three nerves

CN III arises with the Edinger-Westphal nucleus at the superior colliculus level, leaves through the interpeduncular fossa beside the PComm, runs in the lateral wall of the cavernous sinus, and enters the orbit through the superior orbital fissure, its parasympathetic fibers synapsing in the ciliary ganglion. Its palsy turns the eye down and out with ptosis and a dilated pupil;

because pupil fibers ride on the surface, a PComm aneurysm dilates the pupil early, whereas a diabetic infarct spares it. Every CN IV fiber crosses before leaving the back of the midbrain, and its palsy causes hypertropia and extorsion with a head tilt away from the weak side. CN VI arises under the facial colliculus, climbs the clivus, and runs free in the cavernous sinus; its palsy causes esotropia and can be a false localizing sign of raised ICP, and its nucleus is the horizontal gaze center. The MLF links the VI nucleus to the opposite III nucleus, and its lesion causes INO.

When it fails

A PComm aneurysm compresses CN III and blows the pupil, and Weber syndrome combines a same-side CN III palsy with opposite hemiparesis. MS demyelinates the MLF, causing INO, and progressive supranuclear palsy causes a supranuclear vertical gaze palsy with intact nerves.

Relationships to other topics

Olfactory, optic, and vestibulocochlear nerves. CN II and III form the pupillary light reflex, and vestibular input drives the eyes through the MLF.

Trigeminal and facial nerves: face sensation, mastication, and expression

The trigeminal nerve carries sensation from the face through three divisions and supplies the muscles of mastication, with four nuclei extending from the midbrain to C2. The facial nerve supplies facial expression and the stapedius, carries parasympathetic fibers to the lacrimal and salivary glands, and carries taste from the front of the tongue. Together they form the corneal reflex. Upper and lower motor neuron facial palsies differ by forehead involvement, and trigeminal nuclear lesions cause dissociated facial sensory loss. Here, we review these nerves and their failures.

Key points

- CN V has mesencephalic, principal sensory, spinal, and motor nuclei; the spinal nucleus carries facial pain and temperature to C2.
- V1, V2, and V3 leave through the superior orbital fissure, foramen rotundum, and foramen ovale.
- The corneal reflex runs V1 in and VII out.
- The facial nerve passes through the parotid without supplying it.
- A UMN facial palsy spares the forehead; an LMN palsy does not.

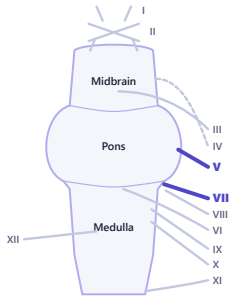


Fig. 1 | The brainstem from the front with the trigeminal and facial nerves highlighted. Brainstem from the front; IV leaves from the back

Trigeminal nerve

CN V has a mesencephalic nucleus for jaw proprioception, a principal sensory nucleus for touch, a spinal nucleus for pain and temperature extending to C2, and a motor nucleus. Its ophthalmic division, through the superior orbital fissure, supplies the forehead and cornea; its maxillary division, through the foramen rotundum, the cheek and upper jaw; and its mandibular division, through the foramen ovale, the lower jaw and front of the tongue, plus the masseter, temporalis, and pterygoids. The angle of the jaw is supplied by C2–C3, and the jaw deviates toward a weak V3.

Trigeminal neuralgia causes brief, electric shock-like pain on one side, usually in V2 or V3, from vascular compression of the root, often by the superior cerebellar artery, and is treated first with carbamazepine or oxcarbazepine.

Facial nerve

The facial nucleus lies in the pons, and its fibers loop over the VI nucleus before leaving at the cerebellopontine angle with CN VIII. At the geniculate ganglion, the greater petrosal nerve leaves for the lacrimal gland; in the facial canal, branches supply the stapedius and form the chorda tympani, which carries taste from the front two-thirds of the tongue and fibers to the submandibular and sublingual glands. Only motor fibers leave the stylomastoid foramen, and the nerve divides into 5 branches within the parotid without supplying it. The corneal reflex runs V1 in and VII out, and loss of the stapedius causes hyperacusis.

Ramsay Hunt syndrome, varicella zoster of the facial nerve with a vesicular rash, causes about 6% of facial palsy.

When it fails

A UMN lesion weakens the lower face and spares the forehead, whereas Bell palsy, an LMN lesion, weakens the whole half of the face with hyperacusis and taste loss. Lateral medullary syndrome damages the spinal trigeminal nucleus, causing same-side facial pain and temperature loss, and vestibular schwannoma can stretch CN VII at the cerebellopontine angle.

Relationships to other topics

Descending tracts. Corticobulbar input to the upper facial nucleus is bilateral, which is why UMN lesions spare the forehead.

Glossopharyngeal, vagus, accessory, and hypoglossal nerves: the lower cranial nerves

The lower cranial nerves leave the medulla and upper cervical cord. CN IX supplies the stylopharyngeus, the parotid, the carotid sinus and body, and the back of the tongue, forming the afferent limb of the gag reflex. CN X supplies the palate, pharynx, and larynx and carries parasympathetic fibers to the heart, lungs, and gut. CN XI supplies the sternocleidomastoid and trapezius, and CN XII the tongue. Their lesions cause dysphagia, hoarseness, weakness of head turning, and tongue deviation, and they localize medullary strokes. Here, we review these nerves and their failures.

Key points

- CN IX is the afferent and CN X the efferent limb of the gag reflex.
- The left recurrent laryngeal nerve hooks under the aortic arch.
- The uvula deviates away from a vagal lesion, and the tongue toward a hypoglossal lesion.
- The right sternocleidomastoid turns the head to the left.
- Lateral medullary syndrome affects IX and X; medial medullary syndrome affects XII.

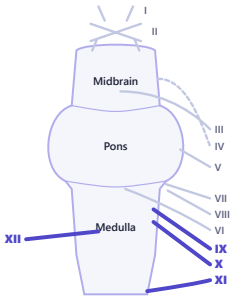


Fig. 1 | The brainstem from the front with the glossopharyngeal, vagus, accessory, and hypoglossal nerves highlighted. Brainstem from the front; IV leaves from the back

The four nerves

CN IX supplies its only muscle, the stylopharyngeus, sends parasympathetic fibers to the parotid through the otic ganglion, carries pressure and oxygen signals from the carotid sinus and body that slow the heart, and carries taste and touch from the back third of the tongue, forming the afferent limb of the gag reflex. CN X, from the nucleus ambiguus, supplies the palate, pharynx, and larynx and forms the efferent limb of the gag, and from the dorsal motor nucleus sends parasympathetic fibers to the heart, lungs, and gut to the splenic flexure; the left recurrent laryngeal nerve hooks under the aortic arch, and the uvula deviates away from a vagal lesion. CN XI arises from C1–C5, rises through the foramen magnum, and leaves through the jugular foramen to supply the sternocleidomastoid and trapezius, and can be injured in posterior triangle node biopsy. CN XII arises in the floor of the fourth ventricle, leaves between the pyramid and olive through the hypoglossal canal, and supplies all tongue muscles except palatoglossus; its lesion deviates the tongue toward the lesion, with wasting and fasciculations.

When it fails

Lateral medullary syndrome damages the nucleus ambiguus, causing dysphagia, hoarseness, and a reduced gag, and medial medullary syndrome damages CN XII, deviating the tongue toward the lesion with opposite hemiparesis. Bilateral UMN lesions cause pseudobulbar palsy, with dysarthria and dysphagia but no wasting, and recurrent laryngeal nerve injury causes hoarseness.

Relationships to other topics

Descending tracts. Corticobulbar fibers reach these motor nuclei, so bilateral UMN lesions cause pseudobulbar palsy.

The respiratory center: rhythm, chemical control, and failures of drive

Breathing is generated by brainstem neurons and tuned to metabolic demand. The dorsal respiratory group in the NTS is mainly inspiratory, the ventral respiratory group is mainly expiratory and contains the preBötzinger complex that generates the rhythm, and pontine centers limit or prolong inspiration. Central chemoreceptors on the ventral medulla sense CSF pH, which follows PaCO2, and supply most of the drive, while carotid and aortic bodies respond mainly to hypoxemia. Here, we review the brainstem groups, chemical and mechanical control, breathing in sleep, and the patterns and failures of drive.

Key points

- PaCO2, sensed as CSF pH by central chemoreceptors, is the main drive to breathe.
- Peripheral chemoreceptors respond mainly to low PaO2; the carotid bodies supply about 15% of drive.
- The preBötzinger complex generates the rhythm and is where opioids depress breathing.
- Cheyne-Stokes breathing follows heart failure and stroke, apneustic breathing pontine lesions, and Biot breathing medullary injury.
- Opioids shift the CO2 response curve right; benzodiazepines and propofol flatten it.

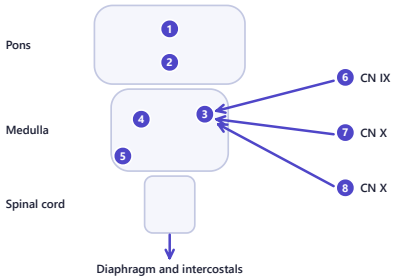


Fig. 1 | The respiratory centers. In the pons, 1, the pneumotaxic center and 2, the apneustic center. In the medulla, 3, the dorsal respiratory group, 4, the ventral respiratory group, and 5, the central chemoreceptors. Sensory input arrives from 6, the carotid bodies through cranial nerve IX, 7, the aortic bodies through cranial nerve X, and 8, lung receptors through cranial nerve X. Output descends the spinal cord to the diaphragm and intercostals. 1, Pneumotaxic center; 2, Apneustic center; 3, Dorsal respiratory group; 4, Ventral respiratory group; 5, Central chemoreceptors; 6, Carotid bodies; 7, Aortic bodies; 8, Lung receptors.

Brainstem groups

The dorsal respiratory group lies in the NTS and is mainly inspiratory. The ventral respiratory group is mainly expiratory, and its rostral part, including the preBötzinger complex, generates the rhythm. In the pons, the pneumotaxic center limits inspiration and the apneustic center prolongs it. Inspiration follows contraction of the diaphragm and external intercostals, which lowers pleural and alveolar pressure, and expiration at rest is passive, from the elastic recoil of the lungs. The resulting rate is about 12 breaths a minute at rest.

Chemical control

Central chemoreceptors on the ventral surface of the medulla and in the retrotrapezoid nucleus sense CSF pH, which falls as CO2 crosses the blood-brain barrier, and supply most of the drive over minutes. Peripheral chemoreceptors respond within seconds, mainly to low PaO2 but also to CO2 and H+; the carotid bodies signal through CN IX and supply about 15% of drive, and the aortic bodies signal through CN X. PaCO2 is the main drive, and oxygen drives the center only in severe hypoxemia; when the receptors disagree, as in hypoxic hypocapnia, the hypoxic response wins.

Between a PaCO2 of 45 and 80 mmHg, ventilation rises 2–5 L/min for each mmHg. Hypoxemia below 60 mmHg and metabolic acidemia shift this curve left, opioids shift it right, and benzodiazepines, propofol, volatile anesthetics, age, and fitness flatten it. Hypoxia causes rapid, shallow breaths, which avoid the work of high volumes, and hypercapnia causes deep, slow breaths, which limit dead space.

Reflexes and sleep

Slowly adapting stretch receptors mediate the Hering-Breuer reflex, in which inflation inhibits inspiration; rapidly adapting irritant receptors cause cough and faster breathing; and juxtacapillary receptors respond to congestion and interstitial edema with faster breathing, all through CN X. In REM sleep, breathing becomes irregular, accessory muscles are paralyzed, upper airway

dilators lose tone, and drive falls and responds less to PaO_2 and $PaCO_2$.

When it fails

Cheyne-Stokes breathing alternates crescendo-decrescendo ventilation with central apnea in cycles of 45–90 seconds and occurs in heart failure, where slow circulation delays the chemoreceptor signal, and in 20–40% of patients acutely after stroke. Biot, or ataxic, breathing has irregular clusters with unpredictable apnea from medullary injury and raised ICP; apneustic breathing has prolonged inspiratory pauses from pontine stroke or TBI; and Kussmaul breathing is deep, rapid, and continuous in metabolic acidosis such as DKA and renal failure. In the Cushing triad, raised ICP compresses the brainstem and causes irregular respirations with widened pulse pressure and bradycardia, as in intracerebral and subarachnoid hemorrhage.

In opioid overdose, μ receptors at the preBötzinger complex blunt the response to CO_2 and hypoxia, causing pinpoint pupils, respiratory depression, and decreased consciousness, and benzodiazepines, inhaled anesthetics, and alcohol also depress drive. In congenital central hypoventilation syndrome, the center ignores $PaCO_2$, so breathing fails in sleep. Central sleep apnea arises from high loop gain in heart failure or low drive from opioids, brainstem stroke, or cervical cord injury and is treated with CPAP or BPAP, and in obesity hypoventilation, retained bicarbonate blunts the central chemoreceptors. With intact drive, an asthma attack causes fast breathing and low CO_2, and a normal CO_2 signals tiring muscles; in COPD, oxygen raises CO_2 through ventilation-perfusion mismatch and the Haldane effect, so saturations of 88–92% are targeted; and in neuromuscular disease, weak muscles cannot respond to an intact drive.

Relationships to other topics

Cardiovascular center. The two centers share the medulla: the ventral respiratory column sits within the rostral ventrolateral medulla, and both receive their afferents in the nucleus tractus solitarius.

Glossopharyngeal, vagus, accessory, and hypoglossal nerves. The carotid bodies signal through CN IX, and the aortic bodies and lung receptors through CN X.

The cardiovascular center: the baroreflex and its failures

Arterial pressure is buffered beat to beat by the baroreflex. Stretch of the carotid sinus and aortic arch opens PIEZO channels in baroreceptor endings, whose signals travel in CN IX and CN X to the NTS. The NTS excites the CVLM, whose GABA neurons inhibit the RVLM, the source of sympathetic drive, and excites cardiac vagal neurons in the nucleus ambiguus, so rising pressure slows the heart and dilates vessels while falling pressure does the reverse. Here, we review the reflex, its other inputs and outputs, and the syndromes of too much reflex, too little reflex, and a medulla cut off or compressed.

Key points

- Baroreceptor afferents from the carotid sinus run in CN IX and from the aortic arch in CN X, ending in the NTS.
- The NTS inhibits sympathetic output through the CVLM and RVLM and drives vagal output through the nucleus ambiguus.
- The baroreflex resets with sustained change, so it buffers only short-term fluctuations.
- Vasovagal syncope and carotid sinus hypersensitivity are excessive reflexes; orthostatic hypotension reflects too little.
- Autonomic dysreflexia follows spinal cord injury at or above T6, and the Cushing reflex follows raised ICP.

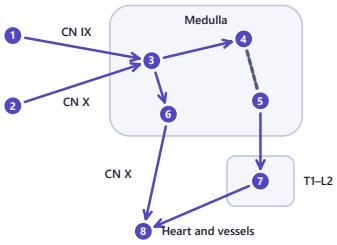


Fig. 1 | The baroreflex. Stretch receptors in 1, the carotid sinus, through cranial nerve IX, and 2, the aortic arch, through cranial nerve X, signal to 3, the nucleus tractus solitarius. It excites 4, the caudal ventrolateral medulla, which inhibits 5, the rostral ventrolateral medulla, the source of drive to 7, sympathetic neurons from T1 to L2. It also excites 6, the nucleus ambiguus, whose vagal fibers slow the heart. Both reach 8, the heart and vessels. 1, Carotid sinus; 2, Aortic arch; 3, NTS; 4, CVLM; 5, RVLM; 6, Nucleus ambiguus; 7, Sympathetic neurons; 8, Heart and vessels.

The baroreflex

Wall stretch opens PIEZO1 and PIEZO2 channels in baroreceptor endings, letting sodium and calcium in; myelinated A fibers track fast, pulsatile change and unmyelinated C fibers report sustained pressure. Carotid sinus afferents travel in the carotid sinus nerve, a branch of CN IX, with cell bodies in the petrosal ganglion, and aortic afferents in CN X, with cell bodies in the nodose ganglion. They release glutamate in the NTS, which also receives chemoreceptor and lung stretch afferents. The NTS excites the CVLM, whose GABA neurons inhibit the RVLM, which drives sympathetic preganglionic neurons and maintains baseline pressure, and it excites cardiac vagal neurons in the nucleus ambiguus and dorsal motor nucleus of the vagus. The ventral respiratory column sits within the RVLM.

When BP rises, baroreceptor firing increases, vagal output rises and sympathetic output falls, and HR and contractility fall while vessels dilate; when BP falls, the reverse occurs. With sustained change, the operating point shifts toward the new pressure, so the baroreflex buffers only short-term change.

Other inputs and outputs

Cardiopulmonary receptors in the atria, ventricles, and pulmonary vessels sense volume: stretch lowers renal sympathetic tone and releases ANP, causing natriuresis, while low volume raises sympathetic tone, RAAS activity, and vasopressin. The area postrema mediates angiotensin II, which raises BP without much change in HR, and the hypothalamus and insular cortex modulate long-term control. Sympathetic preganglionic neurons arise from the lateral gray column from T1 to L2 and release ACh onto nicotinic receptors, and postganglionic neurons release norepinephrine, with adrenaline from the adrenal medulla. Parasympathetic fibers leave in CN III, VII, IX, and X and from S2–S4, synapse near their targets, and release ACh onto muscarinic receptors.

When it fails

Vasovagal syncope, about 50% of syncope, follows standing, heat, pain, or stress, with sympathetic withdrawal causing vasodilation and vagal activity causing bradycardia. Carotid sinus hypersensitivity affects older adults, triggered by tight collars or head turning, and is cardioinhibitory, with asystole of 3

seconds or more, in 70%, vasodepressor, with a systolic fall of 50 mmHg or more, in 10%, and mixed in 20%; monitored carotid sinus massage confirms it, and pacing treats the cardioinhibitory type. Situational syncope follows coughing, micturition, defecation, and swallowing, and a trigger, lightheadedness, brief jerks, and rapid recovery favor syncope over a seizure.

Orthostatic hypotension is a sustained fall of 20 mmHg systolic or 10 mmHg diastolic within 3 minutes of standing, as 300–800 mL pools in the legs and splanchnic veins. It occurs in Parkinson disease, early in multiple system atrophy and dementia with Lewy bodies, and in pure autonomic failure, diabetes, B12 deficiency, and with α- and β-blockers. In POTS, HR rises 30 a minute or more within 10 minutes of standing, or 40 at ages 12–19, without orthostatic hypotension, and afferent baroreflex failure after neck radiation or surgery causes swings between hypertension and hypotension.

In autonomic dysreflexia, after SCI at or above T6, a stimulus below the lesion, usually a full bladder (85%), drives splanchnic vasoconstriction, raising systolic pressure by 20 mmHg or more with headache, while vagal bradycardia and flushing occur only above the lesion. In the Cushing reflex, raised ICP lowers CPP, the difference between MAP and ICP, causing a sympathetic surge with widened pulse pressure, then bradycardia and irregular respirations, a late sign of herniation in intracerebral and subarachnoid hemorrhage.

Relationships to other topics

Respiratory center. Both centers sit in the ventrolateral medulla and receive afferents in the nucleus tractus solitarius, and the Cushing triad disturbs both.

Glossopharyngeal, vagus, accessory, and hypoglossal nerves. Baroreceptor afferents run in CN IX and CN X, and cardiac vagal output leaves through CN X.

Stroke syndromes: localizing the lesion from the cortex to the spinal cord

Stroke symptoms follow the anatomy of the lesion rather than its cause. Cerebral syndromes reflect the ACA, MCA, PCA, and watershed territories, brainstem syndromes produce crossed signs, and spinal cord lesions separate motor, pain, and proprioceptive pathways. Hemorrhagic strokes add features of raised ICP and meningeal irritation. Here, we review these syndromes and their complications, and consider how to distinguish those that resemble each other.

Key points

- Brainstem strokes are crossed: cranial nerve signs are ipsilateral, and limb signs are contralateral.
- ACA favors the leg, whereas MCA favors the face and arm.
- Dominant MCA lesions cause aphasia; non-dominant lesions cause neglect.
- Thunderclap headache with meningeal signs suggests SAH.

Cerebral ischemic syndromes

ACA syndrome causes contralateral leg weakness, greater than arm, contralateral leg sensory loss, urinary incontinence, abulia, and a grasp reflex. MCA syndrome causes contralateral face and arm weakness, greater than leg, gaze toward the lesion, contralateral face and arm sensory loss, and contralateral homonymous hemianopia; dominant lesions add aphasia and Gerstmann syndrome, and non-dominant lesions add extinction, hemispatial neglect, and anosognosia. PCA syndrome usually causes no weakness, with contralateral sensory loss if the thalamus is involved, contralateral homonymous hemianopia with macular sparing, alexia without agraphia (dominant), and prosopagnosia (non-dominant). MCA/ACA watershed syndrome, from low perfusion or ICA stenosis, causes proximal arm and shoulder weakness sparing the hand and face ("man in a barrel" if bilateral), mild proximal arm sensory loss, and transcortical motor aphasia (dominant). MCA/PCA watershed syndrome usually causes no weakness and mild sensory change, with visual field loss, transcortical sensory aphasia (dominant), neglect (non-dominant), and Balint syndrome if bilateral.

Dominant-hemisphere lesions cause characteristic aphasias. Broca aphasia, from the inferior frontal gyrus, is nonfluent with intact comprehension and impaired repetition. Wernicke aphasia, from the posterior superior temporal gyrus, is fluent but paraphasic, with impaired comprehension and repetition. Conduction aphasia, from the arcuate fasciculus, is fluent with relatively intact comprehension but impaired repetition, and global aphasia affects all three. Transcortical motor and sensory aphasias resemble Broca and Wernicke aphasia but preserve repetition.

Lacunar infarcts follow lipohyalinosis or microatheroma of small penetrating arteries, mainly with hypertension and diabetes, and lack cortical signs. Pure motor stroke, the most common at 45%, involves the posterior limb of the internal capsule, corona radiata, or ventral pons; pure sensory stroke involves the thalamus; sensorimotor stroke involves the thalamus and internal capsule; ataxic hemiparesis involves the internal capsule, pons, or corona radiata; and dysarthria-clumsy hand syndrome involves the pons or internal capsule. Amaurosis fugax, a painless, curtain-like loss of vision in one eye that resolves, usually follows emboli from the carotid bifurcation and warns of stroke.

Brainstem strokes

In the midbrain, supplied by P1 paramedian branches and the top of the basilar, Weber syndrome (ventromedial) causes ipsilateral CN III palsy with a dilated pupil and contralateral hemiparesis. Benedikt syndrome (paramedian) causes ipsilateral CN III palsy, contralateral tremor and ataxia, with or without contralateral hemiparesis, and contralateral vibration and proprioception loss, combining Weber and Claude features. Claude syndrome causes ipsilateral CN III palsy and contralateral ataxia, with no weakness. In the pons, ventromedial pontine syndrome, from basilar paramedian branches, causes contralateral hemiparesis, dysarthria, and ataxic hemiparesis, and its caudal form (Raymond syndrome) adds ipsilateral CN VI palsy. Ventral pontine (Millard-Gubler) syndrome causes contralateral hemiplegia, ipsilateral CN VI and CN VII palsy (forehead included), and horizontal diplopia. Inferior medial pontine (Foville) syndrome causes ipsilateral horizontal gaze palsy and CN VII palsy (forehead included), contralateral hemiparesis,

contralateral vibration and proprioception loss, and eyes that deviate toward the weak side; both are supplied by basilar paramedian and short circumferential branches. Lateral pontine (AICA) syndrome causes ipsilateral CN VII palsy (forehead included) and ataxia with preserved strength, crossed pain and temperature loss (ipsilateral face and contralateral body), and ipsilateral hearing loss, vertigo, and Horner syndrome. Marie-Foix syndrome, a lateral pontine syndrome from basilar perforators and the AICA, adds contralateral hemiparesis and contralateral vibration and proprioception loss to ipsilateral limb ataxia, CN VII palsy, hearing loss, vertigo, and nystagmus. Locked-in syndrome, from bilateral ventral pons injury via the basilar artery, causes quadriplegia and anarthria, usually with preserved sensation, consciousness, vertical eye movements, and blinking. In the medulla, lateral medullary (Wallenberg) syndrome, from the vertebral artery or PICA, causes ipsilateral palate, pharynx, and vocal cord weakness and ataxia with preserved strength, crossed pain and temperature loss, and ipsilateral Horner syndrome, vertigo, dysphagia, and hiccups. Medial medullary (Dejerine) syndrome, from the anterior spinal artery, causes contralateral hemiparesis sparing the face, ipsilateral tongue weakness with deviation toward the lesion, and contralateral vibration and proprioception loss.

Spinal cord lesions

Central cord syndrome, from hyperextension injury and syringomyelia, causes arm weakness greater than leg, with the hands worst, cape-like pain and temperature loss, and sacral sparing. Anterior cord syndrome, from the anterior spinal artery, causes bilateral weakness and pain and temperature loss below the lesion, preserved vibration and proprioception, and bladder and bowel dysfunction. Posterior cord syndrome, from the posterior spinal artery, tabes dorsalis, and B12 deficiency, causes sensory ataxia with preserved strength, bilateral vibration and proprioception loss, and a positive Romberg sign. Brown-Séquard syndrome (cord hemisection), from penetrating trauma and compression, causes ipsilateral weakness and vibration and proprioception loss below the lesion, contralateral pain and temperature loss, and ipsilateral Horner syndrome if at or above T1.

Hemorrhagic strokes

Intracerebral hemorrhage causes headache and cortical focal neurologic deficits. Subarachnoid hemorrhage causes thunderclap headache and meningeal signs: photophobia, neck stiffness, and positive Kernig and Brudzinski signs. Both cause raised ICP, with nausea and vomiting, posturing, and Cushing triad, along with loss of consciousness in intracerebral hemorrhage and CN deficits in subarachnoid hemorrhage. Subarachnoid hemorrhage is complicated by rebleeding, highest early before the aneurysm is secured; vasospasm and delayed cerebral ischemia on days 3–14; hydrocephalus; hyponatremia from SIADH or cerebral salt wasting; and seizures.

Complications

Stroke complications are cerebral edema (malignant MCA infarct, with herniation in the first several days), hemorrhagic transformation, aspiration pneumonia from dysphagia, seizures, DVT and PE from immobility, spasticity and contractures, post-stroke depression, and central post-stroke pain, often from a thalamic infarct but possible after any sensory pathway lesion.

Relationships to other conditions

Ischemia and hemorrhage. Both can cause focal deficits, and non-contrast CT head separates them.

Lateral pontine vs lateral medullary. Both cause crossed pain and temperature loss, ataxia, vertigo, and Horner syndrome with preserved strength. CN VII palsy and hearing loss favor the pons; dysphagia, hiccups, and palate weakness favor the medulla.

Midbrain syndromes. All three share CN III palsy. Contralateral hemiparesis means Weber, contralateral ataxia means Claude, and tremor with sensory loss means Benedikt.

Medial medullary vs ventral pontine. Both cause contralateral hemiparesis. Ipsilateral tongue weakness points to the medulla, and CN VI and CN VII palsy to the pons.

Anterior vs posterior cord. Anterior cord lesions lose strength and pain and temperature but preserve vibration and proprioception, whereas posterior cord lesions lose vibration and proprioception but preserve strength.

Acute ischemic stroke: embolic, thrombotic, and low-flow injury and the race to reperfuse

Blocked or inadequate arterial blood flow leads to cerebral ischemia, causing an infarct core surrounded by a salvageable penumbra. Embolus is the most common mechanism; the others are thrombus, global hypoperfusion, vasculitis, and dissection. Management is a race to reperfuse within fixed time windows, followed by prevention tailored to the cause. Here, we review its causes, imaging, acute treatment, and recurrence prevention, and consider how it relates to hemorrhagic stroke.

Key points

- Glucose and non-contrast CT head come first, to rule out hypoglycemia and hemorrhage.
- IV alteplase or tenecteplase is given within 4.5 hours of onset.
- Thrombectomy for large vessel occlusion extends up to 24 hours with penumbra.
- Prevention of recurrence is chosen by cause.

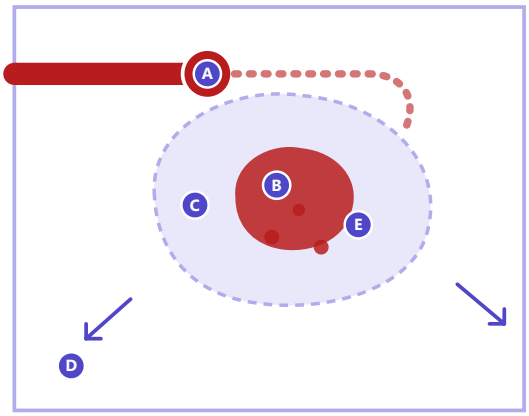


Fig. 1 | An artery blocked by an embolus, with a core of infarcted brain surrounded by a penumbra downstream. Arrows show swelling, and small spots show bleeding into the infarct. A, Arterial occlusion; B, Infarct core; C, Penumbra; D, Cerebral edema; E, Hemorrhagic transformation.

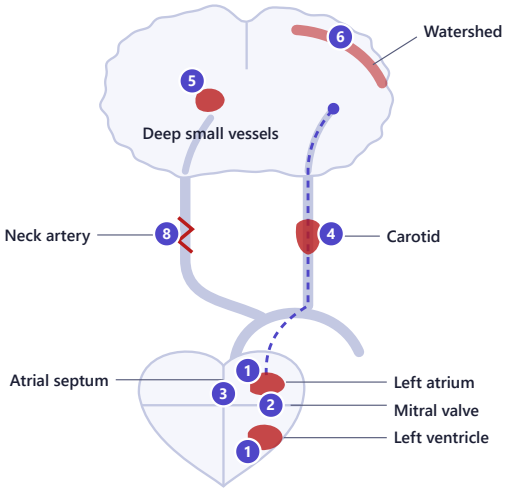


Fig. 2 | Diagram of the heart, aortic arch, neck arteries, and brain showing where each cause of ischemic stroke starts. 1, Cardiac embolus; 2, Infective endocarditis; 3, Paradoxical embolus; 4, Large vessel thrombus; 5, Small vessel thrombus; 6, Global hypoperfusion; 8, Dissection.

Pathophysiology

Blocked or inadequate flow causes cerebral ischemia with decreased oxygen delivery. An embolus or thrombus stops flow downstream. In the infarct core, flow is too low for cells to survive, so injury is irreversible, whereas in the penumbra, flow stops function but not survival, so reperfusion can reverse deficits. Failing ion pumps swell cells, causing cerebral edema, with malignant MCA infarct and herniation in the first several days, and

reperfusion of injured vessels can cause hemorrhagic transformation, with blood within the infarct.

Etiology and mechanisms

Cardiac embolus, an important cause, arises from a left atrial thrombus in A-fib or flutter, mechanical valve, or RHD, or from a left ventricular thrombus in LV aneurysm from MI, anterior or lateral MI, or HFrEF. Infective endocarditis is a separate source of cardiac embolus and a contraindication to thrombolysis. Paradoxical embolus occurs with ASD, patent foramen ovale, and atrial septal aneurysm, where a right-to-left shunt lets a venous thrombus reach arterial circulation. Global hypoperfusion from cardiac arrest, acute respiratory failure, or ICA stenosis with low BP decreases cerebral oxygen delivery, causing watershed infarcts. Large vessel thrombus is atherosclerotic thrombus in the ICA, MCA, and vertebrobasilar arteries. Small vessel thrombus causes lacunar infarcts from the lenticulostriate arteries or small pontine branches, occluded by lipohyalinosis or microatheroma, mainly from hypertension and diabetes. Vasculitis from autoimmune disease, infections, or primary CNS vasculitis causes arterial wall inflammation and can affect any artery. Dissection from trauma, iatrogenic injury, or connective tissue diseases causes an intimal tear with thrombus and embolus.

Diagnosis

Glucose and non-contrast CT head rule out hypoglycemia and hemorrhage. If there is no hemorrhage, CT angiogram of the head and neck looks for large vessel occlusion, and CT perfusion or MRI looks for penumbra if onset is 4.5–24 hours ago or unknown. Cardiac embolus is sought with ECG, telemetry, and echocardiogram, looking for A-fib and thrombus. Paradoxical embolus is sought with bubble echocardiogram, looking for PFO. Large vessel thrombus is assessed by CT angiogram or carotid duplex for ICA stenosis, small vessel thrombus by MRI for lacunar infarct, vasculitis by ESR, CRP, autoimmune serology, and CSF, dissection by CT or MR angiogram of the neck, and infective endocarditis by blood cultures and echocardiogram. Vascular risk factors are assessed with lipids and HbA1c. NIHSS grades stroke severity.

Stroke mimics

Stroke mimics must be recognized without delaying reperfusion. Hypoglycemia causes deficits that resolve with glucose. A seizure with Todd paralysis is favored by a witnessed seizure, tongue biting, and a post-ictal state. Migraine with aura spreads over minutes, with positive symptoms and past similar attacks. Functional neurologic disorder shows inconsistent signs, a positive Hoover sign, and normal imaging. Brain tumor or abscess has a gradual onset, with fever in an abscess, and is seen on CT or MRI with contrast. Bell palsy causes isolated facial weakness that includes the forehead, peripheral vertigo causes isolated vertigo with a reassuring HINTS examination, and systemic illness such as sepsis, hyponatremia, or drugs can unmask an old deficit.

Management

Reperfusion uses IV alteplase or tenecteplase within 4.5 hours of onset, and mechanical thrombectomy for large vessel occlusion up to 24 hours with penumbra. Aspirin is given within 24–48 hours, waiting 24 hours after thrombolysis. Supportive care comprises a swallow screen before oral intake, treating fever and glucose, and DVT prophylaxis. BP is kept below 185/110 for thrombolysis, and otherwise allowed up to 220/120. Thrombolysis contraindications are hemorrhage on CT, head trauma within 3 months, intracranial or spinal surgery within 3 months, active bleeding, platelets under 100000/µL, INR over 1.7, infective endocarditis, aPTT over 40 seconds, therapeutic LMWH within 24 hours, recent DOAC use with possible ongoing effect, suspected aortic dissection, and an intracranial neoplasm with high bleeding risk. Prior ischemic stroke is weighed case by case rather than excluding thrombolysis. After a minor non-cardioembolic stroke (NIHSS 3 or less) or high-risk TIA not treated with thrombolysis, aspirin plus clopidogrel is started within 24 hours and continued for 21 days before changing to one antiplatelet.

Prevention of recurrence

Cardiac embolus is treated with anticoagulation: DOAC, or warfarin for a mechanical valve. Infective endocarditis is treated with ABX. Paradoxical embolus is treated with antiplatelet therapy, with PFO closure if under 60 with no other cause. Large vessel thrombus is treated with antiplatelet therapy and high-intensity statin, with carotid endarterectomy for symptomatic ICA stenosis of 70–99%. Small vessel thrombus is treated with antiplatelet therapy, statin, and BP control. Global hypoperfusion is treated by treating the cause of low perfusion. Vasculitis is treated with immunosuppression and treating any infection. Dissection is treated with antiplatelet therapy or anticoagulation.

In A-fib, the CHA_2DS_2-VASc score estimates stroke risk to guide anticoagulation, with 1 point each for congestive heart failure, hypertension, diabetes, vascular disease, age 65–74, and female sex, and 2 points each for age 75 or older and prior stroke or TIA; a score of 2 or more marks high risk.

Relationships to other conditions

Intracerebral hemorrhage. Non-contrast CT separates them, and BP goals are opposite: ischemic stroke tolerates up to 220/120, whereas hemorrhage is actively lowered. Hemorrhagic transformation links the two.

Subarachnoid hemorrhage. Thunderclap headache with meningeal signs favors SAH, and hemorrhage on CT contraindicates thrombolysis.

Stroke syndromes. Deficits localize the vessel, and watershed syndromes follow low perfusion or ICA stenosis.

Infective endocarditis. It causes ischemic stroke by embolism and contraindicates thrombolysis, and it causes hemorrhage through mycotic aneurysms.

Embolic disease elsewhere. A-fib also embolizes to the SMA, causing acute mesenteric ischemia.

Intracerebral hemorrhage: causes by location, and limiting hematoma growth

Vessel rupture bleeds into the brain parenchyma, forming a hematoma that compresses tissue and raises ICP. Hypertension is the most common cause, and cerebral amyloid angiopathy is a common cause of lobar bleeds in older adults, with location helping to separate them. Management aims to limit hematoma growth, reverse coagulopathy, and control ICP. Here, we review its causes, workup, and treatment, and consider how it relates to ischemic stroke and subarachnoid hemorrhage.

Key points

- Hypertension bleeds in deep structures, especially the basal ganglia; amyloid angiopathy bleeds in the cortex.
- If systolic is 150–220, lower it to a target of 140.
- Reversal is matched to the drug.
- Cerebral venous sinus thrombosis is treated with anticoagulation, even with hemorrhage.

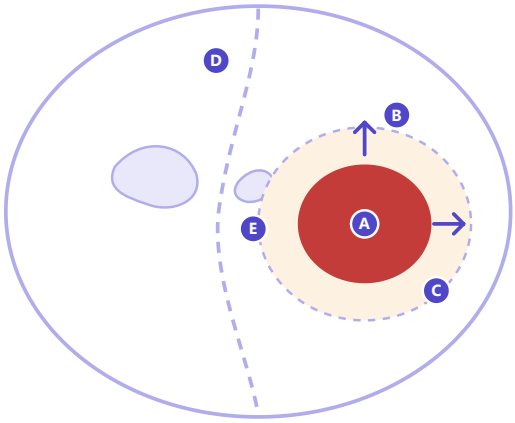


Fig. 1 | A brain in coronal section with a hematoma in one hemisphere. Swelling surrounds the hematoma, arrows show it growing, the midline is pushed across, and the nearby ventricle is compressed. A, Tissue compression; B, Hematoma growth; C, Surrounding edema; D, Raised ICP; E, Hydrocephalus.

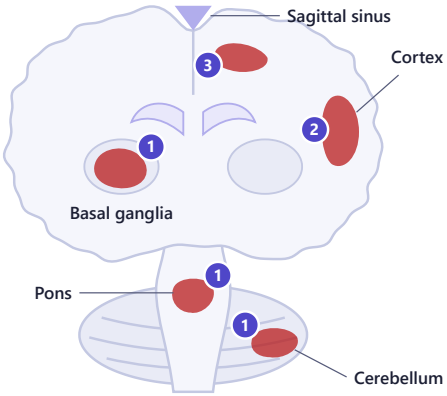


Fig. 2 | Coronal brain diagram: hypertension bleeds in the basal ganglia, pons, and cerebellum; amyloid angiopathy bleeds in the cortex; venous sinus thrombosis bleeds near the cortex under the sagittal sinus. 1, Hypertension; 2, Cerebral amyloid angiopathy; 3, Cerebral venous sinus thrombosis.

Pathophysiology

A weakened vessel ruptures into the brain parenchyma, forming a hematoma. The hematoma presses on brain tissue, causing cortical focal neurologic deficits, and bleeding can continue, causing worsening deficits, which is why BP control and reversal limit it. Injured tissue swells around the hematoma, adding to its mass. This added mass raises ICP, causing headache, nausea and vomiting, loss of consciousness, posturing, and Cushing triad, and blood or mass effect can block CSF flow, causing hydrocephalus that may need an EVD.

Etiology and mechanisms

Hypertension, the most common cause, bleeds in the basal ganglia, thalamus, pons, and cerebellum, and less commonly the cortex. Cerebral amyloid angiopathy, a common cause of lobar bleeds in older adults, bleeds in cortical structures and may present with dementia. Coagulopathy follows warfarin, heparin, DOACs, tPA, aspirin, clopidogrel, and liver disease. Hemorrhagic transformation follows ischemic stroke, through conversion from tPA or reperfusion injury. Arteriovenous malformation is a vascular abnormality of young patients. Mycotic aneurysm follows infective endocarditis vegetation deposits. Cerebral venous sinus thrombosis follows factor V Leiden, decreased AT-III, decreased protein C or S, APS, OCP, and pregnancy, with obstructed venous drainage causing venous infarction or hemorrhage near the cortex. Malignancy, including melanoma, RCC, choriocarcinoma, and thyroid carcinoma, disrupts the BBB. Causes from coagulopathy onward can bleed at any site. Primary ICH, from hypertension or amyloid angiopathy, accounts for 85% of cases and is a diagnosis of exclusion, whereas secondary ICH has an identifiable cause, such as a vascular malformation, tumor, or bleeding diathesis. Chronic hypertension causes lipohyalinosis and Charcot-Bouchard aneurysms in penetrating arterioles.

Clinical presentation

Patients have headache and cortical focal neurologic deficits, with raised ICP causing nausea and vomiting, loss of consciousness, posturing, and Cushing triad.

Diagnosis

Non-contrast CT head shows hyperdense blood in the parenchyma, bloods comprise FBC, PT/INR, aPTT, and LFTs, and CT angiogram is obtained if the patient is young, the bleed is lobar, or there is no hypertension. Hypertension shows a deep location on CT. Cerebral amyloid angiopathy is evaluated by MRI with SWI, showing lobar microbleeds. Coagulopathy is evaluated by those bloods. Hemorrhagic transformation is seen on CT head after stroke or thrombolysis as blood within the infarct. Cerebral venous sinus thrombosis is evaluated by CT or MR venography, with the empty delta sign. Malignancy is evaluated by MRI with contrast, showing an enhancing mass. Arteriovenous malformation is evaluated by CT angiogram, with catheter angiography to confirm. Mycotic aneurysm is evaluated by blood cultures, echocardiogram, and CT angiogram. The ICH score estimates 30-day mortality. Mortality reaches 50% at 30 days, with most deaths in the first 24 hours, and depends on hematoma size and location, GCS on admission, and intraventricular extension or hydrocephalus.

The ICH score predicts 30-day mortality from five items: GCS 3–4 (2 points) or 5–12 (1 point), age 80 or older, infratentorial origin, volume of 30 mL or more, estimated as A × B × C / 2 on CT, and IVH, each 1 point.

Management

Hematoma growth is limited with IV nicardipine, clevidipine, or labetalol, and reversal of any coagulopathy; if systolic is 150–220, it is lowered to a target of 140. ICP is lowered with head of bed at 30°, hypertonic saline or mannitol, and EVD for hydrocephalus. Surgical evacuation is used for a cerebellar bleed over 3 cm with brainstem compression, decreased consciousness, or hydrocephalus. Supportive care comprises ICU admission, treating fever and hyperglycemia, antiseizure medication only for seizures, and intermittent pneumatic compression. Warfarin is reversed with 4-factor PCC and IV vitamin K to an INR below 1.4, with PCC acting faster than FFP, heparin with protamine, dabigatran with idarucizumab, apixaban or rivaroxaban with andexanet alfa or 4-factor PCC, and tPA with cryoprecipitate. For aspirin or clopidogrel, the offending drug is stopped. Liver disease is treated with vitamin K and FFP. By cause, hypertension needs long-term control below 130/80; amyloid angiopathy needs antithrombotics avoided where possible; hemorrhagic transformation needs antithrombotics stopped; venous sinus thrombosis needs anticoagulation, even with hemorrhage; malignancy needs surgery, radiation, or systemic therapy; arteriovenous malformation needs surgical resection, embolization, or radiosurgery; and mycotic aneurysm needs IV ABX for infective endocarditis, with endovascular or surgical repair if ruptured.

Relationships to other conditions

Ischemic stroke. Both cause focal deficits and CT separates them. Thrombolysis and antithrombotics that treat ischemia can cause hemorrhage, and BP goals differ.

Subarachnoid hemorrhage. Blood lies in the parenchyma here and in the subarachnoid space there. AVM and mycotic aneurysm cause both, and thunderclap headache with meningeal signs favors SAH.

Dementia. Cerebral amyloid angiopathy can present with dementia as well as lobar bleeds.

Stroke complications. Hemorrhagic transformation is both a complication of ischemic stroke and a cause of intracerebral hemorrhage.

Subarachnoid hemorrhage: aneurysmal rupture and the threats that follow

Vessel rupture, from trauma or a weakened wall such as an aneurysm, bleeds into the subarachnoid space, causing meningeal irritation and raised ICP. Patients describe a thunderclap headache, and triggers include exercise, straining, and anger. Aneurysms favor the AComm, then the PComm. Severity is graded by Hunt and Hess, and management centers on securing the aneurysm and preventing delayed complications. Here, we review its causes, workup, and treatment, and consider how it relates to other stroke types.

Key points

- Thunderclap headache with meningeal signs is the classic presentation.
- If CT is negative and suspicion remains, lumbar puncture looks for xanthochromia.
- Securing the aneurysm early prevents rebleeding.
- Nimodipine prevents ischemia, and vasospasm occurs on days 3–14.

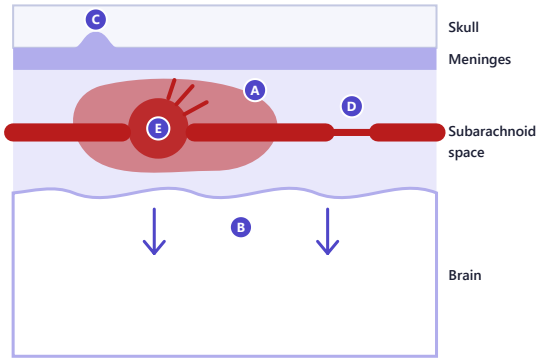


Fig. 1 | A cross-section from skull to brain. An artery in the subarachnoid space has a ruptured aneurysm, and blood spreads through the space. Lettered sites mark irritated meninges, raised pressure on the brain, a blocked arachnoid granulation, a narrowed artery, and the aneurysm that can bleed again. A, Meningeal irritation; B, Raised ICP; C, Hydrocephalus; D, Vasospasm; E, Rebleeding.

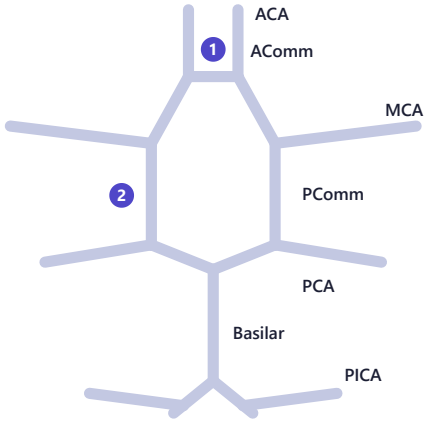


Fig. 2 | Circle of Willis with the two most common aneurysm sites marked: the anterior communicating artery and the posterior communicating artery. About 85% of aneurysms arise in the anterior circulation. 1, AComm; 2, PComm.

Pathophysiology

Trauma or a weakened wall such as an aneurysm leads to vessel rupture, triggered by exercise, straining, or anger, and blood spreads through the CSF of the subarachnoid space. Blood irritates the meninges, causing thunderclap headache, photophobia, and neck stiffness, and added blood volume raises ICP, causing nausea and vomiting, CN deficits, posturing, and Cushing triad. Blood blocks CSF absorption at the arachnoid granulations, causing hydrocephalus, and blood around arteries narrows them on days 3–14, causing vasospasm and delayed cerebral ischemia. The unsecured aneurysm can rupture again, with the highest risk early.

Etiology and mechanisms

Traumatic SAH follows blunt force or penetrating injury. Non-traumatic causes are aneurysms (saccular or berry, giant, fusiform, dissection pseudoaneurysm, and mycotic); hypertension, smoking, and sympathomimetics, with alcohol, OCP, and pregnancy less certain, which cause hemodynamic stress and inflammation at bifurcations and leave a thin or absent tunica media; AVM; and inherited conditions (Ehlers-Danlos syndrome, PKD, fibromuscular dysplasia, coarctation of the aorta, and Marfan syndrome). About 85% of aneurysms arise in the anterior circulation, most often at the AComm, with the PComm also common. Triggers are coffee or cola, physical exercise, sneezing, straining, sexual intercourse, anger, and being startled.

Clinical presentation and complications

Patients have a thunderclap headache, meningeal signs (photophobia, neck stiffness, and positive Kernig and Brudzinski signs), and raised ICP (nausea and vomiting, CN deficits, posturing, and Cushing triad). Complications are rebleeding, highest early before the aneurysm is secured; vasospasm and delayed cerebral ischemia on days 3–14; hydrocephalus; hyponatremia from SIADH or cerebral salt wasting; and seizures.

Diagnosis

Non-contrast CT head shows blood in the subarachnoid space. If CT is negative and suspicion remains, lumbar puncture looks for xanthochromia. CT angiogram finds the source, with catheter angiography if negative. The Hunt and Hess scale grades severity: grade 1 is asymptomatic, or minimal headache and slight neck stiffness; grade 2 is moderate to severe headache and neck stiffness, with no deficit except CN palsy; grade 3 is drowsiness, confusion, or mild focal deficit; grade 4 is stupor, moderate to severe hemiparesis, and possibly early decerebrate rigidity; and grade 5 is deep coma, decerebrate posturing, and a moribund appearance. A higher grade means a worse prognosis. By the Ottawa SAH rule, alert patients over 15 years with a new severe headache peaking within 1 hour are investigated if they have neck pain or stiffness, are aged 40 or older, had a witnessed LOC, developed the headache during exertion, describe a thunderclap headache, or have limited neck flexion. CT within 6 hours of onset has a sensitivity near 100%, so a normal scan can rule out SAH without a lumbar puncture; after 6 hours, sensitivity falls, and a lumbar puncture follows a normal scan.

Management

The aneurysm is secured early by endovascular coiling or surgical clipping. BP is controlled with IV nicardipine or labetalol until then, keeping systolic below 160. Nimodipine prevents ischemia. ICP is lowered with head of bed at 30°, hypertonic saline or mannitol, and EVD for hydrocephalus. Supportive care comprises ICU admission, pain relief, and reversal of any coagulopathy. Rebleeding is prevented by securing the aneurysm; vasospasm is treated with nimodipine and euvolemia, with endovascular therapy if refractory; hydrocephalus is treated with EVD; hyponatremia is treated by treating the cause and avoiding fluid restriction; and seizures are treated with antiseizure medication. By cause, trauma needs supportive care and treatment of associated injuries; vessel wall injury needs hypertension control and stopping smoking, alcohol, and sympathomimetics; AVM needs surgical resection, embolization, or radiosurgery; and inherited conditions need screening for aneurysms. The aneurysm is secured preferably within 24 hours, because rebleeding occurs in 9–17% of patients in the first 72 hours, with the highest risk in the first 6 hours. If repair is delayed, an antifibrinolytic may be given for no more than 72 hours. Until the aneurysm is secured, antithrombotics are stopped, anticoagulation is reversed, and pain is treated with paracetamol, avoiding aspirin and NSAIDs.

Relationships to other conditions

Intracerebral hemorrhage. AVM and mycotic aneurysm cause both. Blood in the subarachnoid space with meningeal signs favors SAH, and focal deficits with parenchymal blood favor ICH.

Ischemic stroke. Vasospasm and delayed cerebral ischemia on days 3–14 link the two, which is why nimodipine is given to prevent ischemia.

Stroke syndromes. Seizures appear in both complication lists, whereas rebleeding and vasospasm are listed only for SAH.

Polycystic kidney disease. PKD is grouped with the inherited conditions that need screening for aneurysms.

Seizures: lowered thresholds, the patterns they produce, and stepwise control

Excess excitation or lost inhibition lowers the seizure threshold, so cortical neurons fire in excessive synchrony. An etiology raises excitability, and triggers such as stress, sleep deprivation, and bright lights and sounds can then push the brain over the threshold. Seizures are focal or generalized, and their features follow the region involved. Management stabilizes the patient, stops prolonged seizures in timed steps, treats the cause, and selects long-term medication by seizure type. Here, we review these mechanisms, causes, patterns, and treatments, and consider how seizures relate to other neurologic conditions.

Key points

- Causes follow VITAMIN DEFG, and triggers act on top of an etiology.
- Focal seizures follow their lobe: motor in the frontal lobe, sensory in the parietal, occipital, and temporal lobes, and autonomic and psychological in the temporal lobe and insula.
- Reticular formation or thalamic involvement brings LOC, often with automatisms, and a post-ictal state.
- Absence seizures have no post-ictal state, and EEG shows a 3 Hz spike-and-wave pattern.
- A seizure lasting 5 minutes or more is status epilepticus and is treated in timed steps.

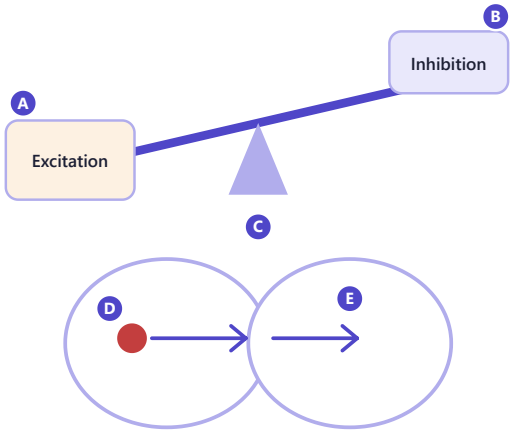


Fig. 1 | A tilted balance with excitation outweighing inhibition. Below it, firing starts at one site in one hemisphere and spreads to the other. A, Excitation; B, Inhibition; C, Synchrony; D, Focal seizure; E, Generalized seizure.

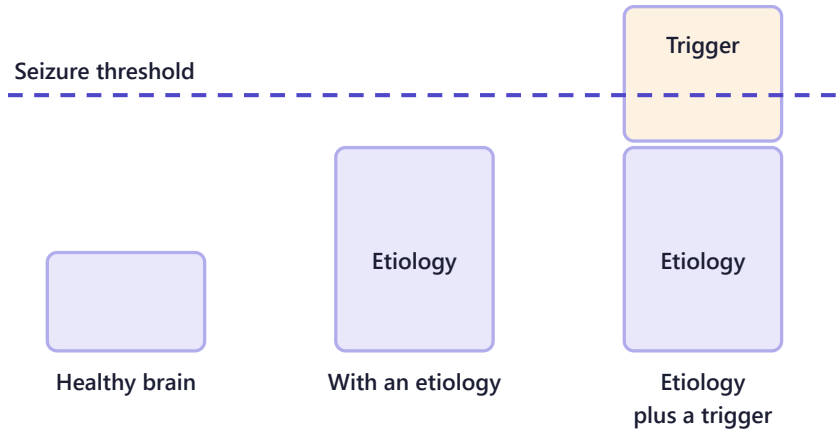


Fig. 2 | Three bars rising toward a dashed seizure threshold. A healthy brain stays far below it. An etiology raises excitability closer to it. A trigger on top of an etiology crosses it, causing a seizure.

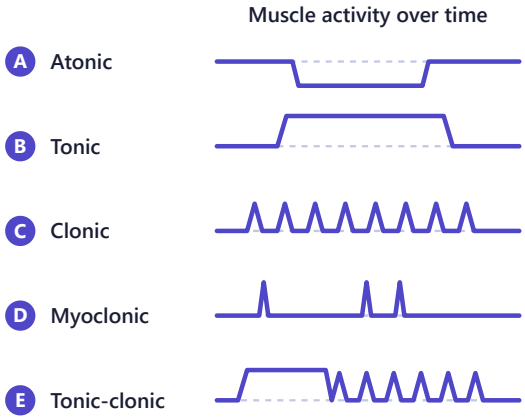


Fig. 3 | Muscle activity over time for five motor seizure types, each against a dashed line of normal tone. A, atonic: tone drops suddenly below the line. B, tonic: tone rises and stays raised. C, clonic: regular, rhythmic jerks. D, myoclonic: a few brief, irregular jerks. E, tonic-clonic: a raised tonic phase, then rhythmic clonic jerks. A, Atonic; B, Tonic; C, Clonic; D, Myoclonic; E, Tonic-clonic.

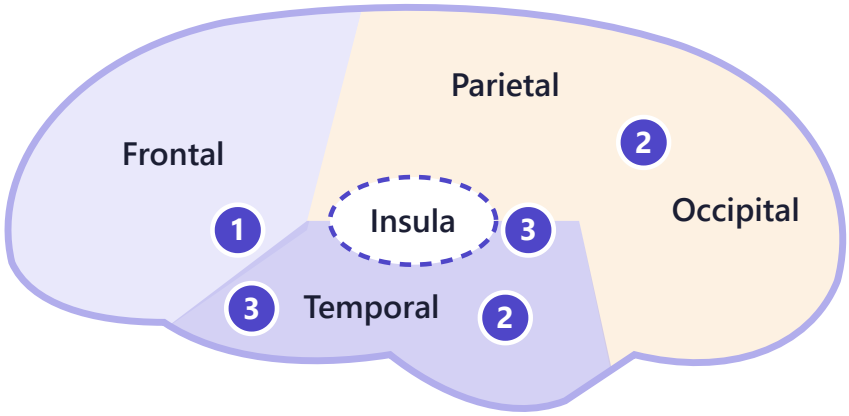


Fig. 4 | A lateral view of the brain. The frontal lobe produces motor seizures, the parietal and occipital lobes produce sensory seizures, the temporal lobe produces sensory, autonomic, and psychological seizures, and the insula produces autonomic and psychological seizures. 1, Motor (frontal); 2, Sensory (parietal, occipital, and temporal); 3, Autonomic and psychological (temporal and insular).

Pathophysiology

Glutamate opens Na+ and Ca2+ channels, causing depolarization, and GABA opens Cl– channels, causing hyperpolarization. When excitation wins, the seizure threshold falls and neurons fire together rather than independently. In a focal seizure, firing starts in one region, so symptoms match that region; a focal seizure that spreads to both hemispheres is a focal to bilateral tonic-clonic seizure. In a generalized seizure, firing rapidly engages networks in both hemispheres.

Etiology

Vascular causes are AIS, ICH, SAH, PRES from increased BP, and anoxic brain injury from global hypoxia. Infectious causes are meningitis, encephalitis, and brain abscess, and traumatic causes are epidural and subdural hematomas. Autoimmune causes are SLE and paraneoplastic syndrome, leading to autoimmune encephalitis. Metabolic causes are decreased vitamin B1, hypoglycemia or hyperglycemia, hyponatremia or hypernatremia, hypocalcemia, hypomagnesemia, hypophosphatemia, increased urea and metabolic waste, increased ammonia, and increased thyroid hormones. Iatrogenic causes are opioids, TCAs, isoniazid, salicylate, cocaine, cefepime, amphetamine, metronidazole, penicillin, bupropion, lithium, and benzodiazepine or alcohol withdrawal. Idiopathic causes are epilepsy, with non-adherence to medications, West syndrome, Lennox-Gastaut syndrome, and juvenile myoclonic epilepsy. Neoplastic causes are glioblastoma, meningioma, and metastatic cancer, and degenerative or demyelinating causes are Alzheimer disease and MS. Eclampsia, fever, and genetic causes complete the list. Triggers are stress, sleep deprivation, and bright lights and sounds.

Febrile seizures occur in children aged 6 months to 5 years with fever of 38 °C or more and no CNS infection. Most are simple, a single generalized seizure of 15 minutes or less; complex febrile seizures have focal features, last 15 minutes or more, or recur within 24 hours. Most resolve without complications, although a subset carries a higher risk of epilepsy.

Clinical presentation

A seizure can pass through an aura, an ictal event, and a post-ictal state. Motor seizures take five forms: atonic seizures are a sudden loss of muscle tone, in a limb or the whole body; tonic seizures are sustained contraction with stiff posturing; clonic seizures are repeated, rhythmic jerking of a body part, from brief contractions alternating with silent periods; myoclonic seizures are sudden, brief, shock-like, irregular jerks; and tonic-clonic seizures are a tonic phase of stiffening, often with a cry, then a clonic phase of rhythmic jerking. Focal motor seizures arise in the frontal lobe and are atonic, tonic, clonic, or myoclonic. In a Jacksonian march, twitching or tingling moves to neighboring body parts in the order of the homunculus, such as from the hand up the arm, as the seizure spreads along the cortex. Focal sensory seizures arise in the parietal, occipital, and temporal lobes and cause pain or paresthesia, and visual, auditory, smell, or taste hallucinations. Focal autonomic and psychological seizures arise in the temporal lobe and insula and cause increased HR and BP, urinary incontinence, salivation, and sweating; fear, dread, anxiety, or pleasure; and déjà vu, depersonalization, and derealization. If the reticular formation or thalamus is involved, there is LOC, often with automatisms such as rapid blinking, chewing, and lip smacking, and a post-ictal state, and if not, there is neither. Auras include déjà vu and nausea.

Generalized seizures and the post-ictal state

A focal seizure that spreads to both hemispheres stays focal, as a focal to bilateral tonic-clonic seizure, the most common seizure type. Motor seizures are atonic, tonic, clonic, myoclonic, or tonic-clonic, with LOC and a post-ictal state, and may feature an involuntary cry or scream, upgaze, pooling of oral secretions, tongue biting, urinary or fecal incontinence, and increased HR and BP. Absence seizures, usually at ages 5–15, are non-motor, with staring, daydreaming, automatisms, and no loss of muscle tone. A post-ictal state follows focal seizures with impairment, generalized motor seizures, and focal to bilateral tonic-clonic seizures, and features confusion, lethargy, and amnesia. Todd paralysis is temporary weakness of a limb or one side, usually the side involved in the seizure, lasting minutes to 1–2 days and resolving without lasting damage; it follows focal or generalized tonic-clonic seizures and can look like a stroke, and mistaking it for one can lead to unneeded thrombolysis.

Diagnosis

Glucose and vital signs come first, so that hypoglycemia is treated. Bloods comprise FBC, electrolytes, urea, LFTs, and drug levels. Non-contrast CT head is the first imaging after a new seizure, EEG classifies the seizure, and MRI finds structural causes. A normal EEG between seizures does not exclude epilepsy. By cause, vascular disease is assessed with CT head, CT angiogram, and MRI for PRES; infection with lumbar puncture, CSF analysis, and blood cultures; trauma with non-contrast CT head; autoimmune disease with ANA and autoimmune encephalitis antibodies in blood and CSF; metabolic disease with glucose, sodium, calcium, magnesium, phosphate, urea, ammonia, and TFTs; drugs and epilepsy with a toxicology screen, antiseizure medication levels, and EEG; tumors with MRI with contrast; degenerative or demyelinating disease with MRI and cognitive assessment; eclampsia with a pregnancy test, BP, and urine protein; fever with temperature and a search for infection; and genetic causes with family history and genetic testing.

Prolactin rises 10–20 minutes after a generalized convulsive seizure but not after PNES; it also rises after convulsive syncope, and a normal level does not confirm PNES.

Seizure mimics

Syncope is favored by a trigger, a prodrome of lightheadedness, brief jerks, and rapid recovery, and is assessed with ECG and orthostatic BP. Psychogenic non-epileptic seizures show closed eyes, side-to-side head movement, asynchronous limb movement, and no post-ictal state, and are confirmed by video EEG. Cardiac arrhythmia is suggested by palpitations, exertional collapse, or a family history of sudden death. TIA causes negative symptoms such as weakness or numbness, without a march, whereas migraine with aura spreads over minutes and is followed by headache. Sleep disorders occur only during sleep, and panic attacks and movement disorders preserve awareness.

Management

During a seizure, the patient is placed in the recovery position with nothing in the mouth, the airway, oxygen, and IV access are secured, and glucose is given if low, with thiamine if alcohol use is likely. Status epilepticus is 5 minutes or more of continuous seizure activity, clinical or on EEG, or recurrent seizures without recovery between them; convulsive status epilepticus has generalized tonic-clonic movements with impaired mental status, nonconvulsive status epilepticus has impaired mental status with seizure activity on EEG but few or no motor signs, and refractory status epilepticus continues despite appropriate antiseizure drugs. At 0–5 minutes, the patient is stabilized and glucose is checked; at 5–20 minutes, IV lorazepam or IM midazolam is given; at 20–40 minutes, IV levetiracetam, valproate, or fosphenytoin is given; and after 40 minutes, the patient needs intubation, ICU care, and an anesthetic infusion with EEG. The cause is treated: the stroke, with BP

lowered in PRES; ABX or antivirals, draining any brain abscess; neurosurgical evacuation of hematomas if needed; corticosteroids and IVIG, treating any underlying tumor; correction of metabolic abnormalities, with thiamine for decreased vitamin B1; stopping a drug, benzodiazepines for withdrawal, and restarting missed medications; surgery, radiation, or chemotherapy; treatment of degenerative or demyelinating disease; magnesium sulfate and delivery in eclampsia; and treatment of fever and its cause. Long-term medication follows seizure type: levetiracetam, lamotrigine, or carbamazepine for focal seizures; valproate, levetiracetam, or lamotrigine for generalized motor seizures; ethosuximide for absence; valproate or levetiracetam for juvenile myoclonic epilepsy; ACTH or vigabatrin for West syndrome; and valproate, lamotrigine, or rufinamide for Lennox-Gastaut syndrome. Refractory epilepsy may need surgery, vagus nerve stimulation, or a ketogenic diet. By mechanism, carbamazepine, phenytoin, lamotrigine, and lacosamide block sodium channels; ethosuximide, and weakly valproate, block T-type calcium channels; benzodiazepines and phenobarbital act at the GABA-A receptor, and vigabatrin blocks GABA breakdown; levetiracetam binds SV2A; and valproate and topiramate have multiple mechanisms. All patients avoid triggers and take medications as prescribed, and valproate is avoided where possible in those who could become pregnant, because it is teratogenic.

Phenytoin causes gingival hyperplasia, nystagmus and ataxia at high levels, SJS, and fetal hydantoin syndrome, and follows zero-order kinetics at high doses. Carbamazepine causes agranulocytosis, aplastic anemia, hyponatremia, and SJS, so patients of Han Chinese ancestry are tested for HLA-B*1502. Valproate raises the risk of neural tube defects about 10-fold and causes hepatotoxicity, especially under age 2.

Relationships to other conditions

Stroke. AIS, ICH, and SAH are vascular causes of seizures, and seizures are listed among stroke complications, so both belong in the first assessment. Non-contrast CT head helps with both, and Todd paralysis after a seizure is the feature most likely to be confused with a stroke deficit.

Focal seizures with impairment vs absence seizures. Both can cause staring and automatisms. An aura and a post-ictal state favor a focal seizure, whereas no post-ictal state and a 3 Hz spike-and-wave pattern on EEG favor absence.

Focal motor vs generalized motor seizures. Both can be atonic, tonic, clonic, or myoclonic. A Jacksonian march favors a focal seizure, whereas LOC with an involuntary cry, upgaze, or tongue biting favors a generalized one, and a focal seizure can spread as a focal to bilateral tonic-clonic seizure.

Epilepsy vs provoked seizures. Metabolic, toxic, and withdrawal causes are corrected directly, whereas epilepsy needs long-term antiseizure medication, and non-adherence is itself a cause.

Multiple sclerosis *and* ***Alzheimer disease***. Both are degenerative or demyelinating causes of seizures, so treatment targets the underlying disease as well.

Eclampsia. A seizure in pregnancy with increased BP and urine protein favors eclampsia, which is treated with magnesium sulfate and delivery rather than standard antiseizure medication.

Multiple sclerosis: immune demyelination and lesions scattered in space and time

Autoreactive T cells release cytokines that bring macrophages across the BBB, where they destroy myelin and oligodendrocytes are injured, forming myelin plaques that slow nerve conduction. Environmental and genetic factors contribute, and the typical patient is a woman aged 20–40 years. Symptoms depend on the site of each plaque, and diagnosis rests on showing dissemination in space, usually with dissemination in time. Management treats acute relapses, modifies the disease, and controls symptoms. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how multiple sclerosis relates to other neurologic conditions.

Key points

- IL-1, IL-6, and TNF-α increase adhesion molecules, vasodilation, vascular permeability, and chemotaxis.
- Macrophages destroy myelin and oligodendrocytes are injured, forming myelin plaques.
- MRI comes first, and the 2024 McDonald criteria need dissemination in space, plus dissemination in time or a substitute.
- CSF oligoclonal bands support the diagnosis if MRI is not diagnostic.
- Acute relapses are treated with IV methylprednisolone, and disease-modifying therapy is long term.

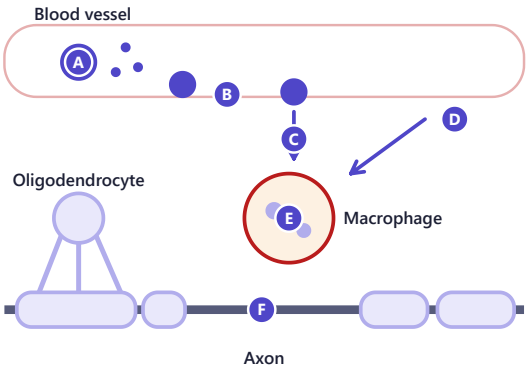


Fig. 1 | A blood vessel above an axon. T cells release cytokines in the vessel, immune cells stick to its wall and cross it, and a macrophage below destroys myelin beside an injured oligodendrocyte, leaving a bare stretch of axon. A, Cytokine release; B, Adhesion molecules; C, Vasodilation and permeability; D, Chemotaxis; E, Phagocytosis; F, Slowed conduction.

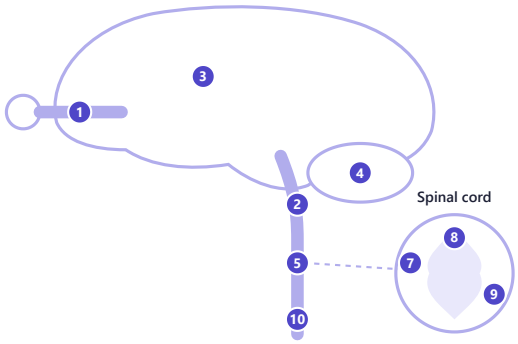


Fig. 2 | A side view of the central nervous system with numbered lesion sites: optic nerve, brainstem, corticobulbar tracts, cerebellum, and cervical cord, beside a cross-section of the spinal cord showing the corticospinal tract, dorsal columns, and spinothalamic tract. 1, Optic neuritis; 2, MLF damage; 3, Pseudobulbar palsy; 4, Cerebellar signs; 5, Lhermitte sign; 7, Upper motor neuron signs; 8, Dorsal column loss; 9, Spinothalamic loss; 10, Bladder dysfunction.

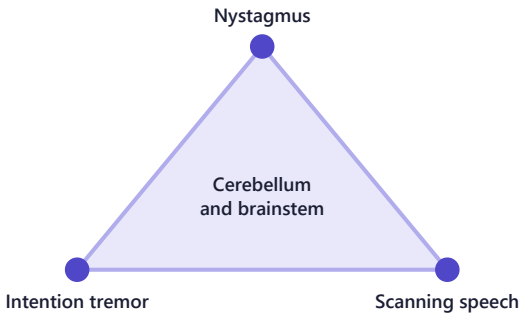


Fig. 3 | A triangle with one sign of the Charcot triad at each tip: nystagmus at the top, intention tremor at the bottom left, and scanning speech at the bottom right, with the cerebellum and brainstem in the middle.

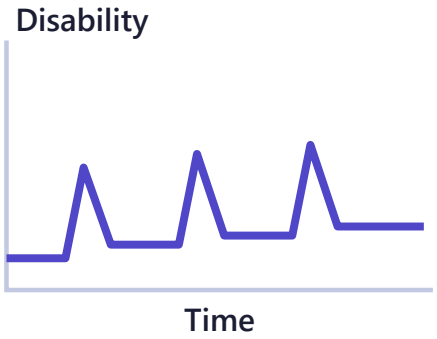


Fig. 4 | Disability over time in relapsing-remitting MS: relapses rise and fall back, leaving small steps. Relapsing-remitting: most people; relapses over days to weeks, with stability between them

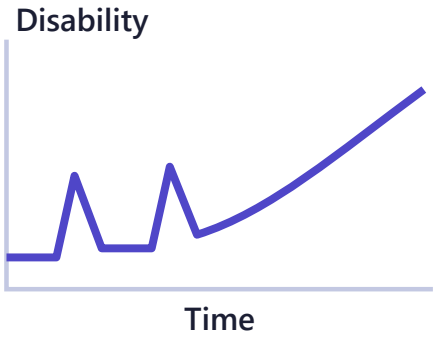


Fig. 5 | Disability over time in secondary progressive MS: early relapses, then a steady climb. Secondary progressive: gradual worsening after 10–15 years, with or without relapses

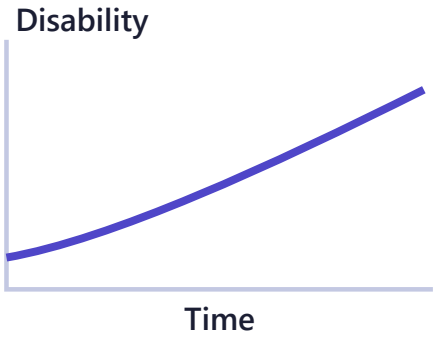


Fig. 6 | Disability over time in primary progressive MS: a steady climb from onset. Primary progressive: a small subset; worsening from onset

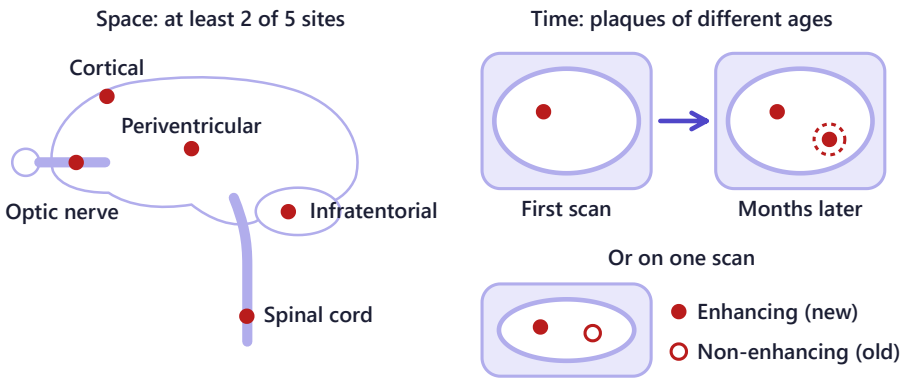


Fig. 7 | Left, space: a side view of the brain, brainstem, spinal cord, and optic nerve with a plaque at each of the 5 sites: cortical or juxtacortical, periventricular, infratentorial, spinal cord, and optic nerve. Right, time: a first scan with one plaque and a scan months later with a new plaque; or one scan with an enhancing, new plaque and a non-enhancing, old plaque.

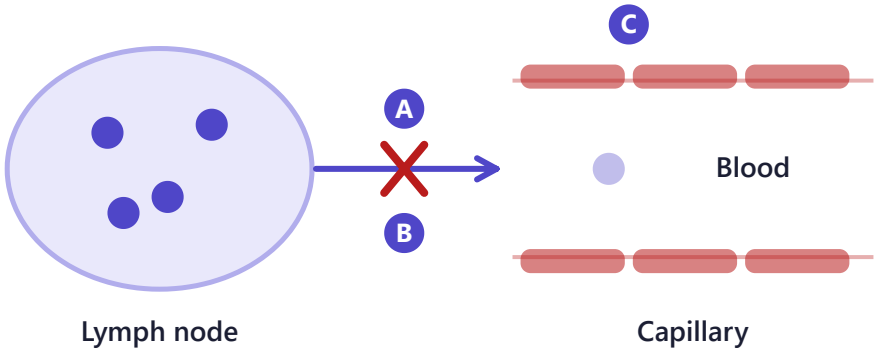


Fig. 8 | A lymph node on the left releases lymphocytes along an arrow toward a capillary on the right. Fingolimod blocks the exit, and the capillary wall is lined with endothelial cells joined tightly together. A, S1P and lymph nodes; B, Fingolimod; C, S1P and capillaries.

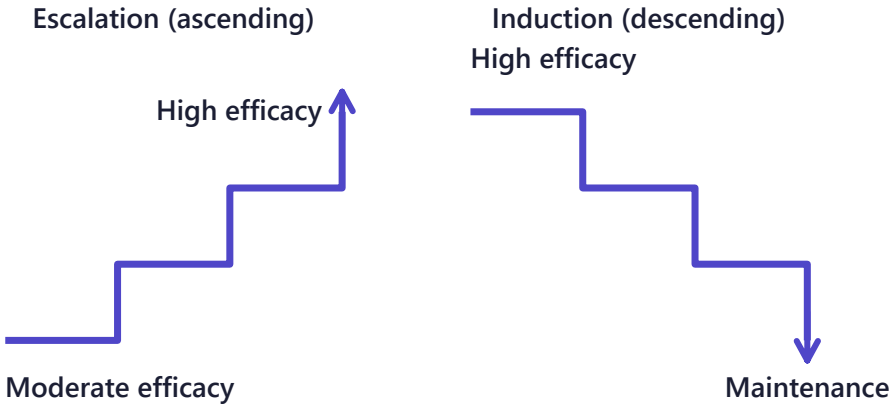


Fig. 9 | Two mirrored staircases. Escalation climbs from moderate-efficacy to high-efficacy drugs. Induction starts with high-efficacy or induction drugs and steps down to maintenance.

Pathophysiology

In a genetically susceptible person, autoreactive T cells release IL-1, IL-6, and TNF-α. These cytokines increase adhesion molecules, so immune cells stick to the vessel wall, and cause vasodilation and increased vascular permeability, so immune cells cross the BBB. Chemotaxis draws macrophages to the site, where they destroy myelin and oligodendrocytes are injured, causing myelin plaque formation. Demyelinated axons conduct poorly, so symptoms depend on the site of the plaque.

Etiology

Environmental factors are EBV, HHV-6, decreased vitamin D, smoking, obesity, and higher latitude. Genetic

and demographic factors are HLA-DR2 and being female and aged 20–40 years.

Clinical presentation

Optic neuritis causes decreased visual acuity, reduced color vision, and RAPD, and MLF damage causes bilateral INO. Pseudobulbar palsy causes decreased chewing, increased jaw jerk, absent facial expressions, dysphagia, dysphonia, increased gag reflex, dysarthria, and spastic tongue. Cerebellar involvement causes ataxia and intention tremor. Lhermitte sign is an electric shock sensation down the spine on neck flexion, and in Uhthoff phenomenon, symptoms worsen with heat. Upper motor neuron signs are muscle weakness, increased tone, hyperreflexia, and positive Babinski sign. Dorsal column damage causes loss of fine touch, vibration, and conscious proprioception, and spinothalamic damage causes loss of pain and temperature. Bladder dysfunction causes incontinence with SNS damage and retention with PNS damage. Most people have a relapsing-remitting course, with relapses over days to weeks and stability between them. Gradual worsening after 10–15 years, with or without relapses, marks the secondary progressive course, and a small subset has a primary progressive course, with worsening from onset. Each course is also described as active or not active, by relapses or new MRI lesions, and progressive courses as progressing or not, and CIS is a first episode that does not yet meet dissemination in time.

The Charcot triad of intention tremor, dysarthria with scanning speech, and nystagmus reflects cerebellar and brainstem involvement, and tremor affects 58–77% of patients.

Diagnosis

MRI brain and spine with gadolinium shows T2 plaques in periventricular, cortical or juxtacortical, infratentorial, spinal cord, and optic nerve sites, with active plaques enhancing. The McDonald criteria diagnose MS when plaques are spread in space, across different sites, and in time, of different ages, once other causes are excluded. The 2024 McDonald criteria need dissemination in space, with plaques in at least 2 of the 5 sites, and dissemination in time, with new plaques over time or enhancing and non-enhancing plaques together. CSF oligoclonal bands, kappa free light chains, 4 or more sites, or 6 plaques with a central vein sign can substitute for time. With only 1 site, the diagnosis needs dissemination in time or positive CSF, plus a central vein sign or a paramagnetic rim plaque. Incidental MRI findings qualify with 2 or more sites, plus dissemination in time, positive CSF, or a central vein sign. The 2017 criteria used 4 sites, without the optic nerve, and always needed dissemination in time or oligoclonal bands. Other causes must be excluded first. Lumbar puncture tests CSF for oligoclonal bands or kappa free light chains if MRI does not meet the criteria. Extra tests are visual evoked potentials, which show slowed conduction in the optic nerve, OCT, which shows an inter-eye difference in retinal thickness, and bloods to exclude mimics: FBC, B12, TFTs, ESR, ANA, syphilis serology, HIV, and AQP4 and MOG antibodies.

MS mimics

NMOSD causes severe optic neuritis and long spinal cord lesions, with AQP4 antibodies, and MOGAD causes bilateral optic neuritis, often with disc swelling, with MOG antibodies. ADEM is one episode after an infection, with encephalopathy, usually in children. B12 deficiency causes dorsal column and corticospinal signs with anemia. SLE is suggested by rash, joint pain, and kidney disease, with ANA, and neurosarcoidosis by lung disease and CN palsies, with CXR and serum ACE. Neurosyphilis, HIV, and Lyme disease are found by serology. Small vessel disease occurs with older age and vascular risk factors, with no oligoclonal bands.

Management

An acute relapse is treated with IV methylprednisolone for 3–7 days, with plasma exchange if refractory. Chronic disease-modifying therapy includes the injectable drugs interferon beta and glatiramer acetate, the oral drugs dimethyl fumarate, teriflunomide, and fingolimod, and the high-efficacy drugs natalizumab, ocrelizumab, and alemtuzumab. Natalizumab can cause PML, so JC virus status is checked first. S1PR1 signaling lets lymphocytes leave lymph nodes for the blood. Fingolimod binds S1PR1, which is then degraded, so lymphocytes stay in lymph nodes and fewer reach the CNS. On endothelial cells, S1PR1 tightens junctions and lowers capillary permeability, although prolonged fingolimod can increase permeability, and S1P receptors on the SA and AV nodes explain fingolimod-induced bradycardia. Siponimod, ozanimod, and ponesimod are more selective. Two strategies guide treatment: escalation, or ascending therapy, starts with moderate-efficacy drugs and moves up if the disease stays active, whereas early high-efficacy treatment uses natalizumab, S1P receptor modulators, or anti-CD20 drugs from the start, and induction, or descending therapy, uses short-term mitoxantrone, alemtuzumab, cladribine, or stem cell transplant before maintenance. Early high-efficacy treatment suits young people with active disease, many T2 lesions, and no irreversible disability. Symptoms are controlled with baclofen for spasticity, oxybutynin for incontinence, catheterization for retention, gabapentin for neuropathic pain, and avoiding heat in Uhthoff phenomenon.

Relationships to other conditions

Stroke syndromes. Both cause focal deficits, including upper motor neuron signs and brainstem signs. Deficits that follow one arterial territory and appear suddenly favor stroke, whereas plaques separated in space and time favor multiple sclerosis.

Posterior cord syndrome. Both cause loss of vibration and proprioception, and B12 deficiency is a cause of posterior cord syndrome, which is why B12 is checked. Low B12 favors deficiency, whereas MRI plaques and CSF oligoclonal bands favor multiple sclerosis.

AQP4 antibody disease and ***MOGAD***. These are mimics that also cause demyelination, so positive antibodies point away from multiple sclerosis.

Seizures. Multiple sclerosis is a demyelinating cause of seizures, so treatment targets the underlying disease as well.

Neuromyelitis optica spectrum disorder: an antibody-mediated astrocytopathy of the optic nerves, spinal cord, and area postrema

Neuromyelitis optica spectrum disorder is an autoimmune disease of the CNS in which AQP4-IgG binds AQP4, the water channel of astrocyte end-feet, activates complement, and destroys astrocytes, with secondary damage to oligodendrocytes and axons. It causes severe optic neuritis, LETM, area postrema syndrome, and brainstem, diencephalic, and cerebral syndromes, and follows a relapsing course in which disability builds with each attack. Diagnosis rests on the 2015 international consensus criteria, which require one core clinical characteristic with AQP4-IgG, or two with supporting MRI findings without it. Attacks are treated with IV methylprednisolone and PLEX, and relapses are prevented with long-term immunosuppression, including B-cell depletion, IL-6 receptor blockade, and complement inhibition. Here, we review its pathophysiology, risk factors, clinical features, diagnosis, and treatment.

Key points

- AQP4-IgG binds AQP4 on astrocyte end-feet and kills astrocytes through complement, so demyelination is secondary.
- Lesions favor regions dense in AQP4: the optic nerves, spinal cord, area postrema, and tissue around the third and fourth ventricles.
- The 2015 criteria require one core clinical characteristic with AQP4-IgG, or two without it, including optic neuritis, myelitis with LETM, or area postrema syndrome.
- Disability builds with each relapse, and a progressive course is very rare.
- Attacks are treated with IV methylprednisolone and often PLEX, long-term immunosuppression follows the first attack, and interferon beta and natalizumab can worsen the disease.

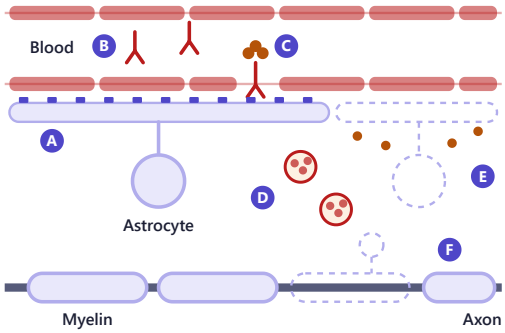


Fig. 1 | A capillary runs across the top, lined by endothelial cells, with a gap in its lower wall. Below it, an astrocyte end-foot carries AQP4 water channels, A. AQP4 antibodies, B, in the blood cross the gap and bind AQP4, where complement, C, gathers. Neutrophils and eosinophils, D, enter the tissue. On the right, the end-foot and its astrocyte are lost and glutamate, E, builds up. Along the bottom, an axon loses a stretch of myelin and its oligodendrocyte, F. A, AQP4; B, AQP4-IgG; C, Complement; D, Neutrophils and eosinophils; E, Glutamate; F, Oligodendrocytes and axons.

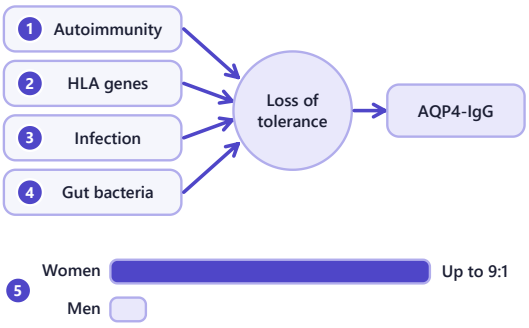


Fig. 2 | Other autoimmune disease, 1, HLA genes, 2, infection or vaccination, 3, and gut bacteria, 4, feed a loss of tolerance to AQP4 that produces AQP4 antibodies. Below, women, 5, outnumber men by up to 9 to 1. 1, Other autoimmune disease; 2, HLA genes; 3, Infection or vaccination; 4, Gut bacteria; 5, Women.

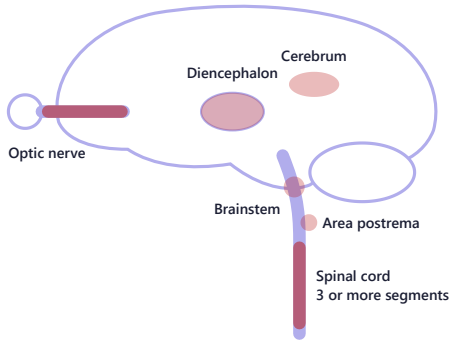


Fig. 3 | A side view of the central nervous system with the six core regions of NMOSD highlighted: the optic nerve, the diencephalon, the cerebrum, the brainstem around the fourth ventricle, the area postrema of the dorsal medulla, and a long spinal cord lesion spanning 3 or more segments.

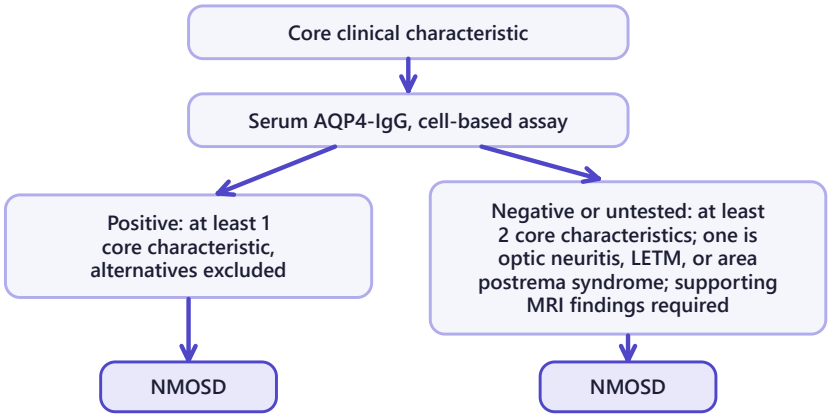


Fig. 4 | The 2015 NMOSD criteria: start with a core clinical characteristic and serum AQP4-IgG on a cell-based assay. If positive, at least 1 core characteristic with alternatives excluded is NMOSD. If negative or untested, at least 2 core characteristics are needed, one being optic neuritis, myelitis with LETM, or area postrema syndrome, with supporting MRI findings. AQP4-IgG or lesions alone, without an attack, are not NMOSD.

Pathophysiology

Antibodies against AQP4, the water channel on astrocyte end-feet, activate complement and destroy astrocytes, and the inflammation that follows damages oligodendrocytes and axons in the optic nerves, spinal cord, and area postrema.

AQP4 is a water channel whose tetramers cluster into orthogonal arrays of particles on the astrocyte end-feet that surround vessels, and it is dense in the optic nerve, spinal cord, and area postrema and other regions around the ventricles. AQP4-IgG is made by plasmablasts, which IL-6 keeps alive, and reaches AQP4 where inflammation opens the BBB; in animal models, the area postrema is one route of entry. The antibody binds the extracellular loops of AQP4, favoring clustered AQP4, and activates the classical complement pathway, whose membrane attack complex kills astrocytes, so lesions lose AQP4 and GFAP. Neutrophils, eosinophils, and macrophages then infiltrate and can cause necrosis.

Binding also pulls AQP4 into the cell together with EAAT2, the glutamate transporter, so glutamate builds up. Oligodendrocytes and axons are damaged as bystanders, so demyelination is secondary and follows astrocyte loss rather than starting it, and axon damage often comes first. AQP4 is also found in the kidney, stomach, and skeletal muscle, but these are mostly spared, perhaps because CD59 limits complement outside the CNS.

Etiology

NMOSD is associated with other autoimmune diseases, including SLE, Sjögren syndrome, myasthenia gravis, autoimmune thyroid disease, and antiphospholipid syndrome, as well as celiac disease and sarcoidosis. HLA-DRB1*03:01 and complement C4 copy number confer risk, in a pattern closer to SLE than to MS. Infection or vaccination precedes 20–30% of attacks, and part of AQP4 resembles a protein of *Clostridium perfringens*, which is overabundant in the gut of some patients, suggesting molecular mimicry.

Women are affected up to 9 times as often as men in adults, and about 3 times as often in children. Onset is usually at 30–40 years, and fewer than 5% of patients are children. NMOSD is more common with Asian or African ancestry, accounting for 1–2% of demyelinating disease in the United States and Italy but 13.7% in India and over 30% in Thailand, and its prevalence is about 0.7–10 per 100000. Smoking, low vitamin D, low physical activity, and a diet low in fruit and vegetables may add risk.

Clinical presentation

Six core clinical characteristics each implicate one region. Optic neuritis is severe, often bilateral or reaching the chiasm, can cause altitudinal field defects, and can leave acuity at 20/200 or worse. Acute myelitis causes a complete cord syndrome, with paraplegia and paroxysmal tonic spasms. Area postrema syndrome, in 16–43% of patients, causes intractable hiccups, nausea, or vomiting and is often seen first by gastroenterologists and misdiagnosed as gastroparesis or reflux. Acute brainstem syndrome arises from lesions around the fourth ventricle, diencephalic syndrome causes symptomatic narcolepsy with thalamic or hypothalamic lesions, and cerebral syndrome occurs with NMOSD-typical brain lesions. Optic neuritis and myelitis can occur together or weeks apart.

Most patients, 80–90%, relapse, about 60% within a year of a first LETM, although relapses can be over a decade apart, and the 5–10% with a single attack are more often without AQP4-IgG. Disability builds with each attack, and a progressive course is very rare. Unlike in MS, relapses increase during pregnancy and after delivery, and recovery is worse with older onset, about 85% at age 20 and 60% at age 60. In relapsing disease, about 60% become blind in one or both eyes and about half develop paraplegia or monoplegia, and respiratory failure in severe attacks raises mortality. Chronic pain, including neuropathic pain, painful attacks, and headache, affects 80–100% of patients, and dysautonomia, often worse than in MS, bladder and bowel dysfunction, depression, and anxiety are common.

Diagnosis

Evaluation starts with a core clinical characteristic, then serum AQP4-IgG on a cell-based assay, MRI of the brain, orbits, and spinal cord, and exclusion of alternatives, with red flags prompting a wider search. Under the 2015 international consensus criteria, one core clinical characteristic suffices with AQP4-IgG once alternatives are excluded. Without AQP4-IgG, or when it is untested, at least 2 core characteristics in different regions are needed, from one or more attacks; one must be optic neuritis, myelitis with LETM, or area postrema syndrome with a dorsal medulla lesion, and supporting MRI findings are required. AQP4-IgG without an attack, or typical lesions without symptoms, is not NMOSD, and seronegative patients are retested after a relapse and before B-cell or antibody-targeted treatment. Area postrema syndrome requires AQP4-IgG and acute or subacute intractable nausea, vomiting, or hiccups, alone or together, for at least 48 hours, or sooner with a new area postrema lesion on MRI, with other causes excluded, and little response to antiemetics is a clue.

Cell-based assays are preferred, with a sensitivity of about 77% and 0.1% false positives, whereas ELISA and immunofluorescence have a sensitivity of 63–64% and more false positives, often at low titer, and titer does not predict severity. MRI typically shows LETM over 3 or more vertebral segments, centered on the gray matter, with swelling, T1 hypointensity, and enhancement, sometimes extending into the brainstem, and bright spotty lesions, although 7–14% of first myelitis attacks are shorter, especially early or in remission. Optic nerve lesions are bilateral, posterior, reach the chiasm, or span over half the nerve's length, and brain lesions sit where AQP4 is dense, around the third and fourth ventricles, in the dorsal medulla, thalamus, hypothalamus, periaqueductal gray, and corpus callosum. Nonspecific white matter lesions occur in up to half, and over 30% meet McDonald criteria, so they do not exclude NMOSD.

CSF OCB are usually absent, and their presence is a red flag, whereas over 50 white cells/μL, in about 35%, or neutrophils or eosinophils favor NMOSD over MS. CSF GFAP is raised for days to weeks after an attack. OCT shows more RNFL thinning than in MS, and visual fields show a central or centrocecal scotoma, with reduced color vision. Short, peripheral cord lesions, Dawson fingers, OCB, and a progressive course favor MS, confirmed with MRI and CSF; disc edema, anterior optic nerve lesions, conus lesions, ADEM, and men or children favor MOGAD, confirmed with MOG-IgG; and response to antiemetics favors gastroparesis, reflux, or food poisoning over area postrema syndrome, which brain MRI can show. LETM also follows infection, granulomatous disease, tumors, paraneoplastic syndromes, ADEM, spinal cord infarction, and dural arteriovenous fistulas.

Management

Attacks are treated with IV methylprednisolone 1 g daily for 2–7 days. Steroids alone rarely restore baseline, so PLEX is often needed, from the start or if steroids fail, and treatment within 14 days of onset more often returns patients to baseline. Antiemetics rarely help area postrema syndrome, but IV methylprednisolone or PLEX stops most attacks within 2 days.

Because each relapse adds disability, long-term immunosuppression starts after the first attack. B cells are depleted with RTX, an anti-CD20 antibody given every 6 months and a first-line drug, or inebilizumab, an anti-CD19 antibody. Satralizumab and tocilizumab block the IL-6 receptor, and eculizumab blocks complement C5 and is given after meningococcal vaccination. AZA with prednisone is also first line, and MMF, methotrexate, and tacrolimus are alternatives. Satralizumab, tocilizumab, eculizumab, and inebilizumab have proven benefit mainly in AQP4-IgG-positive disease. Interferon

beta, natalizumab, and other MS disease-modifying drugs are avoided. With RTX, immunoglobulin levels are checked before treatment and yearly, because low levels raise infection risk. Relapse risk rises in pregnancy, and RTX has been continued safely in case reports.

Relationships to other conditions

Multiple sclerosis. Both are relapsing CNS diseases that affect more women and cause optic neuritis and myelitis, and NMOSD was long considered a variant of multiple sclerosis. AQP4-IgG, LETM over 3 or more segments, bilateral or chiasmal optic neuritis, area postrema syndrome, CSF pleocytosis with neutrophils or eosinophils, and severe attacks with poor recovery favor NMOSD, whereas short, peripheral cord lesions, Dawson fingers, OCB, and a progressive course favor multiple sclerosis, whose interferon beta and natalizumab can worsen NMOSD.

MOGAD. Both are antibody-mediated diseases that cause optic neuritis and LETM, and MOG-IgG was first thought to mark AQP4-negative NMO. AQP4-IgG, area postrema syndrome, posterior and chiasmal optic nerve lesions, other autoimmune disease, and more women favor NMOSD, whereas disc edema, anterior optic nerve lesions, conus lesions, ADEM, children and men, and a monophasic course favor MOGAD, which is a demyelinating disease rather than an astrocytopathy.

Vision loss and red eye. Optic neuritis causes painful vision loss in multiple sclerosis, NMOSD, and MOGAD. Severe, bilateral, or chiasmal optic neuritis with poor recovery favors NMOSD, whereas unilateral optic neuritis with good recovery favors multiple sclerosis.

MOG antibody-associated disease: an antibody-mediated demyelinating disease distinct from MS and NMOSD

MOG antibody-associated disease is an inflammatory demyelinating disease of the CNS in which antibodies target MOG, a glycoprotein on the outermost layer of myelin and on oligodendrocytes. It most often causes optic neuritis, myelitis, or ADEM, alone or in combination, and follows a monophasic or relapsing course without a progressive phase. Diagnosis requires a demyelinating attack, serum MOG-IgG on a live cell-based assay, and exclusion of better diagnoses, including MS. Attacks are treated with corticosteroids, and relapsing disease with immunosuppression. Here, we review its pathophysiology, triggers, clinical syndromes, diagnosis, and treatment.

Key points

- MOG is a transmembrane glycoprotein on the outermost myelin lamellae, with an extracellular, N-terminal, immunoglobulin-like domain.
- MOGAD is distinct from MS and from AQP4-positive NMOSD, an astrocytopathy.
- Optic neuritis, myelitis, and ADEM are the most common syndromes.
- A clear positive titer of 1:100 or more needs no supporting features.
- MS drugs appear ineffective; relapses are prevented with AZA, MMF, IVIG, or corticosteroids.

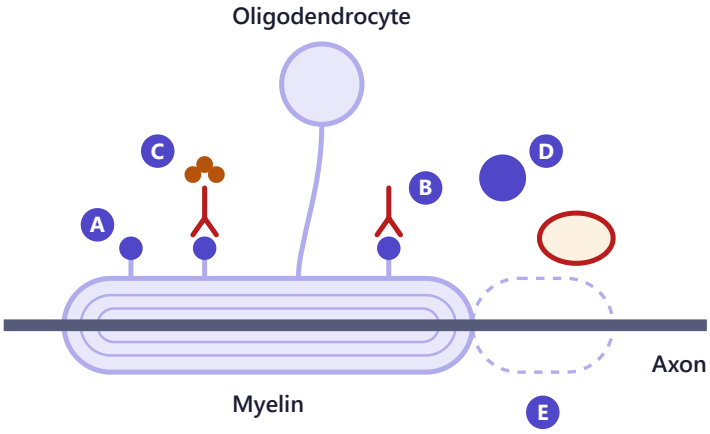


Fig. 1 | An oligodendrocyte wraps an axon in myelin. MOG, A, sits on the outermost layer of myelin with its immunoglobulin-like end facing out. MOG antibodies, B, bind it and recruit complement, C. T cells and macrophages, D, join in, and a stretch of myelin is lost, E. A, MOG; B, MOG-IgG; C, Complement; D, T cells and macrophages; E, Demyelination.

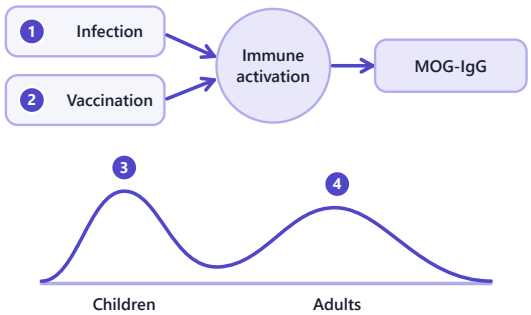


Fig. 2 | Infection, 1, or rarely vaccination, 2, drives immune activation and blood-brain barrier disruption, which lead to MOG antibodies. Below, onset age has two peaks: children, 3, and young to middle-aged adults, 4. 1, Infection; 2, Vaccination; 3, Children; 4, Young to middle-aged adults.

Pathophysiology

Antibodies against MOG on the outer surface of myelin recruit complement and immune cells, causing perivenous and confluent demyelination of the optic nerves, spinal cord, and brain.

MOG is a transmembrane glycoprotein of the outermost myelin lamellae and oligodendrocyte surface, with an extracellular, N-terminal, immunoglobulin-like domain. It is a small part of myelin, but because it faces outward, antibodies reach it easily. IgG1 MOG antibodies activate complement and cell-mediated cytotoxicity, and active lesions contain perivascular lymphocytes and myelin-laden macrophages, with perivenous and confluent white matter and cortical demyelination.

MOG-IgG was first thought to mark AQP4-negative NMO, but MOGAD is now a separate disease. AQP4-positive NMOSD targets AQP4 on astrocytes, an astrocytopathy in which axon damage precedes demyelination, whereas MOGAD targets myelin and oligodendrocytes. MOGAD is monophasic or relapsing, without a progressive phase, and affects both sexes equally, whereas NMOSD and MS affect more women, and MS can be progressive.

Etiology

MOGAD can follow infection, mainly viral, including influenza, Epstein-Barr virus, herpes simplex virus, and SARS-CoV-2, and also Mycoplasma, and, extremely rarely, vaccination. Onset has two peaks: children, who make up about a third of cases, and young to middle-aged adults. Men and women are affected equally, MOGAD is not linked to other autoimmune diseases, unlike AQP4 disease, and pregnancy may protect against relapse.

Clinical presentation

Optic neuritis is the most common syndrome and is often bilateral, with disc edema and a long optic nerve lesion. Myelitis causes weakness, a sensory level, and bladder and bowel dysfunction, with LETM or a conus lesion. ADEM, mainly in children, causes encephalopathy with multifocal deficits. Cortical encephalitis often causes seizures, and brainstem or cerebellar deficits can involve the pons and MCP. These syndromes can occur alone, in sequence, or together.

About half of patients have a single attack. Relapse is more likely in adults and with persistent MOG-IgG, and disability comes from attacks, without a progressive phase.

Diagnosis

Diagnosis requires a demyelinating attack, serum MOG-IgG on a live cell-based assay, and exclusion of better diagnoses, including MS. By the 2023 criteria, a clear positive titer of 1:100 or more needs no supporting features. A low positive titer of 1:10 to below 1:100, a positive result without a titer, or antibodies in CSF only requires negative AQP4-IgG and at least one supporting feature.

Supporting features are bilateral optic neuritis, an optic nerve lesion over half its length, disc edema, perineural optic sheath enhancement, LETM, a conus lesion, an H-sign or central cord lesion, ill-defined, fluffy brain lesions, deep gray matter lesions, and a cortical lesion with meningeal enhancement. Well-circumscribed lesions and CSF OCB favor MS, and area postrema syndrome with AQP4-IgG favors NMOSD. Because MOG-IgG is found in only 0.4% of MS, testing everyone with possible MS yields false positives.

Management

Acute attacks are treated with IV methylprednisolone 1 g daily for 5 days, or oral prednisone 1250 mg, and PLEX or immunoadsorption follows if steroids fail. Relapse prevention usually starts after a second attack, with AZA or MMF bridged by prednisolone for the first months, IVIG at 1–2 g/kg every 4 weeks, or oral corticosteroids, whose long-term risks must be weighed. RTX is less effective than in NMOSD, and MS drugs such as interferons, glatiramer acetate, and fingolimod appear ineffective.

Relationships to other conditions

Multiple sclerosis. Both are demyelinating diseases that cause optic neuritis and myelitis. Bilateral optic neuritis with disc edema, LETM, a conus lesion, fluffy lesions, and MOG-IgG favor MOGAD, whereas well-circumscribed lesions, CSF OCB, and a progressive course favor multiple sclerosis, whose drugs appear ineffective in MOGAD.

NMOSD. Both are antibody-mediated diseases that cause optic neuritis and LETM, and MOG-IgG was first thought to mark AQP4-negative NMO. Disc edema, a conus lesion, ADEM, equal numbers of men and women, and a monophasic course favor MOGAD, whereas AQP4-IgG and area postrema syndrome favor NMOSD, an astrocytopathy in which axon damage precedes demyelination.

Glaucoma: aqueous outflow, intraocular pressure, and optic nerve damage

Glaucoma is an optic neuropathy in which slow drainage of aqueous humor raises IOP and damages the optic nerve. In open-angle glaucoma, aqueous humor exits slowly through a resistant trabecular meshwork, causing painless, gradual peripheral vision loss. In angle closure, the iris blocks the angle, through pupillary block or a thickened iris base after the pupil dilates, and acute angle closure is a medical emergency. Gonioscopy distinguishes the two. Treatment lowers IOP by increasing outflow or decreasing production of aqueous humor. Here, we review the flow of aqueous humor, the mechanisms and features of each type, diagnosis, and treatment.

Key points

- Most aqueous humor drains through the trabecular meshwork, and a minority through the uveoscleral route.
- Open-angle glaucoma is slow exit of aqueous humor through the trabecular meshwork.
- Angle closure follows pupillary block or a thickened iris base after the pupil dilates, and mydriatics can trigger it.
- Gonioscopy shows whether the angle is open or closed.
- Prostaglandin analogs and cholinergic agents increase outflow, and β-blockers, α2-agonists, and carbonic anhydrase inhibitors decrease production.
- Acute angle closure is lowered medically, then treated with laser peripheral iridotomy.

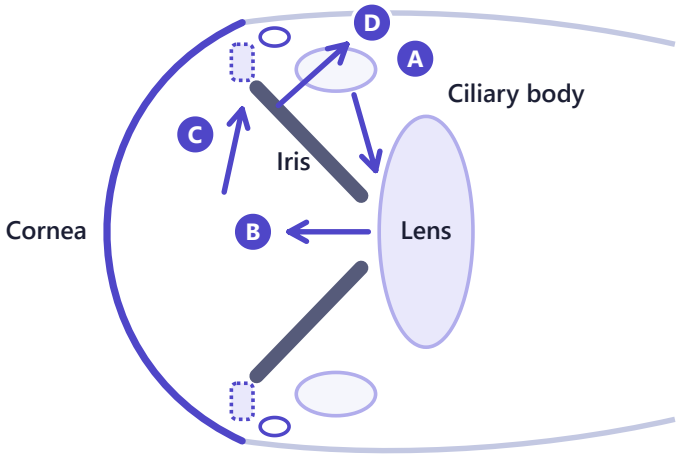


Fig. 1 | A cross-section of the front of the eye. A, the ciliary body secretes aqueous humor behind the iris. B, it flows through the pupil into the anterior chamber. C, most drains through the trabecular meshwork into Schlemm canal at the angle between iris and cornea. D, some leaves through the ciliary muscle, the uveoscleral route. A, Production; B, Through the pupil; C, Trabecular outflow; D, Uveoscleral outflow.

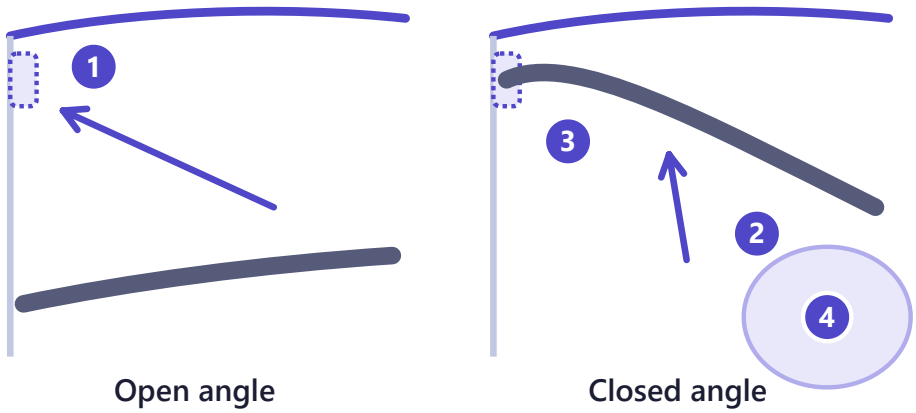


Fig. 2 | Two close-ups of the drainage angle. In open-angle glaucoma, 1, aqueous humor reaches a resistant trabecular meshwork. In angle closure, the iris covers the meshwork: by pupillary block, 2, which bows the iris forward; by a thickened iris base, 3, after the pupil dilates; or by a thick lens pushing forward, 4. 1, Trabecular resistance; 2, Pupillary block; 3, Thickened iris base; 4, Thick lens.

Pathophysiology

Aqueous humor drains too slowly, through a resistant trabecular meshwork in open-angle glaucoma or past an iris that blocks the angle in angle closure, so IOP rises above 21 mmHg and damages the optic nerve. Ganglion cell axons are lost, the cup enlarges, and vision is lost irreversibly, peripheral first. Glaucoma can occur at normal pressure, and raised pressure does not always cause it.

The ciliary body secretes aqueous humor behind the iris, with β2 receptors increasing and α2 receptors decreasing secretion. It flows through the pupil into the anterior chamber. Most drains through the trabecular meshwork into Schlemm canal, the site of resistance in open-angle glaucoma, and a minority leaves through the ciliary muscle by the uveoscleral route, which prostaglandin analogs increase.

Open-angle glaucoma is chronic and painless and makes up most glaucoma. Angle closure, about a third of cases but more often blinding, presses the iris against the trabecular meshwork and can be an acute emergency or chronic.

Etiology

In open-angle glaucoma, aqueous humor exits slowly although the angle is open. Angle closure has three main mechanisms. In pupillary block, the most common, the iris presses on the lens, aqueous humor builds behind it, and the iris bows forward onto the cornea. When the pupil dilates, mydriatics contract the radial dilator muscle, and the bunched iris base narrows the angle. A lens that thickens with age, or swells with cataract, pushes the iris forward.

Risks for open-angle glaucoma are older age, African ancestry, which raises risk 4-fold, family history, myopia, a thin cornea, and type 2 diabetes. Risks for angle closure are age 55–65, female sex, which raises risk 2–4 times, East Asian and Inuit ancestry, hyperopia, and a shallow anterior chamber. Dilating drops, such as atropine and cyclopentolate, topiramate, sulfonamides, and darkness can trigger closure. Secondary glaucoma follows steroids, pseudoexfoliation, neovascularization, trauma, or cataract.

Clinical presentation

Open-angle glaucoma has no early symptoms, and over half of cases are undetected. Vision is lost gradually in the periphery, with arcuate defects, while central vision is kept until late, and the disc shows a cup-to-disc ratio above 0.5 and a notched rim, bilaterally but asymmetrically.

Acute angle closure causes sudden severe eye pain, a red eye, blurred vision, halos, headache, nausea, and vomiting. The pupil is fixed and mid-dilated, with corneal edema, ciliary flush, a shallow anterior chamber, and a firm eye, and IOP is 30–80 mmHg. Glaucoma is the second leading cause of blindness, and acute closure can cause retinal vascular occlusion, ischemic optic neuropathy, and peripheral anterior synechiae.

Diagnosis

Tonometry measures IOP, gonioscopy shows whether the angle is open or closed, fundoscopy and OCT assess the cup and nerve fiber layer, and visual fields show peripheral and arcuate loss. In gonioscopy, a mirrored lens on the cornea shows the angle; in acute angle closure, it is closed for 360°, and the other eye is often narrow. Gonioscopy is repeated after an attack, because IOP can fall from ischemia of the ciliary body even if the angle stays closed. Disc damage at normal pressure prompts assessment of vascular risks for normal-tension glaucoma.

Management

Open-angle glaucoma is treated with drops, a prostaglandin analog first, then laser trabeculoplasty, then trabeculectomy, a tube shunt, or MIGS. Acute angle closure is treated with acetazolamide 500 mg, timolol, and apraclonidine, then IV mannitol 1–2 g/kg if needed, and pilocarpine once IOP is below 40 mmHg. Laser peripheral iridotomy, which lets aqueous humor bypass the pupil, is the treatment of choice, with lens extraction as an alternative.

Treatment lowers IOP, by increasing the outflow of aqueous humor or decreasing its production. Prostaglandin analogs, such as latanoprost, increase uveoscleral outflow and are first line; they lengthen eyelashes and darken the iris. Cholinergic agents, such as pilocarpine, contract the ciliary muscle and constrict the pupil, opening the trabecular meshwork, but are poorly tolerated. α2-agonists, brimonidine and apraclonidine, both decrease production and increase uveoscleral outflow, and brimonidine can cause apnea in children. β-blockers, such as timolol, block ciliary β receptors and are avoided in asthma, COPD, heart block, and bradycardia. Carbonic anhydrase inhibitors, dorzolamide drops or acetazolamide by mouth or IV, reduce bicarbonate and so aqueous humor. Mydriatics and cycloplegics, such as atropine and cyclopentolate, are avoided because they dilate the pupil and can trigger angle closure. Sympathomimetics, such as adrenaline, and phenylephrine and pseudoephedrine in decongestants, also dilate the pupil and can trigger angle closure in predisposed eyes.

Relationships to other conditions

Cataract. A swollen cataract can push the iris forward and close the angle, and a hypermature cataract can leak proteins that clog the trabecular meshwork. A cataract alone causes painless blurring without raised IOP.

Vision loss and red eye. Acute angle closure is a cause of a painful red eye with reduced vision, like bacterial keratitis. A fixed, mid-dilated pupil, halos, and high IOP favor angle closure, whereas a corneal infiltrate after contact lens wear favors keratitis.

Cataract: opacification of the crystalline lens

A cataract is an opacification of the crystalline lens that scatters light and causes gradual, painless vision loss. Age-related nuclear sclerosis is the most common type, and posterior subcapsular cataract affects vision most and progresses to surgery most often. Untreated cataracts can cause phacomorphic or phacolytic glaucoma, and cataract is associated with dementia in people over 65. Diagnosis is clinical, and surgery, usually phacoemulsification with an IOL, is the only treatment. Here, we review the mechanisms, types, causes, complications, and treatment of cataract.

Key points

- Nuclear sclerosis, the most common type, is age-related and causes a myopic shift.
- Posterior subcapsular cataract sits near the nodal point, causes glare, and its growth predicts surgery most strongly.
- Steroids, diabetes, trauma, and uveitis are causes beyond age.
- Phacomorphic glaucoma closes the angle, and phacolytic glaucoma clogs the trabecular meshwork with macrophages.
- Cataract is associated with dementia over age 65, and cataract surgery with lower dementia risk.

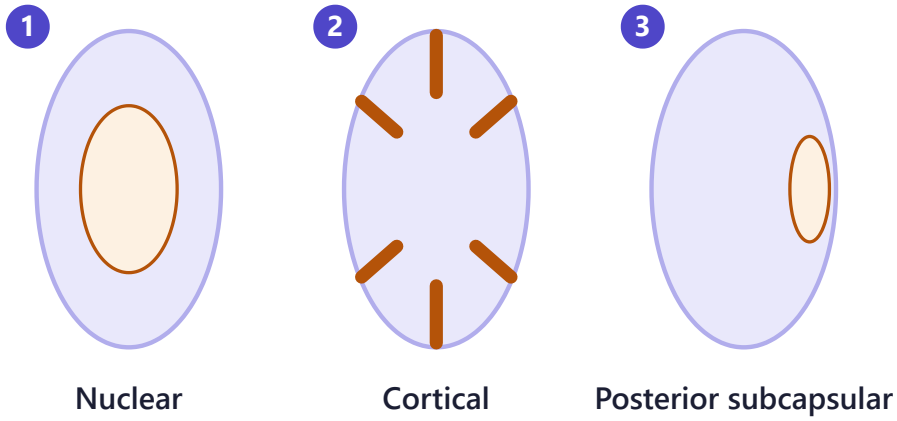


Fig. 1 | Three lenses seen from the side, front to the left. 1, nuclear sclerosis clouds the center. 2, cortical cataract forms wedge-shaped spokes from the edge. 3, posterior subcapsular cataract forms a plaque at the back of the lens. 1, Nuclear sclerosis; 2, Cortical; 3, Posterior subcapsular.

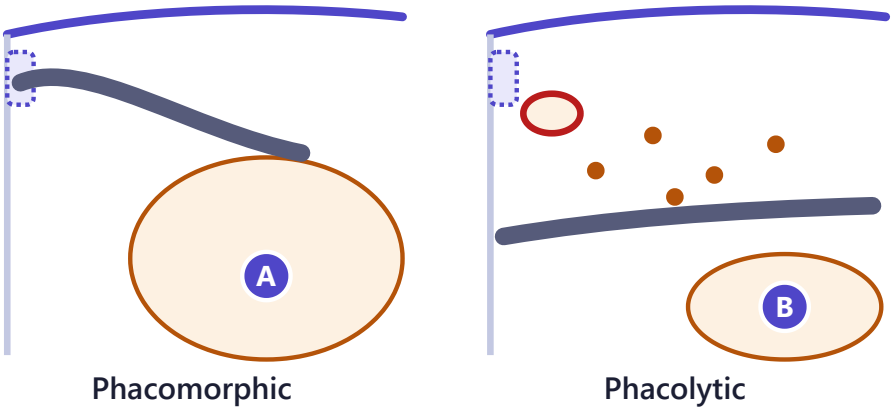


Fig. 2 | Two close-ups of the drainage angle with a cataract. A, phacomorphic glaucoma: a swollen lens pushes the iris forward against the trabecular meshwork. B, phacolytic glaucoma: a hypermature lens leaks proteins, and macrophages that take them up clog the meshwork while the angle stays open. A, Phacomorphic glaucoma; B, Phacolytic glaucoma.

Pathophysiology

Lens proteins change with age or injury, so the crystalline lens opacifies and scatters light, causing glare, halos, and gradual, painless blurring. In nuclear cataract, urochrome pigment collects in the center; in cortical cataract, water collects between lens fibers; and in subcapsular cataract, lens epithelium undergoes fibrous metaplasia. In diabetes, sorbitol accumulates and draws water into the lens.

Etiology

Nuclear sclerosis is the most common type; the center turns yellow-brown, and a myopic shift lets some patients read again without glasses. Cortical cataract forms wedge-shaped spokes from the edge of the lens. Posterior subcapsular cataract forms at the back of the lens, near the eye's nodal point, so it affects vision most, and its growth predicts progression to surgery most strongly.

Age is the most common cause, typically in the fifth and sixth decades. Topical or systemic steroids cause posterior subcapsular cataract, diabetes causes snowflake cortical opacities, trauma is a common cause of unilateral cataract in young adults, and uveitis is the most common cause of secondary cataract. Radiation, myotonic dystrophy, atopic dermatitis, smoking, and alcohol also contribute.

Congenital cataracts follow genetic causes, intrauterine infection, and metabolic disease such as galactosemia; they present with leukocoria and are treated early because they cause amblyopia.

Clinical presentation

Cataract causes gradual, painless blurring, glare and halos from headlights and sunlight, worst with posterior subcapsular cataract, faded, yellowed colors, frequent changes of glasses from the myopic shift, and monocular diplopia as light refracts through clear gaps. Untreated cataract can cause phacomorphic or phacolytic glaucoma. Cataract is associated with dementia in people over 65, and cataract surgery is linked to a lower risk of dementia, with a hazard ratio of 0.71. Cataract is a major cause of blindness worldwide, but it is reversible.

Untreated cataracts can cause lens-induced glaucoma. In phacomorphic glaucoma, a swollen, intumescent lens pushes the iris forward and closes the angle. In phacolytic glaucoma, a hypermature lens leaks proteins, and macrophages clog the trabecular meshwork while the angle stays open. Both are treated by removing the lens.

Diagnosis

Diagnosis is clinical, with visual acuity on a Snellen chart, slit lamp examination before and after dilation to grade type and density, a dimmed red reflex, and examination of the fundus and pupils to exclude retinal or optic nerve disease. An RAPD with a cataract points to retinal or optic nerve disease, because a cataract alone does not cause one. Sudden pain, a red eye, and high IOP with a cataract call for gonioscopy to look for lens-induced glaucoma, and before surgery, tamsulosin, which causes a floppy iris, and anticoagulants are noted.

Management

Surgery is the only treatment. It is offered when vision limits daily activities, such as reading or night driving, when glare is troublesome, or for lens-induced glaucoma. Phacoemulsification breaks up the lens with ultrasound through a small incision, and an IOL replaces it, whereas manual small-incision surgery suits dense, black cataracts.

Relationships to other conditions

Glaucoma. A swollen cataract can close the angle, and a hypermature cataract can clog the trabecular meshwork. Painless blurring with normal IOP favors cataract alone, whereas raised IOP, field loss, or a painful red eye favors glaucoma.

Vision loss and red eye. Cataract is a cause of gradual painless vision loss, like retinitis pigmentosa. Glare and a dimmed red reflex favor cataract, whereas night blindness and tunnel vision favor retinitis pigmentosa.

Alzheimer disease. Cataract is associated with dementia in people over 65, and cataract surgery is linked to a lower risk of dementia.

Vision loss and red eye: telling eye conditions apart by presentation

Eye conditions can be distinguished by how they present: a red eye, painful vision loss, sudden painless vision loss, or gradual painless vision loss. Viral and bacterial conjunctivitis and bacterial keratitis cause a red eye, with vision spared in conjunctivitis. Optic neuritis causes painful vision loss, retinal detachment and wet macular degeneration cause sudden painless loss, and retinitis pigmentosa and dry macular degeneration cause gradual loss. Each sits in a different part of the eye, from the conjunctiva to the optic nerve. Here, we review these eight conditions by presentation.

Key points

- Viral conjunctivitis, usually adenovirus, causes a red eye with watery discharge and normal vision and resolves within 3 weeks.
- Bacterial conjunctivitis causes mucopurulent discharge with lids stuck together, and topical antibiotics shorten it.
- Bacterial keratitis follows contact lens wear and needs scraping and urgent topical antibiotics.
- Optic neuritis causes painful monocular vision loss with an RAPD and is often the first sign of MS.
- Retinal detachment causes flashes, floaters, and a curtain and needs urgent surgery.
- Wet macular degeneration causes central distortion and is treated with anti-VEGF injections.
- Retinitis pigmentosa causes night blindness, then tunnel vision, with bone spicule pigment.
- Dry macular degeneration, most AMD, causes gradual central loss with drusen, and AREDS supplements slow it.

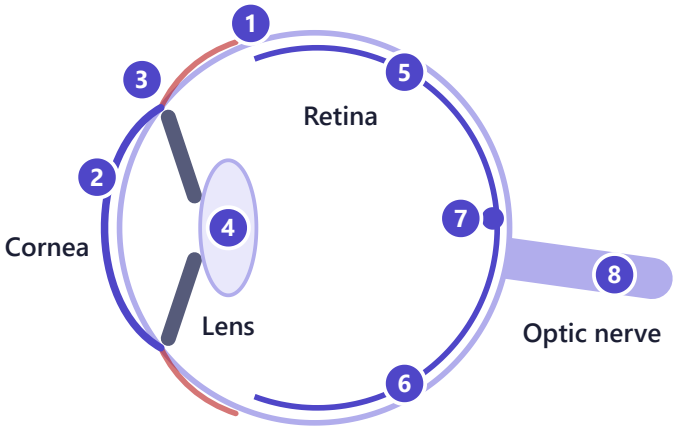


Fig. 1 | A side view of the eye, front to the left, numbering where each condition sits: 1, conjunctiva; 2, cornea; 3, drainage angle; 4, lens; 5, retina; 6, peripheral photoreceptors; 7, macula; 8, optic nerve. 1, Conjunctivitis; 2, Bacterial keratitis; 3, Glaucoma; 4, Cataract; 5, Retinal detachment; 6, Retinitis pigmentosa; 7, Macular degeneration; 8, Optic neuritis.

Red eye

Viral conjunctivitis, caused by adenovirus in up to 90% of cases, causes a sudden red eye with watery discharge, itching, and foreign body sensation, and vision is normal. Follicles and preauricular lymph nodes are typical, often after a respiratory infection. It is highly contagious for 10–14 days and resolves within 3 weeks with artificial tears, cool compresses, and hand hygiene, and antibiotics do not help.

Bacterial conjunctivitis, caused mainly by Staphylococcus aureus in adults and by Haemophilus influenzae, Streptococcus pneumoniae, and Moraxella in children, accounts for most childhood cases. It causes a red eye with mucopurulent discharge and lids stuck together on waking, often bilaterally, with normal vision. It is often self-limited, and topical antibiotics, such as chloramphenicol, polymyxin B with trimethoprim, or a fluoroquinolone, 4 times daily for 5–7 days, shorten it. Hyperacute conjunctivitis from Neisseria gonorrhoeae produces copious purulent discharge, can perforate the

cornea, and needs Gram stain, culture, and urgent treatment.

Bacterial keratitis is a sight-threatening corneal infection, most often after contact lens wear, caused by organisms such as Staphylococcus aureus and Pseudomonas. It causes pain, redness, photophobia, discharge, and reduced vision, with an epithelial defect, a stromal infiltrate, and sometimes hypopyon. Corneal scraping for Gram stain and culture precedes urgent topical antibiotics, moxifloxacin or fortified cefazolin and tobramycin. A painful red eye with halos and a fixed, mid-dilated pupil is acute angle closure instead.

Allergic conjunctivitis follows IgE-mediated mast cell degranulation and causes itching, redness, tearing, and chemosis, often seasonally with atopy, and is treated with allergen avoidance, lubricating drops, and topical antihistamines and mast cell stabilizers. Anterior uveitis causes a painful red eye with blurred vision, photophobia, and ciliary flush, is associated with HLA-B27 disease, juvenile idiopathic arthritis, IBD, sarcoidosis, and Behçet disease, and is treated with topical corticosteroids and cycloplegics.

Painful vision loss

Optic neuritis causes sudden vision loss in one eye, with pain on eye movement that often comes first and reduced color vision. An RAPD is present, and the disc is normal or, in a third, swollen. It affects women more than men, at ages 20–50, and is often the first sign of MS. MRI of the brain and orbits with gadolinium is obtained, and IV methylprednisolone followed by an oral taper speeds recovery without changing the outcome. It is also the most common syndrome of MOGAD, in which it is often bilateral.

Sudden painless vision loss

Retinal detachment is rhegmatogenous, from a tear, in most cases, or tractional, as in diabetic retinopathy, or exudative. Myopia, lattice degeneration, and trauma raise risk. New floaters and flashes precede field loss that spreads from the periphery like a curtain. Urgent surgery, by pneumatic retinopexy, scleral buckle, or vitrectomy, seals the break, and the outcome is best if the macula is still attached.

In wet macular degeneration, VEGF drives new vessels from the choroid through Bruch membrane, and they leak and bleed. It affects 10–15% of patients with AMD, with age, smoking, family history, and hypertension as risks. It causes central distortion and blurring, with a central scotoma on an Amsler grid. OCT shows fluid, fluorescein angiography confirms the vessels, and intravitreal anti-VEGF agents, ranibizumab or aflibercept, are the treatment; without them, 79–90% of eyes become legally blind.

CRAO, usually from carotid or cardiac emboli and occasionally from GCA, causes sudden, severe, painless vision loss with an RAPD and a pale retina with a cherry-red spot, and is treated as a stroke equivalent. CRVO, mostly after age 50 with hypertension, glaucoma, diabetes, or hyperlipidemia, gives a blood and thunder fundus of dilated, tortuous veins and hemorrhages in all quadrants. GCA, after age 50, causes new headache, jaw claudication, and polymyalgia rheumatica, and sudden vision loss from anterior ischemic optic neuropathy; ESR and CRP are followed by temporal artery biopsy, and glucocorticoids are started at once to prevent blindness.

Gradual painless vision loss

In retinitis pigmentosa, gene mutations, inherited in an autosomal recessive, autosomal dominant, or X-linked pattern, kill rods first, then cones; Usher syndrome adds hearing loss. Night blindness comes first, then tunnel vision, and central vision is lost late. The fundus shows bone spicule pigment, narrowed arterioles, and waxy disc pallor, and the ERG is reduced. There is no standard treatment beyond glasses and low-vision support, and gene therapy is emerging.

Dry macular degeneration makes up most AMD. Lipofuscin and lipid deposits, drusen, build up beneath the RPE, and the RPE and photoreceptors then atrophy, ending in geographic atrophy. Age over 60, smoking, family history and complement gene variants, and hypertension raise risk. It causes gradual loss of central vision, with yellow drusen and RPE pigment changes. Antioxidant and zinc supplements, as tested in AREDS, slow progression, smoking is stopped, and an Amsler grid detects conversion to wet AMD. Cataract and open-angle glaucoma also cause gradual painless vision loss.

Papilledema is bilateral optic disc swelling from raised ICP, with headache and transient visual obscurations, from tumors, CSF obstruction, venous sinus thrombosis, meningitis, or idiopathic intracranial hypertension, and calls for urgent imaging. Hypertensive retinopathy shows arteriolar narrowing, AV nicking, flame hemorrhages, hard exudates, and cotton wool spots, with papilledema in malignant hypertension. Horner syndrome, ptosis, miosis, and anhidrosis, follows lesions of the brainstem, lung apex, or internal carotid artery, and Argyll Robertson pupils, small pupils that constrict to near but not light, mark neurosyphilis.

Relationships to other conditions

Glaucoma. Acute angle closure causes a painful red eye like bacterial keratitis, and open-angle glaucoma causes gradual peripheral loss like retinitis pigmentosa. Raised IOP and a large cup favor glaucoma.

Cataract. Cataract causes gradual painless vision loss, like retinitis pigmentosa. Glare and a dimmed red reflex

favor cataract, whereas night blindness and bone spicule pigment favor retinitis pigmentosa.

Multiple sclerosis. Optic neuritis is often the first sign of multiple sclerosis, so MRI of the brain is obtained to look for other lesions.

MOGAD. Optic neuritis is the most common syndrome of MOGAD. Bilateral optic neuritis with disc edema favors MOGAD.

Hearing loss: conductive, sensorineural, and mixed

Hearing loss is conductive when sound is blocked before the cochlea, sensorineural when the cochlea or cochlear nerve fails, and mixed when both occur. Conductive causes include cerumen, effusion, perforation, cholesteatoma, and otosclerosis; sensorineural causes include presbycusis, noise, ototoxic drugs, Ménière disease, and vestibular schwannoma. The Weber and Rinne tests and audiometry separate the types, and tinnitus, vertigo, and asymmetry point to particular causes. Treatment addresses the cause, with hearing aids and cochlear implants for permanent loss. Here, we review the types, causes, clinical clues, diagnosis, and management of hearing loss.

Key points

- Conductive loss makes sound quieter, and sensorineural loss makes it quieter and distorted.
- In conductive loss, the Weber test lateralizes to the affected ear and bone conduction beats air on the Rinne test.
- In sensorineural loss, the Weber test lateralizes to the better ear and air conduction beats bone.
- Presbycusis is bilateral and high-frequency, noise causes a 4 kHz notch, and Ménière disease causes fluctuating low-frequency loss.
- Asymmetric SNHL or unilateral tinnitus calls for MRI to look for vestibular schwannoma.
- Sudden SNHL is an emergency treated with oral steroids.
- GJB2 and GJB6 mutations at DFNB1 are the most common genetic cause, and PAX3 mutations cause Waardenburg syndrome types 1 and 3.

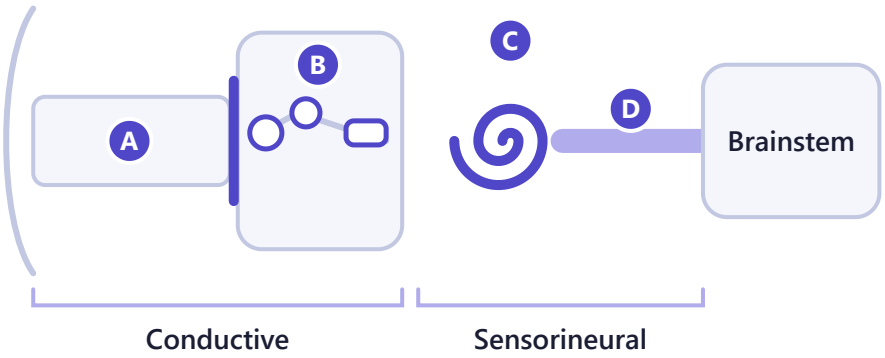


Fig. 1 | Sound passes from the ear canal, A, to the eardrum and ossicles of the middle ear, B, to the hair cells of the cochlea, C, and along the cochlear nerve to the brainstem, D. Loss at A or B is conductive; loss at C or D is sensorineural. A, Ear canal; B, Middle ear; C, Cochlea; D, Cochlear nerve.

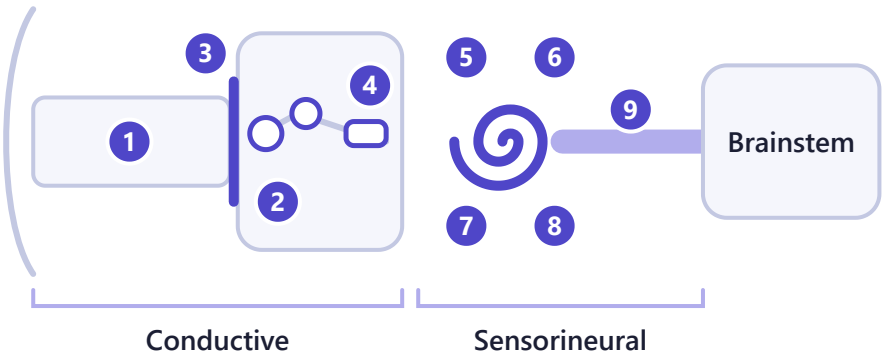


Fig. 2 | Causes of hearing loss placed on the ear: 1, cerumen in the canal; 2, effusion in the middle ear; 3, eardrum perforation or cholesteatoma; 4, otosclerosis fixing the stapes; 5 to 8, presbycusis, noise, ototoxic drugs, and Ménière disease in the cochlea; 9, vestibular schwannoma on the nerve. 1, Cerumen impaction; 2, Otitis media with effusion; 3, Perforation or cholesteatoma; 4, Otosclerosis; 5, Presbycusis; 6, Noise; 7, Ototoxic drugs; 8, Ménière disease; 9, Vestibular schwannoma.

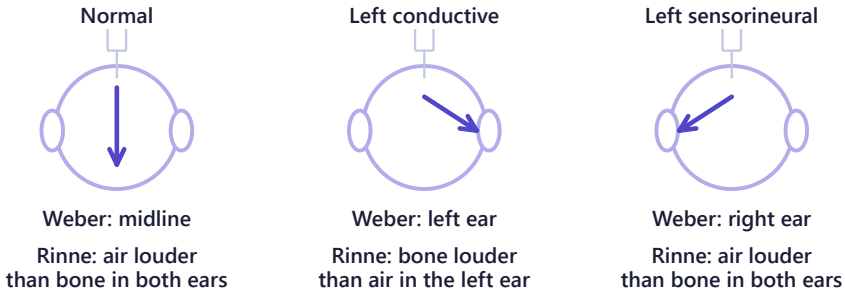


Fig. 3 | Tuning fork tests, facing the patient. Normally, the Weber test is heard in the midline and air conduction is louder than bone in both ears. With left conductive loss, the Weber test lateralizes to the left ear and bone is louder than air on the left. With left sensorineural loss, the Weber test lateralizes to the better, right ear and air is louder than bone in both ears.

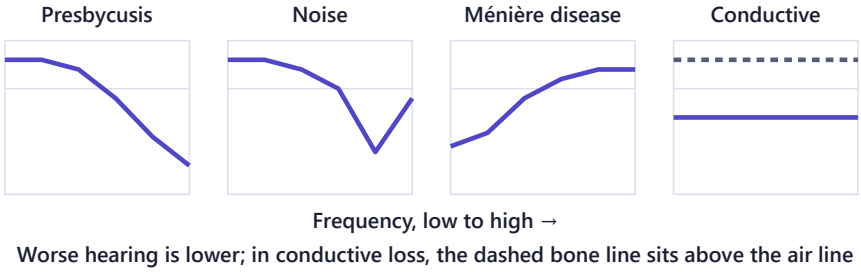


Fig. 4 | Audiogram shapes, with worse hearing lower: presbycusis slopes down at high frequencies; noise leaves a notch near 4 kHz that recovers at 8 kHz; Ménière disease affects low to mid frequencies; and conductive loss leaves a gap between normal bone conduction and reduced air conduction.

Pathophysiology

Sound is collected by the outer ear, amplified by the eardrum and ossicles, transduced by cochlear hair cells, and carried by the cochlear nerve, part of CN VIII, to the brainstem. Loss before the cochlea is conductive and makes sound quieter. Loss at or after the cochlea is sensorineural and makes sound quieter and distorted. Mixed loss involves both, as otosclerosis can.

In conductive loss, bone conduction is normal, leaving an air-bone gap on the audiogram. In sensorineural loss, air and bone conduction are both reduced without a gap. Sensorineural loss is the most common type, and conductive loss is common in children.

Etiology

Conductive loss follows cerumen impaction, foreign bodies, and otitis externa; otitis media with effusion, the most common cause of acquired hearing loss in children; eardrum perforation; cholesteatoma, trapped squamous debris that erodes the middle ear; and otosclerosis, in which spongy bone fixes the stapes. Otosclerosis affects

women twice as often, runs in families in over half of cases, and worsens in pregnancy.

Sensorineural loss follows presbycusis, the most common cause worldwide; noise, which damages hair cell stereocilia; ototoxic drugs, such as aminoglycosides, loop diuretics, and cisplatin; Ménière disease, from endolymphatic hydrops; and vestibular schwannoma, a benign Schwann cell tumor of CN VIII. In children, genetic causes account for over half of cases, and congenital CMV is another cause. Meningitis and diabetes can also cause it, and sudden SNHL is usually idiopathic.

Genetic causes account for about half of childhood hearing loss, 70% of it nonsyndromic and 30% syndromic, and nonsyndromic loss is autosomal recessive in 75–80% of cases, prelingual, severe, and stable. The DFNB1 locus on chromosome 13q12 holds GJB2 and GJB6, which encode connexins 26 and 30, gap junction proteins whose loss lowers the endocochlear potential; DFNB1 mutations are the most common genetic cause of severe to profound congenital deafness, and a GJB6 deletion with one GJB2 mutation also causes it. Waardenburg syndrome follows failed migration of neural crest cells, so melanocytes are missing from the hair, skin, iris, and stria vascularis. Type 1, from PAX3 mutations, is autosomal dominant, with a white forelock, heterochromia, patchy depigmentation, dystopia canthorum, and SNHL in about 20%; type 2, from MITF mutations, lacks dystopia canthorum and causes SNHL in up to 50%; type 3, also from PAX3, adds upper limb anomalies; and type 4, from EDNRB or EDN3 mutations and usually autosomal recessive, adds Hirschsprung disease.

NF2, an autosomal dominant disorder of the merlin gene on chromosome 22, causes bilateral vestibular schwannomas with meningiomas and ependymomas.

Clinical presentation

Presbycusis causes difficulty in noise and loss of high-pitched consonants. Noise causes tinnitus and hyperacusis. Otosclerosis causes progressive conductive loss with a normal eardrum, sometimes with a red promontory, the Schwartze sign. Ménière disease causes fluctuating loss with episodic vertigo, tinnitus, and aural fullness, and vestibular schwannoma causes unilateral loss, tinnitus, and imbalance, rarely with facial palsy. In children, hearing loss is the most common cause of speech delay.

Most tinnitus is subjective, heard only by the patient, and noise trauma is its most common cause, at the frequency lost. Objective tinnitus, which the examiner can hear, can arise from an aneurysm. Unilateral tinnitus with asymmetric hearing suggests vestibular schwannoma, and tinnitus with vertigo and fluctuating loss suggests Ménière disease. High-dose aspirin causes reversible tinnitus, whereas cisplatin and loop diuretics can cause permanent tinnitus. Tinnitus usually accompanies hearing loss, but 20% of people in tinnitus clinics hear normally.

Diagnosis

Otoscopy looks for wax, perforation, effusion, or cholesteatoma. With a 512 Hz tuning fork, the Weber test lateralizes to the affected ear in conductive loss and to the better ear in sensorineural loss, and the Rinne test shows bone louder than air in an ear with conductive loss, whereas air is louder than bone normally and in sensorineural loss.

Audiometry measures air and bone conduction. Presbycusis gives a bilateral, high-frequency, downsloping curve, noise gives a notch at 3–6 kHz, often 4 kHz, with recovery at 8 kHz, Ménière disease gives low- to mid-frequency SNHL, and conductive loss gives an air-bone gap. Sudden SNHL, a loss of over 30 dB at 3 consecutive frequencies within 72 hours, calls for audiometry and MRI of the internal auditory canals, and asymmetric SNHL or unilateral tinnitus calls for MRI with gadolinium to look for vestibular schwannoma. Speech delay in a child always calls for a formal hearing test, and congenital or childhood SNHL calls for GJB2 testing once acquired causes are excluded, with gene-specific testing if a syndrome is suspected.

Management

Conductive loss is treated by its cause: removal or microsuction of cerumen, ventilation tubes for persistent effusion, and hearing aids or stapes surgery for otosclerosis. Sudden SNHL is an emergency treated with oral prednisone 1 mg/kg daily, up to 60 mg, for 7 days, then tapered, because spontaneous recovery is most likely in the first 2 weeks. Chronic SNHL is treated with hearing aids, since lost hearing does not return, and cochlear implants help severe bilateral loss, working best when placed before age 2 in children born deaf. Families with genetic hearing loss receive genetic counseling, since parents of recessive cases are usually unaffected carriers. Ear protection and avoiding or monitoring ototoxic drugs prevent loss, and a small vestibular schwannoma may be observed, whereas surgery or radiation does not restore hearing.

Tinnitus with hearing loss is treated with hearing aids, distress with cognitive behavior therapy and sound therapy, and aspirin-induced tinnitus resolves when aspirin is stopped.

Relationships to other conditions

Vertigo. Ménière disease, labyrinthitis, and vestibular schwannoma cause both hearing loss and vestibular symptoms. Hearing loss with vertigo points to the inner ear or CN VIII, whereas vestibular neuritis and BPPV spare hearing.

Stroke symptoms and complications. Lateral pontine syndromes from the anterior inferior cerebellar artery can cause ipsilateral hearing loss with vertigo. Other brainstem signs favor stroke over an ear cause.

Vertigo: timing, nystagmus, and hearing as guides to the cause

Vertigo is a false sense of motion arising from asymmetric vestibular input, and the VOR turns that asymmetry into nystagmus. Peripheral causes account for over 90% of cases and include BPPV, vestibular neuritis, labyrinthitis, Ménière disease, and vestibular schwannoma; central causes include vestibular migraine and posterior circulation stroke. Timing, triggers, the pattern of nystagmus, and whether hearing loss and tinnitus are present point to the cause, and the HINTS examination separates peripheral from central acute vertigo. Treatment targets the cause. Here, we review the vestibular system, causes of vertigo by timing, nystagmus, diagnosis, and treatment.

Key points

- Peripheral nystagmus is horizontal-torsional, unidirectional, and suppressed by fixation; central nystagmus can change direction or be vertical.
- In acute vertigo, an abnormal head impulse, unidirectional nystagmus, and no skew suggest a peripheral cause.
- BPPV causes episodes under 1 minute with upbeat, torsional nystagmus on the Dix-Hallpike maneuver and is treated with the Epley maneuver.
- Vestibular neuritis spares hearing, whereas labyrinthitis adds hearing loss.
- Ménière disease causes attacks of 20 minutes to 12 hours with fluctuating low-frequency hearing loss, tinnitus, and fullness.
- Vestibular suppressants should be limited to a few days.

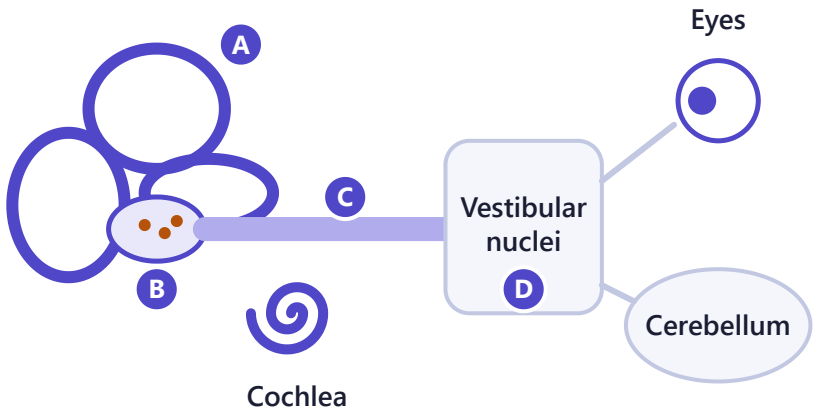


Fig. 1 | The semicircular canals sense rotation, A, and the utricle senses gravity with its otoconia, B. Signals travel along the vestibular nerve, C, to the vestibular nuclei and cerebellum, D, which drive the eyes through the vestibulo-ocular reflex. A, Semicircular canals; B, Utricle; C, Vestibular nerve; D, Vestibular nuclei and cerebellum.

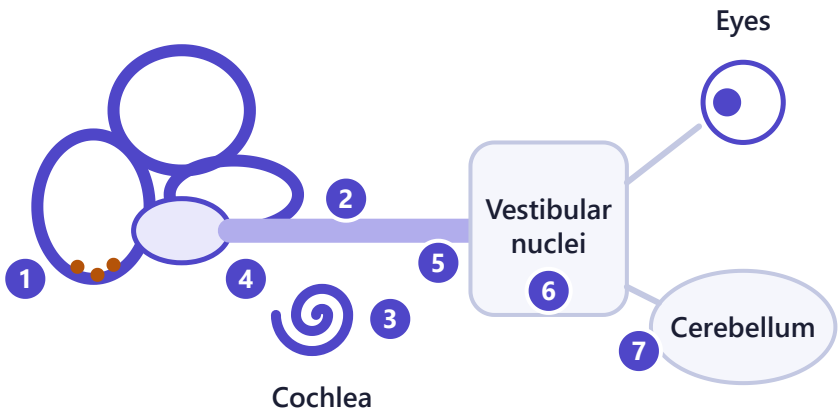


Fig. 2 | Causes of vertigo placed on the vestibular system: 1, BPPV, with otoconia loose in the posterior canal; 2, vestibular neuritis of the vestibular nerve; 3, labyrinthitis of the whole labyrinth, including the cochlea; 4, Ménière disease, with endolymphatic hydrops; 5, vestibular schwannoma near the brainstem; 6, vestibular migraine; 7, stroke of the cerebellum or brainstem. 1, BPPV; 2, Vestibular neuritis; 3, Labyrinthitis; 4, Ménière disease; 5, Vestibular schwannoma; 6, Vestibular migraine; 7, Posterior circulation stroke.

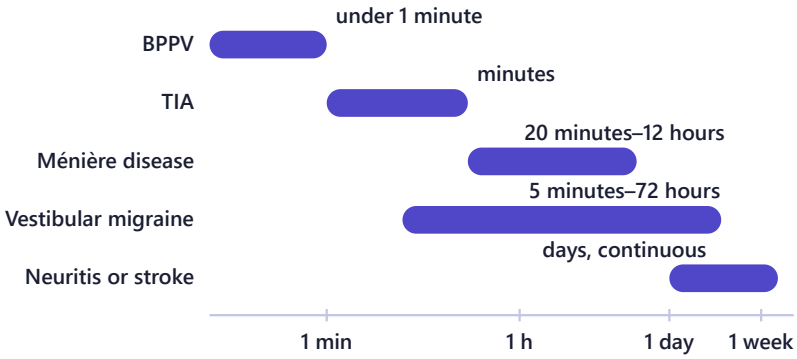


Fig. 3 | How long vertigo lasts, on a scale from seconds to weeks: BPPV under 1 minute; TIA minutes; Ménière disease 20 minutes to 12 hours; vestibular migraine 5 minutes to 72 hours; and vestibular neuritis, labyrinthitis, or posterior circulation stroke days of continuous vertigo.

Pathophysiology

Unequal signals from the two vestibular systems are read as rotation. The VOR drives the eyes in a slow drift toward the weaker side, and corrective fast phases beat away from it, producing nystagmus, which is named for its fast phase and strengthens with gaze toward that phase. The semicircular canals sense rotation: the horizontal canal senses yaw, turning the head side to side, and the anterior and posterior canals, set at 45° to the midline, sense pitching the head forward and backward, working as coplanar push-pull pairs. The utricle and saccule, maculae with otoconia, sense linear acceleration, gravity, and head tilt, the utricle horizontal and the saccule vertical motion, and the vestibular nerve, part of CN VIII, carries signals to the vestibular nuclei and cerebellum.

Peripheral causes account for over 90% of vertigo. Peripheral nystagmus is horizontal-torsional and unidirectional, never purely vertical or torsional, and is suppressed by fixation, with an abnormal head impulse and no skew. Central vertigo, mostly from ischemia of the cerebellum or brainstem, can cause direction-changing, vertical, or purely torsional nystagmus that fixation does not suppress, with a normal head impulse and sometimes skew deviation. Hearing loss points to the ear.

Etiology

BPPV, the most common peripheral cause, follows loose otoconia in a canal, usually the posterior, and age and head injury raise its risk. Vestibular neuritis is presumed viral inflammation of the vestibular nerve, and labyrinthitis inflames the whole labyrinth, usually after a viral infection. Ménière disease follows endolymphatic hydrops, and vestibular schwannoma is a slow-growing tumor of CN VIII. Vestibular migraine is a common central cause, and posterior circulation stroke, most often lateral medullary syndrome, is the cause not to miss. Anticonvulsants, salicylates, and antibiotics can affect vestibular function.

Clinical presentation

Acute, continuous vertigo lasting days with nausea and unsteadiness suggests vestibular neuritis, labyrinthitis, or posterior circulation stroke. Recurrent episodes under

1 minute triggered by rolling over, looking up, or bending suggest BPPV. Spontaneous attacks of 20 minutes to 12 hours with ear symptoms suggest Ménière disease, attacks of 5 minutes to 72 hours with migraine features suggest vestibular migraine, and attacks of minutes suggest TIA. Progressive imbalance with unilateral hearing loss and tinnitus suggests vestibular schwannoma.

Posterior canal BPPV causes upbeat, torsional nystagmus on the Dix-Hallpike maneuver that fatigues with repetition, and lateral canal BPPV causes geotropic or apogeotropic horizontal nystagmus on the head-roll test. Among central patterns, downbeat nystagmus points to the cervicomedullary junction or midline cerebellum, upbeat nystagmus to the pons and midbrain, gaze-evoked nystagmus to sedatives and other drugs, and periodic alternating nystagmus, which reverses every 90–120 seconds, to the brainstem and cerebellum. Complications include falls, chronic dizziness when suppressants are used too long, and missed stroke.

Diagnosis

Vertigo is first separated from presyncope and imbalance, then classified by timing and triggers. Hearing, tinnitus, and nystagmus then help point to the cause. Vestibular neuritis causes spontaneous, unidirectional, horizontal-torsional nystagmus with normal hearing, and labyrinthitis adds unilateral hearing loss and tinnitus. Ménière disease causes fluctuating low-frequency SNHL, tinnitus, and fullness, with peripheral nystagmus during attacks, and vestibular schwannoma causes progressive unilateral SNHL and tinnitus, with imbalance more than vertigo. Stroke usually spares hearing and causes direction-changing or vertical nystagmus with a normal head impulse and skew deviation.

In acute, continuous vertigo, the HINTS examination combines the head impulse, nystagmus, and a test of skew: a normal head impulse, direction-changing nystagmus, or skew suggests a central cause, and HINTS is more sensitive for stroke than MRI in the first 48 hours. In positional vertigo, the Dix-Hallpike maneuver, with the head turned 45° to the tested ear and then hanging back, provokes upbeat, torsional nystagmus in posterior canal BPPV.

A central HINTS result, focal signs, a new headache, or stroke risk calls for MRI and MRA, since CT is less sensitive. Unilateral hearing loss or tinnitus calls for audiometry and MRI with gadolinium, and Ménière disease is diagnosed by 2 or more attacks of 20 minutes to 12 hours with documented low-frequency loss. The absence of focal signs does not exclude a central cause.

Management

Treatment targets the cause, and vestibular suppressants only relieve acute symptoms. BPPV is treated with the Epley maneuver, which repositions otoconia out of the posterior canal. Vestibular neuritis and labyrinthitis are treated with antiemetics, antihistamines such as meclizine, or benzodiazepines for about 3 days, then vestibular rehabilitation, and steroids lack evidence. Ménière disease is treated with less salt, caffeine, and alcohol, thiazides or betahistine, intratympanic steroids, and intratympanic gentamicin if refractory. Vestibular migraine is treated as migraine, stroke with stroke care, and vestibular schwannoma with observation, surgery, or radiation. Vestibular rehabilitation trains the brain to balance with visual and proprioceptive cues, and suppressants beyond a few days delay central compensation.

Relationships to other conditions

Hearing loss. Ménière disease, labyrinthitis, and vestibular schwannoma cause hearing loss and tinnitus with vestibular symptoms. Normal hearing favors vestibular neuritis or BPPV.

Stroke symptoms and complications. Lateral medullary and cerebellar strokes can present with acute vertigo. A normal head impulse, direction-changing nystagmus, skew, or other brainstem signs favor stroke.

Multiple sclerosis. Brainstem demyelination is a central cause of vertigo. Other focal signs and internuclear ophthalmoplegia favor a central lesion.

Alzheimer disease: amyloid plaques, tau tangles, and progressive neuronal loss

Amyloid plaques and neurofibrillary tangles cause gliosis and neuronal loss, leading to progressive dementia. Plaques arise when β-secretase and γ-secretase cleave APP to release Aβ, and tangles arise when hyperphosphorylated tau leaves microtubules and aggregates inside neurons. Age is the main risk factor, and mutations in APP, PSEN1, and PSEN2 cause early-onset disease. Loss of neurons in the entorhinal cortex, hippocampus, and cholinergic and monoaminergic nuclei produces memory loss and later global decline. Diagnosis in life is clinical, with biomarkers, and the ABC score needs brain tissue, usually at autopsy. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how Alzheimer disease relates to other neurologic conditions.

Key points

- α-secretase cuts APP inside the Aβ region, so no Aβ forms, whereas β-secretase followed by γ-secretase releases Aβ.
- PSEN1 and PSEN2 encode the catalytic component of γ-secretase, and their mutations cause early-onset familial disease.
- Aβ begins in the neocortex and moves inward, whereas NFTs begin in the entorhinal cortex and move outward.
- Acetylcholinesterase inhibitors offset the loss of cholinergic neurons from the nucleus basalis of Meynert.
- The ABC score grades Aβ plaques, NFTs, and neuritic plaques, and needs brain tissue, usually at autopsy.

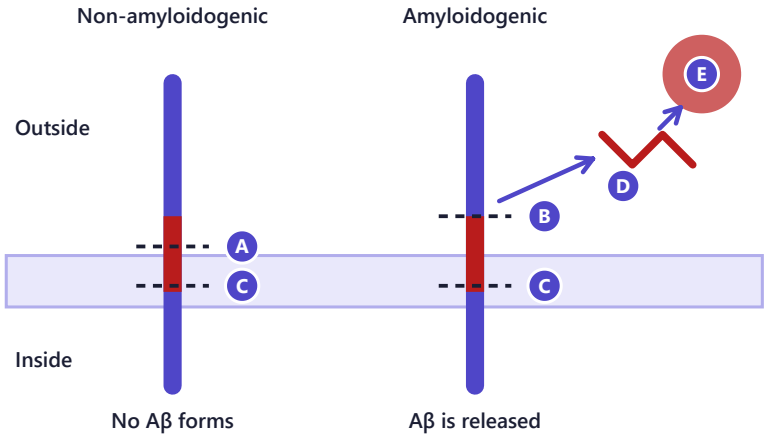


Fig. 1 | Amyloid precursor protein crossing a membrane in two pathways. In the non-amyloidogenic pathway, alpha secretase cuts inside the amyloid beta region, then gamma secretase cuts, and no amyloid beta forms. In the amyloidogenic pathway, beta secretase cuts at the start of the region, then gamma secretase cuts, releasing amyloid beta, which forms fibrils and plaques. A, α-secretase; B, β-secretase; C, γ-secretase; D, Fibrillization; E, Plaque formation.

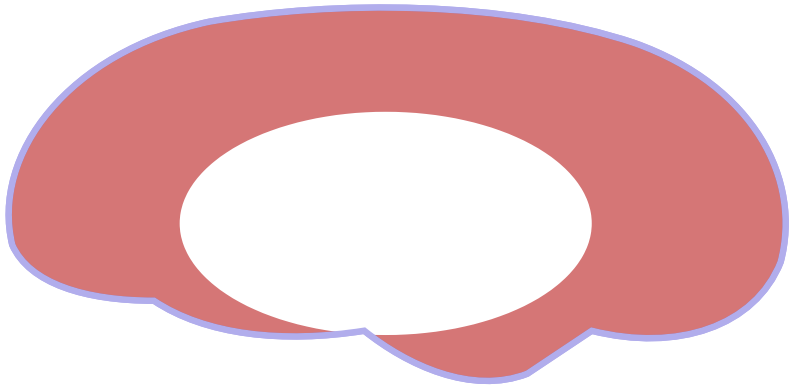


Fig. 2 | A brain with amyloid beta across the neocortex, sparing the deep structures. Aβ: neocortex first, moving inward

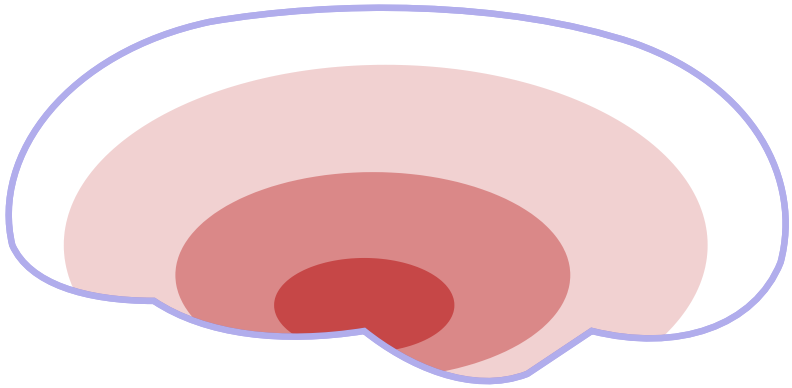


Fig. 3 | A brain with neurofibrillary tangles densest in the entorhinal cortex, fading outward through limbic areas to the neocortex. NFTs: entorhinal cortex first, moving outward

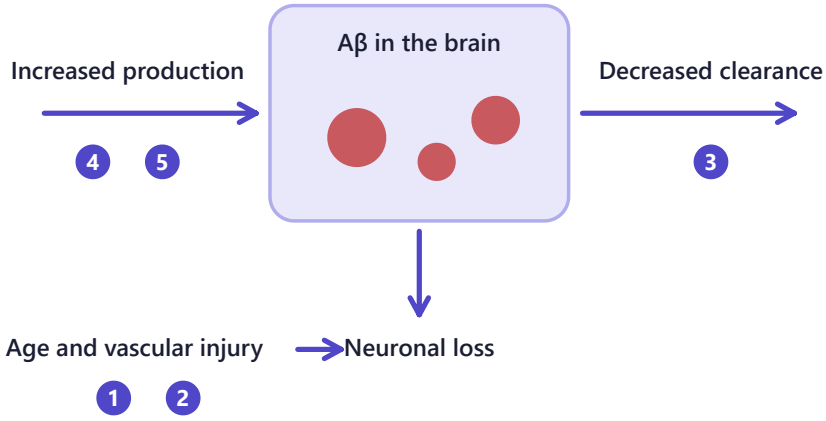


Fig. 4 | Amyloid beta builds up in the brain when production increases or clearance decreases. The build-up leads to neuronal loss, and age and vascular injury add to that loss. 1, Age; 2, Vascular risk factors; 3, APOE ε4; 4, Early-onset mutations; 5, Down syndrome.

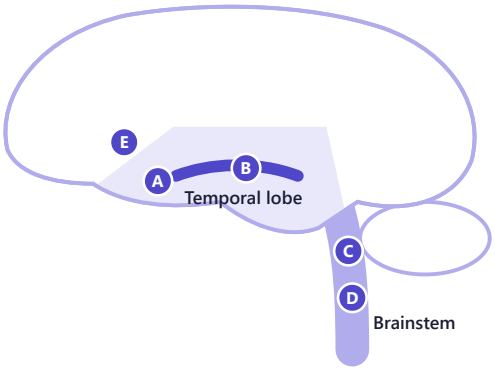


Fig. 5 | A side view of the brain. The entorhinal cortex and hippocampus lie in the medial temporal lobe, the nucleus basalis of Meynert lies in the basal forebrain, and the locus coeruleus and raphe nuclei lie in the brainstem. A, Entorhinal cortex; B, Hippocampus CA1; C, Locus coeruleus; D, Raphe nuclei; E, Nucleus basalis of Meynert.

Pathophysiology

Aβ is produced from APP, forms plaques outside neurons alongside NFTs inside them, and activates microglia and astrocytes, causing gliosis. Neuronal loss then causes atrophy and dementia.

APP is processed by two pathways. In the non-amyloidogenic pathway, α-secretase cuts inside the Aβ region, so no Aβ forms when γ-secretase then cuts within the membrane. In the amyloidogenic pathway, β-secretase cuts at the start of the Aβ region, and the γ-secretase cut releases Aβ. Aβ aggregates into oligomers and then fibrils, which deposit outside neurons as plaques.

Tau is a microtubule-associated protein that stabilizes axons. Hyperphosphorylation detaches tau from microtubules, and tau then aggregates inside neurons as NFTs. Axonal transport fails, and the neurons die.

The two lesions spread differently. Aβ follows the Thal phases, from the neocortex to the allocortex, including the hippocampus, then the diencephalon and striatum, the brainstem, and the cerebellum. NFTs follow the Braak stages, from the entorhinal cortex (I–II) to limbic areas (III–IV) and the neocortex (V–VI). NFTs track symptoms better than plaques do, because they follow the circuits that fail.

Etiology

Age is the main risk factor, and vascular injury from hypertension, diabetes, obesity, and smoking adds to neuronal loss. Decreased Aβ clearance is represented by APOE ε4, which slows clearance and raises the risk of late-onset disease. Increased Aβ production is represented by early-onset mutations in APP, PSEN1, and PSEN2, and by Down syndrome, in which chromosome 21 carries an extra APP gene. The presenilins PSEN1 and PSEN2 are the catalytic component of γ-secretase, and PEN2, APH1, and nicastrin are the other components of the complex.

Clinical presentation

Symptoms follow the structures affected. The entorhinal cortex is the first cortical site of NFTs, causing early memory loss and getting lost, and early atrophy of hippocampus CA1 accompanies short-term memory loss. Noradrenergic loss in the locus coeruleus changes attention and behavior, and serotonergic loss in the raphe nuclei causes depression and sleep changes. Cholinergic neuron loss in the nucleus basalis of Meynert causes memory and attention decline.

Early disease causes short-term memory loss, then executive dysfunction. Middle disease adds language and visuospatial decline, apathy, agitation, and wandering, and late disease causes apraxia, mutism, incontinence, and total dependence. Complications are aspiration pneumonia from dysphagia; infection, with pneumonia the most common cause of death; falls, dehydration, and malnutrition; and agitation, delirium, wandering, and caregiver burden. Life expectancy is about 4–8 years after diagnosis at age 65 or older. Aβ in vessel walls causes cerebral amyloid angiopathy, with lobar ICH.

Diagnosis

Assessment begins with a history that includes a collateral source, followed by cognitive testing with the MMSE or MoCA. Bloods are FBC, TFTs, B12, and electrolytes, and MRI shows medial temporal atrophy. Biomarkers are added if the diagnosis is unclear: CSF shows decreased Aβ42 with increased tau and phosphorylated tau, and amyloid PET shows plaques in life. Genetic testing is considered with early onset and a strong family history.

The ABC score is a neuropathologic score that needs brain tissue, usually at autopsy. A grades Aβ plaques by Thal phase, with A0 for none, A1 for phases 1–2, A2 for phase 3, and A3 for phases 4–5. B grades NFTs by Braak stage, with B0 for none, B1 for stages I–II, B2 for III–IV, and B3 for V–VI. C grades neuritic plaques by CERAD score, with C0 for none, C1 sparse, C2 moderate, and C3 frequent. The three grades combine into a level of

Alzheimer disease neuropathologic change: not, low, intermediate, or high.

Dementia mimics

Delirium is favored by sudden onset, fluctuating attention, and an acute illness, and prompts a search for the cause. Depression shows low mood, poor effort, and awareness of memory problems. Hypothyroidism causes fatigue, cold intolerance, and weight gain, and is found with TFTs, and B12 deficiency causes anemia and neuropathy. Normal pressure hydrocephalus causes gait disturbance and urinary incontinence early, and is assessed with MRI. Dementia with Lewy bodies is favored by visual hallucinations, parkinsonism, and fluctuations. Vascular dementia shows stepwise decline with vascular risk factors, and frontotemporal dementia occurs at a younger age, with early behavior or language change; both are assessed with MRI.

Management

Acetylcholinesterase inhibitors slow ACh breakdown, offsetting cholinergic neuron loss, and give a modest benefit in mild to moderate disease. The drugs are donepezil, rivastigmine, and galantamine, and they can cause bradycardia and GI upset. Memantine is an NMDA receptor antagonist for moderate to severe disease. The anti-amyloid antibodies lecanemab and donanemab clear plaques in early disease, with monitoring for ARIA.

Supportive care covers safety, routine, and caregiver support. Behavior is managed with non-drug approaches first, vascular risk factors are treated, and advance care planning begins early.

Relationships to other conditions

Dementia with Lewy bodies. Both cause progressive dementia. Visual hallucinations, parkinsonism, and fluctuations favor dementia with Lewy bodies, whereas early short-term memory loss with medial temporal atrophy favors Alzheimer disease.

Frontotemporal dementia. Both cause progressive dementia. Early behavior or language change with spared memory favors frontotemporal dementia, whereas early episodic memory loss favors Alzheimer disease.

Intracerebral hemorrhage. Aβ links the two conditions, since Aβ in vessel walls causes cerebral amyloid angiopathy and lobar ICH. Sudden focal deficits with blood on imaging favor hemorrhage, whereas gradual cognitive decline favors Alzheimer disease.

Seizures. Alzheimer disease is a degenerative cause of seizures, so treatment targets the underlying disease as well.

Frontotemporal dementia: frontal and temporal degeneration of behavior and language

Frontotemporal dementia is a group of syndromes caused by frontotemporal lobar degeneration, in which tau, TDP-43, or FUS inclusions kill neurons in the frontal and anterior temporal lobes. The behavioral variant causes disinhibition, apathy, loss of empathy, compulsive behavior, hyperorality, and executive dysfunction with anosognosia, and the language variants cause semantic or nonfluent primary progressive aphasia. About 40% of cases are familial. Diagnosis rests on history, cognitive testing, and frontotemporal atrophy on imaging, and treatment is supportive. Here, we review its pathology, variants, diagnosis, and management.

Key points

- FTLD inclusions contain tau, TDP-43, or, rarely, FUS, and the variant does not reliably predict the protein.
- MAPT causes FTLD-tau, and GRN and C9orf72 cause FTLD-TDP; C9orf72 also causes ALS.
- bvFTD causes disinhibition, apathy, loss of empathy, compulsive behavior, hyperorality, and executive dysfunction, with anosognosia.
- Left anterior temporal atrophy causes the semantic variant, and right anterior temporal atrophy causes prosopagnosia and getting lost.
- SSRIs help behavior, but cholinesterase inhibitors and memantine have no proven benefit.

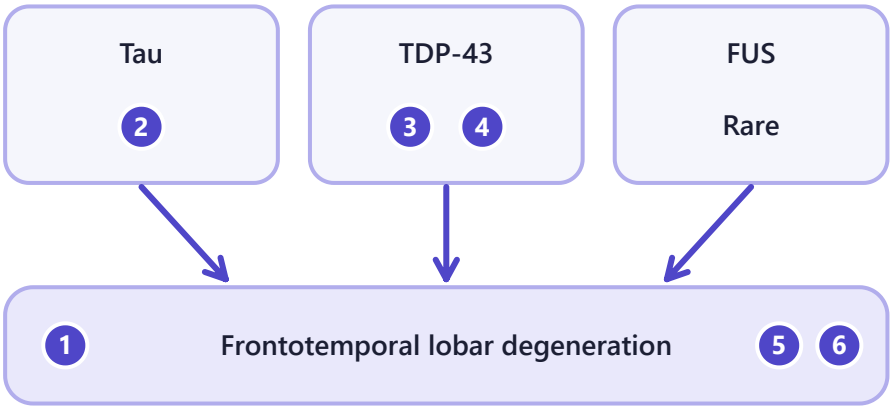


Fig. 1 | Three protein boxes: tau, linked to MAPT, 2; TDP-43, linked to GRN, 3, and C9orf72, 4; and FUS, which is rare. All feed into frontotemporal lobar degeneration, which is usually sporadic, 1, with head trauma, 5, and thyroid disease, 6, as risk factors. 1, Sporadic; 2, MAPT; 3, GRN; 4, C9orf72; 5, Head trauma; 6, Thyroid disease.

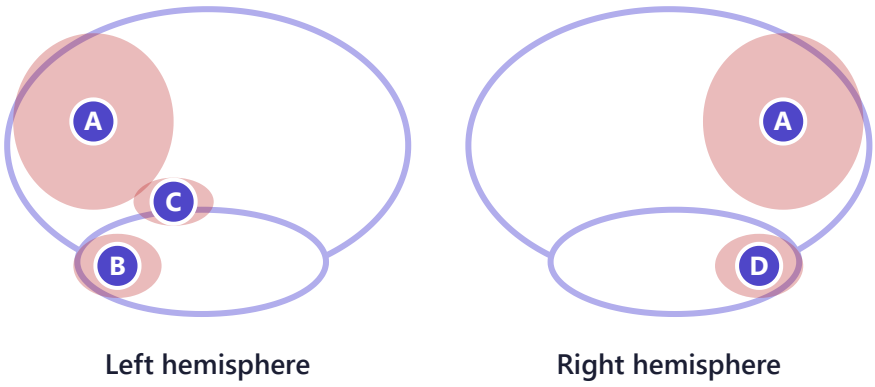


Fig. 2 | Side views of the left and right hemispheres, front toward the outside. A, frontal lobes on both sides: behavioral variant. B, left anterior temporal lobe: semantic variant. C, left inferior frontal and insular cortex: nonfluent variant. D, right anterior temporal lobe: prosopagnosia and getting lost. A, Behavioral variant (bvFTD); B, Semantic variant (svPPA); C, Nonfluent variant (nfvPPA); D, Right temporal variant.

Pathophysiology

Tau, TDP-43, or FUS inclusions kill neurons in the frontal and anterior temporal lobes, producing knife-edge atrophy, so behavior, personality, and language fail while memory is spared early. FTLD-tau includes tau inclusions such as Pick bodies, with ballooned neurons. FTLD-TDP accounts for about half of FTLD, and TDP-43 is also the protein of ALS, whereas FTLD-FUS is rare. The clinical variant does not reliably predict which protein is present.

Etiology

Most cases are sporadic, but about 40% are familial. MAPT, the tau gene, causes FTLD-tau, and GRN and C9orf72 cause FTLD-TDP; the C9orf72 repeat expansion is the most common cause of familial FTD and ALS. Head trauma and thyroid disease raise risk by 3.3-fold and 2.5-fold. Mean onset is at 58 years, making FTD the second most common early-onset dementia, and bvFTD and svPPA are more common in men, nfvPPA in women.

Clinical presentation

The behavioral variant, the most common form, follows frontal involvement. It causes disinhibition, with socially inappropriate and impulsive behavior; apathy, often mistaken for depression; loss of empathy; compulsive behavior, including repetitive movements, rituals, and stereotyped speech; hyperorality, including a sweet tooth, binge eating, and more alcohol; and executive dysfunction, with memory and visuospatial skills spared. Personality changes early, and patients develop anosognosia, losing insight into their changed behavior, and cannot explain why they act as they do.

The semantic variant follows left, dominant, anterior temporal atrophy and causes fluent speech that has lost meaning, with anomia and poor single-word comprehension. The nonfluent variant follows left inferior frontal and insular atrophy and causes effortful, halting, agrammatic speech with apraxia of speech, while word meaning is kept. Right, nondominant, anterior temporal atrophy causes prosopagnosia, getting lost, behavior change, and memory loss.

FTD overlaps with MND, CBD, and PSP. Complications include falls, hypersexuality, unsafe driving and finances, and insomnia and daytime sleepiness, and average survival is 7.5 years.

Diagnosis

Diagnosis begins with an informant history of behavior and personality change and cognitive testing, such as go-no-go, letter fluency, and similarities. MRI or CT shows frontal and anterior temporal atrophy, and SPECT or FDG-PET shows frontotemporal hypoperfusion or hypometabolism, though these findings are not specific. Genetic testing is offered when there is a family history.

Possible bvFTD requires 3 of the 6 behavioral features. The semantic variant requires anomia and impaired single-word comprehension, and the nonfluent variant requires agrammatism or effortful speech with apraxia of speech. Early episodic memory and visuospatial loss favor Alzheimer disease, word-finding pauses with poor repetition favor logopenic PPA, which is usually caused by AD, low mood favors depression, and visual hallucinations, parkinsonism, and fluctuations favor DLB.

Management

No treatment slows FTD, so care targets behavior, safety, and support. SSRIs can improve behavior but not cognition. Cholinesterase inhibitors and memantine have no proven benefit, and donepezil can worsen behavior. Antipsychotics have mixed results, and patients are prone to extrapyramidal effects. Speech therapy helps the PPA variants, physical and occupational therapy

support function and safety, caregivers need education and support, families are offered genetic counseling, and finances, driving, and eating are monitored.

Relationships to other conditions

Alzheimer disease. Both cause progressive dementia, and logopenic PPA is usually caused by AD. Early episodic memory and visuospatial loss favor Alzheimer disease, whereas early behavior or language change with spared memory favors frontotemporal dementia, which does not respond to cholinesterase inhibitors.

Atypical parkinsonism. PSP and CBD are 4R tauopathies that overlap with FTLD. Early parkinsonism, gaze palsy, falls, or asymmetric apraxia favors atypical parkinsonism.

Dementia with Lewy bodies. Both can cause early behavioral and executive change. Visual hallucinations, parkinsonism, and fluctuations favor dementia with Lewy bodies.

Dementia with Lewy bodies: α-synuclein, fluctuating cognition, and antipsychotic sensitivity

Dementia with Lewy bodies is a progressive dementia in which α-synuclein aggregates into Lewy bodies across the brainstem, limbic areas, and cortex, causing cholinergic and dopaminergic deficits. Its core features are fluctuating cognition, recurrent visual hallucinations, RBD, and parkinsonism, and attention, executive function, and visuoperceptual ability fail before memory. Diagnosis is clinical, supported by indicative biomarkers, and treatment targets each symptom, because no treatment slows neuronal loss. Antipsychotics carry a risk of serious reactions. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how dementia with Lewy bodies relates to other neurologic conditions.

Key points

- The four core features are fluctuating cognition, visual hallucinations, RBD, and parkinsonism.
- Probable DLB needs 2 or more core features, or 1 core feature with an indicative biomarker.
- Indicative biomarkers are reduced DAT uptake, low MIBG uptake, and REM sleep without atonia.
- DLB is diagnosed when dementia begins before, with, or within 1 year of parkinsonism.
- Acetylcholinesterase inhibitors come first, and antipsychotics are avoided whenever possible.

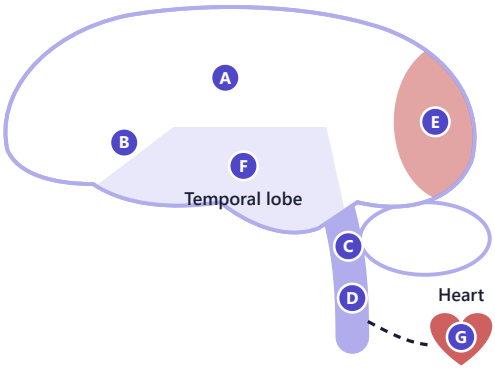


Fig. 1 | A side view of the brain and heart. Lewy bodies affect the cortex, the nucleus basalis of Meynert, the substantia nigra, and the brainstem. The occipital cortex shows reduced metabolism, the medial temporal lobe is relatively preserved, and the heart shows low MIBG uptake. A, Cortex; B, Nucleus basalis of Meynert; C, Substantia nigra; D, Brainstem nuclei; E, Occipital cortex; F, Medial temporal lobe; G, Autonomic nerves.

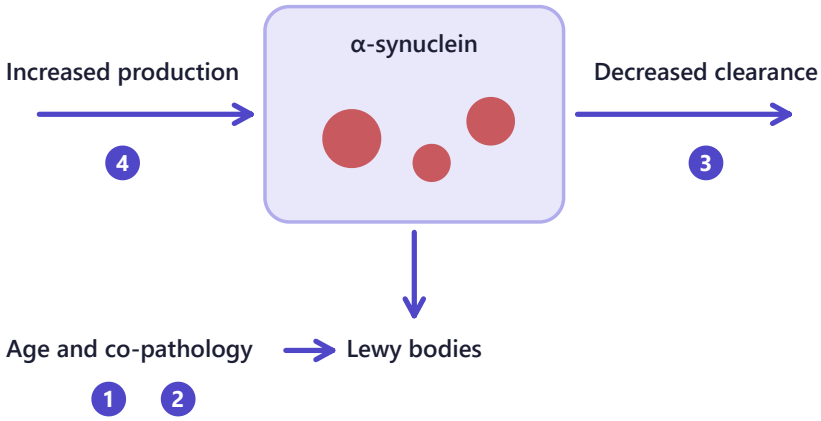


Fig. 2 | Alpha-synuclein builds up in the brain when its production increases or its clearance decreases. The build-up forms Lewy bodies, and age and Alzheimer co-pathology add to the damage. 1, Age; 2, APOE ε4; 3, GBA1 variants; 4, SNCA variants.

Pathophysiology

α-synuclein misfolds and aggregates, and Lewy bodies form as inclusions of α-synuclein and ubiquitin inside neurons across the brainstem, limbic areas, and cortex. Oligomers may be the main toxic form, because Lewy body numbers track neuronal loss poorly.

Each site produces a different effect. Cortical involvement causes attention, executive, and visuospatial decline. Cholinergic loss from the nucleus basalis of Meynert contributes to attention and cognitive decline, and dopaminergic loss from the substantia nigra causes parkinsonism. Brainstem nuclei are also affected, including the locus coeruleus, dorsal raphe, and dorsal motor nucleus of the vagus, with REM sleep without atonia and RBD. The occipital cortex shows reduced metabolism, with visual hallucinations, whereas the medial temporal lobe is relatively preserved, so memory is spared early. Autonomic nerve involvement causes low cardiac MIBG uptake and autonomic dysfunction.

Aβ plaques and tau tangles commonly coexist, and up to 30% of DLB meets the threshold for Alzheimer disease. This co-pathology brings more memory loss, faster progression, and higher mortality.

Etiology

Many studies find DLB more often in men, most people have no family history, and no single cause is known. Incidence rises with age, and APOE ε4 is a risk gene shared with Alzheimer disease. Decreased clearance is represented by GBA1 variants, which affect a lysosomal enzyme and are shared with Parkinson disease. SNCA, the α-synuclein gene, represents increased production, with rare family multiplications.

Clinical presentation

Attention, executive function, and visuoperceptual ability are affected early, and memory impairment becomes evident later. Fluctuating cognition causes pronounced variations in attention and alertness. Visual hallucinations are recurrent, well formed, and detailed. RBD causes acting out of dreams and may precede cognitive decline, and parkinsonism causes bradykinesia, rest tremor, or rigidity.

Supportive features are autonomic dysfunction, with orthostatic hypotension, which is often early, constipation, and urinary incontinence; postural instability, repeated falls, syncope, and transient unresponsiveness; hypersomnia and hyposmia; and delusions, misidentification, other hallucinations, apathy, anxiety, and depression. Complications are antipsychotic reactions, with sudden change in consciousness, impaired swallowing, acute confusion, and worse parkinsonism, as well as traumatic fractures from falls. Life expectancy is 5–8 years after diagnosis, and causes of death include falls, pneumonia, swallowing problems, cardiac complications, and suicide.

Diagnosis

Assessment begins with a history that includes a collateral source, cognitive testing with the MMSE or MoCA, bloods with a chemistry panel, TFTs, B12, syphilis, and HIV, and MRI, which shows relatively preserved medial temporal lobes. Biomarkers are added if the diagnosis is unclear.

Dementia is essential, defined as progressive cognitive decline that interferes with daily activities. Probable DLB needs 2 or more core features, or 1 core feature with 1 or more indicative biomarkers, and biomarkers alone cannot make the diagnosis. Possible DLB is 1 core feature with no indicative biomarker, or an indicative biomarker with no core feature. DLB is less likely if another illness explains the picture, or if parkinsonism is the only core feature and first appears in severe dementia.

Indicative biomarkers are reduced basal ganglia uptake on DAT SPECT or PET, low uptake on MIBG myocardial scintigraphy, and REM sleep without atonia on polysomnography. Supportive biomarkers are relative preservation of the medial temporal lobes on CT or MRI, low occipital activity on SPECT or PET with the cingulate island sign on FDG-PET, and prominent posterior slow-wave activity on EEG. CSF α-synuclein seed amplification assays have over 90% sensitivity and specificity, and only autopsy gives a definite diagnosis. DLB is diagnosed when dementia begins before, with, or within 1 year of parkinsonism, and PDD when dementia begins 1 year or more after established Parkinson disease.

DLB mimics

Alzheimer disease is favored by prominent early memory loss and medial temporal atrophy on MRI. Delirium has a sudden onset, an acute illness, and a reversible cause. Vascular dementia shows stepwise decline with vascular risk factors. Frontotemporal dementia occurs at a younger age, with early behavior or language change, and prion disease causes rapid progression with myoclonus. Drug-induced parkinsonism follows recent dopamine-blocking drug use and is assessed with a medication review. Progressive supranuclear palsy causes early falls and vertical gaze palsy.

Management

No treatment slows neuronal loss, so treatment targets each symptom. Acetylcholinesterase inhibitors, rivastigmine, donepezil, and galantamine, slow ACh breakdown, offsetting cholinergic loss, and improve cognition, function, and hallucinations. Memantine may

add benefit for cognition. Parkinsonism is less responsive than in Parkinson disease and is treated with low-dose levodopa, increased slowly. RBD is treated with melatonin, or cautious bedtime clonazepam, and depression with SSRIs, SNRIs, or mirtazapine. Autonomic dysfunction is treated as in other diseases, with little DLB trial data. Falls call for a safety assessment, bone density, and vitamin D.

Antipsychotics are avoided whenever possible, because serious reactions occur in up to 50% of people. If one is essential, low-dose quetiapine is used, though a small trial was negative, and haloperidol, risperidone, olanzapine, and aripiprazole are generally avoided. Supportive care covers caregiver education, home modifications, physical, occupational, and speech therapy, and early advance care planning.

Relationships to other conditions

Alzheimer disease. Both cause progressive dementia, the two pathologies often coexist, and both are treated with acetylcholinesterase inhibitors. Prominent early memory loss and medial temporal atrophy favor Alzheimer disease, whereas fluctuations, visual hallucinations, RBD, and parkinsonism favor dementia with Lewy bodies.

Typical Parkinson disease. Both involve α-synuclein and parkinsonism. Dementia before, with, or within 1 year of parkinsonism favors dementia with Lewy bodies, whereas dementia 1 year or more after established Parkinson disease is PDD.

Motor signs and syndromes: localizing weakness, abnormal movements, and nerve lesions

Motor signs localize lesions along a chain from the cortex through the brainstem and spinal cord to the peripheral nerve, with the basal ganglia, cerebellum, and dorsal columns tuning movement alongside. UMN lesions cause spastic weakness with brisk reflexes, whereas LMN lesions cause flaccid weakness with atrophy and fasciculations. Facial palsy that spares the forehead is supranuclear, gait patterns point to their lesion, ataxia separates cerebellar from sensory causes, and tremor type, dystonia, chorea, ballismus, myoclonus, and apraxia each have characteristic sites. Peripheral nerve lesions produce wrist drop, claw hand, ape hand, and foot drop. Here, we review these signs and what they localize.

Key points

- UMN lesions cause spasticity, brisk reflexes, and a Babinski sign; LMN lesions cause flaccid weakness, atrophy, and fasciculations.
- Supranuclear facial palsy spares the forehead, whereas nuclear and peripheral palsy weakens the whole half of the face.
- Midline cerebellar lesions cause gait ataxia, hemispheric lesions cause limb ataxia on the same side, and dorsal column lesions cause sensory ataxia with a positive Romberg test.
- Resting tremor suggests PD, postural tremor essential tremor, intention tremor the cerebellum, and rubral tremor a midbrain lesion.
- Hemiballismus classically follows a lesion of the contralateral subthalamic nucleus and tends to resolve.
- Radial nerve injury causes wrist drop, ulnar claw hand, median ape hand, and common peroneal foot drop.

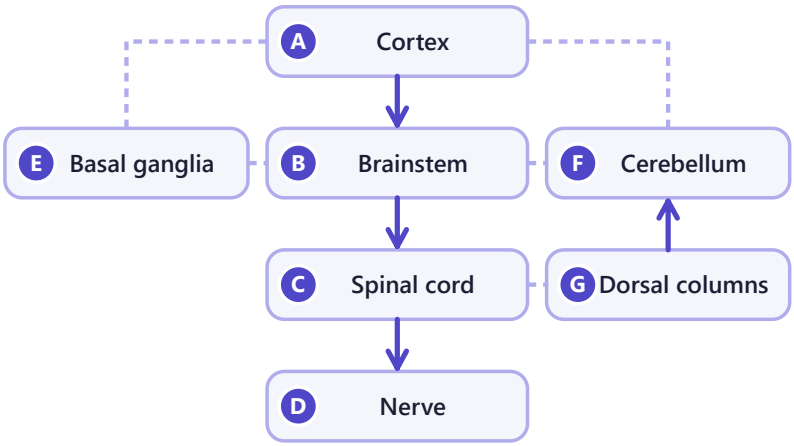


Fig. 1 | The motor system as a chain: cortex, A, through the internal capsule and brainstem, B, to the spinal cord, C, and the peripheral nerve, D. Beside the chain, the basal ganglia, E, and cerebellum, F, tune movement, and the dorsal columns, G, return position sense. A, Cortex and internal capsule; B, Brainstem; C, Spinal cord; D, Peripheral nerve; E, Basal ganglia; F, Cerebellum; G, Dorsal columns.

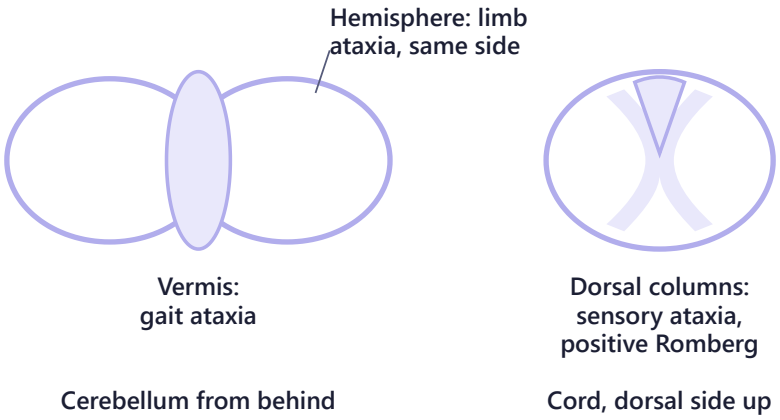


Fig. 2 | Where ataxia comes from: the cerebellum from behind, where a vermis lesion causes gait ataxia and a hemisphere lesion causes limb ataxia on the same side, and the dorsal columns of the cord, whose loss causes sensory ataxia with a positive Romberg test.

Where motor signs arise

Voluntary movement passes from the cortex through the internal capsule and brainstem to the spinal cord and peripheral nerve, while the basal ganglia and cerebellum tune it and the dorsal columns return position sense. UMN lesions, from cortex to cord, cause spastic, velocity-dependent tone, brisk reflexes, a Babinski sign, and weakness worst in arm extensors and leg flexors, as in stroke, MS, and cord injury. LMN lesions, from anterior horn cell to nerve, cause flaccid weakness, reduced reflexes, atrophy, and fasciculations in the pattern of a nerve or root, as in polio, SMA, and nerve injury. A UMN lesion above the pyramidal decussation causes contralateral signs, and one below it, in the cord, causes ipsilateral signs.

Upper motor neuron weakness

Hemiparesis and hemiplegia follow lesions of the contralateral cortex, internal capsule, or brainstem, or of the ipsilateral cord, and cause one-sided weakness with spasticity, brisk reflexes, and a Babinski sign; the face is involved if the lesion lies above the facial nucleus. Stroke, tumor, trauma, infection, and MS are causes, and the gait is hemiparetic. Quadriplegia follows cervical cord or bilateral brainstem lesions, and injury above C5 also paralyzes the diaphragm. Decorticate posturing, with flexed arms and extended legs, follows lesions above the red nucleus, and decerebrate posturing, with extension of all limbs, follows lesions at or below it, in the midbrain or rostral pons; both are usually responses to noxious stimuli and signs of severe brain injury. Central cord syndrome, the most common incomplete injury, weakens the arms more than the legs after neck hyperextension with stenosis.

Supranuclear facial palsy, from a contralateral cortical or corticobulbar lesion, weakens the lower face and spares the forehead, which receives input from both hemispheres; bilateral lesions cause pseudobulbar palsy, with slurred speech, dysphagia, a brisk jaw jerk, and pseudobulbar affect.

Lower motor neuron weakness

Nuclear and peripheral facial palsy, from the ipsilateral facial nucleus or CN VII, weakens the whole half of the face, with hyperacusis, taste loss, and dry eye. Bell palsy causes 38–83% of cases and peaks within 72 hours after ear pain; it is treated with steroids, antivirals, and eye protection, and over 95% recover with prompt treatment. A pontine stroke also weakens the forehead but brings other brainstem signs.

Radial nerve injury at the spiral groove of the humerus, from Saturday night palsy, a humeral shaft fracture, or crutches, causes wrist drop with loss of wrist, finger, and thumb extension and a numb dorsoradial hand; posterior interosseous nerve lesions at the elbow are purely motor. Splinting and rehabilitation are the treatment, with surgery if it does not recover. Ulnar nerve compression, most often at the cubital tunnel and also at the Guyon canal, causes tingling of the ring and little fingers, weak grip and pinch with a Froment sign, and, when severe, intrinsic wasting and clawing.

Median nerve compression, most often in the carpal tunnel, causes night pain and tingling of the thumb, index, middle, and half of the ring finger, with Phalen and Tinel signs; thenar sensation is spared because its branch leaves above the wrist, and severe cases cause thenar wasting and ape hand. Repetitive or vibrating work, pregnancy, diabetes, hypothyroidism, rheumatoid arthritis, and obesity raise risk, and surgical release treats severe or refractory cases. Common peroneal nerve compression at the fibular head, from leg crossing, casts, kneeling, squatting, surgery positioning, or trauma, causes foot drop with weak dorsiflexion and eversion, a numb dorsum of the foot and lateral leg, and a steppage gait. Inversion, supplied by the tibial nerve, is spared, which separates it from L5 radiculopathy, the

most common cause of foot drop. An AFO and conservative care usually suffice.

Erb palsy, a birth traction injury of C5–C6 associated with shoulder dystocia, holds the arm in the waiter's tip position, with the shoulder adducted and internally rotated, elbow extended, forearm pronated, and wrist flexed, and abolishes the biceps reflex. Klumpke palsy, a C8–T1 injury from hyperabduction traction or an apical lung tumor, causes intrinsic hand weakness and clawing with medial forearm sensory loss, and Horner syndrome if sympathetic fibers are involved. Long thoracic nerve injury weakens the serratus anterior and wings the scapula; axillary nerve injury, from anterior dislocation or a proximal humerus fracture, weakens the deltoid and teres minor and numbs the regimental badge area; and musculocutaneous nerve injury weakens elbow flexion and numbs the lateral forearm.

Ataxia

Midline vermis lesions cause gait ataxia, with a broad-based gait and truncal sway, from alcohol, drugs such as phenytoin, lithium, and carbamazepine, or degeneration. Hemispheric lesions cause limb ataxia on the same side, with dysmetria, past-pointing, intention tremor, dysdiadochokinesia, rebound, pendular reflexes, scanning speech, and gaze-evoked nystagmus, from posterior circulation stroke, tumor, abscess, or MS. Sensory ataxia follows dorsal column or large-fiber loss, with lost position sense, a stomping gait, pseudoathetosis, and a positive Romberg test, as in B12 deficiency and tabes dorsalis. The Romberg test detects dorsal column disease, because cerebellar ataxia persists with the eyes open.

Tremor

Resting tremor, easing with movement and unilateral at onset, suggests PD. Postural tremor of both hands at 6–12 Hz, sometimes with head and voice tremor, is essential tremor, which is often familial and treated with propranolol or primidone. Intention tremor grows as the hand nears a target and follows cerebellar lesions. Rubral, or Holmes, tremor is slow, under 4.5 Hz, and coarse, and occurs at rest, in posture, and with intention, because midbrain and cerebellothalamic lesions, usually from stroke, disrupt both cerebellar and dopamine pathways; it begins weeks to 2 years after the lesion.

Dystonia, chorea, and ballismus

Dystonia is sustained co-contraction of agonists and antagonists that twists the body into abnormal postures, worsens with action, and eases with a sensory trick. Cervical dystonia is the most common in adults, with blepharospasm and writer's cramp as other focal forms, and early-onset generalized dystonia is linked to DYT1 and DYT6. Antipsychotics and dopa-responsive dystonia are other causes. Botulinum toxin treats focal dystonia, trihexyphenidyl childhood generalized dystonia, and levodopa dopa-responsive dystonia, with baclofen and benzodiazepines as alternatives.

Hemichorea and hemiballismus classically follow a lesion of the contralateral subthalamic nucleus. Chorea is irregular, flowing, and dance-like, and ballismus is violent, flinging, and high in amplitude. Ischemic stroke is the most common cause over 65, and nonketotic hyperglycemia in type 2 diabetes is another. The movements tend to resolve, and treatment addresses the cause, with haloperidol or risperidone, tetrabenazine, clonazepam, or topiramate. Bilateral chorea with dementia and a family history suggests Huntington disease.

Myoclonus and apraxia

Myoclonus is a sign of sudden, brief, lightning-like jerks. Asterixis is negative myoclonus, brief lapses in tone of the outstretched wrists, as in liver failure. Physiologic myoclonus, such as hiccups, startle, and jerks on falling asleep, needs only reassurance, JME causes morning jerks from puberty, and secondary myoclonus follows hypoxic brain injury, drugs including anticonvulsants, and kidney or liver failure.

Apraxia is loss of learned, skilled movement despite normal strength, sensation, and coordination. Ideational apraxia loses the concept, ideomotor apraxia disconnects concept from output, limb-kinetic apraxia loses fine output, and constructional apraxia impairs drawing and copying. It follows lesions of the dominant parietal and premotor praxis network, occurring in 50–80% of left hemisphere strokes, and is a feature of Alzheimer disease and corticobasal syndrome; physical and occupational therapy are the treatment.

Gait

A spastic hemiparetic gait swings the extended leg in an arc, and a scissor gait, crouched with crossing legs, follows cerebral palsy. A neuropathic steppage gait lifts the foot high and lands on the toes because of foot drop from peroneal, L5, or neuropathic lesions. A cerebellar gait is broad-based and unsteady, a sensory ataxic gait stomps and worsens in the dark, as in B12 deficiency, a festinating gait shuffles with reduced arm swing and freezing in PD, and a waddling gait follows proximal weakness. Most gait disorders are not neurologic, and pain causes an antalgic gait.

Relationships to other conditions

Stroke symptoms and complications. Stroke is the most common cause of hemiparesis, supranuclear facial

palsy, hemiballismus, apraxia, and rubral tremor. A sudden onset in an arterial territory favors stroke.

Typical Parkinson disease. Resting tremor and a festinating gait point to Parkinson disease, whereas postural tremor without bradykinesia favors essential tremor.

Huntington disease. Chorea is the hallmark of Huntington disease. Bilateral chorea with dementia and a family history favors it over hemichorea from a focal lesion.

Atypical parkinsonism. Corticobasal degeneration causes apraxia, dystonia, and myoclonus, and multiple system atrophy causes cerebellar ataxia. Their combination with parkinsonism favors these disorders.

Parkinson disease: α-synuclein, dopamine neuron loss, and dopaminergic treatment

Parkinson disease is a progressive neurodegenerative disorder in which α-synuclein aggregates into Lewy bodies and dopamine neurons of the SNc die, depleting dopamine in the striatum. Genetic factors, such as SNCA and LRRK2, interact with environmental exposures, such as pesticides and MPTP, and with age. Mitochondrial dysfunction, oxidative stress, failure of the ubiquitin-proteasome system, and glutamate excitotoxicity drive neuron loss. Diagnosis is clinical, by the MDS criteria, with DAT SPECT when unclear, and treatment restores dopamine signaling, because no treatment slows the disease. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how Parkinson disease relates to other neurologic conditions.

Key points

- Lewy bodies are α-synuclein-stained inclusions, and the SNc loses its pigment as dopamine neurons die.
- Motor signs appear once 50–60% of nigrostriatal neurons are lost.
- Parkinsonism means bradykinesia plus rest tremor or rigidity.
- The MDS criteria weigh absolute exclusions, red flags, and supportive criteria.
- Levodopa with carbidopa is the most effective treatment, and dopamine agonists cause impulse control disorders.

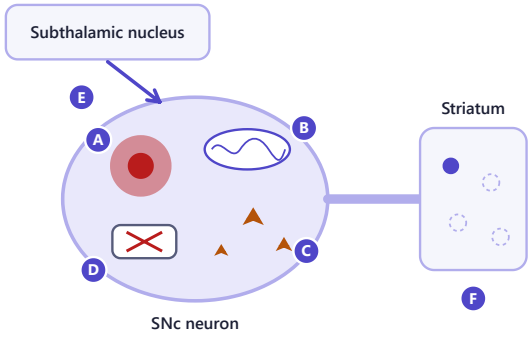


Fig. 1 | A dopamine neuron in the substantia nigra pars compacta projecting to the striatum. Inside the cell body are a Lewy body, a failing mitochondrion, reactive oxygen species, and a blocked proteasome. Glutamate from the subthalamic nucleus floods the neuron, and dopamine release in the striatum falls. A, α-synuclein aggregation; B, Mitochondrial dysfunction; C, Oxidative stress; D, Ubiquitin-proteasome failure; E, Glutamate excitotoxicity; F, Dopamine loss.



Fig. 2 | A midbrain cross-section with a dark band of pigmented substantia nigra on each side. Normal: pigmented substantia nigra



Fig. 3 | A midbrain cross-section in Parkinson disease with pale, depigmented substantia nigra. Parkinson disease: depigmented SNc

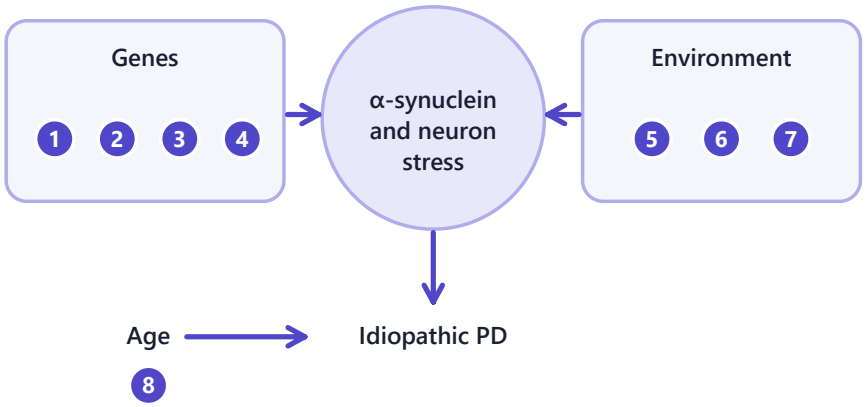


Fig. 4 | Genes on the left and environmental exposures on the right both feed into neuron stress in the center, which leads to idiopathic Parkinson disease; age adds to the risk. 1, SNCA; 2, LRRK2; 3, PRKN, PINK1, and DJ-1; 4, GBA1; 5, Pesticides and herbicides; 6, MPTP; 7, Head injury; 8, Age.

Pathophysiology

Genetic and environmental factors make α-synuclein aggregate into Lewy bodies, and dopamine neurons of the SNc die, leaving too little dopamine in the striatum to drive movement. Motor signs appear once 50–60% of nigrostriatal neurons are lost.

Several mechanisms kill dopamine neurons. Misfolded α-synuclein forms Lewy bodies and Lewy neurites. MPTP and rotenone accumulate in mitochondria and impair them. Reactive oxygen species cause oxidative stress, which DJ-1 normally counters by protecting mitochondria. Failure of the ubiquitin-proteasome system leaves misfolded proteins uncleared, and parkin is a ubiquitin ligase in this system. An overactive subthalamic nucleus floods the SNc with glutamate, causing excitotoxicity.

Loss of neuromelanin-containing dopamine neurons makes the SNc pale. Lewy bodies are α-synuclein-stained inclusions in neuron cell bodies, and Lewy neurites are inclusions in neuron processes. By Braak staging, pathology spreads from the gut, the dorsal motor nucleus of the vagus, and the olfactory bulb to the locus coeruleus, the SNc, and finally the cortex, so constipation, RBD, and anosmia can precede motor signs by years.

Etiology

SNCA, the α-synuclein gene, causes familial PD when extra copies are present. LRRK2 is the most common dominant gene, accounting for 40% of familial cases, and PRKN, PINK1, and DJ-1 serve mitochondrial quality control, while GBA1 encodes a lysosomal enzyme. Environmental factors include pesticides and herbicides, such as rotenone, MPTP, which is taken up by DAT into dopamine neurons, and mild to moderate head injury. Age is a major factor, and PD affects 1% of people over 60.

Most PD is idiopathic: incompletely penetrant genes, such as LRRK2, raise risk most with toxin exposure, and known genes explain only about 3–5% of cases. Drug-induced parkinsonism follows D2 receptor blockade in the striatum, by first-generation antipsychotics, especially haloperidol and fluphenazine, by risperidone

and olanzapine more than quetiapine and clozapine, and by metoclopramide and prochlorperazine, or dopamine depletion by the VMAT2 inhibitors tetrabenazine and reserpine. Its signs are symmetric and widespread after starting the drug, DAT SPECT is normal, and it is most common at ages 60–80. Smoking, caffeine, and physical activity are linked to lower risk.

Clinical presentation

Bradykinesia, slow movement with decreasing amplitude, is the core feature. Rest tremor is unilateral at onset and reemerges at rest but is not required. Rigidity is lead pipe or cogwheel, with reduced arm swing, and postural instability, with festination, freezing, and falls, appears late.

Non-motor features include prodromal constipation, RBD, and anosmia; autonomic orthostatic hypotension, urinary retention, dysphagia, and erectile dysfunction; and depression and anxiety, with dementia and psychosis later. Complications are motor fluctuations, with 'off' periods as levodopa wears off, levodopa-induced dyskinesia, impulse control disorders, mainly with dopamine agonists, and falls.

Diagnosis

Diagnosis is clinical. Parkinsonism means bradykinesia plus rest tremor or rigidity. By the 2015 MDS criteria, clinically probable PD needs parkinsonism, no absolute exclusion criteria, and any red flags balanced by supportive criteria, and clinically established PD needs at least 2 supportive criteria and no red flags.

Absolute exclusion criteria are cerebellar abnormalities, downward vertical supranuclear gaze palsy, bvFTD or primary progressive aphasia, parkinsonism limited to the legs for over 3 years, no response to high-dose levodopa, cortical sensory loss, limb apraxia, or progressive aphasia, and normal DAT imaging. Red flags are a wheelchair within 5 years, no progression over 5 years, early severe bulbar dysfunction, inspiratory stridor, severe autonomic failure within 5 years, recurrent falls within 3 years, disproportionate anterocollis, no non-motor features after 5 years, and unexplained pyramidal signs. Supportive criteria are a clear, dramatic response to dopaminergic therapy, levodopa-induced dyskinesia, rest tremor of a limb, and olfactory loss or cardiac MIBG denervation.

^{123}I-ioflupane SPECT shows presynaptic dopamine transporters in the striatum. Uptake is reduced with dopamine loss, in PD, MSA, PSP, CBD, and DLB, and normal in essential tremor and drug-induced and vascular parkinsonism. A clear response to levodopa supports PD, and a poor response suggests another diagnosis.

Atypical parkinsonism and mimics

Progressive supranuclear palsy causes vertical gaze palsy and early falls. Multiple system atrophy causes early autonomic failure, with cerebellar and pyramidal signs, and corticobasal degeneration causes asymmetric apraxia, cortical sensory loss, and alien limb. Dementia with Lewy bodies brings early dementia and visual hallucinations. Drug-induced parkinsonism follows dopamine-blocking drugs, and essential tremor causes a bilateral postural or action tremor; both have normal DAT imaging.

Management

No treatment slows PD, so treatment restores dopamine signaling and manages symptoms. Levodopa crosses the BBB and becomes dopamine, and carbidopa blocks its peripheral conversion; together they are the most effective treatment. Pramipexole is a dopamine agonist, selective for D3 receptors, rasagiline is a selective MAO-B inhibitor, and entacapone is a COMT inhibitor that prolongs each levodopa dose.

Amantadine, a weak NMDA antagonist, increases dopamine release and blocks its reuptake, easing bradykinesia, rigidity, and tremor, often only briefly. Anticholinergics such as trihexyphenidyl block muscarinic receptors and mainly help tremor, at the cost of dry mouth and other anticholinergic effects. Drug-induced parkinsonism is treated by stopping the causative drug, using an atypical antipsychotic if one is still needed, and anticholinergics such as benztropine.

As neurons are lost, 'on' time shortens, and entacapone or extended-release levodopa is added. Dopamine agonists such as pramipexole are the main trigger of impulse control disorders, including gambling, hypersexuality, compulsive shopping, and binge eating, which call for reducing or stopping the agonist. DBS treats refractory motor fluctuations. Non-motor symptoms are treated with PEG for constipation, SSRIs for depression and anxiety, cholinesterase inhibitors for dementia, pimavanserin or other antipsychotics for psychosis, and cognitive behavior therapy for impulsive behavior.

Relationships to other conditions

Dementia with Lewy bodies. Both involve α-synuclein, Lewy bodies, and parkinsonism. Dementia before, with, or within 1 year of parkinsonism favors dementia with Lewy bodies, whereas dementia 1 year or more after established Parkinson disease is Parkinson disease dementia.

Atypical parkinsonism. Progressive supranuclear palsy, multiple system atrophy, and corticobasal degeneration also cause parkinsonism with reduced DAT

uptake, but they respond poorly to levodopa and add features such as vertical gaze palsy, early autonomic failure, or asymmetric apraxia.

Atypical parkinsonism: multiple system atrophy, progressive supranuclear palsy, and corticobasal degeneration

Atypical parkinsonism describes neurodegenerative disorders that cause parkinsonism with a poor response to levodopa and additional signs. Multiple system atrophy is an α-synucleinopathy with GCIs in oligodendrocytes, whereas progressive supranuclear palsy and corticobasal degeneration are 4R tauopathies. Each damages a characteristic pattern of structures, producing early autonomic failure and cerebellar signs, vertical gaze palsy and early falls, or asymmetric cortical signs. Diagnosis is clinical, supported by MRI patterns, and treatment is symptomatic. Here, we review the proteinopathies, clinical features, imaging, and management of these disorders.

Key points

- α-synuclein collects in PD, DLB, and MSA, 4R tau in PSP and CBD, Aβ with 3R and 4R tau in AD, and TDP-43 in MND and LATE.
- MSA presents with parkinsonism, cerebellar signs, or pyramidal signs, with early autonomic failure.
- PSP causes vertical supranuclear gaze palsy, early falls, akinesia, and cognitive dysfunction.
- CBD causes asymmetric rigidity, dystonia, and myoclonus with apraxia, cortical sensory loss, and alien limb.
- The hot cross bun sign suggests MSA, and midbrain atrophy suggests PSP.
- No treatment slows these disorders, so all three are treated by symptom.

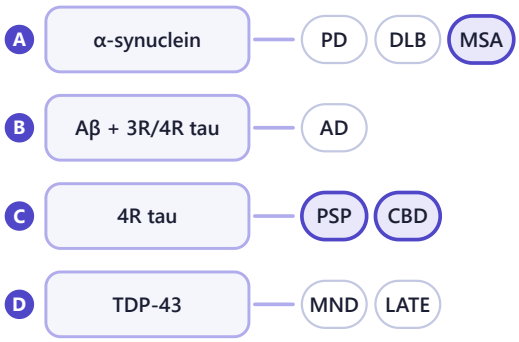


Fig. 1 | Four rows group neurodegenerative diseases by the protein that collects. A, alpha-synuclein: Parkinson disease, dementia with Lewy bodies, and multiple system atrophy. B, amyloid beta with three-repeat and four-repeat tau: Alzheimer disease. C, four-repeat tau: progressive supranuclear palsy and corticobasal degeneration. D, TDP-43: motor neuron disease and LATE. The three atypical parkinsonian disorders are shaded. Shaded: atypical parkinsonism A, α-synuclein; B, Aβ plus 3R and 4R tau; C, 4R tau; D, TDP-43.

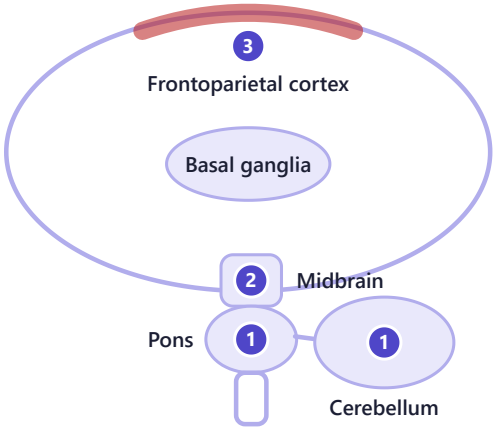


Fig. 2 | A side view of the brain and brainstem. Multiple system atrophy, 1, affects the pons and cerebellum; progressive supranuclear palsy, 2, affects the midbrain; corticobasal degeneration, 3, affects the frontoparietal cortex. All three involve the basal ganglia. 1, Multiple system atrophy; 2, Progressive supranuclear palsy; 3, Corticobasal degeneration.

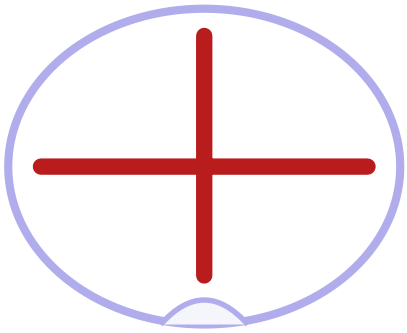


Fig. 3 | An axial slice of the pons with a bright cross through it, the hot cross bun sign. MSA: hot cross bun sign

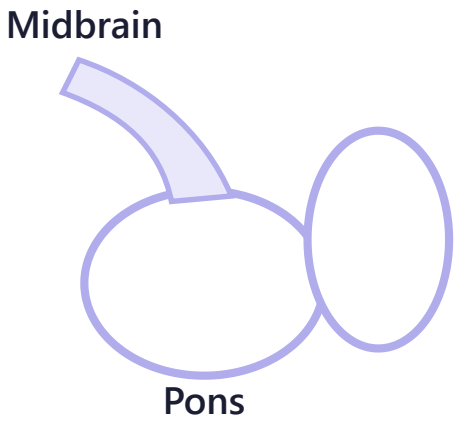


Fig. 4 | A midline side view of the brainstem with a thin, beak-shaped midbrain above a normal pons, the hummingbird sign. PSP: midbrain atrophy

Pathophysiology

An abnormal protein collects in neurons and glia, and cells of the basal ganglia, brainstem, cerebellum, or cortex die, causing parkinsonism that responds poorly to levodopa because striatal targets are lost too.

Neurodegenerative diseases can be grouped by the protein that collects. α-synuclein collects in PD and DLB in neurons and in MSA in oligodendrocytes. AD combines Aβ with tangles that mix 3R and 4R tau. PSP and CBD are 4R tauopathies affecting neurons and astrocytes. TDP-43 collects in MND, LATE, and about half of FTLD.

In MSA, GCIs of α-synuclein in oligodendrocytes accompany widespread neuron loss in the putamen, pons, MCP, and cerebellum, with increased putaminal iron. In PSP, neuronal and glial 4R tau, including tufted astrocytes, affects the basal ganglia and brainstem, with variable cortical involvement. In CBD, neuronal and glial 4R tau, astrocytic plaques, and ballooned neurons affect the posterior frontal and anterior parietal cortex and basal ganglia asymmetrically.

Etiology

All three disorders are sporadic, and their causes are unknown. A family history of parkinsonism or ataxia argues against MSA. In CBD, misfolded tau is thought to spread from cell to cell in a prion-like manner, seeding further pathology.

Clinical presentation

MSA presents with parkinsonism (MSA-P), with bradykinesia, rigidity, tremor, and postural instability, from striatonigral degeneration; with cerebellar signs (MSA-C), including gait and limb ataxia, dysarthria, and nystagmus, from olivopontocerebellar atrophy; or with both, and pyramidal signs, such as hyperreflexia and a Babinski sign, are common. MSA-P accounts for 70–80% of cases in Western populations, whereas MSA-C accounts for 67–84% in Asian populations and begins earlier. Its features include early autonomic failure, with orthostatic hypotension, urinary dysfunction, and erectile dysfunction, a poor levodopa response, dysphagia, hypometric saccades, falls, myoclonus, RBD, which often comes first, and stridor and sleep apnea.

The hallmark of PSP is vertical supranuclear gaze palsy, beginning with slow vertical saccades, with downgaze

affected before upgaze. Postural instability causes early backward falls with a stiff, upright gait, and akinesia is symmetric and axial. Cognitive dysfunction includes apathy, disinhibition, and executive dysfunction. The 2017 MDS criteria describe subtypes: PSP-RS, the classic form, with gaze palsy, early falls, and cognitive change, leads to death about 7 years from onset; PSP-P, which may respond to levodopa, and PSP-PAGF, with gait freezing, allow survival of a decade or more; and cortical variants include PSP-CBS, PSP-bvFTD, and PSP-PNFA.

CBD causes asymmetric limb rigidity or akinesia, dystonia, and myoclonus, with cortical signs: limb apraxia, cortical sensory loss, and alien limb. Nonfluent aphasia and executive dysfunction can occur.

Complications include falls with head injury, dysphagia with aspiration pneumonia, and, in MSA, stridor and sleep apnea from vocal cord paralysis. Patients with PSP become dependent within 3–4 years, and median survival in CBD is about 8 years.

Diagnosis

Diagnosis is clinical: parkinsonism with red flags and a poor or short-lived response to a levodopa trial. MRI shows the pattern of atrophy, and DAT SPECT is reduced in all three, as in PD, so it does not separate them from PD. Only autopsy confirms the protein.

Early autonomic failure, cerebellar or pyramidal signs, and stridor favor MSA, in which MRI shows the hot cross bun sign in the pons, atrophy of the putamen, MCP, pons, and cerebellum, and putaminal iron. Vertical gaze palsy and early falls favor PSP, in which MRI shows midbrain atrophy, the hummingbird sign. Asymmetric apraxia, cortical sensory loss, and alien limb favor CBD, in which MRI shows asymmetric frontoparietal atrophy and FDG-PET shows asymmetric hypometabolism. Probable CBS is asymmetric, with 2 of limb rigidity or akinesia, limb dystonia, and limb myoclonus, plus 2 of orobuccal or limb apraxia, cortical sensory deficit, and alien limb; possible CBS may be symmetric, with 1 of each. CBS is not specific, since PSP, AD, and other pathologies can cause it.

Management

No treatment slows MSA, PSP, or CBD, so all three are treated the same way, by symptom. Parkinsonism is treated with a levodopa trial of up to 900–1200 mg daily for at least a month, tapered if there is no benefit. Focal dystonia is treated with botulinum toxin, with baclofen or clonazepam as alternatives, and falls with physical and occupational therapy and balance training. DBS is not recommended in PSP.

Orthostatic hypotension is treated with midodrine or fludrocortisone, urinary dysfunction with α-blockers, mirabegron, or selective antimuscarinics, and constipation with fiber, fluids, and osmotic laxatives. RBD is treated with melatonin, then low-dose clonazepam, and sialorrhea with botulinum toxin. Early speech pathology referral addresses dysphagia and speech, and cholinesterase inhibitors may help cognition but can worsen motor function in PSP. Care involves neurology and physical, occupational, and speech therapy, with early advance care planning and later hospice.

Relationships to other conditions

Typical Parkinson disease. Both cause parkinsonism with reduced DAT uptake. A clear, lasting levodopa response and rest tremor favor typical Parkinson disease, whereas a poor response with early autonomic failure, gaze palsy, early falls, or asymmetric cortical signs favors atypical parkinsonism.

Dementia with Lewy bodies. Both are α-synucleinopathies when the atypical disorder is MSA. Early dementia and visual hallucinations favor dementia with Lewy bodies.

Frontotemporal dementia. PSP and CBD are tauopathies that overlap with FTLD and can cause behavior and language change. Behavior or language change first, without early parkinsonism, favors frontotemporal dementia.

Huntington disease: CAG repeat expansion, striatal degeneration, and chorea

Huntington disease is an autosomal dominant neurodegenerative disorder caused by an expanded CAG repeat in HTT, which adds a polyglutamine stretch to huntingtin. Medium spiny neurons of the caudate and putamen degenerate, indirect-pathway neurons first, causing chorea, followed by hypokinesia, psychiatric change, and a subcortical dementia. Repeat length predicts penetrance and age of onset, and anticipation follows paternal transmission. Diagnosis rests on genetic testing, and treatment is symptomatic, with VMAT2 inhibitors such as tetrabenazine for chorea. Here, we review the genetics, pathology, clinical course, diagnosis, and treatment of Huntington disease.

Key points

- HD is autosomal dominant: 36–39 CAG repeats have reduced penetrance, and 40 or more almost always cause disease.
- Anticipation occurs mainly with paternal transmission, because repeats lengthen in spermatogenesis.
- Indirect-pathway medium spiny neurons are lost first, causing chorea; direct-pathway loss later causes rigidity.
- Degeneration starts in the caudate and dorsal putamen and spreads dorsomedial to ventrolateral; the Vonsattel grade tracks disability.
- Tetrabenazine inhibits VMAT2, reducing dopamine release and chorea, but can worsen depression.
- Aspiration pneumonia is the most common cause of death, followed by suicide.

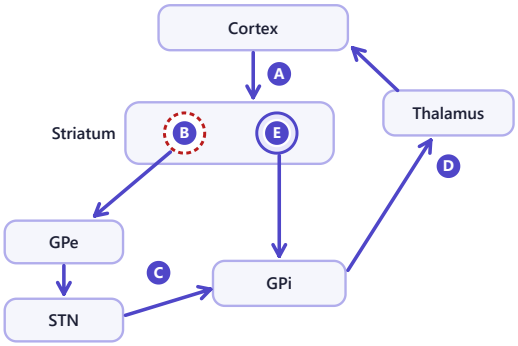


Fig. 1 | The basal ganglia circuit in Huntington disease. A, the cortex drives the striatum. B, indirect-pathway medium spiny neurons, which project to the external globus pallidus, are lost first. C, through the external pallidum and subthalamic nucleus, the internal pallidum then inhibits the thalamus less. D, the released thalamus overdrives the cortex, causing chorea. E, direct-pathway neurons, which project to the internal pallidum, are lost later, causing rigidity and bradykinesia. A, Cortex; B, Indirect-pathway neurons; C, GPe, STN, and GPi; D, Thalamus; E, Direct-pathway neurons.



Fig. 2 | The head of the caudate bulging into the lateral ventricle at each Vonsattel grade: convex at grades 0 to 2, flat at grade 3, and concave at grade 4, as caudate neurons are lost. Caudate head beside the lateral ventricle

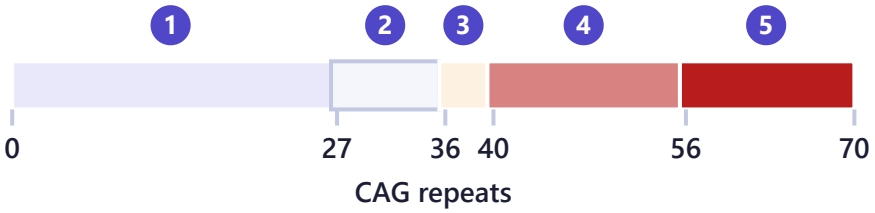


Fig. 3 | A bar of CAG repeat lengths from 0 to 70: 1, 26 or fewer is normal; 2, 27 to 35 is intermediate; 3, 36 to 39 has reduced penetrance; 4, 40 or more is fully penetrant; 5, over 55 causes juvenile disease. 1, 26 or fewer; 2, 27–35, intermediate; 3, 36–39, reduced penetrance; 4, 40 or more; 5, Over 55.

Pathophysiology

An expanded CAG repeat in HTT, on chromosome 4p16.3, adds a polyglutamine stretch to huntingtin, which causes both a toxic gain and a loss of function and forms nuclear inclusions. Over 90% of the GABA medium spiny neurons of the striatum degenerate. Indirect-pathway neurons, which contain enkephalin and project to the GPe, are lost first, so the GPi inhibits the thalamus less and the released thalamus overdrives the cortex, causing chorea. Direct-pathway neurons, projecting to the GPi, are lost later, causing rigidity and bradykinesia.

Degeneration starts in the tail and head of the caudate and the dorsal putamen and proceeds dorsomedial to ventrolateral and caudal to rostral. By Vonsattel grade, grade 0 shows no visible change, grades 1–2 show 50% loss of caudate head neurons with the caudate still convex, and grades 3–4 show a flat, then concave caudate, with 95% of its neurons lost. The cortex, globus pallidus, cerebellum, amygdala, and hippocampus are affected later. The grade tracks clinical disability, and caudate atrophy tracks repeat length and motor score.

Etiology

HD is autosomal dominant, so each child of an affected parent has a 50% chance of inheriting it. Repeats of 26 or fewer are normal, and 27–35 cause no disease but are unstable and can expand in children. Repeats of 36–39 have reduced penetrance, 40 or more almost always cause disease, with longer repeats bringing earlier onset, and over 55, usually over 60, cause juvenile disease. Repeats lengthen during spermatogenesis, so anticipation occurs mainly with paternal transmission. The family history can be negative after early parental death, late onset in the parent, an intermediate allele, or an unrecognized diagnosis, and homozygotes have a similar onset but faster progression.

Prevalence is 2.7 per 100000 worldwide and 10.6–13.7 per 100000 in people of European descent, and mean onset is at 45 years.

Clinical presentation

Chorea starts distally and spreads to the face, trunk, and proximal limbs, and later gives way to bradykinesia, dystonia, and rigidity, with dysarthria, dysphagia, and falls. Psychiatric change often comes first, with irritability, impulsivity, and aggression; apathy is the most common feature, depression is frequent, psychosis appears late, and insight is poor. Cognitive change begins with executive dysfunction and progresses to a subcortical dementia with slow retrieval that spares aphasia and apraxia.

The disease passes from presymptomatic, gene positive without signs, through prodromal, with subtle motor and cognitive signs, apathy, and depression, to manifest, when signs impair daily life. Juvenile disease, before age 20, causes rigidity and bradykinesia (the Westphal variant), dystonia, learning difficulty, and seizures in 30–50% before age 10. Late-onset disease, after 50 in about 25%, causes chorea and gait problems with a milder, longer course. Aspiration pneumonia is the most common cause of death, followed by suicide, which about 25% attempt, and median survival is 15–18 years from onset.

Diagnosis

A family history with chorea and behavioral or cognitive change suggests HD, and genetic testing of CAG repeat length is the gold standard. MRI shows early caudate atrophy, even before symptoms, with cortical and cerebellar atrophy later. Predictive testing is offered to at-risk adults with genetic counseling, and prenatal testing uses chorionic villus sampling at 10–12 weeks or amniocentesis at 15–17 weeks.

Chorea-acanthocytosis, autosomal recessive, causes orofacial chorea worse with eating, tongue biting, myopathy, and onset around 30, with acanthocytes on the blood film. McLeod syndrome, X-linked, causes acanthocytes, hemolysis, and myopathy, with absent Kell antigen. Wilson disease causes orofacial dystonia with parkinsonism, and spinocerebellar ataxias cause prominent ataxia; marked ataxia points away from HD.

Management

No treatment slows HD, so treatment targets symptoms. Tetrabenazine inhibits VMAT2 on presynaptic vesicles, blocking the loading of dopamine and reducing its release, which lessens chorea. Tetrabenazine, deutetrabenazine, and valbenazine are first line but can cause depression, suicidality, fatigue, and akathisia, so mood and suicidal thoughts are monitored. Antipsychotics are used off label, with aripiprazole matching tetrabenazine, at the cost of extrapyramidal effects and tardive dyskinesia.

Rigidity in the Westphal variant is treated with levodopa, dopamine agonists, or amantadine, and focal dystonia with botulinum toxin. Depression, anxiety, and OCD are treated with SSRIs or TCAs, with limited evidence, and no drug helps cognition. Supportive care covers diet, nursing, and equipment, smoking and alcohol are avoided, and patients and families receive counseling and genetic counseling.

Relationships to other conditions

Typical Parkinson disease. Both disturb the basal ganglia, in opposite directions. Loss of indirect-pathway neurons in HD causes excess movement, whereas dopamine loss in Parkinson disease causes too little; the Westphal variant of HD, with rigidity, responds to the same antiparkinsonian drugs.

Alzheimer disease. Both cause dementia. HD causes a subcortical dementia with slow retrieval that spares aphasia and apraxia, whereas Alzheimer disease causes a cortical dementia with early memory loss.

Localizing a neurologic lesion: from signs to level and side

Each neurologic finding points to a level of the nervous system, from cortex to muscle, and to a side, and a lesion is localized where all the findings fit together; when they cannot, several lesions or a diffuse process is more likely. Crossed signs place a lesion in the brainstem, the homunculus separates ACA from MCA territory, and cranial nerve, pupil, eye movement, nystagmus, and visual field patterns pinpoint the level. Cortical and diencephalic signs, headache features, and gait add further clues. Here, we review how to move from the signs to the site.

Key points

- A lesion is localized where every finding fits; findings that cannot fit one site suggest several lesions or a diffuse process.
- A cranial nerve palsy with weakness of the opposite body places the lesion in the brainstem at the level of that nerve.
- Leg weakness more than arm points to the medial ACA territory, face and arm more than leg to the lateral MCA territory, and equal weakness to the internal capsule.
- Central nystagmus changes direction, is vertical or purely torsional, and is not suppressed by fixation.
- A thunderclap headache, a new headache over 50, and papilledema each call for urgent investigation.

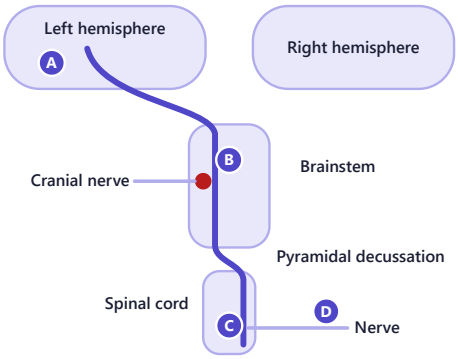


Fig. 1 | The corticospinal tract runs from the cortex, A, through the brainstem, B, where a cranial nerve nucleus sits beside it, and crosses at the pyramidal decussation before descending in the spinal cord, C, to a peripheral nerve, D. A lesion at A or B weakens the opposite body, a lesion at B also hits the cranial nerve on its own side, and a lesion at C weakens the same side below it. A, Cortex or internal capsule; B, Brainstem; C, Spinal cord; D, Nerve or root.

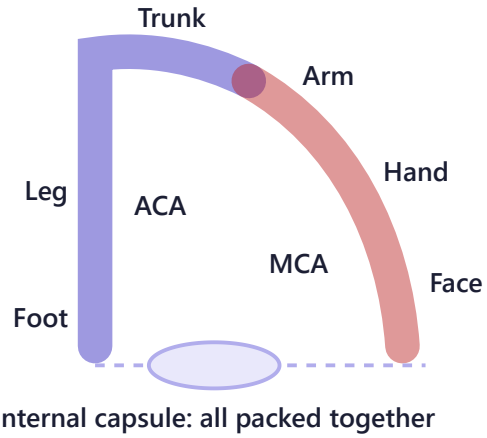


Fig. 2 | A coronal section through the motor and sensory strips of one hemisphere. The leg and foot lie on the medial surface, supplied by the ACA; the trunk sits at the top; and the arm, hand, and face run down the lateral surface, supplied by the MCA. Below, fibers for the face, arm, and leg are packed together in the internal capsule.

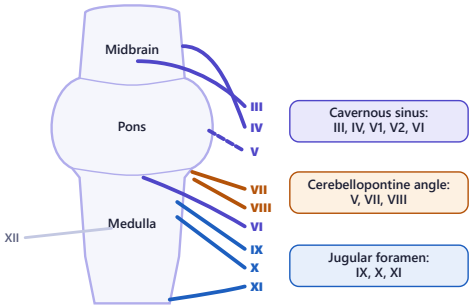


Fig. 3 | The brainstem from the front: III and IV leave the midbrain, V to VIII the pons, and IX to XII the medulla. Nerves that travel together are grouped: III, IV, V1, V2, and VI in the cavernous sinus; V, VII, and VIII at the cerebellopontine angle; and IX, X, and XI through the jugular foramen.

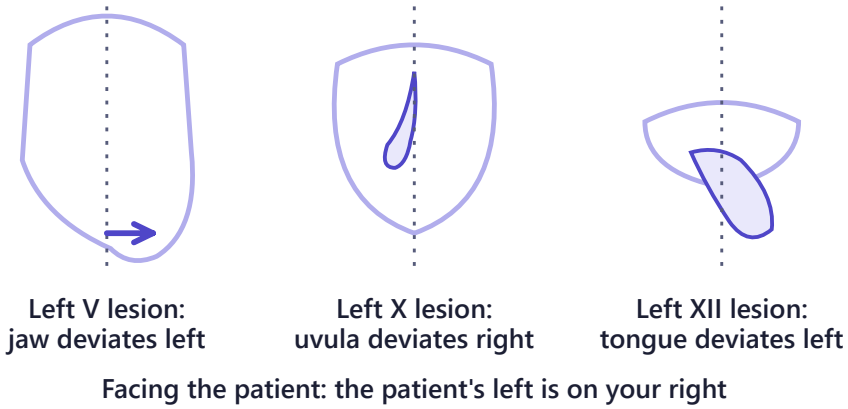


Fig. 4 | Facing the patient, with the patient's left on the viewer's right: with a left trigeminal (V) lesion the jaw deviates to the left, with a left vagal (X) lesion the uvula deviates to the right, away from the lesion, and with a left hypoglossal (XII) lesion the tongue deviates to the left.

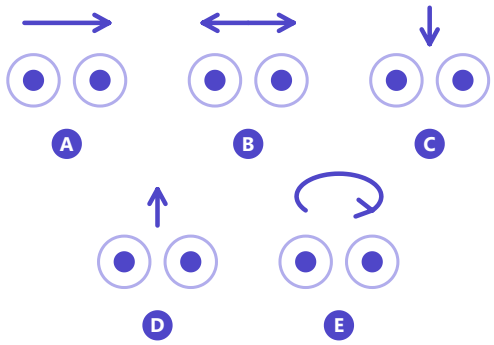


Fig. 5 | Five nystagmus patterns shown on a pair of eyes, with arrows for the fast phase. A, peripheral: horizontal, beating one way. B, direction-changing: beating to whichever side the patient looks. C, downbeat. D, upbeat. E, purely torsional. A, Peripheral; B, Direction-changing; C, Downbeat; D, Upbeat; E, Purely torsional.

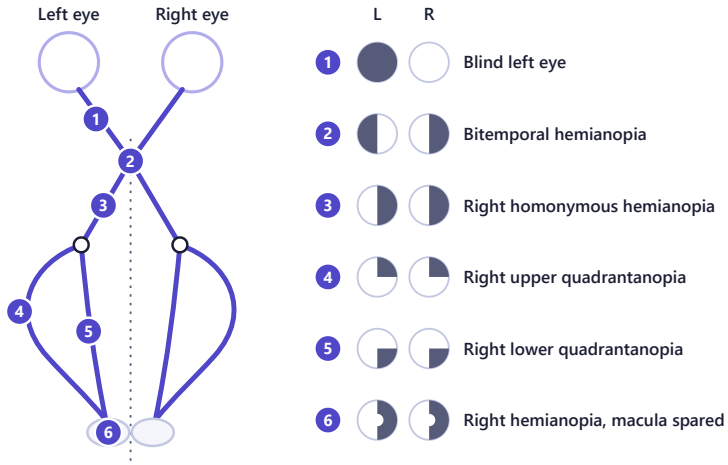


Fig. 6 | The visual pathway from above, with lesions on the left side and the field each one leaves, left and right eye: 1, the left optic nerve blinds the left eye; 2, the chiasm causes bitemporal hemianopia; 3, the left optic tract causes right homonymous hemianopia; 4, the left Meyer loop causes right upper quadrantanopia; 5, the left parietal radiation causes right lower quadrantanopia; 6, the left occipital cortex causes right homonymous hemianopia, with the macula spared.

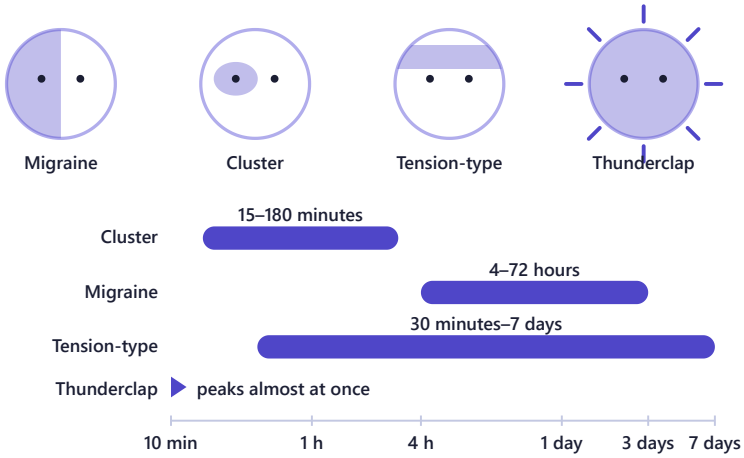


Fig. 7 | Where headaches hurt and how long they last: migraine is one-sided, for 4 to 72 hours; cluster headache is around one eye, for 15 to 180 minutes; tension-type headache is a band around the head, for 30 minutes to 7 days; and thunderclap headache reaches its peak almost at once.

Approach

Each finding points to a level, from cortex to muscle, and a side; when every finding fits one site, the lesion is there, and when they cannot, several lesions or a diffuse process is more likely. Localization proceeds from level to side, then asks whether all findings fit together, before onset and course narrow the cause. Sudden onset in an arterial territory suggests stroke, lesions separated in space and time suggest multiple sclerosis, symptoms that spread over minutes suggest a migraine aura or a seizure march, and gradual onset or fever suggests a tumor or abscess.

Weakness and sensory loss

A cortical or internal capsule lesion weakens the opposite face, arm, and leg, with aphasia, neglect, or field loss if the cortex is involved, and pure motor or sensory loss if the lesion is small and deep. A brainstem lesion causes crossed signs, a cranial nerve palsy on the side of the lesion with weakness of the opposite body. A spinal cord lesion spares the face and weakens the same side below it, with a sensory level, and hemisection adds opposite pain and temperature loss. A nerve or root lesion causes LMN weakness and sensory loss in the pattern of one nerve or dermatome.

In the homunculus, the leg lies on the medial cortex in the ACA territory, and the arm, hand, and face lie on the lateral cortex in the MCA territory, so leg weakness more than arm points to the ACA, face and arm more than leg to the MCA, and equal weakness of face, arm, and leg to the internal capsule, where the fibers are packed together. Symptoms spreading to the next body part are a Jacksonian march, a seizure spreading along the cortex, and proximal arm and shoulder weakness sparing the hand and face is the MCA/ACA watershed pattern, the man-in-a-barrel pattern if bilateral.

UMN lesions cause spasticity, brisk reflexes, and a Babinski sign, with weakness worst in arm extensors and leg flexors, whereas LMN lesions cause flaccid weakness, reduced reflexes, atrophy, and fasciculations in one nerve or root. A lesion above the pyramidal decussation causes signs on the opposite side, and one below it, in the cord, on the same side. The face is involved only by lesions above the facial nucleus, so facial weakness rules out a cord lesion.

Cranial nerves

A CN III palsy with a dilated pupil suggests compression, as by a PComm aneurysm or uncal herniation, whereas one sparing the pupil suggests microvascular ischemia, as in diabetes. A CN IV palsy, with the head tilted away, often follows trauma, and a CN VI palsy can be a false localizing sign of raised ICP or reflect a pontine lesion. Weakness of the whole half of the face, forehead included, localizes to the facial nucleus or nerve on that side, as in Bell palsy, whereas lower facial weakness sparing the forehead localizes to the opposite cortex or corticobulbar tract.

The jaw deviates toward a CN V lesion, the uvula away from a CN X lesion, and the tongue toward a CN XII lesion. CN V, VII, and VIII together point to the cerebellopontine angle, as with a vestibular schwannoma; CN III, IV, V1, V2, and VI together to the cavernous sinus; and CN IX, X, and XI together to the jugular foramen. A brisk jaw jerk with slurred speech and dysphagia is pseudobulbar palsy from bilateral corticobulbar lesions. A cranial nerve palsy with weakness of the opposite body places the lesion in the brainstem, at the level of that nerve.

Eyes and pupils

A small pupil with mild ptosis and anhidrosis is Horner syndrome, from a lesion anywhere along the sympathetic pathway, from hypothalamus and brainstem to cord, lung apex, and carotid. A large pupil with ptosis and an eye turned down and out is a CN III palsy. Pupils that constrict to a near target but not to light show light-near dissociation, from an Adie pupil or the dorsal midbrain, and a pupil that dilates with the swinging light test has an RAPD from an optic nerve lesion, as in optic neuritis.

Double vision that goes when either eye is covered is binocular, from misaligned eyes due to a nerve, muscle, or junction problem, whereas double vision that persists with one eye covered is monocular, from the eye itself. Failure to adduct one eye with nystagmus of the other is INO from an MLF lesion, bilateral in young people with MS and unilateral in older people with stroke. Impaired upgaze, convergence-retraction nystagmus, and light-near dissociation make up Parinaud syndrome of the dorsal midbrain, from a pineal tumor or midbrain stroke. Vertical gaze palsy with early falls suggests progressive supranuclear palsy, and eyes deviated toward one side suggest a hemispheric lesion on that side, as in MCA stroke.

Peripheral nystagmus is horizontal-torsional, beats away from the affected ear in every gaze, and is suppressed by fixation, as in vestibular neuritis or BPPV. Direction-changing nystagmus, which changes with gaze and is not suppressed by fixation, is central, and so is purely torsional nystagmus; downbeat nystagmus points to the cervicomedullary junction or midline cerebellum and upbeat nystagmus to the pons and midbrain.

Visual loss in one eye localizes to the retina or optic nerve, bitemporal hemianopia to the optic chiasm, as with a pituitary tumor, and homonymous hemianopia to the opposite optic tract, radiation, or occipital cortex. Homonymous hemianopia with macular sparing points to the opposite occipital cortex in the PCA territory, and superior quadrantanopia to the opposite temporal lobe through the Meyer loop.

Cortical and diencephalic signs

Nonfluent, effortful speech with intact comprehension localizes to the Broca area in the dominant inferior frontal gyrus, fluent speech with paraphasias and poor comprehension to the Wernicke area in the dominant posterior superior temporal gyrus, and isolated poor repetition to the arcuate fasciculus in conduction aphasia. Agraphia, acalculia, finger agnosia, and left-right disorientation make up Gerstmann syndrome of the dominant angular gyrus, and hemispatial neglect with anosognosia localizes to the nondominant parietal lobe. Alexia without agraphia points to the dominant PCA territory and prosopagnosia to the nondominant PCA territory.

Disinhibition, impulsivity, and tactlessness point to the orbitofrontal cortex; apathy, abulia, and reduced speech to the anterior cingulate cortex, with akinetic mutism if bilateral; and poor planning, fluency, and working memory to the dorsolateral prefrontal cortex. Sensory loss of all types on one side points to the opposite thalamus, as in a pure sensory lacunar stroke, and disordered temperature, appetite, thirst, or sleep points to the hypothalamus. Cortical signs such as aphasia and neglect place a hemiparesis in the cortex rather than the internal capsule or brainstem.

Headache

A headache reaching peak intensity almost at once, the worst ever, is a thunderclap headache, from SAH, RCVS, ICH, dissection, or venous sinus thrombosis, and calls for non-contrast CT, with a lumbar puncture if it is normal after 6 hours. One-sided, pulsating headache lasting 4–72 hours, with nausea and light and sound

sensitivity and worse with activity, is migraine. One-sided orbital pain lasting 15–180 minutes, with tearing or nasal congestion on that side, restlessness, and attacks at night, is cluster headache, three times more common in men and relieved by 100% oxygen within 10 minutes in at least two thirds. Bilateral, band-like, pressing pain for 30 minutes to 7 days without nausea is tension-type headache, the most common primary headache.

Papilledema, pulsatile tinnitus, and CN VI palsy in a young woman with obesity suggest IIH once a mass is excluded, and call for imaging and then a lumbar puncture with opening pressure. A new headache over 50, with jaw claudication, vision loss, or raised ESR and CRP, suggests GCA, which needs prompt high-dose glucocorticoids to prevent blindness. Headache on 15 or more days a month with frequent analgesic use is medication overuse headache.

Gait and balance

A stiff leg swung in an arc is a hemiparetic gait from a UMN lesion, and crouched, crossing legs are a scissor gait from bilateral UMN lesions, as in cerebral palsy. High stepping that lands on the toes is a steppage gait from foot drop due to a peroneal nerve, L5, or neuropathic lesion. A broad-based, unsteady gait is cerebellar, from the vermis, whereas a stomping gait that worsens in the dark with a positive Romberg test is sensory ataxia from the dorsal columns or large fibers. Shuffling with reduced arm swing and freezing is the festinating gait of Parkinson disease, a waddling gait reflects proximal muscle weakness, and gait disturbance with urinary incontinence and cognitive decline suggests NPH.

On the Romberg test, unsteadiness that worsens with the eyes closed indicates lost proprioception, whereas unsteadiness with the eyes open as well is cerebellar. Limb ataxia localizes to the cerebellar hemisphere on the same side.

Relationships to other conditions

Stroke symptoms and complications. That page starts from each arterial territory and lists its deficits, whereas this one starts from the deficit and works back to the territory. A sudden deficit that fits one arterial territory favors stroke over the other causes considered here.

Motor signs and syndromes. Both separate UMN from LMN weakness and read the gait. That page describes each sign in detail, whereas this one combines them with cranial nerve, eye, and cortical findings to place the lesion.

Ruling out mimics. Localization gives the site of a lesion, and the next step is to ask what else could cause the same findings; that page covers the mimics of the main conditions.

Ruling out mimics in stroke, seizure, multiple sclerosis, Alzheimer disease, and Parkinson disease

A presentation suggests a diagnosis, but findings that the diagnosis cannot explain point to a mimic. Treatable and dangerous mimics are excluded first, with the test that best separates them. For stroke, glucose and non-contrast CT come first; for seizure, glucose, bloods, CT, then EEG and MRI; for multiple sclerosis, MRI, CSF, and antibodies; for dementia, delirium, depression, B12, and thyroid function; and for parkinsonism, a medication review, a levodopa trial, and DAT SPECT. Here, we review the signs that point away from each diagnosis and the tests that settle the question.

Key points

- Findings a diagnosis cannot explain point to a mimic, and treatable or dangerous mimics are excluded first.
- A normal early CT does not exclude ischemic stroke, because DWI shows an infarct within minutes.
- A normal EEG during a typical event points to PNES, whereas a normal routine EEG between events does not exclude epilepsy.
- Severe or bilateral optic neuritis, LETM, or intractable hiccups and vomiting point away from MS toward NMOSD.
- DAT SPECT is reduced in Parkinson disease and atypical parkinsonism alike, so it separates them from drug-induced parkinsonism and essential tremor, not from each other.

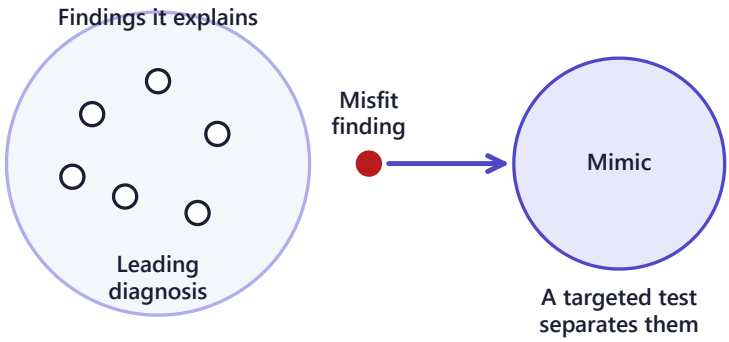


Fig. 1 | Findings the leading diagnosis explains sit inside it; a finding it cannot explain points to a mimic, and a targeted test separates the two.

Approach

A presentation suggests a diagnosis, but findings it cannot explain point to a mimic, so the treatable and dangerous mimics are excluded first with the test that best separates them. The leading diagnosis comes from the presentation, misfit findings are sought, must-not-miss mimics are excluded, and a targeted test separates the remaining possibilities.

Stroke

Evaluation starts with glucose to rule out hypoglycemia and non-contrast CT to rule out hemorrhage, then CT angiogram for LVO, and CT perfusion or MRI at 4.5–24 hours or when onset is unknown. Low glucose with deficits that resolve when it is corrected indicates hypoglycemia. A witnessed seizure, tongue biting, and a post-ictal state suggest a seizure with Todd paralysis, confirmed by EEG. Positive symptoms spreading over minutes with past similar attacks suggest migraine with aura, with MRI if unclear, and inconsistent signs with a positive Hoover sign suggest a functional neurologic disorder, with MRI.

Gradual onset or fever suggests a brain tumor or abscess, shown by CT or MRI with contrast; isolated facial weakness that includes the forehead suggests Bell palsy on clinical examination; isolated vertigo with a reassuring HINTS examination suggests peripheral vertigo, with MRI if unclear; and sepsis, hyponatremia, or new drugs can unmask an old deficit, found with bloods and cultures. A normal early CT does not exclude ischemic stroke, because DWI shows an infarct within minutes, long before CT changes.

Seizure

Evaluation starts with glucose and vital signs, treating hypoglycemia first, then bloods for electrolytes and drug levels, non-contrast CT after a first seizure, and EEG and MRI to classify the seizure and find its cause. A trigger, lightheadedness, brief jerks, and rapid recovery suggest syncope, assessed with ECG and orthostatic BP. Closed eyes, side-to-side head movement, asynchronous limbs, and no post-ictal state suggest PNES, confirmed by video EEG. Palpitations, exertional collapse, or a family history of sudden death suggest a cardiac arrhythmia, assessed with ECG and telemetry, and negative symptoms such as weakness or numbness without a march suggest TIA, assessed with MRI and vascular imaging.

Symptoms spreading over minutes and followed by headache suggest migraine with aura; events only during sleep suggest a sleep disorder, assessed by a sleep study; fear, palpitations, and hyperventilation with awareness preserved suggest a panic attack; and tics or dystonia with awareness preserved suggest a movement disorder, with EEG if unclear. A normal EEG during a typical event points to PNES, whereas a normal routine EEG between events does not exclude epilepsy.

Multiple sclerosis

Evaluation uses MRI brain and spine with gadolinium and the McDonald criteria of space plus time or a substitute, with CSF if MRI falls short and bloods to exclude mimics. Severe or bilateral optic neuritis, LETM, or intractable hiccups and vomiting suggest NMOSD, tested by AQP4-IgG, and bilateral optic neuritis with disc edema or a conus lesion suggests MOGAD, tested by MOG-IgG. One episode after an infection with encephalopathy in a child suggests ADEM, on MRI. Dorsal column and corticospinal signs with anemia suggest B12 deficiency; rash, joint pain, and kidney disease suggest SLE, tested by ANA; and lung disease with CN palsies suggests neurosarcoidosis, tested by CXR and serum ACE.

Risk factors for syphilis, HIV, or Lyme disease call for serology, and older age, vascular risk factors, and no oligoclonal bands suggest small vessel disease, on MRI and CSF. Progressive worsening from onset suggests primary progressive MS or a mimic, since NMOSD rarely progresses, and calls for MRI, CSF, and antibodies.

Alzheimer disease

Evaluation first excludes delirium, with its sudden onset and fluctuating attention, screens for depression, checks B12 and TFTs, and uses MRI for the pattern of atrophy. Sudden onset, fluctuating attention, and an acute illness indicate delirium, which needs a search for its cause, and low mood, poor effort, and awareness of memory problems suggest depression. Fatigue, cold intolerance, and weight gain suggest hypothyroidism, tested by TFTs, and anemia with neuropathy suggests B12 deficiency.

Early gait disturbance and urinary incontinence suggest NPH, on MRI; visual hallucinations, parkinsonism, and fluctuations suggest dementia with Lewy bodies, supported by DAT SPECT; stepwise decline with vascular risk factors suggests vascular dementia, on MRI; younger age with early behavior or language change suggests frontotemporal dementia, on MRI; and rapid progression with myoclonus suggests prion disease, on MRI and CSF. Early memory loss with medial temporal atrophy favors Alzheimer disease, whereas preserved medial temporal lobes favor dementia with Lewy bodies.

Parkinson disease

Evaluation confirms bradykinesia with rest tremor or rigidity, reviews medications for dopamine-blocking drugs, uses a levodopa trial, in which a clear response supports PD, and adds DAT SPECT if still unclear. Vertical gaze palsy and early falls suggest progressive supranuclear palsy, with the hummingbird sign on MRI; early autonomic failure with cerebellar or pyramidal signs suggests multiple system atrophy, with the hot cross bun sign; and asymmetric apraxia, cortical sensory loss, and alien limb suggest corticobasal degeneration, with asymmetric frontoparietal atrophy. Early dementia and visual hallucinations suggest dementia with Lewy bodies.

Symmetric signs after starting a dopamine-blocking drug suggest drug-induced parkinsonism, found by medication review, with a normal DAT SPECT; bilateral postural or action tremor without bradykinesia suggests essential tremor, also with a normal DAT SPECT; and chorea with a family history suggests Huntington disease, confirmed by genetic testing. DAT SPECT is reduced in PD and in atypical parkinsonism alike, so it separates them from drug-induced parkinsonism and essential tremor, not from each other.

Relationships to other conditions

Localizing a lesion. Localization gives the site, and ruling out mimics gives the cause. Findings that cannot fit one site are themselves a clue to a mimic, such as several lesions or a diffuse process.

Choosing imaging and EEG. Most mimics here are settled by a test, and that page explains which imaging sequence or EEG pattern answers each question.

SPECIAL TOPICS

Choosing imaging and EEG in neurology, and the signs to look for

The clinical question sets the test. Non-contrast CT is fast and best for fresh blood, CT angiogram shows vessels, CT perfusion separates ischemic core from salvageable penumbra, and MRI shows tissue in detail, with DWI revealing ischemia within minutes. T1, T2, FLAIR, DWI with ADC, SWI, and gadolinium each make different tissue bright, and conditions carry characteristic signs, from Dawson fingers in MS to the hot cross bun sign in multiple system atrophy. EEG shows electrical activity rather than structure, and nuclear medicine shows dopamine terminals, glucose use, and amyloid. Here, we review which test to order and what to look for.

Key points

- Non-contrast CT rules hemorrhage in or out in minutes but is often normal in the first hours of ischemia; DWI shows an infarct within minutes.
- On CT perfusion, the core has rCBF under 30%, and the penumbra has Tmax over 6 seconds with rCBF 30% or more.
- An acute infarct is bright on DWI and dark on ADC, and an infarct bright on DWI but not yet on FLAIR is under 4.5 hours old.
- SWI shows blood products as dark, revealing microbleeds and hemorrhage.
- Generalized 3 Hz spike-and-wave suggests absence seizures, and a normal EEG during a typical event points to PNES.

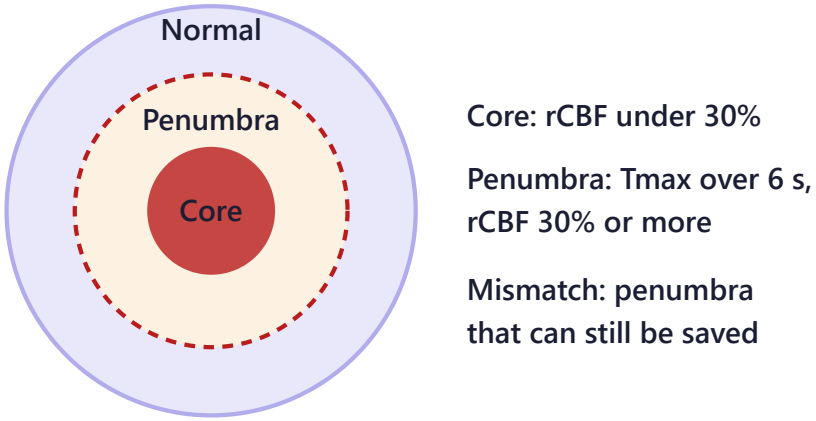


Fig. 1 | CT perfusion shows an ischemic core, with relative cerebral blood flow under 30%, inside a ring of penumbra, where contrast arrives late, with Tmax over 6 seconds, but blood flow is 30% or more. Normal brain lies outside.

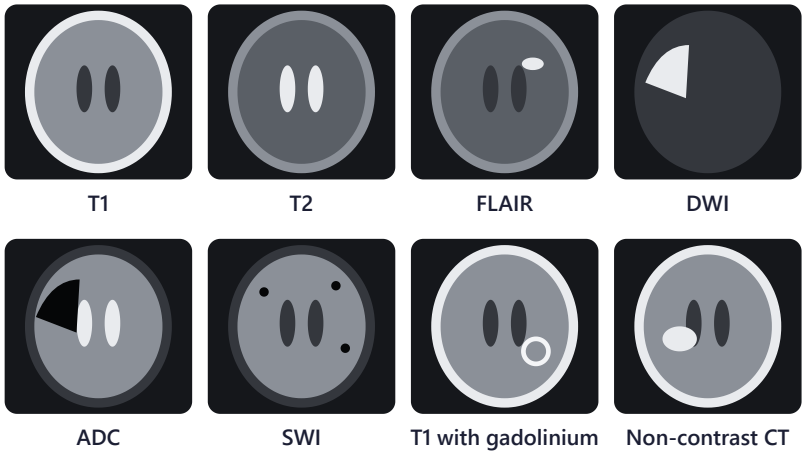


Fig. 2 | Eight schematic axial slices. T1: fluid dark, fat bright. T2: fluid bright. FLAIR: fluid dark, with a bright periventricular plaque. DWI: an acute infarct bright. ADC: the same infarct dark. SWI: blood products as dark dots. T1 with gadolinium: an enhancing ring. Non-contrast CT: fresh blood bright, CSF dark, bone white.



Fig. 3 | Eight classic signs as schematic scans: Dawson fingers, ovoid plaques at right angles to the ventricle, in MS; LETM, a long bright cord lesion, in NMOSD; the empty delta sign, a bright rim around a dark sinus, in venous sinus thrombosis; lobar microbleeds, dark dots at the cortex, in amyloid angiopathy; the hot cross bun sign in the pons in MSA; the hummingbird sign, a thin midbrain, in PSP; boxcar ventricles from caudate atrophy in Huntington disease; and a small, bright hippocampus in mesial temporal lobe epilepsy.

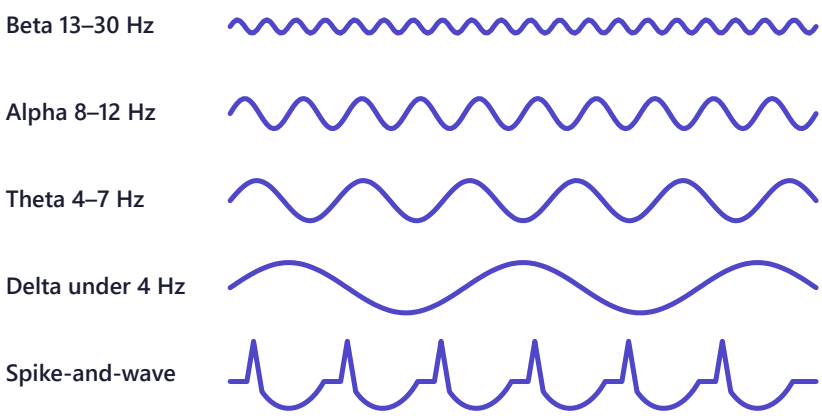


Fig. 4 | Schematic EEG traces over the same time: beta, fast and low, at 13 to 30 Hz; alpha at 8 to 12 Hz; theta at 4 to 7 Hz; delta, slow and high, under 4 Hz; and a 3 Hz spike-and-wave pattern, a sharp spike followed by a slow wave.

Choosing a test

The question sets the test: CT is fast and best for fresh blood, MRI shows tissue in detail and acute ischemia within minutes, angiography shows vessels, and EEG shows electrical activity rather than structure. Suspected stroke calls for non-contrast CT, then CT angiogram, to find hemorrhage and then LVO, and stroke at 4.5–24 hours or of unknown onset calls for CT perfusion or MRI to find salvageable penumbra. A thunderclap headache calls for non-contrast CT, with a lumbar puncture if it is normal after 6 hours, to find SAH, and a lobar ICH in an older person calls for MRI with SWI to find the microbleeds of amyloid angiopathy.

A first seizure calls for non-contrast CT, then MRI and EEG, to find a structural cause and the seizure type, and suspected demyelination calls for MRI brain and spine with gadolinium, with the orbits for NMOSD. Acute vertigo with a central HINTS result calls for MRI and MRA to find posterior circulation stroke, and asymmetric SNHL or one-sided tinnitus for MRI of the internal auditory canals with gadolinium to find a vestibular schwannoma. Cognitive decline calls for MRI, with PET or SPECT if unclear, and parkinsonism of uncertain cause for DAT SPECT.

CT

Non-contrast CT reads density in HU: fat about −50, CSF +15, white matter +25, gray matter +40, and blood +30–45. Acute blood is bright, so hemorrhage is ruled in or out in minutes. Early ischemia under 12 hours often looks normal or shows loss of gray-white differentiation, insular ribbon blurring, or lentiform nucleus effacement, and a hyperdense MCA sign reflects thrombus in the artery and is highly specific for LVO. ASPECTS starts at 10 and loses 1 point for each MCA region with early change, and 6 or more indicates a small to moderate core.

CT angiogram shows vessels from the aortic arch to the vertex with iodinated contrast and finds LVO, dissection, aneurysm, AVM, and vasculitis, and multiphase imaging shows collateral circulation, with good collaterals predicting a larger penumbra. CT perfusion maps CBV, CBF, MTT, and Tmax. The core, with rCBF under 30% of the other side, is likely lost, whereas the penumbra, with Tmax over 6 seconds and rCBF 30% or more, is still salvageable, and the mismatch selects thrombectomy at 6–24 hours and thrombolysis in some late or wake-up strokes.

MRI sequences

T1 makes fat and gadolinium bright and shows anatomy, and T2 makes fluid, edema, and most lesions bright, showing MS plaques and edema. FLAIR makes lesions bright with fluid made dark, showing periventricular plaques, SAH in the sulci, and infarcts after 4.5–6 hours. DWI with ADC shows acute infarct as bright on DWI and dark on ADC within minutes. SWI or gradient echo shows blood products as dark, revealing microbleeds, hemorrhage, and thrombus. T1 with gadolinium shows enhancing lesions, such as active MS plaques, tumors, and abscesses, and MRA shows flowing blood, finding occlusion, dissection, and stenosis without iodinated contrast. On non-contrast CT, by contrast, fresh blood is bright, CSF dark, and bone white.

An infarct bright on DWI but not yet on FLAIR, a DWI-FLAIR mismatch, is under 4.5 hours old, so thrombolysis can be considered when onset is unknown. Old edema or gliosis can also be bright on DWI, T2 shine-through, but true acute infarct is dark on ADC. Gadolinium is avoided in severe renal impairment, because of NSF, and where possible in pregnancy.

Signs by condition

Acute ischemic stroke shows restricted diffusion on DWI, and core and penumbra on CT perfusion; lacunar stroke a small deep infarct on MRI; intracerebral hemorrhage hyperdense blood on non-contrast CT, deep in hypertension; cerebral amyloid angiopathy lobar microbleeds on MRI with SWI; cerebral venous sinus thrombosis the empty delta sign on CT or MR venography; and subarachnoid hemorrhage blood in the subarachnoid space on non-contrast CT, or xanthochromia on lumbar puncture.

Multiple sclerosis shows periventricular Dawson fingers, plaques in 2 of 5 sites, and the central vein sign on MRI with gadolinium; NMOSD shows LETM over 3 or more segments, a dorsal medulla lesion, and a long posterior optic nerve lesion; and MOGAD shows a long anterior optic nerve lesion with disc edema, a conus lesion, and the H-sign.

Alzheimer disease shows medial temporal atrophy on MRI and amyloid plaques on amyloid PET; frontotemporal dementia frontal and anterior temporal atrophy, with hypometabolism there on FDG-PET; and dementia with Lewy bodies preserved medial temporal lobes, reduced DAT uptake, and the cingulate island sign. Parkinson disease shows reduced striatal uptake on DAT SPECT; multiple system atrophy the hot cross bun sign with atrophy of the putamen, MCP, pons, and cerebellum; progressive supranuclear palsy midbrain atrophy, the hummingbird sign; corticobasal degeneration asymmetric frontoparietal atrophy and hypometabolism; and Huntington disease caudate atrophy, even before symptoms. Mesial temporal lobe epilepsy shows hippocampal volume loss with bright T2 and FLAIR signal, and vestibular schwannoma, brain

tumor, and abscess show an enhancing mass on MRI with gadolinium.

EEG

EEG rhythms are beta at 13–30 Hz, alpha at 8–12 Hz, theta at 4–7 Hz, and delta under 4 Hz, and a normal adult at rest has a posterior dominant alpha rhythm at 8–12 Hz. EEG classifies and localizes seizures and assesses encephalopathy, coma, and brain death. A routine 30-minute EEG often misses interictal discharges, so repeated or prolonged recordings find more, and video EEG captures events to separate seizures from PNES.

Generalized 3 Hz spike-and-wave provoked by hyperventilation suggests absence seizures, focal spikes or sharp waves a focal epileptic source, a normal EEG during a typical event PNES, triphasic waves a metabolic encephalopathy such as hepatic or uremic, prominent posterior slow waves dementia with Lewy bodies, and seizure activity without convulsions nonconvulsive status epilepticus.

Nuclear medicine

DAT SPECT shows presynaptic dopamine terminals, reduced in PD and atypical parkinsonism and normal in drug-induced parkinsonism and essential tremor. FDG-PET shows glucose use, low at a seizure focus between seizures, with patterns of dementia, and ictal SPECT shows increased perfusion at a seizure focus when injected within 90 seconds of onset. Amyloid PET shows amyloid plaques in life, in Alzheimer disease.

Relationships to other conditions

Ruling out mimics. That page lists the mimics of each condition and the test that separates them, and this one explains how to read those tests.

Acute ischemic stroke. Imaging decides stroke treatment: non-contrast CT excludes hemorrhage before thrombolysis, CT angiogram finds LVO for thrombectomy, and perfusion or DWI-FLAIR mismatch extends treatment to late or unknown onset.

Antithrombotic drugs and their reversal in cerebrovascular disease

Ischemic stroke is treated by dissolving the clot and prevented by antiplatelets, anticoagulants, and statins, whereas intracranial hemorrhage needs the same drugs stopped and reversed. Thrombolytics act on fibrinolysis, aspirin and clopidogrel on platelets, and heparins, warfarin, and the direct oral anticoagulants on coagulation, and each anticoagulant has a specific reversal. Here, we review these drugs by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- Thrombolysis is given within 4.5 hours with BP below 185/110 and no bleeding contraindication, and aspirin waits 24 hours after it.
- Clopidogrel is a prodrug activated by CYP2C19, so omeprazole and poor metabolizer genotypes reduce its effect.
- Warfarin is used when a mechanical valve rules out a DOAC, has many CYP2C9 interactions, and causes skin necrosis without a heparin bridge.
- 4F-PCC with IV vitamin K reverses warfarin, protamine reverses heparin, idarucizumab reverses dabigatran, and andexanet alfa or 4F-PCC reverses factor Xa inhibitors.
- Statins treat large and small vessel stroke, and CYP3A4 inhibitors raise their levels and cause myopathy.

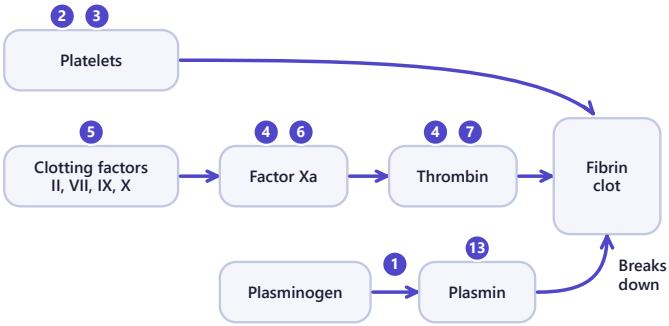


Fig. 1 | Thirteen numbered drug sites on three rows. Platelets: 2, aspirin and 3, clopidogrel. Coagulation, from clotting factors to factor Xa, thrombin, and the fibrin clot: 5, warfarin acts on clotting factor synthesis; 4, heparins act on factor Xa and thrombin; 6, factor Xa inhibitors act on factor Xa; 7, dabigatran acts on thrombin. Fibrinolysis, from plasminogen to plasmin, which breaks down the clot: 1, thrombolytics make plasmin, and 13, antifibrinolytics block it. 1, Thrombolytics; 2, Aspirin; 3, Clopidogrel; 4, Heparins; 5, Warfarin; 6, Factor Xa inhibitors; 7, Dabigatran; 13, Antifibrinolytics and cryoprecipitate.

Dissolving and preventing clots

Thrombolytics, alteplase and tenecteplase, convert fibrin-bound plasminogen to plasmin, which breaks down fibrin so that the clot dissolves. They treat acute ischemic stroke within 4.5 hours of onset, provided BP is below 185/110. Alteplase has a half-life of minutes, so a bolus is followed by an infusion, whereas tenecteplase is more fibrin-specific and lasts longer, so a single bolus suffices. They are contraindicated by hemorrhage on CT; head trauma or intracranial or spinal surgery within 3 months; active bleeding; platelets under 100000/μL, an INR over 1.7, or an aPTT over 40 seconds; therapeutic LMWH within 24 hours or recent DOAC use; infective endocarditis; aortic dissection; and an intracranial neoplasm with high bleeding risk. Their adverse effects are symptomatic ICH, other bleeding, and angioedema, and because antiplatelets and anticoagulants add to the bleeding, aspirin waits 24 hours after thrombolysis.

Aspirin irreversibly acetylates platelet COX-1, blocking TXA2 synthesis, so aggregation is reduced for the 7–10-day life of the platelet; low doses spare endothelial prostacyclin. It is given within 24–48 hours of an ischemic stroke, continued for secondary prevention, and combined with clopidogrel for 21 days after a minor stroke or high-risk TIA. It is contraindicated in active bleeding, aspirin-exacerbated asthma, and children with viral illness, because of Reye syndrome. It causes GI ulcers and bleeding, ICH, tinnitus at high doses, and salicylate toxicity in overdose. Anticoagulants, other antiplatelets, and SSRIs add to the bleeding, and ibuprofen taken first can block aspirin's access to COX-1.

Clopidogrel irreversibly blocks ADP binding to platelet P2Y12 receptors, reducing platelet activation. It is combined with aspirin for 21 days after a minor stroke or high-risk TIA and used for secondary prevention. It is a prodrug activated by CYP2C19, so poor metabolizers respond less, and omeprazole and other CYP2C19 inhibitors reduce its activation. It is contraindicated in active bleeding and causes bleeding and, rarely, TTP.

Heparins activate antithrombin, which inhibits thrombin and factor Xa; LMWH acts mainly on factor Xa. Clots stop propagating, so they are used for DVT prophylaxis and for cerebral venous sinus thrombosis even when hemorrhage is present. UFH is given IV with a short half-life, monitored by aPTT, and fully reversed by protamine, whereas LMWH is given SC, renally cleared, and only partly reversed. They are contraindicated in active bleeding, past HIT, and severe thrombocytopenia, and LMWH in severe renal impairment. They cause bleeding, HIT, hyperkalemia, and osteoporosis with long use, and antiplatelets, NSAIDs, and other anticoagulants add to the bleeding.

Warfarin blocks vitamin K epoxide reductase (VKORC1), so factors II, VII, IX, and X and proteins C and S are made in inactive forms, and anticoagulation develops over several days as existing factors clear. It prevents cardioembolic stroke when a mechanical valve rules out a DOAC. It is monitored by INR, with a target of 2–3, and metabolized by CYP2C9, so its interactions are many: metronidazole, TMP-SMX, fluconazole, and amiodarone raise the INR, rifampin, carbamazepine, and phenytoin lower it, and so does dietary vitamin K. It is teratogenic and contraindicated in pregnancy and active bleeding. It causes bleeding and, because protein C falls early, skin necrosis, so heparin bridges at the start.

Factor Xa inhibitors, apixaban and rivaroxaban, bind factor Xa directly, without antithrombin, and anticoagulate within hours. They are the first choice to prevent cardioembolic stroke in A-fib. They are given in fixed doses without routine monitoring, are metabolized by CYP3A4, carried by P-gp, and partly renally cleared, so strong CYP3A4 and P-gp inhibitors raise their levels, and rifampin, carbamazepine, and phenytoin lower them. They are contraindicated with mechanical heart valves, active bleeding, and severe renal impairment, and they cause bleeding, with less ICH than warfarin.

Dabigatran binds thrombin directly, without antithrombin, and anticoagulates within hours. It prevents cardioembolic stroke in A-fib. It is a prodrug, mostly renally cleared, and a P-gp substrate, so verapamil and amiodarone raise its levels and rifampin lowers them. It is contraindicated with mechanical heart valves and severe renal impairment and causes bleeding and dyspepsia.

Reversing anticoagulation

IV vitamin K restores hepatic synthesis of active factors II, VII, IX, and X, so the INR falls over hours. It treats warfarin-related ICH with 4F-PCC and the coagulopathy of liver disease with FFP. It acts too slowly to be used alone in ICH but keeps working after PCC wears off. It opposes warfarin, and IV doses rarely cause anaphylactoid reactions.

Clotting factor replacement uses 4F-PCC, which replaces factors II, VII, IX, and X, or FFP, which replaces all factors, and corrects the INR immediately. 4F-PCC treats warfarin-related ICH, to an INR below 1.4, and apixaban or rivaroxaban when andexanet alfa is unavailable, and FFP treats liver disease. PCC is a small volume and acts faster than FFP, which needs large volumes and thawing. PCC contains heparin, so recent thrombosis and HIT are contraindications, and both cause thrombosis, with volume overload and transfusion reactions from FFP.

Protamine is a positively charged protein that binds and neutralizes heparin within minutes, fully reversing UFH and partly reversing LMWH, and it treats heparin-related ICH. Too much protamine is itself an anticoagulant. A previous protamine reaction is a contraindication, and it causes hypotension, pulmonary hypertension, and anaphylaxis, especially with fish allergy or protamine-containing insulin.

Idarucizumab is an antibody fragment that binds dabigatran far more tightly than thrombin does, reversing it immediately, and it treats dabigatran-related ICH. It is given IV, acts only on dabigatran, and can cause thrombosis once anticoagulation is reversed.

Andexanet alfa acts as a decoy factor Xa that binds apixaban and rivaroxaban, rapidly reversing their effect in ICH, with 4F-PCC as the alternative. It is short acting, so a bolus is followed by an infusion. It causes thrombosis, including stroke and MI, and it also binds heparin-antithrombin, impairing later heparin.

Antifibrinolytics and cryoprecipitate keep or rebuild clots: tranexamic acid blocks plasminogen binding to fibrin, and cryoprecipitate replaces fibrinogen. Cryoprecipitate treats ICH after tPA, and an antifibrinolytic is used in SAH when aneurysm repair is delayed, for no more than 72 hours. TXA is renally cleared, so the dose is lowered in renal impairment, and it is contraindicated in active thrombosis. Both cause thrombosis, including cerebral ischemia, which PCC and hormonal contraceptives make more likely, and high doses of TXA can cause seizures.

Preventing recurrence

Statins inhibit HMG-CoA reductase, which slows cholesterol synthesis and raises hepatic LDL receptors, lowering LDL and stabilizing plaque. High-intensity statins such as atorvastatin treat large and small vessel ischemic stroke long term. Most, including atorvastatin and simvastatin, are metabolized by CYP3A4. They are contraindicated in pregnancy and active liver disease, and they cause myalgia and, rarely, myopathy and rhabdomyolysis with raised CK, raised transaminases,

and new diabetes; myopathy is more likely with CYP3A4 inhibitors such as clarithromycin and azoles, gemfibrozil, and niacin.

PCC acts faster than FFP, so 4F-PCC with IV vitamin K is the choice for warfarin-related ICH.

Relationships to other conditions

Acute ischemic stroke *and* ***intracerebral hemorrhage***. The same drugs act in opposite directions: thrombolytics, antiplatelets, and anticoagulants treat and prevent ischemic stroke, whereas ICH needs them stopped and reversed. BP targets differ too, below 185/110 for thrombolysis but a systolic of 140 after ICH.

Cardioembolic *and* ***large vessel stroke***. Cardioembolic stroke is prevented with a DOAC, or warfarin with a mechanical valve, whereas large and small vessel stroke is prevented with an antiplatelet and a high-intensity statin, with dual antiplatelets for 21 days after a minor stroke.

Lowering blood pressure and intracranial pressure after brain hemorrhage

After intracerebral and subarachnoid hemorrhage, blood pressure is lowered to limit bleeding, delayed ischemia is prevented, and raised intracranial pressure is lowered. Dihydropyridine calcium channel blockers and labetalol control blood pressure, nimodipine prevents ischemia after SAH, and osmotic therapy lowers intracranial pressure. Here, we review these drugs by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- After ICH, a systolic BP of 150–220 is lowered to 140, and in SAH systolic is kept below 160 until the aneurysm is secured.
- Nimodipine is given to every patient with aneurysmal SAH for 21 days and improves outcome more than vasospasm on imaging.
- Hypotension is nimodipine's main hazard, and it can itself cause ischemia.
- Mannitol and hypertonic saline lower ICP within minutes by drawing water out of the brain.

Lowering blood pressure

Dihydropyridine CCBs, nicardipine and clevidipine, block L-type calcium channels in arterial smooth muscle, relaxing arterioles, so BP falls quickly and can be titrated, with little effect on the heart. They lower a systolic BP of 150–220 to 140 after ICH and keep systolic below 160 in SAH until the aneurysm is secured. Both are IV infusions, and clevidipine is broken down by blood esterases and lasts minutes. Severe aortic stenosis is a contraindication, and clevidipine is also avoided with soy or egg allergy and disordered lipid metabolism. They cause reflex tachycardia, headache, flushing, and hypotension, which other antihypertensives worsen.

Labetalol blocks α1 and β receptors, lowering resistance without reflex tachycardia, and is used in ICH and SAH and to lower BP before thrombolysis. It is given as IV boluses or an infusion and metabolized in the liver. It is contraindicated in asthma, heart block, bradycardia, and decompensated heart failure, and it causes bradycardia, orthostatic hypotension, bronchospasm, and hepatotoxicity, with additive AV block alongside verapamil and diltiazem.

Nimodipine blocks L-type calcium channels with a preference for cerebral arteries. Every patient with aneurysmal SAH receives it by mouth or tube for 21 days to prevent ischemia from vasospasm, and it reduces delayed ischemic deficits, improving outcome more than vasospasm on imaging. It is lipophilic, so it crosses the BBB, and it has high first-pass metabolism by CYP3A4, so CYP3A4 inhibitors raise its levels and inducers such as carbamazepine and phenytoin lower them. It is contraindicated with strong CYP3A4 inhibitors, and the contents of the oral capsule must never be given IV. Its main adverse effect is hypotension, which can itself cause ischemia.

Lowering ICP

Osmotic therapy with mannitol or hypertonic saline raises plasma osmolality, drawing water out of the brain across the BBB and lowering ICP within minutes. It treats raised ICP in ICH and SAH, and mannitol, at 1–2 g/kg IV, also treats acute angle closure. Mannitol is renally excreted and acts as an osmotic diuretic, whereas hypertonic saline expands volume. Mannitol is contraindicated in anuria, heart failure, and hypovolemia and causes hypovolemia, acute kidney injury, and electrolyte shifts, with additive kidney injury alongside other nephrotoxic drugs; hypertonic saline is contraindicated in hypernatremia and heart failure and causes hypernatremia, volume overload, and osmotic demyelination if sodium was chronically low.

In SAH, nimodipine prevents ischemia, whereas nicardipine and labetalol control BP.

Relationships to other conditions

Intracerebral hemorrhage *and* ***subarachnoid hemorrhage***. Both lower BP and ICP with the same drugs, but their targets differ, a systolic of 140 after ICH and below 160 in SAH until the aneurysm is secured, and only SAH adds nimodipine to prevent delayed ischemia.

The cardiovascular center. Raised ICP drives the Cushing triad of hypertension, bradycardia, and irregular breathing, so osmotic therapy treats the cause of that hypertension rather than the BP itself.

Antiseizure drugs: mechanisms, choice by seizure type, and hazards

Antiseizure drugs either reduce excitation, by blocking sodium or T-type calcium channels, binding SV2A, or blocking glutamate receptors, or increase inhibition, by enhancing GABA-A receptors or blocking GABA breakdown. The choice depends on seizure type and on hazards such as teratogenicity, serious rashes, and CYP interactions. Here, we review the drugs named on these pages, with the treatments for status epilepticus, eclampsia, infantile spasms, and metabolic causes.

Key points

- Status epilepticus starts with IV lorazepam or IM midazolam, followed by IV levetiracetam, valproate, or fosphenytoin.
- Phenytoin follows zero-order kinetics at high doses, carbamazepine induces its own metabolism, and both induce CYP enzymes.
- HLA-B*1502 is tested before carbamazepine in people of Han Chinese ancestry, and valproate doubles lamotrigine levels, raising the risk of SJS.
- Ethosuximide is first line for absence seizures, vigabatrin causes permanent visual field loss, and topiramate causes kidney stones and acute angle closure.
- Valproate is avoided in those who could become pregnant where possible, because it raises the risk of neural tube defects about 10 times.

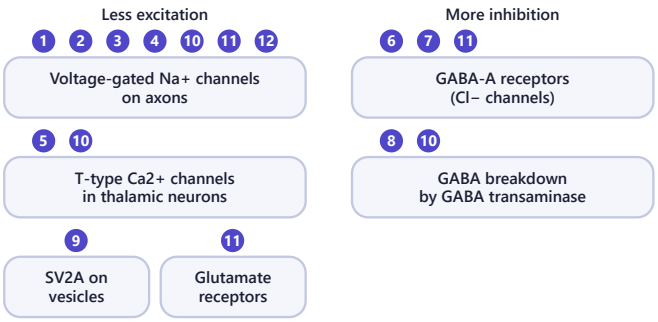


Fig. 1 | Numbered drug sites in two columns. Less excitation: voltage-gated sodium channels, 1, phenytoin, 2, carbamazepine, 3, lamotrigine, 4, lacosamide, 10, valproate, 11, topiramate, and 12, rufinamide; T-type calcium channels in thalamic neurons, 5, ethosuximide and 10, valproate; SV2A on synaptic vesicles, 9, levetiracetam; and glutamate receptors, 11, topiramate. More inhibition: GABA-A receptors, 6, benzodiazepines, 7, barbiturates, and 11, topiramate; and GABA breakdown, 8, vigabatrin and 10, valproate. 1, Phenytoin and fosphenytoin; 2, Carbamazepine and oxcarbazepine; 3, Lamotrigine; 4, Lacosamide; 5, Ethosuximide; 6, Benzodiazepines; 7, Barbiturates; 8, Vigabatrin; 9, Levetiracetam; 10, Valproate; 11, Topiramate; 12, Rufinamide.

Sodium channel blockers

Phenytoin and its prodrug fosphenytoin stabilize inactivated sodium channels, limiting repetitive firing. Fosphenytoin is given IV for status epilepticus after a benzodiazepine, and phenytoin treats focal seizures. Phenytoin follows zero-order kinetics at high doses, so small dose changes cause large changes in level; it is highly protein bound, valproate displaces it, and it induces CYP3A4, lowering levels of warfarin, oral contraceptives, and DOACs. Fosphenytoin is water soluble and can be given faster. Pregnancy is a contraindication, as are sinus bradycardia and heart block for IV use. Phenytoin causes gingival hyperplasia, hirsutism, nystagmus and ataxia at high levels, SJS, megaloblastic anemia, osteomalacia, and fetal hydantoin syndrome, and IV phenytoin causes hypotension, arrhythmia, and purple glove syndrome.

Carbamazepine and oxcarbazepine block sodium channels in the inactivated state. They treat focal seizures and are first line for trigeminal neuralgia. Carbamazepine induces CYP3A4, including its own metabolism, so its levels fall over the first weeks, and it lowers levels of oral contraceptives, warfarin, and DOACs; oxcarbazepine induces less, and CYP3A4 inhibitors such as clarithromycin raise carbamazepine levels. HLA-B*1502 predicts SJS and is tested in people of Han Chinese ancestry, and marrow suppression and MAOI use are also contraindications. They cause agranulocytosis and aplastic anemia, hyponatremia from SIADH, more often with oxcarbazepine, SJS and TEN, diplopia and ataxia, hepatotoxicity, and teratogenicity.

Lamotrigine blocks sodium channels and reduces glutamate release. It treats focal and generalized seizures and Lennox-Gastaut syndrome. It is glucuronidated and started slowly to lower the risk of rash, because it causes SJS and TEN, especially with fast titration or with valproate, which doubles its levels; a previous serious rash is a contraindication. Estrogen contraceptives and

enzyme inducers lower its levels, and it also causes dizziness, diplopia, and ataxia.

Lacosamide enhances slow inactivation of sodium channels and treats focal seizures. It is given by mouth or IV and mostly renally cleared. It prolongs the PR interval, which adds to other drugs that slow conduction, so second- or third-degree heart block is a contraindication, and it also causes dizziness and diplopia.

Rufinamide prolongs the inactive state of sodium channels, reducing drop attacks in Lennox-Gastaut syndrome. It is taken with food and has little CYP metabolism, though valproate raises its levels. It shortens the QT interval, so familial short QT syndrome is a contraindication, and it causes somnolence and hypersensitivity.

Calcium channels and vesicles

Ethosuximide blocks the T-type calcium currents in thalamic neurons that drive thalamocortical spike-wave rhythms, stopping absence seizures, for which it is first line. It has a long half-life and is metabolized by CYP3A4, so valproate raises and inducers lower its levels. It is avoided with succinimide hypersensitivity, and it causes GI upset, fatigue, headache, and, rarely, SJS and blood dyscrasias.

Levetiracetam binds SV2A on synaptic vesicles, reducing transmitter release. It treats focal and generalized seizures and juvenile myoclonic epilepsy and is given IV for status epilepticus after a benzodiazepine. It is renally cleared without CYP metabolism, so it has few interactions, and the dose is lowered in renal impairment. It causes irritability, aggression, depression, suicidality, and fatigue.

GABA

Benzodiazepines, including lorazepam, midazolam, diazepam, and clonazepam, increase how often GABA-A chloride channels open, hyperpolarizing neurons, which sedates and controls seizures. IV lorazepam or IM midazolam is first line for status epilepticus; benzodiazepines also treat alcohol and benzodiazepine withdrawal and acute vertigo, and clonazepam treats RBD and dystonia. Lorazepam has no active metabolites, so it suits liver disease, diazepam is long acting, and midazolam acts fast IM or intranasally. They are contraindicated in respiratory depression, severe sleep apnea, and acute angle-closure glaucoma, and they cause sedation, respiratory depression, falls and confusion in older adults, tolerance, dependence, and withdrawal seizures. Opioids and alcohol add to the respiratory depression, and flumazenil reverses them.

Barbiturates, phenobarbital and primidone, increase how long GABA-A chloride channels stay open. Phenobarbital treats refractory status epilepticus, and primidone, which is metabolized to phenobarbital, treats essential tremor. They are strong CYP inducers that lower levels of warfarin, oral contraceptives, and other antiseizure drugs, and they add to CNS depression. Acute intermittent porphyria and severe respiratory disease are contraindications, and they cause sedation, respiratory depression, and cognitive slowing.

Vigabatrin irreversibly blocks GABA transaminase, raising brain GABA, and it treats West syndrome, with ACTH as the alternative. It is renally cleared, and its effect outlasts the drug because the enzyme must be remade; it lowers phenytoin levels. It causes permanent concentric visual field loss, so vision is monitored and pre-existing field loss is a contraindication, as well as sedation.

Several mechanisms

Valproate blocks sodium channels and, weakly, T-type calcium channels while raising GABA, giving broad seizure control. It treats generalized seizures, juvenile myoclonic epilepsy, and Lennox-Gastaut syndrome and is given IV for status epilepticus after a benzodiazepine. It is highly protein bound and inhibits hepatic enzymes, raising lamotrigine and phenobarbital levels, and carbapenems sharply lower its own levels. It is avoided in pregnancy and those who could become pregnant where possible, in liver disease, and in urea cycle and mitochondrial disorders such as POLG mutations. It raises the risk of neural tube defects about 10 times and causes hepatotoxicity, especially under age 2, pancreatitis, tremor, weight gain, hair loss, thrombocytopenia, and hyperammonemia.

Topiramate blocks sodium channels and glutamate receptors while enhancing GABA-A receptors, and it also inhibits carbonic anhydrase. It treats focal and generalized seizures, is mostly renally cleared, and at higher doses lowers estrogen contraceptive levels. Metabolic acidosis is a contraindication, and it causes cognitive slowing and word-finding difficulty, paresthesia, weight loss, kidney stones, metabolic acidosis, and acute angle-closure glaucoma.

Special situations

ACTH raises cortisol and also acts directly on melanocortin receptors in the brain, stopping infantile spasms and improving the EEG, and it is first line for West syndrome with vigabatrin. It is injected over weeks. Uncontrolled hypertension and active infection are contraindications, and it causes hypertension, hyperglycemia, irritability, infection, and cushingoid features.

Magnesium sulfate antagonizes calcium and blocks NMDA receptors, depressing neuromuscular and CNS excitability, and it prevents and treats eclamptic seizures, with delivery. It is given IV and renally cleared, so it accumulates in renal impairment, and it potentiates neuromuscular blockers. Myasthenia gravis and heart block are contraindications. Toxicity first removes reflexes, then causes respiratory depression and cardiac arrest, and calcium gluconate reverses it.

Thiamine and glucose restore the cofactor of glucose metabolism and the glucose itself. Glucose is given for seizures with low glucose, with thiamine when alcohol use is likely, and thiamine treats seizures from decreased vitamin B1 and prevents Wernicke encephalopathy. Thiamine is given IV because absorption is poor with alcohol use, and IV doses rarely cause anaphylaxis.

Valproate is avoided in those who could become pregnant where possible, because it is teratogenic.

Relationships to other conditions

Seizures. Long-term treatment follows the seizure type: levetiracetam, lamotrigine, or carbamazepine for focal seizures; valproate, levetiracetam, or lamotrigine for generalized motor seizures; ethosuximide for absence; valproate or levetiracetam for juvenile myoclonic epilepsy; ACTH or vigabatrin for West syndrome; and valproate, lamotrigine, or rufinamide for Lennox-Gastaut syndrome.

Neurons *and* ***synapses***. Sodium channel blockers act on the action potential itself, whereas benzodiazepines, barbiturates, and vigabatrin act at the inhibitory GABA synapse, and levetiracetam at transmitter release.

Immunotherapy for multiple sclerosis, NMOSD, and MOGAD

Acute attacks of demyelinating disease are treated with high-dose corticosteroids, and relapses are prevented by drugs ranging from moderate-efficacy injectables and oral agents to high-efficacy antibodies and induction therapy. NMOSD responds to B-cell depletion, IL-6 receptor blockade, complement inhibition, and broad immunosuppression, and some MS drugs worsen it. Here, we review these drugs by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- MS relapses are treated with IV methylprednisolone for 3–7 days, and NMOSD and MOGAD attacks with 1 g daily.
- Natalizumab risks PML from JC virus, so JC virus status is checked first, and S1P receptor modulators cause first-dose bradycardia and AV block.
- B-cell depleters need HBV screening, eculizumab needs meningococcal vaccination, and alemtuzumab causes autoimmune thyroid disease, ITP, and anti-GBM disease.
- Interferon beta, natalizumab, and other MS drugs can worsen NMOSD, which is treated with rituximab, IL-6 receptor blockers, eculizumab, or azathioprine.

Acute attacks

Corticosteroids, including methylprednisolone, prednisone, and prednisolone, enter the nucleus through the glucocorticoid receptor, decreasing inflammatory gene transcription and BBB leakiness, so relapses recover faster. IV methylprednisolone is given for 3–7 days in MS relapse and 1 g daily in NMOSD and MOGAD attacks, oral steroids bridge in MOGAD, prednisone 1 mg/kg treats sudden SNHL, and steroids treat autoimmune encephalitis. High-dose IV pulses act over days, and prednisone is activated in the liver to prednisolone. Active untreated infection is a contraindication. Pulses cause hyperglycemia, insomnia, mood change, and hypertension, and long use causes osteoporosis, infection, cataracts, adrenal suppression, and avascular necrosis. NSAIDs add to the GI bleeding, and CYP3A4 inducers lower their levels.

IVIG, pooled IgG, saturates FcRn, speeding autoantibody clearance, while blocking Fc receptors and complement, so antibody-mediated injury lessens. It prevents MOGAD relapses at 1–2 g/kg every 4 weeks and treats autoimmune encephalitis. It is infused IV and its effect lasts weeks, and it blunts responses to live vaccines. IgA deficiency with anti-IgA antibodies is a contraindication, and it causes headache and aseptic meningitis, thrombosis, renal failure with sucrose-containing products, hemolysis, and anaphylaxis.

Moderate efficacy

Interferon beta acts on type I interferon receptors, shifting cytokines toward anti-inflammatory signals and reducing T-cell migration across the BBB, so relapses are fewer. It treats relapsing MS by SC or IM injection but is avoided in NMOSD. Severe depression and decompensated liver disease are contraindications, and it causes flu-like symptoms, injection site reactions, depression, raised transaminases, and cytopenias, with additive marrow suppression.

Glatiramer acetate resembles myelin basic protein and acts as a decoy, shifting T cells toward regulatory types and reducing relapses. It treats relapsing MS and is safe in pregnancy. It is injected SC daily or three times weekly and has few interactions. Hypersensitivity to it or mannitol is a contraindication, and it causes injection site reactions, lipoatrophy, and brief flushing, chest tightness, and palpitations after injection.

Dimethyl fumarate activates the Nrf2 antioxidant pathway, reducing oxidative stress and inflammation, and treats relapsing MS, taken by mouth twice daily. Lymphopenia is a contraindication, and it causes flushing, which aspirin before doses reduces, GI upset, lymphopenia, and, rarely, PML.

Teriflunomide inhibits DHODH in pyrimidine synthesis, slowing proliferation of activated lymphocytes, and treats relapsing MS. It has a very long half-life, and cholestyramine speeds its elimination; it affects the INR on warfarin variably. It is teratogenic, so pregnancy is a contraindication, as is severe liver disease, and it causes hepatotoxicity, hair thinning, and diarrhea.

High efficacy

S1P receptor modulators, fingolimod, siponimod, ozanimod, and ponesimod, bind S1PR1 on lymphocytes, which is internalized, trapping lymphocytes in lymph nodes so fewer reach the CNS. They treat relapsing MS but are avoided in NMOSD. They are oral; fingolimod is a prodrug activated by phosphorylation, and the newer drugs are more selective. MI, unstable angina, stroke, TIA, or decompensated heart failure within 6 months and heart block without a pacemaker are contraindications. They cause first-dose bradycardia and AV block, which beta-blockers and calcium channel blockers worsen, macular edema, infections including VZV, raised transaminases, and rebound disease on stopping.

Natalizumab blocks α4 integrin on lymphocytes, stopping them from crossing the BBB, and is highly effective in preventing relapses of highly active relapsing MS, given IV every 4 weeks; it is avoided in NMOSD. It can cause PML from JC virus, so JC virus status is checked first, past PML and immunosuppression are contraindications, and other immunosuppressants raise the risk. It also causes infusion reactions and hepatotoxicity.

B-cell depleters, ocrelizumab and rituximab against CD20 and inebilizumab against CD19, deplete B cells through antibody- and complement-mediated killing, reducing relapses and slowing disability. Ocrelizumab treats relapsing and primary progressive MS; rituximab every 6 months is first line in NMOSD, as is inebilizumab, which also removes plasmablasts carrying CD19; and rituximab treats MOGAD, though less effectively. They are given IV every 6 months. Active HBV, screened for first, and active infection are contraindications. They cause infusion reactions, infections, HBV reactivation, and low immunoglobulins, which are checked before treatment and yearly, and vaccines are given beforehand because responses are poor.

Induction

Alemtuzumab depletes T and B cells through CD52, and they repopulate over months, giving a long-lasting immune reset in highly active MS. It is given as two IV courses a year apart, and HIV infection is a contraindication. Months to years later it causes autoimmune thyroid disease, ITP, and anti-GBM disease, and it also causes infusion reactions and stroke.

Mitoxantrone intercalates into DNA and inhibits topoisomerase II, killing dividing immune cells, and is used for short-term induction in aggressive MS. It is given IV with a capped lifetime dose. Reduced left ventricular function is a contraindication, and it causes cardiotoxicity and therapy-related leukemia.

Cladribine, a purine analog, accumulates in lymphocytes and kills them, selectively depleting lymphocytes in highly active MS. It is taken as two short courses of tablets a year apart. Active malignancy, HIV, TB, and pregnancy are contraindications, and it causes lymphopenia, infections including herpes zoster, and possibly malignancy.

NMOSD and broad immunosuppression

IL-6 receptor blockers, satralizumab and tocilizumab, block IL-6 signaling, lowering plasmablast survival and AQP4-IgG production, so NMOSD relapses are fewer in patients with AQP4-IgG. Satralizumab is given SC monthly and tocilizumab IV or SC. Because IL-6 suppresses CYP enzymes, blocking it can lower levels of CYP3A4 substrates. Active infection including TB is a contraindication, and they cause infections, raised transaminases and cholesterol, and neutropenia.

Eculizumab blocks cleavage of complement C5, preventing the membrane attack complex and so antibody-mediated astrocyte injury in NMOSD with AQP4-IgG. It is given IV every 2 weeks after loading. It causes meningococcal infection, which can be fatal, so vaccination comes first, and lack of vaccination or active meningococcal infection is a contraindication.

Broad immunosuppressants block lymphocyte proliferation in different ways: azathioprine blocks purine synthesis, MMF does so through IMPDH, methotrexate blocks folate metabolism, and tacrolimus blocks calcineurin. Azathioprine with prednisone is first line in NMOSD, and these drugs prevent MOGAD relapses, reducing relapses after months. Their onset is slow, so steroids bridge; TPMT is tested before azathioprine, and allopurinol raises its levels; tacrolimus levels are monitored, and CYP3A4 inhibitors raise them. MMF and methotrexate are contraindicated in pregnancy, and azathioprine with low TPMT activity. All cause marrow suppression, infection, and malignancy; azathioprine also causes hepatotoxicity and pancreatitis, and tacrolimus nephrotoxicity, tremor, and diabetes.

Interferon beta, natalizumab, and other MS drugs can worsen NMOSD, so the diagnosis is settled before choosing one.

Relationships to other conditions

Multiple sclerosis *and* ***NMOSD***. Both treat attacks with IV methylprednisolone and plasma exchange, but their preventive drugs differ. MS uses interferons, glatiramer, oral drugs, S1P receptor modulators, natalizumab, and anti-CD20 drugs, by escalation or early high efficacy, whereas NMOSD uses B-cell depletion, IL-

6 receptor blockers, eculizumab, or azathioprine, and interferon beta and natalizumab can worsen it.

MOGAD. Relapse prevention usually starts after a second attack, with azathioprine or MMF under a prednisolone bridge, IVIG, or oral corticosteroids; rituximab is less effective than in NMOSD, and MS drugs such as interferons, glatiramer acetate, and fingolimod appear ineffective.

Drugs for spasticity, bladder dysfunction, orthostatic hypotension, and sleep behavior

Neurological disease often leaves symptoms that need their own treatment, whatever the cause. Baclofen and botulinum toxin relax overactive muscles, antimuscarinics, mirabegron, and α-blockers treat bladder dysfunction, midodrine and fludrocortisone raise standing blood pressure, and melatonin treats REM sleep behavior disorder. Here, we review these drugs by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- Abrupt withdrawal of baclofen causes seizures, hallucinations, and hyperthermia.
- Botulinum toxin treats focal dystonia and sialorrhea and is avoided in myasthenia gravis and Lambert-Eaton syndrome.
- Oxybutynin crosses the BBB and causes confusion in older adults, and it opposes cholinesterase inhibitors.
- Midodrine and fludrocortisone both cause supine hypertension, and melatonin is tried before clonazepam for RBD.

Muscles

Baclofen activates GABA-B receptors in the spinal cord, reducing excitatory transmitter release onto motor neurons and so spasticity. It treats spasticity in MS and dystonia in atypical parkinsonism. It is renally cleared, so the dose is lowered in renal impairment, and it is given by mouth or intrathecally. It causes sedation, weakness, and dizziness, adds to other CNS depressants, and its abrupt withdrawal causes seizures, hallucinations, and hyperthermia.

Botulinum toxin cleaves SNARE proteins in cholinergic nerve terminals, blocking ACh release, which causes local weakness or less secretion for about 3 months, until terminals sprout. Injected locally, it treats focal dystonia, including in HD, and sialorrhea in atypical parkinsonism. It is avoided with infection at the site and in myasthenia gravis and Lambert-Eaton syndrome. It can weaken nearby muscles, causing dysphagia or ptosis, and aminoglycosides strengthen the block.

Bladder

Antimuscarinics, such as oxybutynin, block ACh at M3 receptors on the detrusor, reducing uninhibited contractions. They treat urge incontinence in MS, and selective antimuscarinics treat urinary dysfunction in atypical parkinsonism. Oxybutynin crosses the BBB, whereas selective drugs cause less confusion. Urinary retention, angle-closure glaucoma, and myasthenia gravis are contraindications, and they cause dry mouth, constipation, retention, and confusion in older adults; they add to the anticholinergic burden and oppose cholinesterase inhibitors.

Mirabegron activates β3 receptors on the detrusor, relaxing it and enlarging bladder capacity, and treats urinary dysfunction in atypical parkinsonism. It is a moderate CYP2D6 inhibitor, raising levels of substrates such as metoprolol. Severe uncontrolled hypertension is a contraindication, and it causes hypertension and urinary retention.

α-blockers, such as tamsulosin, block α1 receptors in the bladder neck and prostate, relaxing smooth muscle at the outlet so the bladder empties more easily, and treat urinary dysfunction in atypical parkinsonism. Tamsulosin is selective for α1A. Orthostatic hypotension is a contraindication, and they cause orthostatic hypotension, dizziness, and IFIS during cataract surgery, with additive hypotension alongside PDE-5 inhibitors and other antihypertensives.

Blood pressure and sleep

Midodrine is a prodrug of an α1 agonist that constricts arterioles and veins, raising standing BP, and treats orthostatic hypotension in atypical parkinsonism. It is short acting and given during the day only. Severe heart disease, urinary retention, and supine hypertension are contraindications, and it causes supine hypertension, piloerection, scalp itching, and urinary retention, with additive bradycardia alongside beta-blockers.

Fludrocortisone activates mineralocorticoid receptors in the kidney, increasing sodium and water retention, which expands plasma volume and raises BP, and it treats orthostatic hypotension in atypical parkinsonism. It is long acting and taken once daily. Heart failure is a contraindication, and it causes hypokalemia, which adds

to digoxin toxicity and diuretic losses, edema, supine hypertension, and heart failure.

Melatonin activates MT1 and MT2 receptors in the suprachiasmatic nucleus, reducing RBD episodes, and is first line for RBD before clonazepam. It has a short half-life and is taken at bedtime. It causes morning drowsiness and headache, and fluvoxamine raises its levels.

Relationships to other conditions

Multiple sclerosis. Chronic symptoms are treated alongside disease-modifying therapy: baclofen for spasticity, oxybutynin for incontinence and catheterization for retention, and gabapentin for neuropathic pain.

Atypical parkinsonism *and* ***DLB***. No treatment slows MSA, PSP, or CBD, so autonomic and other symptoms are treated directly: midodrine or fludrocortisone for orthostatic hypotension, α-blockers, mirabegron, or selective antimuscarinics for urinary dysfunction, botulinum toxin for dystonia and sialorrhea, and melatonin and then low-dose clonazepam for RBD, as in DLB.

Drugs for dementia, myasthenia, and mood in neurological disease

Cholinesterase inhibitors and memantine give modest symptomatic benefit in Alzheimer disease and the Lewy body dementias, while anti-amyloid antibodies slow decline in early Alzheimer disease. Pyridostigmine treats myasthenia gravis, and antidepressants treat the depression, anxiety, and behavioral change that accompany neurodegenerative disease. Here, we review these drugs by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- Cholinesterase inhibitors help Alzheimer disease, DLB, and dementia in Parkinson disease, but not FTD, where donepezil can worsen behavior.
- Cholinesterase inhibitors cause bradycardia and syncope, which beta-blockers worsen.
- Anti-amyloid antibodies need confirmed amyloid and MRI monitoring because of ARIA, which is more common in APOE4 carriers.
- SSRIs cause serotonin syndrome with MAOIs, including high-dose rasagiline.

Dementia

Cholinesterase inhibitors, donepezil, rivastigmine, and galantamine, slow ACh breakdown by AChE, offsetting the loss of cholinergic neurons, with modest benefit in cognition and function and fewer hallucinations in DLB. They treat mild to moderate Alzheimer disease, DLB, and dementia in PD, but not FTD, where donepezil can worsen behavior. Donepezil has a long half-life and is taken once daily, and rivastigmine comes as a patch. Sick sinus syndrome or heart block without a pacemaker calls for caution. They cause nausea, diarrhea, bradycardia, syncope, and vivid dreams and can worsen motor function in PSP; anticholinergics oppose them, and beta-blockers add to the bradycardia.

Memantine blocks NMDA channels noncompetitively, limiting excitotoxicity, with modest benefit in moderate to severe Alzheimer disease, and it may add benefit in DLB. It is renally cleared, so the dose is lowered in renal impairment. It causes dizziness, headache, confusion, and constipation, and its effects add to those of other NMDA antagonists such as ketamine and amantadine.

Anti-amyloid antibodies, lecanemab and donanemab, bind amyloid β and clear plaques, slowing decline in early Alzheimer disease with confirmed amyloid. They are IV infusions with MRI monitoring. Anticoagulation, many microbleeds, and cerebral amyloid angiopathy are contraindications, and anticoagulants and thrombolytics raise the risk of hemorrhage. They cause ARIA, edema or microhemorrhage, more often in APOE4 carriers, and infusion reactions.

Pyridostigmine blocks AChE at the neuromuscular junction, prolonging ACh at nicotinic receptors and strengthening contractions, and it is first line for myasthenia gravis before immunosuppression. It is quaternary, so it does not cross the BBB. Mechanical bowel or urinary obstruction is a contraindication. It causes cramps, diarrhea, salivation, and bradycardia, excess causes cholinergic crisis, and anticholinergics oppose it.

Mood

SSRIs, such as sertraline and citalopram, block the serotonin transporter, so more serotonin stays in the synapse. They treat behavior in FTD, though not cognition, and depression and anxiety in PD, DLB, and HD. They take weeks to work fully, and citalopram prolongs the QT interval at higher doses. MAOI use within 2 weeks is a contraindication. They cause GI upset, sexual dysfunction, hyponatremia, bleeding, and serotonin syndrome, which follows their use with MAOIs, including high-dose rasagiline, and tramadol, and antiplatelets and anticoagulants add to the bleeding.

Mirtazapine blocks presynaptic α2 receptors, increasing norepinephrine and serotonin release, and also blocks 5-HT2 and 5-HT3 receptors, giving an antidepressant effect with sedation and weight gain. It treats depression in DLB and is sedating at low doses. MAOI use is a contraindication, with which it causes serotonin syndrome, and it causes sedation, weight gain, and, rarely, agranulocytosis.

Relationships to other conditions

Alzheimer disease *and* ***DLB***. Both respond to cholinesterase inhibitors, often better in DLB, where they also reduce hallucinations, and memantine is added in

both. DLB adds a severe sensitivity to antipsychotics, whereas anti-amyloid antibodies are used only in early Alzheimer disease with confirmed amyloid.

Frontotemporal dementia. Unlike Alzheimer disease, FTD does not respond to cholinesterase inhibitors or memantine, so SSRIs treat behavior, and antipsychotics give mixed results, with patients prone to extrapyramidal effects.

Drugs for Parkinson disease, chorea, and tremor

Parkinson disease is treated by raising dopamine signaling, with levodopa, dopamine agonists, and inhibitors of dopamine and levodopa breakdown, while amantadine and anticholinergics add benefit. Chorea in Huntington disease is treated by lowering dopamine signaling, with VMAT2 inhibitors or antipsychotics, and essential tremor responds to propranolol. Here, we review these drugs by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- Levodopa with carbidopa is the most effective drug for PD; carbidopa blocks peripheral DDC, and wearing off develops as neurons are lost.
- Dopamine agonists such as pramipexole are the main trigger of impulse control disorders.
- VMAT2 inhibitors are first line for chorea but cause depression and suicidality.
- Antipsychotics cause parkinsonism, most with haloperidol and fluphenazine and least with quetiapine and clozapine, and are avoided in DLB.

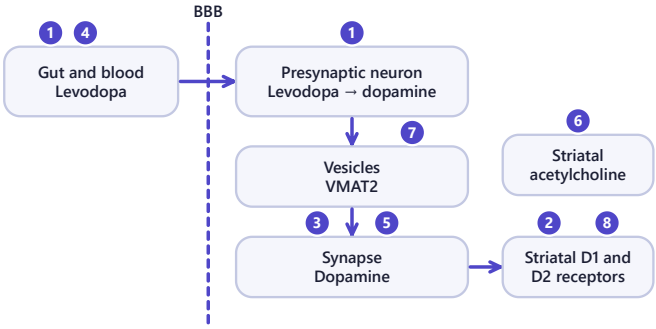


Fig. 1 | Numbered drug sites along the path of levodopa. In the gut and blood, 1, carbidopa blocks peripheral conversion and 4, COMT inhibitors block breakdown. Across the blood-brain barrier, 1, levodopa becomes dopamine in the presynaptic neuron. 7, VMAT2 inhibitors block loading into vesicles. In the synapse, 3, MAO-B inhibitors slow breakdown and 5, amantadine increases release. At striatal receptors, 2, dopamine agonists stimulate and 8, antipsychotics block. 6, anticholinergics block striatal acetylcholine. 1, Levodopa with carbidopa; 2, Dopamine agonists; 3, MAO-B inhibitors; 4, COMT inhibitors; 5, Amantadine; 6, Anticholinergics; 7, VMAT2 inhibitors; 8, Antipsychotics.

Raising dopamine

Levodopa with carbidopa raises striatal dopamine: levodopa crosses the BBB and becomes dopamine, while carbidopa blocks peripheral DDC, so less is converted outside the brain. Bradykinesia and rigidity lessen. It is the most effective drug for PD, is tried in atypical parkinsonism, is used at low doses in DLB, and treats rigidity in HD. Its short half-life means that, as neurons are lost, on time shortens, which is wearing off, and dietary protein competes with it for absorption and transport across the BBB. Nonselective MAOIs within 2 weeks, with which it causes hypertensive crisis, and angle-closure glaucoma are contraindications. It causes nausea, orthostatic hypotension, dyskinesia, hallucinations, and motor fluctuations, and D2 antagonists oppose it.

Dopamine agonists, such as pramipexole, stimulate striatal D2 and D3 receptors directly, reducing bradykinesia. They treat PD, especially early or in younger patients, and rigidity in HD. They are longer acting than levodopa and renally cleared. Psychosis and impulse control disorder are contraindications, and they cause impulse control disorders, including gambling, hypersexuality, shopping, and binge eating, as well as sleep attacks, hallucinations, and edema; antipsychotics oppose them.

MAO-B inhibitors, such as rasagiline, irreversibly inhibit MAO-B, the main form in the brain, slowing dopamine breakdown and prolonging its effect. They treat PD early or as an add-on, and they are selective at usual doses, so dietary tyramine is safe. Other MAOIs, meperidine, and tramadol are contraindicated, and with SSRIs, SNRIs, TCAs, meperidine, and tramadol they cause serotonin syndrome. They cause insomnia, orthostatic hypotension, and worse dyskinesia with levodopa.

COMT inhibitors, such as entacapone, block peripheral COMT, so each dose of levodopa lasts longer, and they treat wearing off. They only work with levodopa, whose exposure they raise. Nonselective MAOIs are contraindicated, and they cause dyskinesia, diarrhea, and orange urine.

Amantadine increases dopamine release and blocks its reuptake, with weak NMDA antagonism, easing bradykinesia, rigidity, tremor, and dyskinesia, often only briefly. It treats PD and rigidity in HD. It is renally cleared and accumulates in renal impairment, so severe renal impairment is a contraindication. It causes livedo

reticularis, ankle edema, confusion, and hallucinations, with additive anticholinergic and NMDA effects.

Anticholinergics, trihexyphenidyl and benztropine, block striatal muscarinic receptors, restoring the balance between dopamine and ACh and reducing tremor and dystonia. Trihexyphenidyl treats tremor in younger patients with PD, and benztropine treats drug-induced parkinsonism and acute dystonia. They cross the BBB, so dementia and older age, as well as angle-closure glaucoma and urinary retention, are contraindications. They cause dry mouth, constipation, urinary retention, blurred vision, and confusion, add to the anticholinergic burden, and oppose cholinesterase inhibitors.

Lowering dopamine

VMAT2 inhibitors, tetrabenazine, deutetrabenazine, and valbenazine, block loading of dopamine into presynaptic vesicles, so less is released and chorea lessens, and they are first line for chorea in HD. They are metabolized by CYP2D6, so inhibitors such as fluoxetine raise their levels, and deutetrabenazine lasts longer, needing fewer doses. Untreated depression or suicidality, MAOIs, and liver impairment are contraindications. They cause depression and suicidality, fatigue, akathisia, parkinsonism, and QT prolongation, which adds to that of other drugs.

Antipsychotics, including haloperidol, risperidone, olanzapine, aripiprazole, quetiapine, and clozapine, block D2 receptors in mesolimbic and striatal pathways, and the newer drugs also block 5-HT2A, reducing psychosis and chorea but causing parkinsonism when striatal block is strong. They treat chorea in HD off label, and quetiapine or clozapine treats psychosis in PD. Extrapyramidal effects are most likely with haloperidol and fluphenazine, then risperidone and olanzapine, and least with quetiapine and clozapine. They are avoided in DLB, where serious reactions occur in up to 50%, and in dementia-related psychosis in older adults. They cause drug-induced parkinsonism, acute dystonia, akathisia, tardive dyskinesia, NMS, QT prolongation, and metabolic syndrome, and clozapine causes agranulocytosis; they oppose levodopa and dopamine agonists, and QT prolongation is additive.

Pimavanserin is an inverse agonist at 5-HT2A receptors that does not block dopamine, so it reduces psychosis without worsening parkinsonism, and it treats psychosis in PD. It has a long half-life and is metabolized by CYP3A4, so strong CYP3A4 inhibitors raise its levels. QT prolongation is a contraindication, and it causes peripheral edema, confusion, nausea, and QT prolongation.

Tremor

Propranolol blocks β receptors, damping peripheral and central tremor oscillation and reducing tremor amplitude, and it treats essential tremor, with primidone as the alternative. It is lipophilic, crosses the BBB, and has high first-pass metabolism. Asthma, bradycardia, heart block, and decompensated heart failure are contraindications, and it causes bradycardia, fatigue, bronchospasm, depression, and masked hypoglycemia, with additive AV block alongside verapamil and diltiazem.

Dopamine agonists such as pramipexole are the main trigger of impulse control disorders.

Relationships to other conditions

Parkinson disease *and* ***Huntington disease***. PD is treated by raising dopamine signaling, whereas the chorea of HD is treated by lowering it, and each treatment can cause the other disease's signs: levodopa causes dyskinesia, and dopamine blockers and depleters cause parkinsonism. Rigidity in the Westphal variant of HD is treated like PD, with levodopa, dopamine agonists, or amantadine.

Drug-induced parkinsonism *and* ***the basal ganglia***. Drug-induced parkinsonism follows striatal D2 block or dopamine depletion, so it is treated by stopping the drug, switching to an atypical antipsychotic if one is still needed, and giving an anticholinergic such as benztropine, while levodopa helps little.

Analgesics and anesthetics acting on the pain pathway

Analgesics act at every level of the pain pathway, from prostaglandin synthesis at the site of injury to transmission in the dorsal horn and descending inhibition from the brainstem, while anesthetics block conduction or depress the brain. Opioids, paracetamol, NSAIDs, SNRIs and TCAs, gabapentinoids, ketamine, local anesthetics, and propofol each have characteristic hazards, especially respiratory depression. Here, we review their targets, uses, properties, contraindications, adverse effects, and interactions.

Key points

- Opioids shift the CO_2 response curve right, whereas benzodiazepines and propofol flatten it, and together they add to respiratory depression.
- Paracetamol replaces aspirin and NSAIDs for pain relief in SAH.
- SNRIs, TCAs, and gabapentinoids treat neuropathic pain, and TCAs are lethal in overdose.
- Local anesthetic toxicity starts with perioral numbness and seizures and progresses to arrhythmia and cardiac arrest.

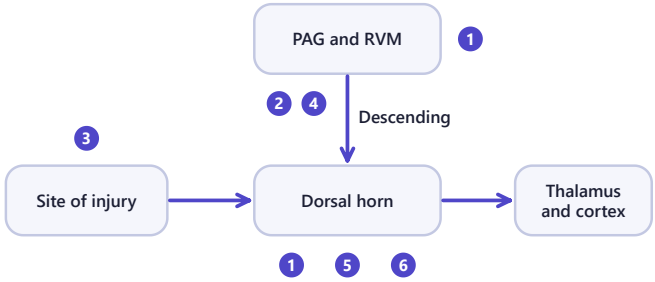


Fig. 1 | Six numbered drug sites. 3, NSAIDs act at the site of injury. 1, opioids, 5, gabapentinoids, and 6, ketamine act in the dorsal horn, and opioids also act in the periaqueductal gray and rostral ventromedial medulla. 2, paracetamol, and 4, SNRIs and TCAs, strengthen descending inhibition. 1, Opioids; 2, Paracetamol; 3, NSAIDs; 4, SNRIs and TCAs; 5, Gabapentinoids; 6, Ketamine.

Analgesics

Opioids, such as morphine and fentanyl, act on μ-opioid receptors in the PAG, RVM, dorsal horn, and peripheral afferents, opening potassium channels and closing N-type calcium channels, and in the PAG they release descending inhibition, giving analgesia for severe pain. Morphine metabolites accumulate in renal failure, and fentanyl is lipophilic and potent, with patches lasting 3 days. Respiratory depression and paralytic ileus are contraindications. They cause respiratory depression, shifting the CO_2 response curve right, as well as constipation, nausea, miosis, sedation, tolerance, and dependence; benzodiazepines and alcohol add to the respiratory depression, and naloxone reverses them.

Paracetamol inhibits central COX and strengthens descending serotonergic inhibition, also acting on endocannabinoid pathways, giving analgesia and antipyresis with little anti-inflammatory effect. It relieves pain in SAH, in place of aspirin and NSAIDs, and treats fever after stroke. It is metabolized in the liver, and a small fraction becomes a toxic metabolite detoxified by glutathione, so overdose causes hepatotoxicity, chronic alcohol raises that risk, and severe liver disease is a contraindication.

NSAIDs, such as ibuprofen, aspirin, and celecoxib, decrease prostaglandin synthesis through COX-1 and COX-2, and aspirin binds irreversibly, giving analgesic, antipyretic, and anti-inflammatory effects; celecoxib selects COX-2. They treat pain and inflammation but are avoided in SAH. Because it spares COX-1, celecoxib causes fewer ulcers but raises cardiovascular risk. Peptic ulcer or bleeding, renal impairment, and late pregnancy are contraindications, and celecoxib is avoided in cardiovascular disease and sulfa allergy. They cause GI ulcers and bleeding, kidney injury, and fluid retention, anticoagulants and SSRIs add to the bleeding, and they raise lithium levels.

SNRIs and TCAs, such as duloxetine and amitriptyline, block serotonin and norepinephrine reuptake, raising both in the spinal cord and strengthening descending inhibition. They treat neuropathic pain and depression in DLB and HD, and duloxetine treats fibromyalgia. Amitriptyline also blocks muscarinic, H1, and α1 receptors and sodium channels. MAOI use is a contraindication, with which they cause serotonin syndrome, as with other serotonergic drugs, and TCAs are avoided after recent MI and in heart block. Duloxetine causes nausea and hypertension, and TCAs

cause anticholinergic effects, sedation, orthostatic hypotension, and QRS widening and are lethal in overdose.

Gabapentinoids, gabapentin and pregabalin, bind the α2δ-1 subunit of voltage-gated calcium channels, reducing calcium entry and excitatory transmitter release and relieving neuropathic pain, including in MS. They are renally cleared unchanged, so the dose is lowered in renal impairment. They cause sedation, dizziness, edema, and misuse, and they add to the respiratory depression of opioids.

Anesthetics

Ketamine blocks NMDA glutamate receptors noncompetitively, giving analgesia and dissociative anesthesia with breathing preserved. It acts fast and raises sympathetic tone, so uncontrolled hypertension is a contraindication, as is psychosis. It causes emergence hallucinations, hypertension, tachycardia, and increased secretions, and its effects add to those of other NMDA antagonists.

Local anesthetics, such as lidocaine, block voltage-gated sodium channels from inside the channel, so conduction fails and pain signals stop, giving local and regional anesthesia. Small, myelinated pain fibers block first, adrenaline prolongs the block by vasoconstriction, and amides are metabolized in the liver, so beta-blockers and cimetidine raise lidocaine levels. Amide allergy is rare. Systemic toxicity begins with perioral numbness and seizures and progresses to arrhythmia and cardiac arrest.

Propofol potentiates GABA-A receptors, giving rapid sedation and anesthesia for surgery and ICU sedation, with fast onset and offset through redistribution. Egg or soy allergy calls for caution. It causes respiratory depression, flattening the CO_2 response curve, hypotension, and propofol infusion syndrome, and opioids and benzodiazepines add to the respiratory depression.

Opioids shift the CO_2 response curve right, whereas benzodiazepines and propofol flatten it.

Relationships to other conditions

Pain pathways. NSAIDs act at the site of injury, opioids, gabapentinoids, and ketamine in the dorsal horn, and opioids, paracetamol, SNRIs, and TCAs strengthen descending inhibition from the PAG and RVM.

The respiratory center *and* ***SAH***. Opioids, benzodiazepines, and propofol depress the respiratory center in different ways and add to one another. In SAH, analgesia uses paracetamol, avoiding aspirin and NSAIDs, which add to bleeding.

Drugs for glaucoma, macular degeneration, and conjunctivitis, and those that cause angle closure

Glaucoma drugs lower intraocular pressure either by increasing outflow of aqueous humor or by decreasing its production, and their adverse effects follow from local and systemic absorption. Anti-VEGF agents treat wet macular degeneration, topical antibiotics treat bacterial conjunctivitis, and several drugs that dilate the pupil can trigger acute angle closure. Here, we review these drugs by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- Prostaglandin analogs are first line for open-angle glaucoma and darken the iris and lengthen eyelashes.
- Timolol drops are absorbed systemically and are avoided in asthma, COPD, heart block, and bradycardia.
- Acute angle closure is treated with acetazolamide, timolol, and apraclonidine, with mannitol if needed, and pilocarpine once IOP is below 40 mmHg.
- Mydriatics, sympathomimetics, anticholinergics, H1 antagonists, and topiramate can trigger angle closure.

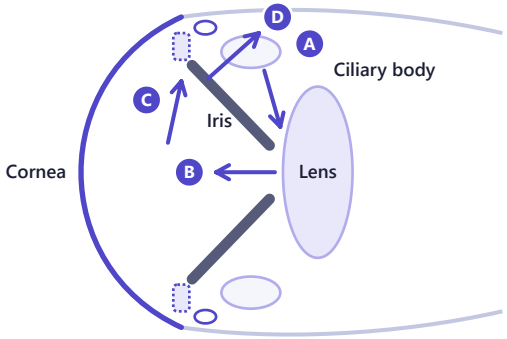


Fig. 1 | A cross-section of the front of the eye. A, the ciliary body secretes aqueous humor behind the iris. B, it flows through the pupil into the anterior chamber. C, most drains through the trabecular meshwork into Schlemm canal at the angle between iris and cornea. D, some leaves through the ciliary muscle, the uveoscleral route.

Increasing outflow

Prostaglandin analogs, such as latanoprost, act on FP receptors in the ciliary muscle to increase uveoscleral outflow, lowering IOP. They are first line for open-angle glaucoma, as once-daily drops. Active uveitis calls for caution, and they darken the iris and periorbital skin, lengthen eyelashes, and redden the conjunctiva.

Cholinergic agents, such as pilocarpine, act on M3 receptors in the ciliary muscle and iris sphincter, contracting the ciliary muscle and constricting the pupil, which opens the trabecular meshwork and increases outflow. Pilocarpine treats acute angle closure once IOP is below 40 mmHg, because an ischemic iris sphincter does not respond, and open-angle glaucoma, though it is poorly tolerated, needing drops several times a day. Uveitis is a contraindication, and it causes brow ache, dim vision from miosis, and myopia.

Decreasing production

β-blockers, such as timolol, block β receptors on the ciliary epithelium, reducing aqueous production, and treat open-angle glaucoma and acute angle closure. The drops are absorbed systemically through the nasolacrimal duct, causing bradycardia and bronchospasm, so asthma, COPD, heart block, and bradycardia are contraindications, and oral beta-blockers and calcium channel blockers add to the bradycardia.

α2-agonists, brimonidine and apraclonidine, act on α2 receptors on the ciliary epithelium, decreasing aqueous production while increasing uveoscleral outflow. They treat open-angle glaucoma, and apraclonidine treats acute angle closure. Brimonidine crosses the BBB and causes apnea and CNS depression in children, so infants and young children should not receive it, and MAOI use is a contraindication. They also cause allergic conjunctivitis and dry mouth.

Carbonic anhydrase inhibitors, dorzolamide and acetazolamide, inhibit carbonic anhydrase in the ciliary epithelium, reducing bicarbonate and so aqueous production. Acetazolamide 500 mg, by mouth or IV, treats acute angle closure, and dorzolamide drops treat open-angle glaucoma. Sulfa allergy calls for caution, and acetazolamide is avoided in hypokalemia and severe liver disease. Acetazolamide causes paresthesia, metabolic acidosis, hypokalemia, and kidney stones, and

topiramate and high-dose salicylates add to the acidosis; dorzolamide causes a bitter taste and stinging.

Other eye drugs

Anti-VEGF agents, ranibizumab and aflibercept, bind VEGF and stop new leaky vessels, reducing fluid and bleeding under the retina in wet AMD. They are injected into the eye every 1–3 months. Eye or periocular infection is a contraindication, and they cause endophthalmitis, raised IOP, and retinal detachment.

Topical antibiotics, including chloramphenicol, polymyxin B with trimethoprim, and fluoroquinolones such as moxifloxacin, kill or stop bacteria on the ocular surface by inhibiting protein synthesis or DNA gyrase, shortening bacterial conjunctivitis. They are used 4 times daily for 5–7 days and act locally, causing stinging and local allergy.

Drugs that can trigger angle closure

Mydriatics and cycloplegics, atropine and cyclopentolate, and sympathomimetics, adrenaline, phenylephrine, and pseudoephedrine, dilate the pupil and can trigger angle closure, as can anticholinergics and H1 antagonists. Topiramate triggers it differently, by swelling the ciliary body so that the lens and iris are pushed forward.

Pilocarpine waits until IOP is below 40 mmHg, because an ischemic iris sphincter does not respond.

Relationships to other conditions

Open-angle glaucoma *and* ***acute angle closure***. Open-angle glaucoma starts with a prostaglandin analog, then laser trabeculoplasty and surgery, whereas acute angle closure is an emergency treated with acetazolamide, timolol, and apraclonidine, mannitol if needed, and pilocarpine, followed by laser peripheral iridotomy, the treatment of choice.

Vision loss and red eye. Wet AMD is treated with intravitreal anti-VEGF agents, without which most eyes become legally blind, and bacterial conjunctivitis, often self-limited, is shortened by topical antibiotics.

Drugs for vertigo and Ménière disease

Vestibular suppressants relieve acute vertigo but delay central compensation, so they are used only briefly, while Ménière disease is treated by reducing endolymphatic hydrops and, when refractory, by destroying vestibular hair cells. Here, we review meclizine and other suppressants, betahistine, thiazides, and intratympanic gentamicin by target, mechanism, use, properties, contraindications, adverse effects, and interactions.

Key points

- Vestibular suppressants are given for about 3 days in vestibular neuritis and labyrinthitis, because longer use delays central compensation.
- Ménière disease is treated with less salt, caffeine, and alcohol, thiazides or betahistine, and intratympanic gentamicin if refractory.
- Aminoglycosides, loop diuretics, and cisplatin are ototoxic, and high-dose aspirin causes reversible tinnitus.

Vertigo and Ménière disease

Vestibular suppressants, such as meclizine, as well as benzodiazepines and antiemetics, block histamine and ACh at H1 and muscarinic receptors in the vestibular nuclei, reducing vestibular signaling and so vertigo and nausea. They are given for about 3 days in vestibular neuritis and labyrinthitis and have little role in BPPV, because beyond a few days they delay central compensation. Angle-closure glaucoma and urinary retention are contraindications. They cause sedation, which alcohol and other CNS depressants worsen, dry mouth, and confusion in older adults.

Betahistine is a weak agonist at H1 receptors and an antagonist at H3 receptors; it improves inner ear microcirculation and reduces vestibular nucleus firing, so attacks of vertigo are fewer, and it treats Ménière disease. Its short half-life means several doses a day, and antihistamines oppose it. Pheochromocytoma is a contraindication, and it causes headache and GI upset.

Thiazides, such as hydrochlorothiazide, block NCC in the distal convoluted tubule, lowering body fluid and endolymphatic hydrops, and treat Ménière disease alongside less salt, caffeine, and alcohol. They are taken once daily. Sulfa allergy calls for caution, and gout is a contraindication. They cause hypokalemia, hyponatremia, hypercalcemia, hyperuricemia, and hyperglycemia, raise lithium levels, and their hypokalemia adds to digoxin toxicity.

Intratympanic gentamicin is an aminoglycoside injected through the eardrum that destroys vestibular hair cells, a chemical labyrinthectomy, for Ménière disease not controlled otherwise. Bilateral disease is a contraindication, and it causes hearing loss and imbalance.

Aminoglycosides, loop diuretics, and cisplatin are ototoxic, and high-dose aspirin causes reversible tinnitus.

Relationships to other conditions

Vertigo. Treatment targets the cause: the Epley maneuver for BPPV, brief suppressants and then vestibular rehabilitation for vestibular neuritis and labyrinthitis, and the measures above for Ménière disease; suppressants only relieve acute symptoms.

Hearing loss. Drugs both treat and cause it: oral prednisone treats sudden SNHL, an emergency, whereas aminoglycosides, loop diuretics, and cisplatin cause sensorineural loss that may be permanent, and intratympanic gentamicin deliberately destroys vestibular function at the risk of hearing loss.

Drugs as causes of neurological disease

Many drugs provoke neurological disease, from seizures and parkinsonism to hearing loss, intracranial hemorrhage, and acute angle closure, and recognizing them often points to the diagnosis and the treatment. Here, we review the drugs named on these pages as causes and the conditions they produce.

Key points

- Opioids, TCAs, isoniazid, salicylate, cocaine, cefepime, amphetamine, metronidazole, penicillin, bupropion, and lithium provoke seizures, as does benzodiazepine or alcohol withdrawal.
- Drug-induced parkinsonism follows striatal D2 block or dopamine depletion by tetrabenazine and reserpine.
- Antipsychotics cause serious reactions in up to 50% of people with DLB, and dopamine agonists cause impulse control disorders.
- Interferon beta and natalizumab can worsen NMOSD, and natalizumab causes PML.

Drugs and the conditions they cause

Opioids, TCAs, isoniazid, salicylate, cocaine, cefepime, amphetamine, metronidazole, penicillin, bupropion, and lithium provoke seizures, as does withdrawal from benzodiazepines or alcohol.

Drug-induced parkinsonism follows striatal D2 block by first-generation antipsychotics, especially haloperidol and fluphenazine, by risperidone and olanzapine more than quetiapine and clozapine, and by metoclopramide and prochlorperazine, or dopamine depletion by tetrabenazine and reserpine. Antipsychotics cause serious reactions in up to 50% of people with DLB, and dopamine agonists cause impulse control disorders.

Aminoglycosides, loop diuretics, and cisplatin cause sensorineural hearing loss that may be permanent, whereas high-dose aspirin causes reversible tinnitus. Interferon beta and natalizumab can worsen NMOSD, and natalizumab can cause PML from JC virus. Warfarin, heparin, DOACs, tPA, and antiplatelets cause ICH from coagulopathy, and mydriatics, sympathomimetics, and topiramate can trigger acute angle closure.

Relationships to other conditions

Drug-induced parkinsonism *and* ***Parkinson disease***. Drug-induced parkinsonism gives symmetric, widespread signs after the drug is started, with a normal DAT SPECT, because the dopaminergic neurons lost in Parkinson disease are intact, and it is treated by stopping the causative drug.

Provoked seizures. Iatrogenic seizures are treated by stopping the drug, benzodiazepines for withdrawal, and restarting missed antiseizure medications.

Abbreviations

4F-PCC Four-factor prothrombin complex concentrate

5-HT2A Serotonin receptor 2A

ABC Amyloid, Braak, CERAD

ABX Antibiotics

ACA Anterior cerebral artery

ACE Angiotensin-converting enzyme

ACh Acetylcholine

AChE Acetylcholinesterase

AChR Acetylcholine receptor

AComm Anterior communicating artery

ACTH Adrenocorticotropic hormone

AD Alzheimer disease

ADC Apparent diffusion coefficient

ADEM Acute disseminated encephalomyelitis

ADP Adenosine diphosphate

AFO Ankle-foot orthosis

AICA Anterior inferior cerebellar artery

AIS Acute ischemic stroke

ALS Amyotrophic lateral sclerosis

AMD Age-related macular degeneration

ANA Antinuclear antibody

ANP Atrial natriuretic peptide

APH1 Anterior pharynx-defective 1

APOE Apolipoprotein E

APOE4 Apolipoprotein E ε4 allele

APP Amyloid precursor protein

APS Antiphospholipid syndrome

aPTT Activated partial thromboplastin time

AQP4 Aquaporin-4

AQP4-IgG Aquaporin-4 immunoglobulin G antibody

AREDS Age-Related Eye Disease Study

ARIA Amyloid-related imaging abnormalities; Amyloid-related imaging abnormalities: edema or microhemorrhage

ASD Atrial septal defect

ASPECTS Alberta Stroke Program Early CT Score

AT-III Antithrombin III

ATP Adenosine triphosphate

AV Atrioventricular; Arteriovenous

AVM Arteriovenous malformation

AZA Azathioprine

Aβ42 Amyloid beta 42, the 42-amino-acid form

B12 Vitamin B12 (cobalamin)

BBB Blood-brain barrier

BP Blood pressure

BPAP Bilevel positive airway pressure

BPPV Benign paroxysmal positional vertigo

bvFTD Behavioral variant frontotemporal dementia

C5 Complement component 5

C9orf72 Chromosome 9 open reading frame 72 gene

CAG Cytosine-adenine-guanine trinucleotide

CBD Corticobasal degeneration

CBF Cerebral blood flow

CBS Corticobasal syndrome

CBV Cerebral blood volume

CCBs Calcium channel blockers

CD19 A B-cell surface protein, also on plasmablasts

CD20 A B-cell surface protein

CD52 A surface protein on lymphocytes

CD59 A complement regulatory protein

CERAD Consortium to Establish a Registry for Alzheimer Disease

CGRP Calcitonin gene-related peptide

CIS Clinically isolated syndrome

CK Creatine kinase

CMV Cytomegalovirus

CN Cranial nerve

CN III Cranial nerve III (oculomotor)

CN VI Cranial nerve VI (abducens)

CN VII Cranial nerve VII (facial)

CN VIII Cranial nerve VIII (vestibulocochlear)

CN X Cranial nerve X (the vagus nerve)

CNS Central nervous system

COMT Catechol-O-methyltransferase

COPD Chronic obstructive pulmonary disease

COX Cyclooxygenase

COX-1 Cyclooxygenase 1

COX-2 Cyclooxygenase 2

CPAP Continuous positive airway pressure

CPP Cerebral perfusion pressure

CRAO Central retinal artery occlusion

CRP C-reactive protein

CRPS Complex regional pain syndrome

CRVO Central retinal vein occlusion

CSF Cerebrospinal fluid

CT Computed tomography

CVLM Caudal ventrolateral medulla

CXR Chest x-ray

CYP Cytochrome P450

CYP2C19 Cytochrome P450 2C19

CYP2C9 Cytochrome P450 2C9

CYP2D6 Cytochrome P450 2D6

CYP3A4 Cytochrome P450 3A4

D1 Dopamine receptor 1

D2 Dopamine receptor 2

D3 Dopamine receptor 3

DAT Dopamine transporter

DBS Deep brain stimulation

DDC DOPA decarboxylase

DFNB1 Deafness, autosomal recessive, locus 1

DHODH Dihydroorotate dehydrogenase

DJ-1 Parkinson disease protein 7 gene

DKA Diabetic ketoacidosis

DLB Dementia with Lewy bodies

DNA Deoxyribonucleic acid

DOAC Direct oral anticoagulant

DOACs Direct oral anticoagulants

DRG Dorsal root ganglion

DVT Deep vein thrombosis

DWI Diffusion-weighted imaging

DYT1 Dystonia 1 gene (TOR1A)

DYT6 Dystonia 6 gene (THAP1)

EAAT2 Excitatory amino acid transporter 2

EBV Epstein-Barr virus

ECG Electrocardiogram

EDN3 Endothelin 3 gene

EDNRB Endothelin receptor type B gene

EEG Electroencephalogram

ELISA Enzyme-linked immunosorbent assay

ERG Electroretinogram

ESR Erythrocyte sedimentation rate

EVD External ventricular drain

FBC Full blood count

FcRn Neonatal Fc receptor, which recycles IgG

FDG Fluorodeoxyglucose

FDG-PET Fluorodeoxyglucose positron emission tomography

FFP Fresh frozen plasma

FLAIR Fluid-attenuated inversion recovery

FP Prostaglandin F receptor

FTD Frontotemporal dementia

FTLD Frontotemporal lobar degeneration

FUS Fused in sarcoma protein

GABA Gamma-aminobutyric acid

GABA-A GABA type A receptor, a chloride channel

GABA-B GABA type B receptor

GBA1 Glucosylceramidase beta 1 gene

GCA Giant cell arteritis

GCIs Glial cytoplasmic inclusions

GCS Glasgow Coma Scale

GFAP Glial fibrillary acidic protein

GI Gastrointestinal

GJB2 Gap junction beta 2 gene (connexin 26)

GJB6 Gap junction beta 6 gene (connexin 30)

GPe Globus pallidus externus

GPi Globus pallidus internus

GRN Granulin gene

H1 Histamine receptor 1

H3 Histamine receptor 3

HbA1c Glycated hemoglobin

HBV Hepatitis B virus

HD Huntington disease

HFrEF Heart failure with reduced ejection fraction

HHV-6 Human herpesvirus 6

HINTS Head impulse, nystagmus, and test of skew

HIT Heparin-induced thrombocytopenia

HIV Human immunodeficiency virus

HLA Human leukocyte antigen

HLA-B*1502 Human leukocyte antigen B*15:02 allele; A human leukocyte antigen allele that predicts carbamazepine SJS

HLA-B27 Human leukocyte antigen B27

HMG-CoA 3-hydroxy-3-methylglutaryl coenzyme A

HR Heart rate

HTT Huntingtin gene

HU Hounsfield unit

IBD Inflammatory bowel disease

ICA Internal carotid artery

ICH Intracerebral hemorrhage

ICP Intracranial pressure

ICU Intensive care unit

IFIS Intraoperative floppy iris syndrome

IgE Immunoglobulin E

IgG1 Immunoglobulin G subclass 1

IIH Idiopathic intracranial hypertension

IL-1 Interleukin 1

IL-6 Interleukin 6

IM Intramuscular

IMPDH Inosine monophosphate dehydrogenase

INO Internuclear ophthalmoplegia

INR International normalized ratio

IOL Intraocular lens

IOP Intraocular pressure

ITP Immune thrombocytopenia

IV Intravenous

IVH Intraventricular hemorrhage

IVIG Intravenous immunoglobulin

JME Juvenile myoclonic epilepsy

LATE Limbic-predominant age-related TDP-43 encephalopathy

LDL Low-density lipoprotein

LETM Longitudinally extensive transverse myelitis

LFTs Liver function tests

LGN Lateral geniculate nucleus

LMN Lower motor neuron

LMWH Low-molecular-weight heparin

LOC Loss of consciousness

LRRK2 Leucine-rich repeat kinase 2 gene

LV Left ventricular

LVO Large vessel occlusion

M1 Muscarinic acetylcholine receptor 1; Primary motor cortex

M3 Muscarinic acetylcholine receptor 3

MAO-B Monoamine oxidase B

MAOI Monoamine oxidase inhibitor

MAOIs Monoamine oxidase inhibitors

MAP Mean arterial pressure

MAPT Microtubule-associated protein tau gene

MCA Middle cerebral artery

MCP Middle cerebellar peduncle

MDS International Parkinson and Movement Disorder Society

MGN Medial geniculate nucleus

MI Myocardial infarction

MIBG Metaiodobenzylguanidine

MIGS Minimally invasive glaucoma surgery

MITF Melanocyte-inducing transcription factor gene

MLF Medial longitudinal fasciculus

MMF Mycophenolate mofetil

MMSE Mini-Mental State Examination

MND Motor neuron disease

MoCA Montreal Cognitive Assessment

MOG Myelin oligodendrocyte glycoprotein

MOG-IgG Myelin oligodendrocyte glycoprotein immunoglobulin G antibody

MOGAD MOG antibody-associated disease

MPTP 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine

MR Magnetic resonance

MRA Magnetic resonance angiography

MRI Magnetic resonance imaging

MS Multiple sclerosis

MSA Multiple system atrophy

MSA-C Multiple system atrophy, cerebellar type

MSA-P Multiple system atrophy, parkinsonian type

MT1 Melatonin receptor 1

MT2 Melatonin receptor 2

MTT Mean transit time

MuSK Muscle-specific kinase

NCC Sodium-chloride cotransporter

NF2 Neurofibromatosis type 2

NFTs Neurofibrillary tangles

nfvPPA Nonfluent variant primary progressive aphasia

NGF Nerve growth factor

NIHSS National Institutes of Health Stroke Scale

NMDA N-methyl-D-aspartate; N-methyl-D-aspartate glutamate receptor

NMO Neuromyelitis optica

NMOSD Neuromyelitis optica spectrum disorder

NMS Neuroleptic malignant syndrome

NPH Normal pressure hydrocephalus

Nrf2 Nuclear factor erythroid 2-related factor 2

NSAIDs Nonsteroidal anti-inflammatory drugs

NSF Nephrogenic systemic fibrosis

NTS Nucleus tractus solitarius

OCB Oligoclonal bands

OCD Obsessive-compulsive disorder

OCP Oral contraceptive pill

OCT Optical coherence tomography

P2Y12 The platelet ADP receptor

PaCO2 Arterial partial pressure of carbon dioxide

PAG Periaqueductal gray

PaO2 Arterial partial pressure of oxygen

PAX3 Paired box 3 gene

PCA Posterior cerebral artery

PCC Prothrombin complex concentrate

PComm Posterior communicating artery

PD Parkinson disease

PDD Parkinson disease dementia

PDE-5 Phosphodiesterase 5

PE Pulmonary embolism

PEG Polyethylene glycol

PEN2 Presenilin enhancer 2

PET Positron emission tomography

PFO Patent foramen ovale

PICA Posterior inferior cerebellar artery

PINK1 PTEN-induced kinase 1 gene

PKD Polycystic kidney disease

PLEX Plasma exchange

PML Progressive multifocal leukoencephalopathy

PNES Psychogenic nonepileptic seizures

PNS Peripheral nervous system; Parasympathetic nervous system

POLG DNA polymerase gamma, whose mutations cause mitochondrial disease

POTS Postural orthostatic tachycardia syndrome

PPA Primary progressive aphasia

PR The interval from the start of the P wave to the start of the QRS complex

PRES Posterior reversible encephalopathy syndrome

PRKN Parkin gene

PSEN1 Presenilin 1

PSEN2 Presenilin 2

PSP Progressive supranuclear palsy

PSP-bvFTD Progressive supranuclear palsy, behavioral variant frontotemporal dementia

PSP-CBS Progressive supranuclear palsy, corticobasal syndrome

PSP-P Progressive supranuclear palsy, parkinsonism predominant

PSP-PAGF Progressive supranuclear palsy, pure akinesia with gait freezing

PSP-PNFA Progressive supranuclear palsy, progressive nonfluent aphasia

PSP-RS Progressive supranuclear palsy, Richardson syndrome

PT Prothrombin time

QRS The ventricular depolarization complex on an ECG

QT The interval from the start of the QRS complex to the end of the T wave

RAAS Renin-angiotensin-aldosterone system

RAPD Relative afferent pupillary defect

RBD REM sleep behavior disorder

rCBF Relative cerebral blood flow

RCC Renal cell carcinoma

RCVS Reversible cerebral vasoconstriction syndrome

REM Rapid eye movement

RHD Rheumatic heart disease

RNFL Retinal nerve fiber layer

RPE Retinal pigment epithelium

RTX Rituximab

RVLM Rostral ventrolateral medulla

RVM Rostral ventromedial medulla

S1 Primary somatosensory cortex; First sacral spinal segment

S1P Sphingosine 1-phosphate; Sphingosine-1-phosphate

S1PR1 Sphingosine 1-phosphate receptor 1; Sphingosine-1-phosphate receptor 1

SA Sinoatrial

SAH Subarachnoid hemorrhage

SARS-CoV-2 Severe acute respiratory syndrome coronavirus 2

SC Subcutaneous

SCI Spinal cord injury

SIADH Syndrome of inappropriate antidiuretic hormone secretion; Syndrome of inappropriate antidiuretic hormone

SJS Stevens-Johnson syndrome

SLE Systemic lupus erythematosus

SMA Supplementary motor area; Superior mesenteric artery; Spinal muscular atrophy

SNARE Soluble NSF attachment protein receptor; Proteins that fuse vesicles with the cell membrane

SNc Substantia nigra pars compacta

SNCA Synuclein alpha gene

SNHL Sensorineural hearing loss

SNr Substantia nigra pars reticulata

SNRIs Serotonin-norepinephrine reuptake inhibitors

SNS Sympathetic nervous system

SPECT Single-photon emission computed tomography

SSRIs Selective serotonin reuptake inhibitors

STN Subthalamic nucleus

SV2A Synaptic vesicle protein 2A; Synaptic vesicle glycoprotein 2A

svPPA Semantic variant primary progressive aphasia

SWI Susceptibility-weighted imaging

TB Tuberculosis

TBI Traumatic brain injury

TCAs Tricyclic antidepressants

TDP-43 TAR DNA-binding protein 43

TEN Toxic epidermal necrolysis

TFTs Thyroid function tests

TIA Transient ischemic attack

TMP-SMX Trimethoprim-sulfamethoxazole

TNF Tumor necrosis factor

tPA Tissue plasminogen activator

TPMT Thiopurine methyltransferase

TRP Transient receptor potential

TTP Thrombotic thrombocytopenic purpura

TXA Tranexamic acid

TXA2 Thromboxane A2

UFH Unfractionated heparin

UMN Upper motor neuron

V1 Primary visual cortex; Ophthalmic division of the trigeminal nerve

VA Ventral anterior nucleus of the thalamus

VEGF Vascular endothelial growth factor

VKORC1 Vitamin K epoxide reductase, the target of warfarin

VL Ventral lateral nucleus of the thalamus

VMAT2 Vesicular monoamine transporter 2

VOR Vestibulo-ocular reflex

VPL Ventral posterolateral nucleus of the thalamus

VPM Ventral posteromedial nucleus of the thalamus

VZV Varicella-zoster virus

References

Neurons and action potentials: structure, channels, and conduction

1. Ludwig PE, Reddy V, Varacallo MA. Neuroanatomy, Neurons. 2023. In: *StatPearls*. StatPearls Publishing. NBK441977
2. Chrysafides SM, Bordes SJ, Sharma S. Physiology, Resting Potential. 2023. In: *StatPearls*. StatPearls Publishing. NBK538338
3. Grider MH, Jessu R, Kabir R. Physiology, Action Potential. 2023. In: *StatPearls*. StatPearls Publishing. NBK538143
4. Jozsa F, Lui F. Neuroanatomy, Neuron Action Potential. 2026. In: *StatPearls*. StatPearls Publishing. NBK546639
5. Hernandez CM, Richards JR. Physiology, Sodium Channels. 2023. In: *StatPearls*. StatPearls Publishing. NBK545257
6. Ludwig PE, Das JM. Histology, Glial Cells. 2023. In: *StatPearls*. StatPearls Publishing. NBK441945
7. Gawdi R, Shumway KR, Emmady PD. Physiology, Blood Brain Barrier. 2023. In: *StatPearls*. StatPearls Publishing. NBK557721
8. Hall WA, Munakomi S, Nadi M. Axonotmesis. 2026. In: *StatPearls*. StatPearls Publishing. NBK562304
9. Nadi M, Dabbas W, Das JM. Peripheral Nerve Injury. 2026. In: *StatPearls*. StatPearls Publishing. NBK549848

Synapses and neurotransmitters: release, receptors, and clearance

1. Caire MJ, Reddy V, Varacallo MA. Physiology, Synapse. 2023. In: *StatPearls*. StatPearls Publishing. NBK526047
2. Sheffler ZM, Reddy V, Pillarisetty LS. Physiology, Neurotransmitters. 2023. In: *StatPearls*. StatPearls Publishing. NBK539894
3. Sam C, Bordoni B. Physiology, Acetylcholine. 2023. In: *StatPearls*. StatPearls Publishing. NBK557825
4. Jewett BE, Sharma S. Physiology, GABA. 2023. In: *StatPearls*. StatPearls Publishing. NBK513311
5. Khalil B, Marwaha K, Bollu PC. Physiology, Neuromuscular Junction. 2025. In: *StatPearls*. StatPearls Publishing. NBK470413
6. Pana A, Saggu BM. Dystonia. 2023. In: *StatPearls*. StatPearls Publishing. NBK448144
7. Beloor Suresh A, Asuncion RMD. Myasthenia Gravis. 2023. In: *StatPearls*. StatPearls Publishing. NBK559331
8. Jayarangaiah A, Lui F, Theetha Kariyanna P. Lambert-Eaton Myasthenic Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK507891
9. Howes OD, Kapur S. The dopamine hypothesis of schizophrenia: version III—the final common pathway. *Schizophr Bull*. 2009;35(3):549–562. PMC2669582

Ascending tracts: dorsal columns, spinothalamic tracts, and spinocerebellar tracts

1. Al-Chalabi M, Reddy V, Alsalman I. Neuroanatomy, Posterior Column (Dorsal Column). 2023. In: *StatPearls*. StatPearls Publishing. NBK507888
2. Navarro-Orozco D, Miller R, Bollu PC. Neuroanatomy, Medial Lemniscus (Reils Band, Reils Ribbon). 2025. In: *StatPearls*. StatPearls Publishing. NBK526040
3. Al-Chalabi M, Reddy V, Gupta S. Neuroanatomy, Spinothalamic Tract. 2023. In: *StatPearls*. StatPearls Publishing. NBK507824
4. Koh M, Markovich B. Neuroanatomy, Spinocerebellar Dorsal Tract. 2023. In: *StatPearls*. StatPearls Publishing. NBK556013
5. Whitman PA, Launico MV, Adigun OO. Anatomy, Skin, Dermatomes. 2023. In: *StatPearls*. StatPearls Publishing. NBK535401

Descending tracts: corticospinal, corticobulbar, and brainstem motor tracts

1. AbuHasan Q, Munakomi S. Neuroanatomy, Pyramidal Tract. 2023. In: *StatPearls*. StatPearls Publishing. NBK545314
2. Natali AL, Reddy V, Bordoni B. Neuroanatomy, Corticospinal Cord Tract. 2023. In: *StatPearls*. StatPearls Publishing. NBK535423
3. Bhardwaj N, Yadala S. Neuroanatomy, Corticobulbar Tract. 2023. In: *StatPearls*. StatPearls Publishing. NBK555891
4. Lee J, Muzio MR. Neuroanatomy, Extrapyramidal System. 2022. In: *StatPearls*. StatPearls Publishing. NBK554542

Basal ganglia pathways: direct, indirect, and the role of dopamine

1. Young CB, Reddy V, Sonne J. Neuroanatomy, Basal Ganglia. 2023. In: *StatPearls*. StatPearls Publishing. NBK537141
2. Lee J, Muzio MR. Neuroanatomy, Extrapyramidal System. 2022. In: *StatPearls*. StatPearls Publishing. NBK554542
3. Rocha Cabrero F, De Jesus O. Hemiballismus. 2023. In: *StatPearls*. StatPearls Publishing. NBK559127

Pain pathways and descending modulation: from nociceptor to cortex and back

1. Chen J, Kandle PF, Murray IV, Fitzgerald LA, Sehdev JS. Physiology, Pain. 2023. In: *StatPearls*. StatPearls Publishing. NBK539789
2. Kendroud S, Fitzgerald LA, Murray IV, Hanna A. Physiology, Nociceptive Pathways. 2022. In: *StatPearls*. StatPearls Publishing. NBK470255
3. Trachsel LA, Munakomi S, Cascella M. Pain Theory. 2023. In: *StatPearls*. StatPearls Publishing. NBK545194
4. Mokhtar M, Singh P. Neuroanatomy, Periaqueductal Gray. 2023. In: *StatPearls*. StatPearls Publishing. NBK554391
5. Dhaliwal A, Gupta M. Physiology, Opioid Receptor. 2023. In: *StatPearls*. StatPearls Publishing. NBK546642
6. Regina AC, Goyal A, Mechanic OJ. Opioid Toxicity. 2025. In: *StatPearls*. StatPearls Publishing. NBK470415
7. Biswas T, Patel P, Molla M. Duloxetine. 2026. In: *StatPearls*. StatPearls Publishing. NBK549806
8. Moraczewski J, Awosika AO, Aedma KK. Tricyclic Antidepressants. 2023. In: *StatPearls*. StatPearls Publishing. NBK557791
9. Yasaei R, Katta S, Patel P, Saadabadi A. Gabapentin. 2024. In: *StatPearls*. StatPearls Publishing. NBK493228
10. Orhurhu VJ, Roberts JS, Ly NK, Cohen SP. Ketamine in Acute and Chronic Pain Management. 2023. In: *StatPearls*. StatPearls Publishing. NBK539824

Olfactory, optic, and vestibulocochlear nerves: the special senses

1. Sonne J, Omole AE, Lopez-Ojeda W. Neuroanatomy, Cranial Nerve. 2025. In: *StatPearls*. StatPearls Publishing. NBK470353
2. Helwany M, Bordoni B. Neuroanatomy, Cranial Nerve 1 (Olfactory). 2023. In: *StatPearls*. StatPearls Publishing. NBK556051
3. Smith AM, Czyz CN. Neuroanatomy, Cranial Nerve 2 (Optic). 2022. In: *StatPearls*. StatPearls Publishing. NBK507907
4. Ireland AC, Carter IB. Neuroanatomy, Optic Chiasm. 2023. In: *StatPearls*. StatPearls Publishing. NBK542287
5. Bordoni B, Omole AE, Daly DT. Neuroanatomy, Cranial Nerve 8 (Vestibulocochlear). 2026. In: *StatPearls*. StatPearls Publishing. NBK537359
6. Dichter SL, Khan Suheb MZ, Shubert GS. Argyll Robertson Pupil. 2024. In: *StatPearls*. StatPearls Publishing. NBK537179

Oculomotor, trochlear, and abducens nerves: moving the eyes

1. Sonne J, Omole AE, Lopez-Ojeda W. Neuroanatomy, Cranial Nerve. 2025. In: *StatPearls*. StatPearls Publishing. NBK470353
2. Joyce C, Le PH, Peterson DC. Neuroanatomy, Cranial Nerve 3 (Oculomotor). 2023. In: *StatPearls*. StatPearls Publishing. NBK537126
3. Kim SY, Motlagh M, Naqvi IA. Neuroanatomy, Cranial Nerve 4 (Trochlear). 2023. In: *StatPearls*. StatPearls Publishing. NBK537244
4. Nguyen V, Reddy V, Varacallo MA. Neuroanatomy, Cranial Nerve 6 (Abducens). 2022. In: *StatPearls*. StatPearls Publishing. NBK430711
5. Khan Z, Bollu PC. Horner Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK500000

Trigeminal and facial nerves: face sensation, mastication, and expression

1. Sonne J, Omole AE, Lopez-Ojeda W. Neuroanatomy, Cranial Nerve. 2025. In: *StatPearls*. StatPearls Publishing. NBK470353
2. Huff T, Weisbrod LJ, Daly DT. Neuroanatomy, Cranial Nerve 5 (Trigeminal). 2024. In: *StatPearls*. StatPearls Publishing. NBK482283
3. Hohman MH, Warner MJ, Varacallo MA. Bell Palsy. 2024. In: *StatPearls*. StatPearls Publishing. NBK482290
4. Shankar Kikkeri N, Nagalli S. Trigeminal Neuralgia. 2024. In: *StatPearls*. StatPearls Publishing. NBK554486
5. Hohman MH, Mazzoni T. Facial Nerve Palsy. 2026. In: *StatPearls*. StatPearls Publishing. NBK549815

Glossopharyngeal, vagus, accessory, and hypoglossal nerves: the lower cranial nerves

1. Sonne J, Omole AE, Lopez-Ojeda W. Neuroanatomy, Cranial Nerve. 2025. In: *StatPearls*. StatPearls Publishing. NBK470353
2. Thomas K, Minutello K, Das JM. Neuroanatomy, Cranial Nerve 9 (Glossopharyngeal). 2022. In: *StatPearls*. StatPearls Publishing. NBK539877
3. Kenny BJ, Bordoni B. Neuroanatomy, Cranial Nerve 10 (Vagus Nerve). 2022. In: *StatPearls*. StatPearls Publishing. NBK537171
4. Bordoni B, Varacallo MA. Neuroanatomy, Cranial Nerve 11 (Accessory). 2025. In: *StatPearls*. StatPearls Publishing. NBK507722
5. Kim SY, Naqvi IA. Neuroanatomy, Cranial Nerve 12 (Hypoglossal). 2022. In: *StatPearls*. StatPearls Publishing. NBK532869
6. Reese V, Das JM, Al Khalili Y. Cranial Nerve Testing. 2023. In: *StatPearls*. StatPearls Publishing. NBK585066

The respiratory center: rhythm, chemical control, and failures of drive

1. Brinkman JE, Toro F, Sharma S. Physiology, Respiratory Drive. 2023. In: *StatPearls*. StatPearls Publishing. NBK482414
2. Benner A, Lewallen NF, Sharma S. Physiology, Carbon Dioxide Response Curve. 2023. In: *StatPearls*. StatPearls Publishing. NBK538146
3. Rahman M, Tadi P. Neuroanatomy, Pons. 2023. In: *StatPearls*. StatPearls Publishing. NBK560589
4. Iordanova R, Reddivari AKR. Neuroanatomy, Medulla Oblongata. 2023. In: *StatPearls*. StatPearls Publishing. NBK551589
5. Chapa-Rodriguez A, Para K, Sankari A, Bollu PC. Cheyne-Stokes Respirations. 2026. In: *StatPearls*. StatPearls Publishing. NBK448165
6. Rana AM, Sankari A. Central Sleep Apnea. 2023. In: *StatPearls*. StatPearls Publishing. NBK578199
7. Sankari A, Antoine MH, Bollu PC, Kaul P. Obesity-Hypoventilation Syndrome. 2025. In: *StatPearls*. StatPearls Publishing. NBK542216
8. Regina AC, Goyal A, Mechanic OJ. Opioid Toxicity. 2025. In: *StatPearls*. StatPearls Publishing. NBK470415
9. Dinallo S, Waseem M. Cushing Reflex. 2023. In: *StatPearls*. StatPearls Publishing. NBK549801

The cardiovascular center: the baroreflex and its failures

1. Denault D, Khalil B. Physiology, Baroreceptors. 2026. In: *StatPearls*. StatPearls Publishing. NBK538172
2. Iordanova R, Reddivari AKR. Neuroanatomy, Medulla Oblongata. 2023. In: *StatPearls*. StatPearls Publishing. NBK551589
3. Mirza M, Das JM. Neuroanatomy, Area Postrema. 2023. In: *StatPearls*. StatPearls Publishing. NBK544249
4. LeBouef T, Yaker Z, Whited L. Physiology, Autonomic Nervous System. 2023. In: *StatPearls*. StatPearls Publishing. NBK538516
5. Grossman SA, Badireddy M. Syncope. 2025. In: *StatPearls*. StatPearls Publishing. NBK442006
6. Kharsa A, Ahmed I, Wadhwa R. Carotid Sinus Hypersensitivity. 2025. In: *StatPearls*. StatPearls Publishing. NBK559059
7. Sánchez-Manso JC, Gujarathi R, Varacallo MA. Autonomic Dysfunction. 2023. In: *StatPearls*. StatPearls Publishing. NBK430888
8. Bilgin Badur N, Winkle MJ, Leslie SW. Autonomic Dysreflexia. 2025. In: *StatPearls*. StatPearls Publishing. NBK482434
9. Dinallo S, Waseem M. Cushing Reflex. 2023. In: *StatPearls*. StatPearls Publishing. NBK549801

Stroke syndromes: localizing the lesion from the cortex to the spinal cord

1. Matos Casano HA, Benjamin R, Ciofoaia GA. Anterior Cerebral Artery Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK537333
2. Benjamin R, Galuska MA. Middle Cerebral Artery Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK556132
3. Benjamin R, Tadi P, Singh J. Posterior Cerebral Artery Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK532296
4. Dogariu OA, Dogariu I, Vasile CM, et al. Diagnosis and treatment of watershed strokes: a narrative review. *J Med Life*. 2023;16(6):842–850. PMC10478671
5. Altabakhi I, Sun CE. Gerstmann Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK519528
6. Parvathaneni A, Das JM. Balint Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK544347
7. D Barbosa AC, Asuncion RMD, Emmady PD. Alexia. 2024. In: *StatPearls*. StatPearls Publishing. NBK557669
8. Rocha Cabrero F, De Jesus O. Prosopagnosia. 2023. In: *StatPearls*. StatPearls Publishing. NBK559324
9. Gowda SN, Munakomi S. Brainstem Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK560896
10. Munakomi S, Das JM. Weber Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK559158
11. Burr P, Mandalaneni K. Benedikt Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK560509
12. Sakuru R, Elnahry AG, Lui F, Bollu PC. Millard-Gubler Syndrome. 2024. In: *StatPearls*. StatPearls Publishing. NBK532907
13. Khazaal O, Marquez DL, Naqvi IA. Foville Syndrome. 2024. In: *StatPearls*. StatPearls Publishing. NBK544268
14. Das JM, Anosike K, Asuncion RMD. Locked-in Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK559026
15. Easton-Carr R, Lui F, Das JM. Lateral Medullary Syndrome (Wallenberg Syndrome). 2026. In: *StatPearls*. StatPearls Publishing. NBK551670
16. Siddik AB, Singh J, Gupta V. Medial Medullary Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK560590
17. Margetis K, Munakomi S, Gillis CC. Central Cord Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK441932
18. Khan MM, Munakomi S, Dalal K. Anterior Cord Syndrome (Ventral Cord Syndrome). 2025. In: *StatPearls*. StatPearls Publishing. NBK541011

19. Shams S, Farrell CL, Arain A. Brown-Séquard Syndrome. 2024. In: *StatPearls*. StatPearls Publishing. NBK538135
20. Anosike KC, Rajaram Manoharan SVR. Central Post-Stroke Pain Syndrome. 2024. In: *StatPearls*. StatPearls Publishing. NBK604196
21. Singh J, Lui F. Acute Stroke: Evaluation and Management. 2026. In: *StatPearls*. StatPearls Publishing. NBK535369
22. Ziu E, Khan Suheb MZ, Mesfin FB. Subarachnoid Hemorrhage. 2023. In: *StatPearls*. StatPearls Publishing. NBK441958
23. Benjamin R, Karsonovich T. Intracerebral Hemorrhagic Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK553103
24. Le H, Lui F, Lui MY. Aphasia. 2024. In: *StatPearls*. StatPearls Publishing. NBK559315
25. Lui F, Wroten M. Wernicke Aphasia. 2025. In: *StatPearls*. StatPearls Publishing. NBK441951
26. Acharya AB, Lui F, Maani CV. Conduction Aphasia. 2024. In: *StatPearls*. StatPearls Publishing. NBK537006
27. Gore M, Bansal K, Lui F, Asuncion RMD. Lacunar Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK563216
28. Kaur K, Najem K, Margolin E. Amaurosis Fugax. 2025. In: *StatPearls*. StatPearls Publishing. NBK470528

Acute ischemic stroke: embolic, thrombotic, and low-flow injury and the race to reperfuse

1. Singh J, Lui F. Acute Stroke: Evaluation and Management. 2026. In: *StatPearls*. StatPearls Publishing. NBK535369
2. Lui F, Khan Suheb MZ, Patti L. Ischemic Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK499997
3. Baig MU, Patel P. Thrombolytic Therapy. 2026. In: *StatPearls*. StatPearls Publishing. NBK557411
4. Gore M, Bansal K, Lui F, Asuncion RMD. Lacunar Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK563216
5. Kimyaghalam A, Lui F. Carotid Artery Stenosis. 2026. In: *StatPearls*. StatPearls Publishing. NBK442025
6. DaCosta M, Tadi P, Surowiec SM. Carotid Endarterectomy. 2023. In: *StatPearls*. StatPearls Publishing. NBK470582
7. Goodfriend SD, Tadi P, Koury R. Carotid Artery Dissection. 2022. In: *StatPearls*. StatPearls Publishing. NBK430835
8. Hampton T, Alsaleem M, Murphy-Lavoie HM. Patent Foramen Ovale. 2022. In: *StatPearls*. StatPearls Publishing. NBK493151
9. Randhawa S, Mehta JL, Dhar G. Percutaneous patent foramen ovale closure: stroke and beyond. *Curr Cardiol Rev*. 2024;20(3):77–86. PMC11284695
10. Park J, Hwang JH. Where are we now with decompressive hemicraniectomy for malignant middle cerebral artery infarction? *J Cerebrovasc Endovasc Neurosurg*. 2013;15(2):61–66. PMC3704996
11. Hohman MH, Warner MJ, Varacallo MA. Bell Palsy. 2024. In: *StatPearls*. StatPearls Publishing. NBK482290
12. Peeling JL, Muzio MR. Functional Neurologic Disorder. 2023. In: *StatPearls*. StatPearls Publishing. NBK551567
13. Shankar Kikkeri N, Nagalli S. Migraine With Aura. 2024. In: *StatPearls*. StatPearls Publishing. NBK554611
14. Xu SY, Li ZX, Wu XW, et al. Frequency and pathophysiology of post-seizure Todd's paralysis. *Med Sci Monit*. 2020;26:e920751. PMC7075081
15. Gallù M, Marrone G, Legramante JM, De Lorenzo A, et al. Female sex as a thromboembolic risk factor in the era of nonvitamin K antagonist oral anticoagulants. *Cardiovasc Ther*. 2020;2020:1743927. PMC7336195
16. Vinson DR, Warton EM, Mark DG, Ballard DW, et al. Thromboprophylaxis for patients with high-risk atrial fibrillation and flutter discharged from the emergency department. *West J Emerg Med*. 2018;19(2):346–360. PMC5851510

Intracerebral hemorrhage: causes by location, and limiting hematoma growth

1. Hosking A, Samarasekera N, Moullaali TJ, et al. Predisposing factors, pathologies, and precipitating factors causing intracerebral hemorrhage. *Stroke*. 2026;57(4):1086–1096. PMC13003937
2. Veltkamp R, Purrucker J. Management of spontaneous intracerebral hemorrhage. *Curr Neurol Neurosci Rep*. 2017;17(10):80. PMC5590024
3. Brouwers HB, Greenberg SM. Hematoma expansion following acute intracerebral hemorrhage. *Cerebrovasc Dis*. 2013;35(3):195–201. PMC3743539
4. Witsch J, Siegerink B, Nolte CH, et al. Prognostication after intracerebral hemorrhage: a review. *Neurol Res Pract*. 2021;3(1):22. PMC8091769
5. Seiffge D, Polymeris A, Pfeilschifter W, et al. Reversal of anticoagulation in patients with intracerebral haemorrhage related to oral anticoagulants: state of the evidence. *Eur Stroke J*. 2025;10(1 Suppl):14–23. PMC12098327
6. Sweidan AJ, Singh NK, Conovaloff JL, et al. Coagulopathy reversal in intracerebral haemorrhage. *Stroke Vasc Neurol*. 2020;5(1):29–33. PMC7213499
7. Goldstein JN, Marrero M, Masrur S, et al. Management of thrombolysis-associated symptomatic intracerebral hemorrhage. *Arch Neurol*. 2010;67(8):965–969. PMC3690951
8. Singh J, Lui F. Acute Stroke: Evaluation and Management. 2026. In: *StatPearls*. StatPearls Publishing. NBK535369
9. American Heart Association. Reversal of thrombolysis-associated intracranial hemorrhage. Blog post, 2017. ahajournals.org
10. Kozberg MG, Perosa V, Gurol ME, van Veluw SJ. A practical approach to the management of cerebral amyloid angiopathy. *Int J Stroke*. 2021;16(4):356–369. PMC9097498
11. Łupicka A, Kowalczyk W, Cyman B, et al. Should we be afraid of radiotherapy for hemorrhagic brain metastases? A narrative review. *Ther Adv Med Oncol*. 2024;16:17588359241289203. PMC11481081
12. Yogendrakumar V, Lun R, Hutton B, et al. Comparing pharmacological venous thromboembolism prophylaxis to intermittent pneumatic compression in acute intracerebral haemorrhage: protocol for a systematic review and network meta-analysis. *BMJ Open*. 2018;8(11):e024405. PMC6231584
13. Benjamin R, Karsonovich T. Intracerebral Hemorrhagic Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK553103
14. Singh J, Hegazy MI. Cerebral Amyloid Angiopathy. 2025. In: *StatPearls*. StatPearls Publishing. NBK556105
15. Singh J, Munakomi S, Baruffi S. Cerebral Venous Sinus Thrombosis. 2025. In: *StatPearls*. StatPearls Publishing. NBK459315
16. Fischer MA, Rjepaj C, Steckbeck R, Margetis K. Cerebellar Hemorrhage. 2026. In: *StatPearls*. StatPearls Publishing. NBK541076
17. Baskaran J, Lopez RA, Cassagnol M. Prothrombin Complex Concentrate. 2024. In: *StatPearls*. StatPearls Publishing. NBK539716
18. Pinto VL, Adeyinka A. Increased Intracranial Pressure. 2025. In: *StatPearls*. StatPearls Publishing. NBK482119
19. Dinallo S, Waseem M. Cushing Reflex. 2023. In: *StatPearls*. StatPearls Publishing. NBK549801
20. Hall WA, Majeed H, Ahmad F. Mycotic Aneurysm. 2025. In: *StatPearls*. StatPearls Publishing. NBK560736
21. Karsonovich T, Farrell CL, Hall WA. Intracranial Arteriovenous Malformations. 2026. In: *StatPearls*. StatPearls Publishing. NBK531479
22. Rodríguez-Fernández S, Castillo-Lorente E, Guerrero-Lopez F, Rodríguez-Rubio D, et al. Validation of the ICH score in patients with spontaneous intracerebral haemorrhage admitted to the intensive care unit in Southern Spain. *BMJ Open*. 2018;8(8):e021719. PMC6091906

Subarachnoid hemorrhage: aneurysmal rupture and the

threats that follow

1. Claassen J, Park S. Spontaneous subarachnoid haemorrhage. *Lancet*. 2022;400(10355):846–862. PMC9987649
2. Marcolini E, Hine J. Approach to the diagnosis and management of subarachnoid hemorrhage. *West J Emerg Med*. 2019;20(2):203–211. PMC6404699
3. Patel S, Parikh A, Okorie ON. Subarachnoid hemorrhage in the emergency department. *Int J Emerg Med*. 2021;14(1):31. PMC8117305
4. Perry JJ, Stiell IG, Sivilotti ML, et al. Sensitivity of computed tomography performed within six hours of onset of headache for diagnosis of subarachnoid haemorrhage: prospective cohort study. *BMJ*. 2011;343:d4277. PMC3138338
5. Cossu G, Messerer M, Oddo M, Daniel RT. To look beyond vasospasm in aneurysmal subarachnoid haemorrhage. *Biomed Res Int*. 2014;2014:628597. PMC4055362
6. Vlak MH, Rinkel GJ, Greebe P, et al. Trigger factors for rupture of intracranial aneurysms in relation to patient and aneurysm characteristics. *J Neurol*. 2012;259(7):1298–1302. PMC3390687
7. Naamansen AB, Larsen CC, Johannsson B, et al. Small ruptured intracranial aneurysms are overrepresented at the anterior and posterior communicating artery: results of a multiple regression analysis. *Surg Neurol Int*. 2022;13:288. PMC9282757
8. Rautalin IM, Asikainen A, Korja M. Modifiable risk factors for subarachnoid hemorrhage: narrative review with an emphasis on common controversies and epidemiologic pitfalls. *Neurology*. 2024;103(11):e210052. PMC11627175
9. Ziu E, Khan Suheb MZ, Mesfin FB. Subarachnoid Hemorrhage. 2023. In: *StatPearls*. StatPearls Publishing. NBK441958
10. Okal F, Foster DM. Cerebral Aneurysm. 2026. In: *StatPearls*. StatPearls Publishing. NBK507902
11. Dugas C, Jamal Z, Bollu PC. Xanthochromia. 2024. In: *StatPearls*. StatPearls Publishing. NBK526048
12. Das JM, Zito PM. Nimodipine. 2024. In: *StatPearls*. StatPearls Publishing. NBK534870
13. Pinto VL, Adeyinka A. Increased Intracranial Pressure. 2025. In: *StatPearls*. StatPearls Publishing. NBK482119
14. Dinallo S, Waseem M. Cushing Reflex. 2023. In: *StatPearls*. StatPearls Publishing. NBK549801
15. Karsonovich T, Farrell CL, Hall WA. Intracranial Arteriovenous Malformations. 2026. In: *StatPearls*. StatPearls Publishing. NBK531479
16. Ravi U, Daley SF. Ehlers-Danlos Syndromes. 2026. In: *StatPearls*. StatPearls Publishing. NBK549814

Seizures: lowered thresholds, the patterns they produce, and stepwise control

1. Beniczky S, Trinka E, Wirrell E, et al. Updated classification of epileptic seizures: position paper of the International League Against Epilepsy. *Epilepsia*. 2025;66(6):1804–1823. PMC12169392
2. Glauser T, Shinnar S, Gloss D, et al. Evidence-based guideline: treatment of convulsive status epilepticus in children and adults: report of the Guideline Committee of the American Epilepsy Society. *Epilepsy Curr*. 2016;16(1):48–61. PMC4749120
3. Kapur J, Elm J, Chamberlain JM, et al. Randomized trial of three anticonvulsant medications for status epilepticus. *N Engl J Med*. 2019;381(22):2103–2113. PMC7098487
4. Heuser K, Olsen KB, Ulvin LB, Gjerstad L, Taubøll E. Modern treatment of status epilepticus in adults. In: Czuczwar SJ, ed. *Epilepsy*. Exon Publications; 2022. NBK580620
5. Manford M. Recent advances in epilepsy. *J Neurol*. 2017;264(8):1811–1824. PMC5533817
6. Xu SY, Li ZX, Wu XW, et al. Frequency and pathophysiology of post-seizure Todd's paralysis. *Med Sci Monit*. 2020;26:e920751. PMC7075081
7. Englot DJ, Blumenfeld H. Consciousness and epilepsy: why are complex-partial seizures complex? *Prog Brain Res*. 2009;177:147–170. PMC2901990
8. Thundiyil JG, Kearney TE, Olson KR. Evolving epidemiology of drug-induced seizures reported to a Poison Control Center System. *J Med Toxicol*. 2007;3(1):15–19. PMC3550124
9. Keyser A, De Bruijn SF. Epileptic manifestations and vitamin B1 deficiency. *Eur Neurol*. 1991;31(3):121–125. PMID 2044623
10. Go CY, Mackay MT, Weiss SK, et al. Evidence-based guideline update: medical treatment of infantile spasms. *Neurology*. 2012;78(24):1974–1980. PMC3369510
11. Lovik K, Murr NI. Seizure. 2026. In: *StatPearls*. StatPearls Publishing. NBK430765
12. Ighodaro ET, Maini K, Arya K, Sharma S. Focal Onset Seizure. 2023. In: *StatPearls*. StatPearls Publishing. NBK500005
13. Kumar A, Ighodaro ET, Sharma S. Focal Impaired Awareness Seizure. 2024. In: *StatPearls*. StatPearls Publishing. NBK519030
14. Kodankandath TV, Theodore D, Samanta D. Generalized Tonic-Clonic Seizure. 2023. In: *StatPearls*. StatPearls Publishing. NBK554496
15. Albuja AC, Ighodaro ET, Khan GQ. Absence Seizure. 2024. In: *StatPearls*. StatPearls Publishing. NBK499867
16. Wylie T, Sandhu DS, Murr NI. Status Epilepticus. 2023. In: *StatPearls*. StatPearls Publishing. NBK430686
17. Subbarao BS, Silverman A, Eapen BC. Seizure Medications. 2023. In: *StatPearls*. StatPearls Publishing. NBK482269
18. Lui F, Murr NI. Psychogenic Nonepileptic Seizures. 2024. In: *StatPearls*. StatPearls Publishing. NBK441871
19. Grossman SA, Badireddy M. Syncope. 2025. In: *StatPearls*. StatPearls Publishing. NBK442006
20. Amrutkar CV, Benjamin R. Juvenile Myoclonic Epilepsy. 2026. In: *StatPearls*. StatPearls Publishing. NBK537109
21. Amrutkar CV, Lui F. Lennox-Gastaut Syndrome. 2025. In: *StatPearls*. StatPearls Publishing. NBK532965
22. Rahman M, Awosika AO, Nguyen H. Valproic Acid. 2024. In: *StatPearls*. StatPearls Publishing. NBK559112
23. Anand A, Sun CE. Autoimmune Encephalitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK578203
24. Magley M, Hinson MR. Eclampsia. 2024. In: *StatPearls*. StatPearls Publishing. NBK554392
25. Mandalaneni K, Rayi A. Vagus Nerve Stimulator. 2023. In: *StatPearls*. StatPearls Publishing. NBK562175
26. Chamarthi VS, Samanta D, Chamarthi PP. Febrile Seizure. 2026. In: *StatPearls*. StatPearls Publishing. NBK448123
27. Gupta M, Tripp J. Phenytoin. 2023. In: *StatPearls*. StatPearls Publishing. NBK551520
28. Maan JS, Duong TVH, Saadabadi A. Carbamazepine. 2023. In: *StatPearls*. StatPearls Publishing. NBK482455
29. Sanders AE, Zafar N, Sharma S. Myoclonus. 2024. In: *StatPearls*. StatPearls Publishing. NBK537015
30. Caprara ALF, Tharwat Ali H, Elrefaey A, Elejla SA, et al. Somatosensory auras in epilepsy: a narrative review of the literature. *Medicines (Basel)*. 2023;10(8):49. PMC10456342
31. Fotedar N, Acar A, Hakami S, Praditukrit K, et al. Semiology and neurophysiology of clonic seizures: a report of 39 patients. *Neurol Clin Pract*. 2024;14(2):e200252. PMC10996907

Multiple sclerosis: immune demyelination and lesions scattered in space and time

1. Reich DS, Lucchinetti CF, Calabresi PA. Multiple sclerosis. *N Engl J Med*. 2018;378(2):169–180. PMC6942519
2. Samara A, Ontaneda D. Evolving the diagnosis of multiple sclerosis: a new landscape in light of the 2024 McDonald criteria. *Biomedicines*. 2025;13(11). PMC12650676
3. Solomon AJ, Brownlee WJ, Viswanathan S, et al. Application of the 2024 McDonald criteria: earlier diagnosis and avoiding misdiagnosis of multiple sclerosis in a global setting. *Neurology*. 2026;107(6):e218425. PMC13528897

4. Biström M, Jons D, Engdahl E, et al. Epstein-Barr virus infection after adolescence and human herpesvirus 6A as risk factors for multiple sclerosis. *Eur J Neurol*. 2021;28(2):579–586. PMC7839468
5. Gontika MP, Anagnostouli MC. Anti-myelin oligodendrocyte glycoprotein and human leukocyte antigens as markers in pediatric and adolescent multiple sclerosis: on diagnosis, clinical phenotypes, and therapeutic responses. *Mult Scler Int*. 2018;2018:8487471. PMC6282147
6. Rochfort KD, Collins LE, Murphy RP, et al. Downregulation of blood-brain barrier phenotype by proinflammatory cytokines involves NADPH oxidase-dependent ROS generation: consequences for interendothelial adherens and tight junctions. *PLoS One*. 2014;9(7):e101815. PMC4081725
7. Gonzalez Caldito N. Role of tumor necrosis factor-alpha in the central nervous system: a focus on autoimmune disorders. *Front Immunol*. 2023;14:1213448. PMC10360935
8. Chen JJ, Flanagan EP, Jitprapaikulsan J, et al. Myelin oligodendrocyte glycoprotein antibody-positive optic neuritis: clinical characteristics, radiologic clues, and outcome. *Am J Ophthalmol*. 2018;195:8–15. PMC6371779
9. Freeman L, Longbrake EE, Coyle PK, et al. High-efficacy therapies for treatment-naïve individuals with relapsing-remitting multiple sclerosis. *CNS Drugs*. 2022;36(12):1285–1299. PMC9645316
10. Lublin FD, Reingold SC, Cohen JA, et al. Defining the clinical course of multiple sclerosis: the 2013 revisions. *Neurology*. 2014;83(3):278–286. PMC4117366
11. Edan G, Le Page E. Escalation versus induction/high-efficacy treatment strategies for relapsing multiple sclerosis: which is best for patients? *Drugs*. 2023;83(15):1351–1363. PMC10582148
12. Weigel C, Bellaci J, Spiegel S. Sphingosine-1-phosphate and its receptors in vascular endothelial and lymphatic barrier function. *J Biol Chem*. 2023;299(6):104775. PMC10220486
13. Lobaina M, Shanina E. Fingolimod. 2024. In: *StatPearls*. StatPearls Publishing. NBK604198
14. Kumar SJ, Biswas DA. Anatomical aspects of neurogenic bladder and the approach in its management: a narrative review. *Cureus*. 2022;14(11):e31165. PMC9725102
15. Tafti D, Ehsan M, Xixis KL. Multiple Sclerosis. 2024. In: *StatPearls*. StatPearls Publishing. NBK499849
16. Guier CP, Kaur K, Stokkermans TJ. Optic Neuritis. 2025. In: *StatPearls*. StatPearls Publishing. NBK557853
17. Feroze KB, Wang J. Internuclear Ophthalmoplegia. 2023. In: *StatPearls*. StatPearls Publishing. NBK441970
18. Saleem F, Munakomi S. Pseudobulbar Palsy. 2023. In: *StatPearls*. StatPearls Publishing. NBK553160
19. Teoli D, Rocha Cabrero F, Smith T, Ghassemzadeh S. Lhermitte Sign. 2023. In: *StatPearls*. StatPearls Publishing. NBK493237
20. Panginikkod S, Rayi A, Rocha Cabrero F, Rukmangadachar LA. Uhthoff Phenomenon. 2022. In: *StatPearls*. StatPearls Publishing. NBK470244
21. Leslie SW, Tadi P, Tayyeb M. Neurogenic Bladder and Neurogenic Lower Urinary Tract Dysfunction. 2023. In: *StatPearls*. StatPearls Publishing. NBK560617
22. Baiano C, Zeppieri M. Visual Evoked Potential. 2023. In: *StatPearls*. StatPearls Publishing. NBK582128
23. Shumway CL, Patel BC, Tripathy K, De Jesus O. Neuromyelitis Optica Spectrum Disorder (NMOSD). 2024. In: *StatPearls*. StatPearls Publishing. NBK572108
24. Anilkumar AC, Foris LA, Tadi P. Acute Disseminated Encephalomyelitis. 2024. In: *StatPearls*. StatPearls Publishing. NBK430934
25. Ankar A, Kumar A. Vitamin B12 Deficiency. 2024. In: *StatPearls*. StatPearls Publishing. NBK441923
26. Bowers W, Blum MA. Systemic Lupus Erythematosus. 2026. In: *StatPearls*. StatPearls Publishing. NBK535405
27. Sukumaran S, Lui F. Neurosarcoidosis. 2026. In: *StatPearls*. StatPearls Publishing. NBK534768
28. Bokhari SRA, Safdar A, Rout P. Sarcoidosis. 2026. In: *StatPearls*. StatPearls Publishing. NBK430687
29. Sergent SR, Ashurst JV. Plasmapheresis. 2023. In: *StatPearls*. StatPearls Publishing. NBK560566
30. Babaesfahani A, Khanna NR, Patel P, Kuns B. Natalizumab. 2024. In: *StatPearls*. StatPearls Publishing. NBK534201
31. Ghanavatian S, Derian A. Baclofen. 2024. In: *StatPearls*. StatPearls Publishing. NBK526037
32. Dwyer J, Tafuri SM, LaGrange CA. Oxybutynin. 2023. In: *StatPearls*. StatPearls Publishing. NBK499985
33. Yasaei R, Katta S, Patel P, Saadabadi A. Gabapentin. 2024. In: *StatPearls*. StatPearls Publishing. NBK493228
34. Singh R, Pandey S. Movement disorder in demyelinating disease: tracing the Charcot's foot print. *Ann Indian Acad Neurol*. 2022;25(5):821–831. PMC9764914

Neuromyelitis optica spectrum disorder: an antibody-mediated astrocytopathy of the optic nerves, spinal cord, and area postrema

1. Shumway CL, Patel BC, Tripathy K, De Jesus O. Neuromyelitis Optica Spectrum Disorder (NMOSD). 2024. In: *StatPearls*. StatPearls Publishing. NBK572108
2. Wingerchuk DM, Banwell B, Bennett JL, Cabre P, et al. International consensus diagnostic criteria for neuromyelitis optica spectrum disorders. *Neurology*. 2015;85(2):177–189. PMC4515040
3. Holroyd KB, Manzano GS, Levy M. Update on neuromyelitis optica spectrum disorder. *Curr Opin Ophthalmol*. 2020;31(6):462–468. PMC7771018
4. Abe Y, Yasui M. Aquaporin-4 in neuromyelitis optica spectrum disorders: a target of autoimmunity in the central nervous system. *Biomolecules*. 2022;12(4):591. PMC9030581
5. Fujihara K, Bennett JL, de Seze J, Haramura M, et al. Interleukin-6 in neuromyelitis optica spectrum disorder pathophysiology. *Neurol Neuroimmunol Neuroinflamm*. 2020;7(5):e841. PMC7455314
6. Shosha E, Dubey D, Palace J, Nakashima I, et al. Area postrema syndrome: frequency, criteria, and severity in AQP4-IgG-positive NMOSD. *Neurology*. 2018;91(17):e1642–e1651. PMC6205685

MOG antibody-associated disease: an antibody-mediated demyelinating disease distinct from MS and NMOSD

1. Ambrosius W, Michalak S, Kozubski W, Kalinowska A. Myelin oligodendrocyte glycoprotein antibody-associated disease: current insights into the disease pathophysiology, diagnosis and management. *Int J Mol Sci*. 2020;22(1):100. PMC7795410
2. Höftberger R, Guo Y, Flanagan EP, Lopez-Chiriboga AS, et al. The pathology of central nervous system inflammatory demyelinating disease accompanying myelin oligodendrocyte glycoprotein autoantibody. *Acta Neuropathol*. 2020;139(5):875–892. PMC7181560
3. Longbrake E. Myelin oligodendrocyte glycoprotein-associated disorders. *Continuum (Minneap Minn)*. 2022;28(4):1171–1193. PMC9523511
4. Al-Ani A, Chen JJ, Costello F. Myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD): current understanding and challenges. *J Neurol*. 2023;270(8):4132–4150. PMC10165591
5. Shumway CL, Patel BC, Tripathy K, De Jesus O. Neuromyelitis Optica Spectrum Disorder (NMOSD). 2024. In: *StatPearls*. StatPearls Publishing. NBK572108

Glaucoma: aqueous outflow, intraocular pressure, and optic nerve damage

1. Dietze J, Blair K, Zeppieri M, Havens SJ. Glaucoma. 2024. In: *StatPearls*. StatPearls Publishing. NBK538217
2. Tripathy K, Zeppieri M. Open Angle Glaucoma. 2024. In: *StatPearls*. StatPearls Publishing. NBK441887
3. Khazaeni B, Zeppieri M, Khazaeni L. Acute Angle-Closure Glaucoma. 2026. In: *StatPearls*. StatPearls Publishing. NBK430857
4. Machiele R, Motlagh M, Zeppieri M, Patel BC. Intraocular Pressure. 2024. In: *StatPearls*. StatPearls Publishing. NBK532237

5. Tripathy K, Patel P, Geetha R. Latanoprost. 2024. In: *StatPearls*. StatPearls Publishing. NBK540978
6. Barnes J, Moshirfar M. Timolol. 2024. In: *StatPearls*. StatPearls Publishing. NBK545176
7. Aslam S, Gupta V. Carbonic Anhydrase Inhibitors. 2023. In: *StatPearls*. StatPearls Publishing. NBK557736
8. Pakala RS, Brown KN, Preuss CV. Cholinergic Medications. 2023. In: *StatPearls*. StatPearls Publishing. NBK538163
9. Dalal R, Grujic D. Epinephrine. 2024. In: *StatPearls*. StatPearls Publishing. NBK482160

Cataract: opacification of the crystalline lens

1. Nizami AA, Gurnani B, Gulani AC. Cataract. 2024. In: *StatPearls*. StatPearls Publishing. NBK539699
2. Moshirfar M, Milner D, Patel BC. Cataract Surgery. 2023. In: *StatPearls*. StatPearls Publishing. NBK559253
3. Shah SS, Gurnani B. Lens-Induced Glaucoma. 2026. In: *StatPearls*. StatPearls Publishing. NBK574524
4. Indaram M, Agrón E, Clemons TE, Sperduto RD, et al. Changes in lens opacities on the Age-Related Eye Disease Study grading scale predict progression to cataract surgery and vision loss: Age-Related Eye Disease Study report no. 34. *Ophthalmology*. 2015;122(5):888–896. PMC4414714
5. Lee CS, Gibbons LE, Lee AY, Yanagihara RT, et al. Association between cataract extraction and development of dementia. *JAMA Intern Med*. 2022;182(2):134–141. PMC8649913
6. Yu HN, Ying GS. A narrative review of the association between cataracts and dementia. *Eye Brain*. 2025;17:151–156. PMC12702263
7. Ray A, Gurnani B. Pediatric Cataract. 2026. In: *StatPearls*. StatPearls Publishing. NBK572080

Vision loss and red eye: telling eye conditions apart by presentation

1. Solano D, Fu L, Czyz CN. Viral Conjunctivitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK470271
2. Hyde MR, Le JK. Bacterial Conjunctivitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK546683
3. Gurnani B, Kaur K. Bacterial Keratitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK574509
4. Guier CP, Kaur K, Stokkermans TJ. Optic Neuritis. 2025. In: *StatPearls*. StatPearls Publishing. NBK557853
5. Blair K, Czyz CN. Retinal Detachment. 2024. In: *StatPearls*. StatPearls Publishing. NBK551502
6. Hobbs SD, Tripathy K, Pierce K. Wet Age-Related Macular Degeneration (AMD). 2024. In: *StatPearls*. StatPearls Publishing. NBK572147
7. Ruia S, Kaufman EJ. Macular Degeneration. 2023. In: *StatPearls*. StatPearls Publishing. NBK560778
8. O'Neal TB, Tripathy K, Luther EE. Retinitis Pigmentosa. 2024. In: *StatPearls*. StatPearls Publishing. NBK519518
9. Gurnani B, Le PH. Allergic Conjunctivitis. 2026. In: *StatPearls*. StatPearls Publishing. NBK448118
10. Duplechain A, Conrady CD, Patel BC, Baker S. Uveitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK540993
11. Tripathy K, Shah SS, Waymack JR. Central Retinal Artery Occlusion. 2024. In: *StatPearls*. StatPearls Publishing. NBK470354
12. Blair K, Czyz CN. Central Retinal Vein Occlusion. 2023. In: *StatPearls*. StatPearls Publishing. NBK525985
13. Ameer MA, Vaqar S, Savadkar A, Khazaeni B. Giant Cell Arteritis (Temporal Arteritis). 2024. In: *StatPearls*. StatPearls Publishing. NBK459376
14. Asuncion RMD, Margolin E. Papilledema. 2025. In: *StatPearls*. StatPearls Publishing. NBK538295
15. Tripathy K, Arsiwalla T. Hypertensive Retinopathy. 2025. In: *StatPearls*. StatPearls Publishing. NBK525980
16. Khan Z, Bollu PC. Horner Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK500000
17. Dichter SL, Khan Suheb MZ, Shubert GS. Argyll Robertson Pupil. 2024. In: *StatPearls*. StatPearls Publishing. NBK537179

Hearing loss: conductive, sensorineural, and mixed

1. Anastasiadou S, Al Khalili Y. Hearing Loss. 2023. In: *StatPearls*. StatPearls Publishing. NBK542323
2. Sooriyamoorthy T, De Jesus O. Conductive Hearing Loss. 2023. In: *StatPearls*. StatPearls Publishing. NBK563267
3. Tanna RJ, Lin JW, De Jesus O. Sensorineural Hearing Loss. 2023. In: *StatPearls*. StatPearls Publishing. NBK565860
4. Cheslock M, De Jesus O. Presbycusis. 2023. In: *StatPearls*. StatPearls Publishing. NBK559220
5. Hohman MH, Khan MAB. Otosclerosis. 2024. In: *StatPearls*. StatPearls Publishing. NBK560671
6. Priya JS, Hohman MH. Noise Exposure and Hearing Loss. 2023. In: *StatPearls*. StatPearls Publishing. NBK594247
7. Matz O, Peterson DC. Tinnitus. 2023. In: *StatPearls*. StatPearls Publishing. NBK430809
8. Wahid NWB, Hogan CJ, Attia M. Weber Test. 2023. In: *StatPearls*. StatPearls Publishing. NBK526135
9. Belardo K, Kong EL, Sutton AE. Rinne Test. 2026. In: *StatPearls*. StatPearls Publishing. NBK431071
10. Sheikh MM, De Jesus O. Vestibular Schwannoma. 2023. In: *StatPearls*. StatPearls Publishing. NBK562312
11. Young A, Ng M. Genetic Hearing Loss. 2023. In: *StatPearls*. StatPearls Publishing. NBK580517
12. Sutton AE, Goldman J. Syndromic Sensorineural Hearing Loss. 2025. In: *StatPearls*. StatPearls Publishing. NBK526088
13. Winters R, Masood S. Waardenburg Syndrome. 2025. In: *StatPearls*. StatPearls Publishing. NBK560879
14. Pandya A, O'Brien A, Kovasala M, Bademci G, et al. Analyses of del(GJB6-D13S1830) and del(GJB6-D13S1834) deletions in a large cohort with hearing loss: caveats to interpretation of molecular test results in multiplex families. *Mol Genet Genomic Med*. 2020;8(4):e1171. PMC7196463
15. Pitton Rissardo J, Margetis K. Neurofibromatosis Type 2. 2026. In: *StatPearls*. StatPearls Publishing. NBK470350

Vertigo: timing, nystagmus, and hearing as guides to the cause

1. Dy JS, Freeman AM. Vertigo in Clinical Practice: Evidence-Based Diagnosis and Treatment. 2025. In: *StatPearls*. StatPearls Publishing. NBK482356
2. Koshi EJ, Sutton AE. Benign Paroxysmal Positional Vertigo. 2025. In: *StatPearls*. StatPearls Publishing. NBK470308
3. Smith T, Rider J, Cen S, Borger J. Vestibular Neuronitis. 2023. In: *StatPearls*. StatPearls Publishing. NBK549866
4. Barkwill D, Winters R, Arora R. Labyrinthitis. 2025. In: *StatPearls*. StatPearls Publishing. NBK560506
5. Bowers E, Winters R. Meniere Disease. 2026. In: *StatPearls*. StatPearls Publishing. NBK536955
6. Kaur K, Rocha Cabrero F, Deibel JP. Nystagmus Types. 2025. In: *StatPearls*. StatPearls Publishing. NBK539711
7. Lui F, Foris LA, Tadi P. Central Vertigo. 2024. In: *StatPearls*. StatPearls Publishing. NBK441861
8. Hilton DB, Lui F, Shermetaro C. Migraine-Associated Vertigo. 2024. In: *StatPearls*. StatPearls Publishing. NBK507859
9. Talmud JD, Coffey R, Hsu NM, Edemekong PF. Dix-Hallpike Maneuver. 2023. In: *StatPearls*. StatPearls Publishing. NBK459307
10. Nguyen CT, Basso M. Epley Maneuver. 2024. In: *StatPearls*. StatPearls Publishing. NBK563287
11. Sheikh MM, De Jesus O. Vestibular Schwannoma. 2023. In: *StatPearls*. StatPearls Publishing. NBK562312
12. Casale J, Browne T, Murray IV, Gupta G. Physiology, Vestibular System. 2023. In: *StatPearls*. StatPearls Publishing. NBK532978

13. Simakurthy S, Tripathy K. Oculovestibular Reflex. 2026. In: *StatPearls*. StatPearls Publishing. NBK542281

Alzheimer disease: amyloid plaques, tau tangles, and progressive neuronal loss

1. Knopman DS, Amieva H, Petersen RC, et al. Alzheimer disease. *Nat Rev Dis Primers*. 2021;7(1):33. PMC8574196
2. Hyman BT, Phelps CH, Beach TG, et al. National Institute on Aging-Alzheimer's Association guidelines for the neuropathologic assessment of Alzheimer's disease. *Alzheimers Dement*. 2012;8(1):1–13. PMC3266529
3. Alafuzoff I, Thal DR, Arzberger T, et al. Assessment of beta-amyloid deposits in human brain: a study of the BrainNet Europe Consortium. *Acta Neuropathol*. 2009;117(3):309–320. PMC2910889
4. Serrano-Pozo A, Frosch MP, Masliah E, et al. Neuropathological alterations in Alzheimer disease. *Cold Spring Harb Perspect Med*. 2011;1(1):a006189. PMC3234452
5. Haass C, Kaether C, Thinakaran G, et al. Trafficking and proteolytic processing of APP. *Cold Spring Harb Perspect Med*. 2012;2(5):a006270. PMC3331683
6. Wolfe MS. Structure, mechanism and inhibition of gamma-secretase and presenilin-like proteases. *Biol Chem*. 2010;391(8):839–847. PMC2997569
7. Bekris LM, Yu CE, Bird TD, et al. Genetics of Alzheimer disease. *J Geriatr Psychiatry Neurol*. 2010;23(4):213–227. PMC3044597
8. Igarashi KM. Entorhinal cortex dysfunction in Alzheimer's disease. *Trends Neurosci*. 2023;46(2):124–136. PMC9877178
9. Kerchner GA, Hess CP, Hammond-Rosenbluth KE, et al. Hippocampal CA1 apical neuropil atrophy in mild Alzheimer disease visualized with 7-T MRI. *Neurology*. 2010;75(15):1381–1387. PMC3013485
10. Simic G, Stanic G, Mladinov M, et al. Does Alzheimer's disease begin in the brainstem? *Neuropathol Appl Neurobiol*. 2009;35(6):532–554. PMC2787819
11. Arendt T, Brückner MK, Morawski M, et al. Early neurone loss in Alzheimer's disease: cortical or subcortical? *Acta Neuropathol Commun*. 2015;3:10. PMC4359478
12. Holland N, Robbins TW, Rowe JB. The role of noradrenaline in cognition and cognitive disorders. *Brain*. 2021;144(8):2243–2256. PMC8418349
13. Ferreira-Vieira TH, Guimaraes IM, Silva FR, et al. Alzheimer's disease: targeting the cholinergic system. *Curr Neuropharmacol*. 2016;14(1):101–115. PMC4787279
14. Rabinovici GD, Selkoe DJ, Schindler SE, et al. Donanemab: appropriate use recommendations. *J Prev Alzheimers Dis*. 2025;12(5):100150. PMC12180672
15. Lui F, Tsao JW. Alzheimer Disease. 2024. In: *StatPearls*. StatPearls Publishing. NBK499922
16. Rukmangadachar LA, Bollu PC. Amyloid Beta Peptide. 2025. In: *StatPearls*. StatPearls Publishing. NBK459119
17. Singh J, Hegazy MI. Cerebral Amyloid Angiopathy. 2025. In: *StatPearls*. StatPearls Publishing. NBK556105
18. Ramírez Echeverría MDL, Schoo C. Delirium. 2026. In: *StatPearls*. StatPearls Publishing. NBK470399
19. Patil N, Rehman A, Anastasopoulou C, Jialal I. Hypothyroidism. 2024. In: *StatPearls*. StatPearls Publishing. NBK519536
20. Ankar A, Kumar A. Vitamin B12 Deficiency. 2024. In: *StatPearls*. StatPearls Publishing. NBK441923
21. Margetis K, Das JM, Biagioni MC. Idiopathic Normal Pressure Hydrocephalus. 2026. In: *StatPearls*. StatPearls Publishing. NBK542247
22. Sanders AE, Schoo C, Kalish VB. Vascular Dementia. 2023. In: *StatPearls*. StatPearls Publishing. NBK430817

Frontotemporal dementia: frontal and temporal degeneration of behavior and language

1. Irwin DJ, Cairns NJ, Grossman M, McMillan CT, et al. Frontotemporal lobar degeneration: defining phenotypic diversity through personalized medicine. *Acta Neuropathol*. 2015;129(4):469–491. PMC4369168
2. Bonner MF, Ash S, Grossman M. The new classification of primary progressive aphasia into semantic, logopenic, or nonfluent/agrammatic variants. *Curr Neurol Neurosci Rep*. 2010;10(6):484–490. PMC2963791
3. Ulugut Erkoyun H, Groot C, Heilbron R, Nelissen A, et al. A clinical-radiological framework of the right temporal variant of frontotemporal dementia. *Brain*. 2020;143(9):2831–2843. PMC9172625
4. Tsai RM, Boxer AL. Treatment of frontotemporal dementia. *Curr Treat Options Neurol*. 2014;16(11):319. PMC4920050
5. Khan I, De Jesus O. Frontotemporal Lobe Dementia. 2023. In: *StatPearls*. StatPearls Publishing. NBK559286
6. Pippin MM, Gupta V. Pick Disease. 2023. In: *StatPearls*. StatPearls Publishing. NBK562226
7. Kiymaz T, Khan Suheb MZ, Lui F, De Jesus O. Primary Progressive Aphasia. 2024. In: *StatPearls*. StatPearls Publishing. NBK563145

Dementia with Lewy bodies: α-synuclein, fluctuating cognition, and antipsychotic sensitivity

1. McKeith IG, Boeve BF, Dickson DW, et al. Diagnosis and management of dementia with Lewy bodies: fourth consensus report of the DLB Consortium. *Neurology*. 2017;89(1):88–100. PMC5496518
2. Erskine D, Taylor JP. Current strategies in the management of dementia with Lewy bodies and future directions based on disease pathophysiology. *Mol Neurodegener*. 2025;20(1):78. PMC12220670
3. Karantzoulis S, Galvin JE. Update on dementia with Lewy bodies. *Curr Transl Geriatr Exp Gerontol Rep*. 2013;2(3):196–204. PMC4219734
4. Rongve A, Witoelar A, Ruiz A, et al. GBA and APOE ε4 associate with sporadic dementia with Lewy bodies in European genome wide association study. *Sci Rep*. 2019;9(1):7013. PMC6504850
5. Stefani A, Antelmi E, Arnaldi D, et al. From mechanisms to future therapy: a synopsis of isolated REM sleep behavior disorder as early synuclein-related disease. *Mol Neurodegener*. 2025;20(1):19. PMC11817540
6. Chiu SY, Wyman-Chick KA, Ferman TJ, et al. Sex differences in dementia with Lewy bodies: focused review of available evidence and future directions. *Parkinsonism Relat Disord*. 2023;107:105285. PMC10024862
7. Kong W, Satoh K, Shimamura MI, et al. Seed amplification assays for Parkinson's disease: a review of α-synuclein assays in body fluids and tissues. *J Neurochem*. 2026;170(5):e70435. PMC13179529
8. Ferreira-Vieira TH, Guimaraes IM, Silva FR, et al. Alzheimer's disease: targeting the cholinergic system. *Curr Neuropharmacol*. 2016;14(1):101–115. PMC4787279
9. Haider A, Spurling BC, Sánchez-Manso JC. Lewy Body Dementia. 2023. In: *StatPearls*. StatPearls Publishing. NBK482441
10. Lui F, Tsao JW. Alzheimer Disease. 2024. In: *StatPearls*. StatPearls Publishing. NBK499922
11. Ramírez Echeverría MDL, Schoo C. Delirium. 2026. In: *StatPearls*. StatPearls Publishing. NBK470399
12. Sanders AE, Schoo C, Kalish VB. Vascular Dementia. 2023. In: *StatPearls*. StatPearls Publishing. NBK430817
13. Khan I, De Jesus O. Frontotemporal Lobe Dementia. 2023. In: *StatPearls*. StatPearls Publishing. NBK559286
14. Hall WA, Masood W. Creutzfeldt-Jakob Disease. 2025. In: *StatPearls*. StatPearls Publishing. NBK507860
15. Shrimanker I, Tadi P, Schoo C, Sánchez-Manso JC. Parkinsonism. 2024. In: *StatPearls*. StatPearls Publishing. NBK542224

16. Agarwal S, Gilbert R. Progressive Supranuclear Palsy. 2023. In: *StatPearls*. StatPearls Publishing. NBK526098
17. Alzheimer's Association. Dementia with Lewy bodies. alz.org
18. Lewy Body Dementia Association. Treatment guidance and medication considerations in LBD. lbda.org

Motor signs and syndromes: localizing weakness, abnormal movements, and nerve lesions

1. Emos MC, Agarwal S. Neuroanatomy, Upper Motor Neuron Lesion. 2023. In: *StatPearls*. StatPearls Publishing. NBK537305
2. Javed K, Daly DT. Neuroanatomy, Lower Motor Neuron Lesion. 2023. In: *StatPearls*. StatPearls Publishing. NBK539814
3. Margetis K, Das JM, Emmady PD. Spinal Cord Injuries. 2025. In: *StatPearls*. StatPearls Publishing. NBK560721
4. Hohman MH, Mazzoni T. Facial Nerve Palsy. 2026. In: *StatPearls*. StatPearls Publishing. NBK549815
5. Hohman MH, Warner MJ, Varacallo MA. Bell Palsy. 2024. In: *StatPearls*. StatPearls Publishing. NBK482290
6. Pitton Rissardo J, Lui F. Gait Disturbances. 2026. In: *StatPearls*. StatPearls Publishing. NBK560610
7. Nori SL, Das JM. Steppage Gait. 2023. In: *StatPearls*. StatPearls Publishing. NBK547672
8. Forbes Kaprive J, Munakomi S, Cronovich HA. Romberg Test. 2023. In: *StatPearls*. StatPearls Publishing. NBK563187
9. Ataullah AHM, Singla R, Naqvi IA. Cerebellar Dysfunction. 2024. In: *StatPearls*. StatPearls Publishing. NBK562317
10. Unverdi M, Alsayouri K. Neuroanatomy, Cerebellar Dysfunction. 2023. In: *StatPearls*. StatPearls Publishing. NBK545251
11. Rocha Cabrero F, Singh J. Holmes Tremor. 2026. In: *StatPearls*. StatPearls Publishing. NBK562149
12. Agarwal S, Biagioni MC. Essential Tremor. 2023. In: *StatPearls*. StatPearls Publishing. NBK499986
13. Pana A, Saggu BM. Dystonia. 2023. In: *StatPearls*. StatPearls Publishing. NBK448144
14. Rocha Cabrero F, De Jesus O. Hemiballismus. 2023. In: *StatPearls*. StatPearls Publishing. NBK559127
15. Sanders AE, Zafar N, Sharma S. Myoclonus. 2024. In: *StatPearls*. StatPearls Publishing. NBK537015
16. Gowda SN, Hodis B, Kolton Schneider L. Apraxia. 2026. In: *StatPearls*. StatPearls Publishing. NBK585110
17. DeCastro A, Cyphers G, Sun CE. Wrist Drop. 2026. In: *StatPearls*. StatPearls Publishing. NBK532993
18. Lleva JMC, Munakomi S, Sun CE, Chang KV. Ulnar Neuropathy. 2025. In: *StatPearls*. StatPearls Publishing. NBK534226
19. Stretanski MF, Dydyk AM, Cascella M. Median Nerve Injury. 2025. In: *StatPearls*. StatPearls Publishing. NBK553109
20. Sevy JO, Sina RE, Varacallo MA. Carpal Tunnel Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK448179
21. Lezak B, Massel DH, Varacallo MA. Peroneal Nerve Injury. 2024. In: *StatPearls*. StatPearls Publishing. NBK549859
22. Nori SL, Stretanski MF. Foot Drop. 2025. In: *StatPearls*. StatPearls Publishing. NBK554393
23. Bisht S, Daley SF, Madhani NB. Erb Palsy. 2026. In: *StatPearls*. StatPearls Publishing. NBK513260
24. Merryman J, Varacallo MA. Klumpke Palsy. 2023. In: *StatPearls*. StatPearls Publishing. NBK531500
25. Lung K, Lui F. Anatomy, Thorax, Long Thoracic Nerve. 2023. In: *StatPearls*. StatPearls Publishing. NBK535396
26. Tessler J, Singh S, Talati R. Axillary Nerve Injury. 2026. In: *StatPearls*. StatPearls Publishing. NBK539895
27. Desai SS, Arbor TC, Varacallo MA. Anatomy, Shoulder and Upper Limb, Musculocutaneous Nerve. 2023. In: *StatPearls*. StatPearls Publishing. NBK534199
28. Knight J, Decker LC. Decerebrate and Decorticate Posturing. 2023. In: *StatPearls*. StatPearls Publishing. NBK559135

Parkinson disease: α-synuclein, dopamine neuron loss, and dopaminergic treatment

1. Fox SH, Luca DG, Postuma RB, et al. Revisiting the 2015 MDS diagnostic criteria for Parkinson disease: insights from autopsy-confirmed cases. *NPJ Parkinsons Dis*. 2025;11(1):360. PMC12748991
2. Hopfner F, Höglinger G, German Parkinson's Guidelines Group, et al. Definition and diagnosis of Parkinson's disease: guideline "Parkinson's disease" of the German Society of Neurology. *J Neurol*. 2024;271(11):7102–7119. PMC11561062
3. Kwakye GF, McMinimy RA, Aschner M. Disease-toxicant interactions in Parkinson's disease neuropathology. *Neurochem Res*. 2017;42(6):1772–1786. PMC5344789
4. Muddapu VR, Mandali A, Chakravarthy VS, et al. A computational model of loss of dopaminergic cells in Parkinson's disease due to glutamate-induced excitotoxicity. *Front Neural Circuits*. 2019;13:11. PMC6397878
5. Erskine D, Taylor JP. Current strategies in the management of dementia with Lewy bodies and future directions based on disease pathophysiology. *Mol Neurodegener*. 2025;20(1):78. PMC12220670
6. Akdemir ÜÖ, Bora Tokçaer A, Atay LÖ. Dopamine transporter SPECT imaging in Parkinson's disease and parkinsonian disorders. *Turk J Med Sci*. 2021;51(2):400–410. PMC8203173
7. Carbone F, Djamshidian A. Impulse control disorders in Parkinson's disease: an overview of risk factors, pathogenesis and pharmacological management. *CNS Drugs*. 2024;38(6):443–457. PMC11098885
8. Zafar S, Lui F, Yaddanapudi SS. Parkinson Disease. 2025. In: *StatPearls*. StatPearls Publishing. NBK470193
9. Vertes AC, Beato MR, Sonne J, Khan Suheb MZ. Parkinson-Plus Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK585113
10. Bhatti KS, Sun CE, Walker I. Corticobasal Degeneration. 2025. In: *StatPearls*. StatPearls Publishing. NBK611986
11. Agarwal S, Gilbert R. Progressive Supranuclear Palsy. 2023. In: *StatPearls*. StatPearls Publishing. NBK526098
12. Agarwal S, Biagioni MC. Essential Tremor. 2023. In: *StatPearls*. StatPearls Publishing. NBK499986
13. Gandhi KR, Saadabadi A. Levodopa (L-Dopa). 2023. In: *StatPearls*. StatPearls Publishing. NBK482140
14. Leyden E, Tadi P. Carbidopa. 2023. In: *StatPearls*. StatPearls Publishing. NBK554552
15. Singh R, Parmar M. Pramipexole. 2023. In: *StatPearls*. StatPearls Publishing. NBK557539
16. Patel P, Saadabadi A. Monoamine Oxidase Inhibitors (MAOIs). 2025. In: *StatPearls*. StatPearls Publishing. NBK539848
17. Fariba KA, Gokarakonda SB. Impulse Control Disorders. 2023. In: *StatPearls*. StatPearls Publishing. NBK562279
18. Fariba KA, Gupta V. Deep Brain Stimulation. 2023. In: *StatPearls*. StatPearls Publishing. NBK557847
19. Chang C, Ramphul K. Amantadine. 2023. In: *StatPearls*. StatPearls Publishing. NBK499953
20. Jilani TN, Sabir S, Patel P, Sharma S. Trihexyphenidyl. 2024. In: *StatPearls*. StatPearls Publishing. NBK519488
21. Shrimanker I, Tadi P, Schoo C, Sánchez-Manso JC. Parkinsonism. 2024. In: *StatPearls*. StatPearls Publishing. NBK542224

Atypical parkinsonism: multiple system atrophy, progressive supranuclear palsy, and corticobasal degeneration

1. Jellinger KA. Multiple system atrophy: an oligodendroglioneural synucleinopathy. *J Alzheimers Dis*. 2018;62(3):1141–1179. PMC5870010
2. Sugiyama A, Yokota H, Yamanaka Y, Mukai H, et al. Vertical pons hyperintensity and hot cross bun sign in cerebellar-type multiple system atrophy and spinocerebellar ataxia type 3. *BMC Neurol*. 2020;20(1):157. PMC7184719

3. Frey KA. Molecular imaging of extrapyramidal movement disorders with dementia: the 4R tauopathies. *Semin Nucl Med*. 2021;51(3):275–285. PMC7988291
4. Dregni AJ, Duan P, Xu H, Changolkar L, et al. Fluent molecular mixing of Tau isoforms in Alzheimer's disease neurofibrillary tangles. *Nat Commun*. 2022;13(1):2967. PMC9142584
5. Irwin DJ, Cairns NJ, Grossman M, McMillan CT, et al. Frontotemporal lobar degeneration: defining phenotypic diversity through personalized medicine. *Acta Neuropathol*. 2015;129(4):469–491. PMC4369168
6. Nelson PT, Fardo DW, Wu X, Aung KZ, et al. Limbic-predominant age-related TDP-43 encephalopathy (LATE-NC): co-pathologies and genetic risk factors provide clues about pathogenesis. *J Neuropathol Exp Neurol*. 2024;83(6):396–415. PMC11110076
7. Fecek C, Nagalli S. Shy-Drager Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK560502
8. Agarwal S, Gilbert R. Progressive Supranuclear Palsy. 2023. In: *StatPearls*. StatPearls Publishing. NBK526098
9. Bhatti KS, Sun CE, Walker I. Corticobasal Degeneration. 2025. In: *StatPearls*. StatPearls Publishing. NBK611986
10. Vertes AC, Beato MR, Sonne J, Khan Suheb MZ. Parkinson-Plus Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK585113

Huntington disease: CAG repeat expansion, striatal degeneration, and chorea

1. Ajitkumar A, Lui F, De Jesus O. Huntington Disease. 2025. In: *StatPearls*. StatPearls Publishing. NBK559166
2. Lui F, Merical B, Sánchez-Manso JC. Chorea. 2025. In: *StatPearls*. StatPearls Publishing. NBK430923
3. Ramakrishnan S, Gupta V. Trinucleotide Repeat Disorders. 2024. In: *StatPearls*. StatPearls Publishing. NBK559254
4. Reiner A, Dragatsis I, Dietrich P. Genetics and neuropathology of Huntington's disease. *Int Rev Neurobiol*. 2011;98:325–372. PMC4458347
5. Galvan L, André VM, Wang EA, Cepeda C, et al. Functional differences between direct and indirect striatal output pathways in Huntington's disease. *J Huntingtons Dis*. 2012;1(1):17–25. PMC4116084

Localizing a neurologic lesion: from signs to level and side

1. Sánchez JR, Asuncion RMD. Neurologic Exam. 2025. In: *StatPearls*. StatPearls Publishing. NBK557589
2. Gowda SN, Munakomi S. Brainstem Stroke. 2026. In: *StatPearls*. StatPearls Publishing. NBK560896
3. Raju H, Tadi P. Neuroanatomy, Somatosensory Cortex. 2022. In: *StatPearls*. StatPearls Publishing. NBK555915
4. Najem K, Asuncion RMD, Margolin E. Diplopia. 2024. In: *StatPearls*. StatPearls Publishing. NBK441905
5. Payne WN, Blair K, Barrett MJ. Anisocoria. 2023. In: *StatPearls*. StatPearls Publishing. NBK470384
6. Khan Z, Bollu PC. Horner Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK500000
7. Feroze KB, Wang J. Internuclear Ophthalmoplegia. 2023. In: *StatPearls*. StatPearls Publishing. NBK441970
8. Feroze KB, Patel BC. Parinaud Syndrome. 2023. In: *StatPearls*. StatPearls Publishing. NBK441892
9. Kaur K, Rocha Cabrero F, Deibel JP. Nystagmus Types. 2025. In: *StatPearls*. StatPearls Publishing. NBK539711
10. Altabakhi I, Sun CE. Gerstmann Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK519528
11. Jabbari B, Lui F. Frontal Lobe Syndrome. 2026. In: *StatPearls*. StatPearls Publishing. NBK532981
12. Casipit CG, Anastasopoulou C. Hypothalamic Dysfunction. 2026. In: *StatPearls*. StatPearls Publishing. NBK560743
13. Sekhon S, Sharma R, Cascella M. Thunderclap Headache. 2023. In: *StatPearls*. StatPearls Publishing. NBK560629
14. Pescador Ruschel MA, De Jesus O. Migraine Headache. 2024. In: *StatPearls*. StatPearls Publishing. NBK560787
15. Kandel SA, Mandiga P. Cluster Headache. 2023. In: *StatPearls*. StatPearls Publishing. NBK544241
16. Shah N, Asuncion RMD, Hameed S. Muscle Contraction Tension Headache. 2024. In: *StatPearls*. StatPearls Publishing. NBK562274
17. Hoffman H, Klovenski V. Idiopathic Intracranial Hypertension. 2026. In: *StatPearls*. StatPearls Publishing. NBK536924
18. Ameer MA, Vaqar S, Savadkar A, Khazaeni B. Giant Cell Arteritis (Temporal Arteritis). 2024. In: *StatPearls*. StatPearls Publishing. NBK459376
19. Aleksenko D, Lui F, Sánchez-Manso JC. Medication Overuse Headache. 2025. In: *StatPearls*. StatPearls Publishing. NBK470171
20. Pitton Rissardo J, Lui F. Gait Disturbances. 2026. In: *StatPearls*. StatPearls Publishing. NBK560610
21. Forbes Kaprive J, Munakomi S, Cronovich HA. Romberg Test. 2023. In: *StatPearls*. StatPearls Publishing. NBK563187

Ruling out mimics in stroke, seizure, multiple sclerosis, Alzheimer disease, and Parkinson disease

1. Simon K, Tafti D, Shafaat O, Sotoudeh H. Stroke Imaging. 2026. In: *StatPearls*. StatPearls Publishing. NBK546635
2. Emmady PD, Asuncion RMD, Anilkumar AC. EEG Abnormal Waveforms. 2025. In: *StatPearls*. StatPearls Publishing. NBK557655
3. Lovik K, Murr NI. Seizure. 2026. In: *StatPearls*. StatPearls Publishing. NBK430765
4. Lui F, Murr NI. Psychogenic Nonepileptic Seizures. 2024. In: *StatPearls*. StatPearls Publishing. NBK441871
5. Wingerchuk DM, Banwell B, Bennett JL, Cabre P, et al. International consensus diagnostic criteria for neuromyelitis optica spectrum disorders. *Neurology*. 2015;85(2):177–189. PMC4515040
6. Shrimanker I, Tadi P, Schoo C, Sánchez-Manso JC. Parkinsonism. 2024. In: *StatPearls*. StatPearls Publishing. NBK542224

Choosing imaging and EEG in neurology, and the signs to look for

1. Simon K, Tafti D, Shafaat O, Sotoudeh H. Stroke Imaging. 2026. In: *StatPearls*. StatPearls Publishing. NBK546635
2. Pai A, Shetty R, Hodis B, Chowdhury YS. Magnetic Resonance Imaging Physics. 2023. In: *StatPearls*. StatPearls Publishing. NBK564320
3. DenOtter TD, Schubert J. Hounsfield Unit. 2023. In: *StatPearls*. StatPearls Publishing. NBK547721
4. Rayi A, Murr NI. Electroencephalogram. 2022. In: *StatPearls*. StatPearls Publishing. NBK563295
5. Emmady PD, Asuncion RMD, Anilkumar AC. EEG Abnormal Waveforms. 2025. In: *StatPearls*. StatPearls Publishing. NBK557655
6. Reeves RA, Karsonovich T, Gorniak R. Neuroimaging In Epilepsy Surgery. 2023. In: *StatPearls*. StatPearls Publishing. NBK557740
7. Wylie T, Sandhu DS, Murr NI. Status Epilepticus. 2023. In: *StatPearls*. StatPearls Publishing. NBK430686

Antithrombotic drugs and their reversal in cerebrovascular disease

1. Baig MU, Patel P. Thrombolytic Therapy. 2026. In: *StatPearls*. StatPearls Publishing. NBK557411
2. Patel S, Singh R, Preuss CV, Patel N. Warfarin. 2024. In: *StatPearls*. StatPearls Publishing. NBK470313
3. Agrawal A, Manna B. Apixaban. 2024. In: *StatPearls*. StatPearls Publishing. NBK507910
4. Baskaran J, Lopez RA, Cassagnol M. Prothrombin Complex Concentrate. 2024. In: *StatPearls*. StatPearls Publishing. NBK539716
5. Applefield D, Krishnan S. Protamine. 2023. In: *StatPearls*. StatPearls Publishing. NBK547753

Lowering blood pressure and intracranial pressure after brain hemorrhage

1. Das JM, Zito PM. Nimodipine. 2024. In: *StatPearls*. StatPearls Publishing. NBK534870
2. Patel P, Patel R, Thorell W. Mannitol. 2024. In: *StatPearls*. StatPearls Publishing. NBK470392

Antiseizure drugs: mechanisms, choice by seizure type, and hazards

1. Gupta M, Tripp J. Phenytoin. 2023. In: *StatPearls*. StatPearls Publishing. NBK482444
2. Maan JS, Duong TvH, Saadabadi A. Carbamazepine. 2023. In: *StatPearls*. StatPearls Publishing. NBK507852
3. Betchel NT, Fariba KA, Saadabadi A. Lamotrigine. 2023. In: *StatPearls*. StatPearls Publishing. NBK470442
4. Hanrahan B, Carson RP. Ethosuximide. 2024. In: *StatPearls*. StatPearls Publishing. NBK544244
5. Kumar A, Maini K, Kadian R. Levetiracetam. 2023. In: *StatPearls*. StatPearls Publishing. NBK499890
6. Singh R, Carson RP. Vigabatrin. 2024. In: *StatPearls*. StatPearls Publishing. NBK557579
7. Rahman M, Awosika AO, Nguyen H. Valproic Acid. 2024. In: *StatPearls*. StatPearls Publishing. NBK559112
8. Fariba KA, Saadabadi A. Topiramate. 2024. In: *StatPearls*. StatPearls Publishing. NBK554530

Immunotherapy for multiple sclerosis, NMOSD, and MOGAD

1. Babaesfahani A, Patel P, Bajaj T. Glatiramer. 2024. In: *StatPearls*. StatPearls Publishing. NBK541007
2. Lobaina M, Shanina E. Fingolimod. 2024. In: *StatPearls*. StatPearls Publishing. NBK604198
3. Babaesfahani A, Khanna NR, Patel P, Kuns B. Natalizumab. 2024. In: *StatPearls*. StatPearls Publishing. NBK534201
4. Flynn JP, Gerriets V. Ocrelizumab. 2023. In: *StatPearls*. StatPearls Publishing. NBK547750
5. Hanif N, Anwer F. Rituximab. 2024. In: *StatPearls*. StatPearls Publishing. NBK564374
6. Rocha Cabrero F, Morrison EH. Cladribine. 2023. In: *StatPearls*. StatPearls Publishing. NBK545307
7. Preuss CV, Anjum F. Tocilizumab. 2024. In: *StatPearls*. StatPearls Publishing. NBK570644

Drugs for spasticity, bladder dysfunction, orthostatic hypotension, and sleep behavior

1. Ghanavatian S, Derian A. Baclofen. 2024. In: *StatPearls*. StatPearls Publishing. NBK580550
2. Padda IS, Tadi P. Botulinum Toxin. 2023. In: *StatPearls*. StatPearls Publishing. NBK557387
3. Dwyer J, Tafuri SM, LaGrange CA. Oxybutynin. 2023. In: *StatPearls*. StatPearls Publishing. NBK499985
4. Rahman M, Anjum F. Fludrocortisone. 2024. In: *StatPearls*. StatPearls Publishing. NBK564331

Drugs for dementia, myasthenia, and mood in neurological disease

1. Kumar A, Gupta V, Sharma S. Donepezil. 2023. In: *StatPearls*. StatPearls Publishing. NBK513257
2. Kuns B, Rosani A, Patel P, Varghese D. Memantine. 2024. In: *StatPearls*. StatPearls Publishing. NBK500025

Drugs for Parkinson disease, chorea, and tremor

1. Gandhi KR, Saadabadi A. Levodopa (L-Dopa). 2023. In: *StatPearls*. StatPearls Publishing. NBK482140
2. Singh R, Parmar M. Pramipexole. 2023. In: *StatPearls*. StatPearls Publishing. NBK557539
3. Cusick E, Gupta V. Pimavanserin. 2023. In: *StatPearls*. StatPearls Publishing. NBK557712

Analgesics and anesthetics acting on the pain pathway

1. Gerriets V, Patel P, Nappe TM. Acetaminophen. 2024. In: *StatPearls*. StatPearls Publishing. NBK441917
2. Thour A, Alag P, Patel P, Marwaha R. Amitriptyline. 2026. In: *StatPearls*. StatPearls Publishing. NBK537225
3. Rosenbaum SB, Gupta V, Patel P, Palacios JL. Ketamine. 2024. In: *StatPearls*. StatPearls Publishing. NBK470357
4. Beecham GB, Patel P. Lidocaine. 2026. In: *StatPearls*. StatPearls Publishing. NBK539881
5. Folino TB, Muco E, Safadi AO, Parks LJ. Propofol. 2023. In: *StatPearls*. StatPearls Publishing. NBK430884

Drugs for glaucoma, macular degeneration, and conjunctivitis, and those that cause angle closure

1. Tripathy K, Patel P, Geetha R. Latanoprost. 2024. In: *StatPearls*. StatPearls Publishing. NBK540978
2. Barnes J, Moshirfar M. Timolol. 2024. In: *StatPearls*. StatPearls Publishing. NBK545176
3. Aslam S, Gupta V. Carbonic Anhydrase Inhibitors. 2023. In: *StatPearls*. StatPearls Publishing. NBK557736

Drugs for vertigo and Ménière disease

1. Houston BT, Patel P, Chowdhury YS. Meclizine. 2024. In: *StatPearls*. StatPearls Publishing. NBK560645

Drugs as causes of neurological disease

1. Lovik K, Murr NI. Seizure. 2026. In: *StatPearls*. StatPearls Publishing. NBK430765
2. Shrimanker I, Tadi P, Schoo C, Sánchez-Manso JC. Parkinsonism. 2024. In: *StatPearls*. StatPearls Publishing. NBK542224
3. Anastasiadou S, Al Khalili Y. Hearing Loss. 2023. In: *StatPearls*. StatPearls Publishing. NBK542323
4. Shumway CL, Patel BC, Tripathy K, De Jesus O. Neuromyelitis Optica Spectrum Disorder (NMOSD). 2024. In: *StatPearls*. StatPearls Publishing. NBK572108